

Representation Frontiers for the Fabius–Rvachev System

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Provenance

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- fabius_rvachev_report_package/fabius_rvachev_representations.tex
SHA-256 66480cdb437c0032823729c520ed18d9fd11c1b4a71f2a2d319155768434dc53
- Fabius_Rvachev_Polyphase_Representation_Report/fabius_rvachev_polyphase_representations.tex
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- Fabius_Rvachev_Thue_Morse_Representation_Frontiers/fabius_rvachev_thue_morse_representation_frontiers.tex
SHA-256 7485978f335c5b376dd382b738e0d8e42c15cbb9784b374e264db6f264ea29a0
- rvachev_fabius_representations_2026/rvachev-fabius-representations.tex
SHA-256 61f8453e78cceab6a62503c6d74e93db373afa1b8a42ac79dd794f393d1a7148
- Fabius_Rvachev_Multiresolution_Representations/Fabius_Rvachev_Multiresolution_Representations.tex
SHA-256 832c790e8bd247ea1569c7ef1ac09017a9c083d7c8f5ed17d57035e2aa82175c

Parts I–III are the first consolidation wave; Parts IV–VIII are the five second-wave drafts (the last five items), first merged into the companion volume *Representation Frontiers: The Second Wave* and folded into this volume on 28 August 2026. The fold restored per-part arabic section numbering (the standalone second-wave volume let \appendix lettering run across part boundaries, so its Parts II–V rendered with colliding appendix letters), restored the members’ full part titles, and deduplicated colliding macro definitions; all edits are marked % ed. : in the source.

Part I

Resolvent, Continued-Fraction, and Transform Representations of the Fabius–Rvachev System

Abstract

This report audits the LaTeX corpus in the repository subtree `Analysis/FabiusFunction/docs`, including its living expositions, consolidated frontier volume, active draft families, archive lineages, and paper transcriptions [5]. The repository already contains an unusually broad theory of the Fabius function, Rvachev’s up-function, their dyadic values, Thue–Morse expansions, Fourier products, moment and cumulant algebra, inverse asymptotics, and spectral decay. The objective here is therefore not to reproduce that material, but to organize it as a representation atlas and then add constructions that are absent from the audited corpus.

The principal new object is the Cauchy–Stieltjes transform

$$R(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} dx.$$

We derive its hyperbolic-product Laplace representation, its one-sided Fourier–Laplace representation, an exact logarithmic fixed-point identity, the functional-differential law

$$R'(z) = 2(R(2z + 1) - R(2z - 1)),$$

and an all-orders Thue–Morse derivative tree. Finite dyadic convolution laws yield an explicit 2^N -term logarithmic formula for R_N , whose boundary jump is exactly the finite Thue–Morse spline for the density. This gives a new “logarithm-to-spline” bridge.

We also construct the Jacobi continued fraction of R , equivalently the Stieltjes continued fraction of the ordinary even-moment generating function. Its coefficients are positive rational numbers; the first forty are computed exactly. The first coefficients are

$$\beta_1 = \frac{1}{9}, \quad \beta_2 = \frac{32}{225}, \quad \beta_3 = \frac{619}{3675}, \quad \beta_4 = \frac{835232}{4640643}.$$

The Denisov–Rakhmanov theorem proves $\beta_n \rightarrow 1/4$, while exact computation through $n = 40$ supports strict monotonicity and a logarithmically corrected convergence law. Further new representations include a trace-class Fredholm determinant for the Fourier image, three equivalent meromorphic logarithmic-derivative expansions, a digamma/Hurwitz- zeta hierarchy, exact one-sided Laplace and Fourier transforms of up and F , and pushforward transform identities for the inverse Fabius function. A commented Python program, exact CSV data, numerical checks, and figures accompany the report.

1 Executive summary and novelty boundary

1.1 What was audited

The repository contains several kinds of LaTeX source that must not be conflated:

1. a living primary synthesis of the formalized and analytic theory;
2. a large consolidated non-formalized frontier volume;
3. current draft reports on arithmetic rays, half-integer spectra, self-sampling, inverse asymptotics, Newton/exponent-sequence hierarchies, Fourier decay, and related topics;
4. archived earlier lineages, retained because some contain historically distinctive formulas even when superseded;
5. seven paper-source transcriptions or corrected transcriptions.

The repository’s own pinned audit ledger records 67 LaTeX paths at the following commit:

32d6d36c51d803289e6d6a0dc0c37753766eba47.

The live tree inspected on 27 August 2026 contains additional current drafts, notably the self-sampling, inverse, and Newton/Rvachev reports. Appendix B records the pinned ledger and the live additions used in this study.

1.2 Non-duplication protocol

Every formula in the atlas is tagged conceptually as follows.

Tag	meaning
K	already explicit in the audited repository or classical source; included for orientation
D	an immediate transform or recombination of known identities; useful but not claimed as new
N	apparently absent from the audited corpus; proved or conjectured here

The word “new” throughout this report means *new relative to the audited repository*. No claim of global bibliographic priority is made without a separate literature search. In particular, the current frontier report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as an open direction. The present report constructs that fraction and develops its consequences.

1.3 Main new results

The most substantial additions are:

1. the Cauchy transform R and its moment, Laplace-product, Fourier–Laplace, boundary-value, and logarithmic fixed-point representations;
2. the exact resolvent equation and its all-orders Thue–Morse derivative tree;
3. a finite-level 2^N -term logarithmic formula converging to R , paired by boundary jumps with the finite Thue–Morse spline density;

4. the Jacobi J -fraction of R , the contracted Stieltjes S -fraction of the even moments, exact rational coefficients through order 40, Gaussian quadrature convergents, and the theorem $\beta_n \rightarrow 1/4$;
5. a Fredholm determinant realization of the Fourier image and a trace-resolvent interpretation of its logarithmic derivative;
6. cotangent, tangent, valuation-weighted partial-fraction, digamma, and Hurwitz-zeta representations of the logarithmic derivatives of the Fourier image;
7. exact one-sided Laplace and Fourier transforms of up , exact transforms of F on $[0, 1]$, and a master pushforward identity for F^{-1} .

2 Normalization and structural dictionary

2.1 The functions

Let $F : [0, 1] \rightarrow [0, 1]$ denote the bounded Fabius function [4, 2], extended by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$. On the unit interval it satisfies

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x), \quad (1)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (2)$$

The Rvachev function is written

$$\rho(x) := \text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (3)$$

Then ρ is even, strictly positive on $(-1, 1)$, normalized by $\int \rho = 1$, flat at ± 1 , and satisfies

$$\rho'(x) = 2(\rho(2x+1) - \rho(2x-1)). \quad (4)$$

2.2 Fourier convention and basic transforms

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} \, dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i \xi x} \, d\xi. \quad (5)$$

Set

$$\Phi(\xi) := \widehat{\rho}(\xi), \quad \text{sinc}_{\pi}(z) := \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_{\pi}(0) = 1. \quad (6)$$

The classical products are

$$\Phi(z) = \prod_{n=0}^{\infty} \text{sinc}_{\pi}\left(\frac{z}{2^n}\right) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi z}{2^m}\right)^m. \quad (7)$$

The scaling law is

$$\Phi(z) = \text{sinc}_{\pi}(z) \Phi(z/2). \quad (8)$$

2.3 The probability model

Let $U_1, U_2, \dots$ be independent random variables, uniform on $[-1, 1]$, and set

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (9)$$

Then X has density ρ . Equivalently,

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (10)$$

where $U \sim \text{Unif}[-1, 1]$ and X' is an independent copy of X . The moment generating function is

$$M(t) := \mathbb{E}e^{tX} = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} = \Phi\left(\frac{it}{2\pi}\right). \quad (11)$$

The even moments are

$$\mu_{2n} := \mathbb{E}X^{2n} = \int_{-1}^1 x^{2n} \rho(x) \, dx, \quad \mu_{2n+1} = 0. \quad (12)$$

Finally, if

$$Y = \frac{X + 1}{2}, \quad (13)$$

then F is the distribution function of Y , and

$$F'(x) = 2\rho(2x - 1), \quad 0 < x < 1. \quad (14)$$

3 Audited representation atlas

This section compresses the principal representation families already present in the repository and places the new formulas beside them. Proofs of the new entries begin in Section 4.

3.1 Real-space and probability representations of ρ

Infinite convolution (K). Since $2^{-k}U_k$ is uniform on $[-2^{-k}, 2^{-k}]$,

$$\rho = \bigstar_{k=1}^{\infty} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]} \right). \quad (15)$$

This is the real-space counterpart of the sinc product.

Infinite cube integral (D). For every bounded continuous test function g ,

$$\int_{-1}^1 g(x) \rho(x) \, dx = \int_{[-1, 1]^{\mathbb{N}}} g\left(\sum_{k \geq 1} 2^{-k} u_k\right) \prod_{k \geq 1} \frac{du_k}{2}. \quad (16)$$

Formally, this may be written as a Dirac-delta integral for $\rho(x)$.

Finite Thue–Morse splines (K). Put

$$X_N = \sum_{k=1}^N 2^{-k} U_k, \quad a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \quad \varepsilon_j = (-1)^{\text{wt}_2(j)}. \quad (17)$$

The density ρ_N of X_N is

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (x + a_{N,j})_+^{N-1}. \quad (18)$$

For each fixed derivative order, $\rho_N \rightarrow \rho$ uniformly after N is sufficiently large relative to that order. The repository develops several centered and one-sided variants with explicit error control.

Fourier inversion and cosine form (K). Because Φ is real and even,

$$\rho(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (19)$$

$$\rho(x) = 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (20)$$

The audited corpus also contains Poisson-summation formulas, Legendre expansions, periodized cosine series, dyadic spline limits, and endpoint transseries. They are not rederived here.

3.2 Product, zero, gamma, and exponential representations of Φ

Canonical zero product (K). Let $\nu_2(n)$ be the exponent of 2 in n . The nonzero integer n is a zero of multiplicity $1 + \nu_2(|n|)$, and

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}. \quad (21)$$

Gamma product (D). Euler’s reflection identity gives

$$\Phi(z) = \prod_{k=0}^{\infty} \frac{1}{\Gamma(1 + z/2^k) \Gamma(1 - z/2^k)}. \quad (22)$$

This is the natural entry point for the Barnes and multiple-gamma compressions studied in current repository drafts.

Formal version and boundary. At formal source checkpoint 71ab6f6728fceb753c88d8b0573077a59acf2682, `Fabius.shiftedReciprocalGamma_mul_neg` proves the total pointwise reciprocal reflection formula and `Fabius.dyadicReciprocalGamma_mul_neg` multiplies it over the dyadic scales to give (22); the exact Gamma-denominator form is `Fabius.rvachevFourierProduct_eq_one_div_dyadicGamma_mul`. These declarations live in `GeometricReciprocalGamma.lean`. They do not formalize the Barnes or multiple-gamma compressions mentioned in the following sentence.

Spectral-zeta exponential (K). For $|z| < 1$,

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 2^{-2m})} z^{2m}. \quad (23)$$

The coefficient sequence is the spectral zeta function

$$Z_{\Phi}(s) = \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (24)$$

Fredholm determinant (N). Section 9 upgrades (21) to

$$\Phi(z) = \det(I - z^2 \mathcal{K}), \quad (25)$$

for an explicit positive trace-class diagonal operator $\mathcal{K}$. This makes (24) literally the trace zeta function of $\mathcal{K}$.

3.3 Moment, cumulant, and Bell-polynomial representations

The moment generating function has the convergent Taylor series

$$M(t) = \sum_{n=0}^{\infty} \mu_{2n} \frac{t^{2n}}{(2n)!}. \quad (26)$$

Its logarithm is

$$\log M(t) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(4^m - 1)} t^{2m}, \quad (27)$$

so all moments are complete Bell polynomials in explicit Bernoulli/Mersenne cumulants. These formulas, together with reciprocal Bell expansions for $1/\Phi$, are developed extensively in the repository.

3.4 Transform representations of F and F^{-1}

The CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(\xi) \frac{\sin(2\pi(2x-1)\xi)}{\xi} d\xi, \quad 0 < x < 1. \quad (28)$$

Section 7 adds closed Laplace and Fourier transforms of F as a function, as well as transforms of its derivative measure and its inverse.

3.5 A compact atlas table

Table 1: Representation atlas. Tags refer to the audited-corpus novelty boundary.

Object	Representation family	Tag	Location
ρ	fold of F , compact-support DDE, infinite convolution, random dyadic sum	K	§2–3
ρ	finite Thue–Morse truncated-power splines	K	(18)
ρ	Fourier inversion, cosine integral, Poisson and Legendre series	K	(19)
ρ	boundary jump of a finite logarithmic resolvent sum	N	Thm. 6.1
Φ	sinc product and weighted cosine product	K	(7)
Φ	canonical integer-zero product	K	(21)
Φ	gamma/Barnes products	K/D	(22)
Φ	zeta-exponential series	K	(23)
Φ	trace-class Fredholm determinant	N	Thm. 9.2

Object	Representation family	Tag	Location
Φ'/Φ	cotangent dyadic sum	N	(63)
Φ'/Φ	weighted tangent dyadic sum	N	(64)
Φ'/Φ	valuation-weighted partial fractions	N	(65)
$(\log \Phi)^{(r)}$	digamma and Hurwitz-zeta hierarchy	N	Thm. 8.2
M	hyperbolic-sinc product, moment series, Bell cumulants	K	(11)–(27)
M	positive Fredholm determinant	N	(74)
R	moment Laurent series	N	(31)
R	Laplace transform of the hyperbolic product	N	(32)
R	one-sided Fourier–Laplace integral	N	(33)
R	logarithmic fixed-point expectation	N	(36)
R	dyadic functional-differential equation	N	(37)
$R^{(n)}$	finite Thue–Morse affine orbit of R	N	(38)
R	finite 2^N -logarithm limit	N	(44)
R	Jacobi continued fraction	N	(85)
R	Gaussian-quadrature partial fractions	N	(94)
$\sum \mu_{2n} t^n$	Stieltjes continued fraction and product integral	N	(87)
F	CDF sine-product integral	D	(28)
F	exact Laplace transform on $[0, 1]$	N	(48)
F	exact Fourier transform on $[0, 1]$	N	(49)
$\rho _{[0,1]}$	exact one-sided Laplace/Fourier/sine transforms	N	Thm. 7.2
F^{-1}	pushforward expectation, Laplace, Fourier, and Cauchy transforms	N	Thm. 7.3

4 The Cauchy–Stieltjes transform

Definition 4.1. For $z \in \mathbb{C} \setminus [-1, 1]$, define

$$R(z) = \int_{-1}^1 \frac{\rho(x)}{z - x} dx = \mathbb{E} \frac{1}{z - X}. \quad (29)$$

Remark 4.2 (Formal Cauchy–Stieltjes core). `CauchyTransform.lean` defines the report-oriented transform as `rvachevCauchyTransform`; `Mathlib`’s `resolventTransform` uses $(x - z)^{-1}$, so the formal definition is its negative. The declarations `rvachevCauchyTransform_apply` and `rvachevCauchyTransform_eq_integral_rvachevUp` give respectively the measure and density forms of (29), while `analyticOn_rvachevCauchyTransform` and `hasDerivAt_rvachevCauchyTransform` prove holomorphy off the complexified support interval and the measure-form first derivative

$$R'(z) = - \int (z - x)^{-2} d\mu_{\text{up}}(x).$$

Expanding $\mu_{\text{up}} = \rho(x) dx$ recovers the corresponding density formula. The total Lean definition follows the Bochner-integral convention at points where the kernel is not integrable; the analytic Cauchy-transform interpretation is asserted only off the cut.

The companion `CauchyCDF.lean` proves the atom-exact compact-support formula

$$\int \frac{d\mu(x)}{z - x} = \mu(\mathbb{R}) \frac{1}{z - b} - \int_a^b \frac{\mu((-\infty, t])}{(z - t)^2} dt.$$

Its generic declaration `integral_inv_sub_eq_mass_smul_sub_intervalIntegral_measureReal_Iic` assumes only a finite measure, $a \leq b$, almost-everywhere support on $[a, b]$, and z outside the complexified interval. The declaration `integral_inv_sub_eq_sub_intervalInte`

`gral_cdf` gives the probability normalization, and `fabiusStieltjesTransform_eq_inv_sub_one_sub_intervalIntegral_fabiusReal` specializes it on `fabiusStieltjesDomain` to

$$S(z) = \frac{1}{z-1} - \int_0^1 \frac{F(t)}{(z-t)^2} dt.$$

Closed lower rays retain atoms exactly. These declarations do not supply the higher resolvent-power, logarithmic, inverse-quantile, or boundary formulas below.

The transform R is odd because ρ is even. It is an anti-Herglotz function: when $\operatorname{Im} z > 0$,

$$\operatorname{Im} R(z) = -\operatorname{Im} z \int_{-1}^1 \frac{\rho(x)}{|z-x|^2} dx < 0. \quad (30)$$

In particular, R has no zeros off the cut. For real $z > 1$, $R(z) > 0$; for $z < -1$, $R(z) < 0$.

Theorem 4.3 (Four complementary representations of R). *The following formulas hold.*

(i) For $|z| > 1$,

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_{2n}}{z^{2n+1}}. \quad (31)$$

(ii) For $\operatorname{Re} z > 1$,

$$R(z) = \int_0^{\infty} e^{-zt} \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} dt. \quad (32)$$

(iii) For $\operatorname{Im} z > 0$,

$$R(z) = -2\pi i \int_0^{\infty} e^{2\pi i z \xi} \Phi(\xi) d\xi. \quad (33)$$

For $\operatorname{Im} z < 0$, take the conjugate counterpart.

(iv) For $-1 < x < 1$, the non-tangential boundary values satisfy

$$\rho(x) = -\frac{1}{\pi} \lim_{y \downarrow 0} \operatorname{Im} R(x + iy), \quad (34)$$

$$\operatorname{PV} \int_{-1}^1 \frac{\rho(t)}{x-t} dt = \lim_{y \downarrow 0} \operatorname{Re} R(x + iy) = 2\pi \int_0^{\infty} \Phi(\xi) \sin(2\pi x \xi) d\xi. \quad (35)$$

Proof. For $|z| > 1$, expand $(z-X)^{-1} = z^{-1} \sum_{m \geq 0} (X/z)^m$, use $|X| \leq 1$, and then symmetry. This gives (31).

For $\operatorname{Re} z > 1$,

$$\frac{1}{z-X} = \int_0^{\infty} e^{-(z-X)t} dt.$$

Fubini's theorem is justified because $X \leq 1$. Taking expectation and using (11) yields (32).

For $\operatorname{Im} z > 0$,

$$\frac{1}{z-x} = -2\pi i \int_0^{\infty} e^{2\pi i (z-x)\xi} d\xi.$$

The exponential damping from $\operatorname{Im} z > 0$, together with the rapid decay of Φ , permits Fubini and gives (33). Finally, Plemelj's boundary formula gives (34); taking real parts of (33) gives (35). $\square$

Remark 4.4. The four formulas emphasize different structures. The Laurent series exposes exact rational moments; the Laplace integral exposes the hyperbolic product; the Fourier–Laplace integral exposes spectral decay; and the boundary values recover both ρ and its Hilbert transform. Thus R is a single analytic object joining the moment, product, and inversion theories.

5 A dyadic functional equation for the resolvent

5.1 Logarithmic fixed-point identity

Theorem 5.1 (Logarithmic fixed point). *For real $z > 1$,*

$$R(z) = \int_{-1}^1 \rho(y) \log\left(\frac{2z+1-y}{2z-1-y}\right) dy. \quad (36)$$

The identity extends by analytic continuation to any simply connected subdomain of $\mathbb{C} \setminus [-1, 1]$, with the logarithm chosen by continuation from large positive z .

Proof. Use the fixed point (10) and condition on X' :

$$\begin{aligned} R(z) &= \mathbb{E} \left[\int_{-1/2}^{1/2} \frac{du}{z-u-X'/2} \right] \\ &= \mathbb{E} \log \left(\frac{z+1/2-X'/2}{z-1/2-X'/2} \right), \end{aligned}$$

which is (36) after multiplying numerator and denominator by 2. $\square$

5.2 Functional-differential equation

Theorem 5.2 (Resolvent equation). *For every $z \in \mathbb{C} \setminus [-1, 1]$,*

$$R'(z) = 2(R(2z+1) - R(2z-1)). \quad (37)$$

Proof. Differentiate (36) first for $z > 1$:

$$\begin{aligned} R'(z) &= \mathbb{E} \left(\frac{2}{2z+1-X'} - \frac{2}{2z-1-X'} \right) \\ &= 2(R(2z+1) - R(2z-1)). \end{aligned}$$

Both sides are holomorphic on $\mathbb{C} \setminus [-1, 1]$, so the identity theorem gives the stated continuation. $\square$

This is the transform-side analogue of the Rvachev equation (4). It is not merely a Fourier rephrasing: it acts on the spectral parameter of a Cauchy transform and therefore interacts directly with moments, Padé approximation, and orthogonal polynomials.

5.3 All-order Thue–Morse derivative tree

Theorem 5.3 (Thue–Morse derivative orbit). *For every integer $n \geq 0$,*

$$R^{(n)}(z) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (38)$$

Consequently,

$$\int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx = \frac{(-1)^n 2^{n(n+1)/2}}{n!} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (39)$$

Proof. The case $n = 0$ is trivial and $n = 1$ is (37). Assume the formula at order n . Differentiating contributes a factor 2^n , and then applying (37) contributes another factor 2. The exponent therefore changes by $n + 1$, from $n(n + 1)/2$ to $(n + 1)(n + 2)/2$.

For the shifts, write $a_{n,j} = 2^n - 1 - 2j$. The two children are

$$2a_{n,j} + 1 = 2^{n+1} - 1 - 2(2j), \quad 2a_{n,j} - 1 = 2^{n+1} - 1 - 2(2j + 1).$$

The signs obey $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$, exactly matching the subtraction in (37). This proves (38). Formula (39) follows from

$$R^{(n)}(z) = (-1)^n n! \int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx.$$

□

Proposition 5.4 (Generic affine-difference orbit; formalized). *Over a commutative ring, let*

$$(\Delta_{b,c}f)(z) = f(bz + c) - f(bz - c), \quad \xi_{n,T}(z) = b^n z + c \sum_{k=0}^{n-1} b^k - 2c \sum_{k \in T} b^k.$$

For every function with values in an additive commutative group,

$$(\Delta_{b,c}^n f)(z) = \sum_{T \subseteq \{0, \dots, n-1\}} (-1)^{|T|} f(\xi_{n,T}(z)). \quad (40)$$

Moreover, suppose the ring is a nontrivially normed field, let S be open and invariant under $z \mapsto bz \pm c$, and let f take values in a normed vector space. If $f'(z) = a \Delta_{b,c}f(z)$ for $z \in S$, then no separate smoothness assumption is needed: for every $n \geq 0$ and $z \in S$,

$$f^{(n)}(z) = a^n b^{\binom{n}{2}} (\Delta_{b,c}^n f)(z). \quad (41)$$

Constant scalar multiplication commutes with every iterate. Taking S to be the whole field gives the global form, while $b = 2, c = 1$ gives

$$(\Delta_{2,1}^n f)(z) = \sum_{j=0}^{2^n-1} \varepsilon_j f(2^n z + 2^n - 1 - 2j). \quad (42)$$

Lean proof and boundary. The following declarations in `AffineDifferenceOrbit.lean` prove the statements above:

```
affineDifference_iterate_const_smul
affineDifference_iterate_apply
iteratedDeriv_eq_affineDifference_iterate_on
iteratedDeriv_eq_affineDifference_iterate
affineDifference_iterate_two_one_apply.
```

The first formula is a Boolean-cube induction. The second differentiates that finite operator orbit on the invariant open set; its one-step derivative hypothesis bootstraps all regularity needed. The last formula reindexes subsets by binary support, so its signs are precisely the Thue–Morse signs.

Thus the conditional passage from (37) to (38) is formalized in greater generality. On the density side, `iteratedDeriv_rvachevUp_eq_thueMorse_sum` applies the same engine to the already formalized differential equation for ρ , globally on the real line. The transform definition, density formula, natural-domain holomorphy, first Cauchy-kernel derivative, and affine identity $S(z) = 2R(2z - 1)$ are formalized in `CauchyTransform.lean`.

The five public theorems in `CauchyRenormalization.lean` now close the one-step and derivative-orbit obligations. The declarations `mapsTo_rvachevCauchyDomain_two_mul_add_one` and `mapsTo_rvachevCauchyDomain_two_mul_sub_one` prove that both dyadic branches preserve the slit domain. For every bounded Fabius solution and every $z \in \mathbb{C} \setminus [-1, 1]$, `hasDerivAt_rvachevCauchyTransform_affineDifference` and `deriv_rvachevCauchyTransform` prove (37) directly, without using the logarithmic fixed point. Finally, `iteratedDeriv_rvachevCauchyTransform_eq_thueMorse_sum` instantiates the generic open-domain engine and proves exactly the derivative formula (38) for every $n \in \mathbb{N}$. The corresponding unit-interval relation is now a direct named theorem, not merely a corollary through the affine bridge: `hasDerivAt_fabiusStieltjesTransform_refinement` proves $S'(z) = 4(S(2z) - S(2z - 1))$ throughout the unit slit domain. The bridge itself has an all-power version `fabiusStieltjesPower_eq_two_pow_mul_rvachevCauchyPower`, and `fabiusStieltjesPower_succ` and `rvachevCauchyPower_succ` give the two adjacent-order recurrences throughout their complex slit domains.

The real $z > 1$ instance of the logarithmic identity (36) is a direct theorem of `StieltjesLogFixedPoint.lean`, namely `integral_inv_sub_eq_integral_log`; it is not a complex-logarithm or branch continuation theorem. `StieltjesResolventHierarchy.lean` gives the real positive integer hierarchy, while `StieltjesGeneralizedOrder.lean` lowers real $\alpha > 1$ at real $z > 1$. The real and complex exterior moment/Laurent series, finite-height Herglotz–Poisson formulas, integrated interval Stieltjes–Perron inversion, and initial exact Jacobi data are likewise direct named results in the corresponding `Stieltjes*.lean` and `OrthogonalPolynomialJacobi.lean` modules. The later four-module tranche—`PolynomialMomentGramDeterminant.lean`, `GramStieltjesNaturality.lean`, `RvachevRationalJacobi.lean`, and `FabiusLegendreHankelDeterminant.lean`—supplies polynomial-basis determinant transport, scalar naturality, the exact all-degree rational centered Jacobi system, and its Legendre Gram specialization. Its exhaustive 41-name inventory and its zero-based-index convention $n \leftrightarrow \beta_{n+1}$ are recorded at the finite-algebra boundary below. Continued-fraction identification and asymptotics are not part of that tranche.

The bounded-measure Laurent status is stronger than those two named specializations. `MeasureCauchyMomentLaurent.lean` contains exactly one public definition and fifteen public theorems. For every finite real measure, every scalar field $\mathbb{K}$ through Lean’s $\{K\} \text{ [RCLike } K]$ interface, arbitrary center $c \in \mathbb{K}$, radius $R \geq 0$, and almost-everywhere relative-ball bound $\|(x : \mathbb{K}) - c\| \leq R$, it defines

$$m_n^{(c)} = \int ((x : \mathbb{K}) - c)^n d\mu(x)$$

and, when $R < \|z - c\|$, proves the exact finite identity

$$\int \frac{d\mu(x)}{z - x} = \sum_{k < N} \frac{m_k^{(c)}}{(z - c)^{k+1}} + (z - c)^{-N} \int \frac{((x : \mathbb{K}) - c)^N}{z - x} d\mu(x).$$

The same module proves measurability and the required kernel/remainder integrability, the mass-scaled moment estimate, the sharp uniform remainder bound

$$\left\| \int \frac{d\mu(x)}{z - x} - \sum_{k < N} \frac{m_k^{(c)}}{(z - c)^{k+1}} \right\| \leq \mu(\mathbb{R})(\|z - c\| - R)^{-1} \left(\frac{R}{\|z - c\|} \right)^N,$$

and the resulting `Summable`, `tsum`, and `HasSum` forms; complex wrappers connect the direct integral to `measureCauchyTransform`. The earlier up-measure statements in `StieltjesMomentLaurent.lean` and `StieltjesCauchyTransform.lean` are the $c = 0, R = 1$ instances. The compatibility declarations `ae_norm_sub_zero_le_one_rvachevMeasure` and `measureCauchyMoment_rvachevMeasure_zero` supply respectively the relative-ball hypothesis and the coefficient identification with `upMoment`. None of these declarations proves the coefficient recursion or the compact moment-determinacy/uniqueness argument used later in this report.

Still open are the complex logarithmic continuation, complex order, pointwise or nontangential Sokhotski–Plemelj/Hilbert-principal-value formulas, the full J-fraction/Padé package, and the separate conversion of the all-order derivative orbit into the explicit higher-kernel identity (39). $\square$

Taking boundary imaginary parts in (38) recovers the familiar iterated derivative formula for ρ . The new point is that the same finite Thue–Morse orbit controls every higher Cauchy kernel, not only the density derivatives.

5.4 Uniqueness in the Stieltjes class

Proposition 5.5. *Among Cauchy transforms of probability measures supported on $[-1, 1]$, R is the unique solution of (37) satisfying*

$$R(z) = \frac{1}{z} + \mathcal{O}(z^{-3}) \quad (z \rightarrow \infty). \quad (43)$$

Proof. Let $G(z) = \sum_{r \geq 0} m_r z^{-r-1}$ be the Laurent expansion at infinity of any such Cauchy transform. The normalization gives $m_0 = 1$ and $m_1 = 0$. Substitute this expansion into (37). In the coefficient of z^{-r-2} , the contribution containing m_r is

$$-(r+1)m_r + \frac{r+1}{2^r}m_r = -(r+1)(1-2^{-r})m_r,$$

while every other contribution contains only $m_0, \dots, m_{r-2}$. Since the displayed factor is nonzero for $r \geq 1$, the differential equation determines all moments recursively. The transform R is one solution, so G and R have identical moments. Probability measures on the compact interval $[-1, 1]$ are determinate by their moments (equivalently, polynomials are dense in $C[-1, 1]$); hence the two measures and their Cauchy transforms coincide. $\square$

6 Finite logarithmic resolvents and the spline boundary jump

The finite random sum X_N has two exact representations: its density is a Thue–Morse spline, while its Cauchy transform is a Thue–Morse sum of logarithms. The second formula appears to be absent from the audited corpus.

Theorem 6.1 (Finite logarithmic formula). *Let*

$$R_N(z) = \mathbb{E} \frac{1}{z - X_N}, \quad X_N = \sum_{k=1}^N 2^{-k} U_k.$$

For $z \notin [-1 + 2^{-N}, 1 - 2^{-N}]$, with logarithms continued consistently from $z = +\infty$,

$$R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}). \quad (44)$$

Moreover, $R_N \rightarrow R$ locally uniformly on $\mathbb{C} \setminus [-1, 1]$, so

$$R(z) = \lim_{N \rightarrow \infty} \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}). \quad (45)$$

Finally, the upper boundary jump of (44) is exactly the spline (18):

$$\rho_N(x) = -\frac{1}{\pi} \text{Im } R_N(x + i0). \quad (46)$$

Proof. For $a > 0$, convolution with the uniform density on $[-a, a]$ sends a Cauchy transform $G'(z)$ to

$$\frac{1}{2a} \int_{-a}^a G'(z - u) \, du = \frac{G(z + a) - G(z - a)}{2a}. \quad (47)$$

Start from the point mass at zero, whose Cauchy transform is $1/z$. An N -fold antiderivative is

$$H_N(z) = \frac{z^{N-1} \operatorname{Log} z}{(N-1)!}; \quad H_N^{(N)}(z) = \frac{1}{z}.$$

Applying (47) successively with $a_k = 2^{-k}$ gives

$$R_N(z) = \frac{1}{\prod_{k=1}^N 2a_k} \sum_{\sigma \in \{\pm 1\}^N} \left(\prod_{k=1}^N \sigma_k \right) H_N \left(z + \sum_{k=1}^N \sigma_k a_k \right).$$

Since

$$\frac{1}{\prod_{k=1}^N 2a_k} = 2^{N(N-1)/2},$$

and binary subsets identify the shifts and signs with $a_{N,j}$ and ε_j , this is (44).

The residual tail $X - X_N$ has support in $[-2^{-N}, 2^{-N}]$, so $X_N \rightarrow X$ almost surely and uniformly in support. The kernels $(z - x)^{-1}$ are uniformly continuous on compact subsets away from $[-1, 1]$; hence $R_N \rightarrow R$ locally uniformly.

For the boundary value, $\operatorname{Im} \operatorname{Log}(x + a + i0) = \pi$ when $x + a < 0$ and zero when $x + a > 0$. The alternating finite difference annihilates every polynomial of degree less than N , so the sum of the full powers $(x + a_{N,j})^{N-1}$ is zero. Replacing the negative-part sum by minus the positive-part sum yields exactly (18). This proves (46). $\square$

Remark 6.2 (Two complementary approximation geometries). The finite-convolution approximant R_N is a sum of logarithms and preserves the dyadic construction. The Jacobi approximants in Section 10 are rational functions and preserve moments. Both converge locally uniformly off the support, but their poles, branch cuts, and error mechanisms are completely different. This duality suggests hybrid algorithms: use finite convolution to resolve endpoint geometry and a short Jacobi fraction to compress the remaining moment error.

7 Exact transforms of F , one-sided up, and F^{-1}

7.1 The Laplace and Fourier transforms of F

Theorem 7.1 (Transform of the Fabius function). *For complex s , with the value at $s = 0$ understood by continuity,*

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} M(s/2) - e^{-s}}{s}. \quad (48)$$

Equivalently, for real or complex ξ ,

$$\int_0^1 F(x) e^{-2\pi i \xi x} \, dx = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (49)$$

At $\xi = 0$, the value is $1/2$.

Proof. The distributional Laplace transform of Y is

$$\int_0^1 e^{-sx} \, dF(x) = \mathbb{E}e^{-sY} = e^{-s/2} M(s/2). \quad (50)$$

Integration by parts gives

$$\int_0^1 e^{-sx} \, dF(x) = e^{-s} + s \int_0^1 e^{-sx} F(x) \, dx,$$

which is (48). Substitute $s = 2\pi i\xi$ and use $M(\pi i\xi) = \Phi(\xi/2)$. $\square$

Separating real and imaginary parts gives the explicit sine and cosine transforms

$$\int_0^1 F(x) \cos(2\pi\xi x) \, dx = \frac{\sin(2\pi\xi) - \Phi(\xi/2) \sin(\pi\xi)}{2\pi\xi}, \quad (51)$$

$$\int_0^1 F(x) \sin(2\pi\xi x) \, dx = \frac{\Phi(\xi/2) \cos(\pi\xi) - \cos(2\pi\xi)}{2\pi\xi}. \quad (52)$$

7.2 One-sided transforms of the up-function

On $[0, 1]$, $\rho(x) = F(1 - x) = 1 - F(x)$. Therefore the previous theorem immediately yields the following closed forms.

Theorem 7.2 (One-sided transform package). *For complex s , with the removable value at zero,*

$$\boxed{\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} M(s/2)}{s}.} \quad (53)$$

Consequently,

$$\boxed{\int_0^1 e^{-2\pi i\xi x} \rho(x) \, dx = \frac{1 - e^{-\pi i\xi} \Phi(\xi/2)}{2\pi i\xi}.} \quad (54)$$

The half-line cosine and sine transforms are

$$\int_0^1 \rho(x) \cos(2\pi\xi x) \, dx = \frac{1}{2} \Phi(\xi), \quad (55)$$

$$\int_0^1 \rho(x) \sin(2\pi\xi x) \, dx = \frac{1 - \Phi(\xi/2) \cos(\pi\xi)}{2\pi\xi}. \quad (56)$$

The sum of (53) at s and $-s$ recovers $M(s)$ precisely because

$$M(s) = \frac{2 \sinh(s/2)}{s} M(s/2), \quad (57)$$

which is the hyperbolic continuation of (8).

7.3 The inverse Fabius function as a transport map

Let $I = F^{-1}$ on $[0, 1]$. Since I pushes Lebesgue measure forward to the law of Y , every integral of a composition with I reduces to the up-law.

Theorem 7.3 (Inverse pushforward master identity). *For every Borel function g for which one side is integrable,*

$$\int_0^1 g(I(u)) \, du = \int_0^1 g(x) \, dF(x) = \int_{-1}^1 g\left(\frac{x+1}{2}\right) \rho(x) \, dx. \quad (58)$$

In particular,

$$\int_0^1 e^{-sI(u)} \, du = e^{-s/2} M(s/2), \quad (59)$$

$$\int_0^1 e^{-2\pi i \xi I(u)} \, du = e^{-\pi i \xi} \Phi(\xi/2), \quad (60)$$

$$\int_0^1 \frac{du}{z - I(u)} = 2R(2z - 1), \quad z \in \mathbb{C} \setminus [0, 1], \quad (61)$$

$$\int_0^1 I(u)^n \, du = 2^{-n} \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mu_{2r}. \quad (62)$$

Proof. If U is uniform on $[0, 1]$, then $I(U)$ has CDF F , hence the same law as Y . This proves (58). The remaining formulas follow by taking $g(x) = e^{-sx}$, $e^{-2\pi i \xi x}$, $(z - x)^{-1}$, and x^n , respectively, and then using (13) and symmetry of X . $\square$

Remark 7.4. Formula (61) transports the entire Jacobi-fraction theory of R to the inverse Fabius function. Thus a function whose pointwise inverse asymptotics involve Lambert W nevertheless has a globally simple Cauchy transform when sampled with uniform measure in its range variable.

8 Logarithmic derivatives of the Fourier image

8.1 Three meromorphic expansions

Theorem 8.1 (Cotangent, tangent, and zero-set formulas). *Away from the integer zero set of Φ ,*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} \left[\frac{\pi}{2^k} \cot\left(\frac{\pi z}{2^k}\right) - \frac{1}{z} \right], \quad (63)$$

$$= -\pi \sum_{m=1}^{\infty} \frac{m}{2^m} \tan\left(\frac{\pi z}{2^m}\right), \quad (64)$$

$$= -2z \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^2 - z^2}. \quad (65)$$

All three series converge locally normally on compact subsets avoiding the poles.

Proof. Differentiate the logarithm of each of the three products (7) and (21). For large k , the bracket in (63) and the term in (64) are $\mathcal{O}(4^{-k})$ uniformly on compact sets. The terms in (65) are bounded by a constant times $(1 + \nu_2(n))/n^2$, which is summable by (24) at $s = 2$. $\square$

The three forms have different numerical regimes. The tangent sum is usually the fastest away from its poles; the cotangent sum tracks the original sinc factors; and the zero-set sum exposes multiplicities and supports rigorous rational tail bounds.

8.2 Digamma and Hurwitz-zeta hierarchy

The valuation weight has the divisor decomposition

$$1 + \nu_2(n) = \sum_{k=0}^{\infty} \mathbf{1}_{2^k | n}. \quad (66)$$

This turns the zero sum into a dyadic superposition of classical special functions.

Theorem 8.2 (Digamma and Hurwitz-zeta representations). *Let $\psi = \Gamma'/\Gamma$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} 2^{-k} \left[\psi\left(1 - \frac{z}{2^k}\right) - \psi\left(1 + \frac{z}{2^k}\right) \right]. \quad (67)$$

For every integer $r \geq 2$,

$$\frac{d^r}{dz^r} \log \Phi(z) = (r-1)! \sum_{k=0}^{\infty} 2^{-kr} \left[-\zeta\left(r, 1 - \frac{z}{2^k}\right) + (-1)^{r-1} \zeta\left(r, 1 + \frac{z}{2^k}\right) \right]. \quad (68)$$

Proof. From (21), for $r \geq 1$, paired symmetrically when $r = 1$,

$$(\log \Phi)^{(r)}(z) = (r-1)! \sum_{n \geq 1} (1 + \nu_2(n)) \left[-\frac{1}{(n-z)^r} + \frac{(-1)^{r-1}}{(n+z)^r} \right]. \quad (69)$$

Insert (66), write $n = 2^k m$, and sum over $m \geq 1$. For $r \geq 2$, the inner sums are Hurwitz zeta functions. For $r = 1$, the convergent paired sum is the standard digamma difference. $\square$

Corollary 8.3 (Local jet data at nonzero zeros). *Let $N \neq 0$ be an integer and $d_N = 1 + \nu_2(|N|)$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \frac{d_N}{z - N} + H_N(z), \quad (70)$$

where H_N is holomorphic near N , and every derivative of H_N can be written as an absolutely convergent valuation-weighted Hurwitz-zeta sum. Thus the complete local Taylor jet of $\Phi(z)/(z - N)^{d_N}$ is algorithmically explicit.

9 A Fredholm determinant for the Fourier product

The canonical product can be realized as an ordinary Fredholm determinant, not merely as a formal spectral product.

Definition 9.1. Let

$$\mathcal{I} = \{(n, r) : n \geq 1, 1 \leq r \leq 1 + \nu_2(n)\}, \quad (71)$$

and define the positive diagonal operator $\mathcal{K}$ on $\ell^2(\mathcal{I})$ by

$$\mathcal{K}e_{n,r} = \frac{1}{n^2} e_{n,r}. \quad (72)$$

Theorem 9.2 (Fredholm determinant representation). *The operator $\mathcal{K}$ is trace class and*

$$\Phi(z) = \det(I - z^2 \mathcal{K}), \quad (73)$$

$$M(t) = \det\left(I + \frac{t^2}{4\pi^2} \mathcal{K}\right). \quad (74)$$

Furthermore,

$$\mathrm{tr}(\mathcal{K}^m) = \frac{\zeta(2m)}{1 - 2^{-2m}}, \quad (75)$$

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \, \mathrm{tr}(\mathcal{K}(I - z^2\mathcal{K})^{-1}). \quad (76)$$

Proof. By (24),

$$\mathrm{tr} \mathcal{K} = \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^2} = \frac{\zeta(2)}{1 - 2^{-2}} < \infty,$$

so $\mathcal{K}$ is trace class. Its Fredholm determinant is the product over its eigenvalues, with multiplicity:

$$\det(I - z^2\mathcal{K}) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1 + \nu_2(n)} = \Phi(z).$$

Substituting $z = it/(2\pi)$ gives (74). Formula (75) is again (24), and differentiating the Fredholm logarithm gives (76). $\square$

Remark 9.3. This representation converts several scalar identities into operator identities. For example, the zeta expansion (23) is the standard trace expansion

$$\log \det(I - z^2\mathcal{K}) = - \sum_{m \geq 1} \frac{z^{2m}}{m} \mathrm{tr}(\mathcal{K}^m).$$

It also suggests perturbation theory for generalized exponent sequences: changing the multiplicity sequence changes only the diagonal spectral multiplicities.

10 Jacobi continued fractions and exact rational coefficients

10.1 Exact moment recurrence

Proposition 10.1 (Moment recurrence). *The even moments satisfy $\mu_0 = 1$ and, for $n \geq 1$,*

$$(4^n - 1)\mu_{2n} = \sum_{j=1}^n \binom{2n}{2j} \frac{\mu_{2n-2j}}{2j+1}. \quad (77)$$

In particular, every μ_{2n} is rational.

Proof. Raise (10) to the $2n$ -th power and take expectations:

$$4^n \mu_{2n} = \sum_{j=0}^n \binom{2n}{2j} \mathbb{E} U^{2j} \mu_{2n-2j}.$$

Since $\mathbb{E} U^{2j} = 1/(2j+1)$, the $j=0$ term is μ_{2n} . Move it to the left. $\square$

10.2 Monic orthogonal polynomials

Let P_n be the monic orthogonal polynomials for the probability measure $\rho(x) \, dx$, and let

$$h_n = \int_{-1}^1 P_n(x)^2 \rho(x) \, dx. \quad (78)$$

Evenness forces the diagonal Jacobi parameters to vanish, so

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \quad \beta_n = \frac{h_n}{h_{n-1}} > 0. \quad (79)$$

The first polynomials are

$$P_0(x) = 1, \quad (80)$$

$$P_1(x) = x, \quad (81)$$

$$P_2(x) = x^2 - \frac{1}{9}, \quad (82)$$

$$P_3(x) = x^3 - \frac{19}{75}x, \quad (83)$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \quad (84)$$

10.3 The Jacobi and Stieltjes fractions

Theorem 10.2 (Jacobi continued fraction). *The Cauchy transform has the locally uniform continued fraction*

$$R(z) = \frac{1}{z - \frac{\beta_1}{z - \frac{\beta_2}{z - \frac{\beta_3}{z - \ddots}}}}. \quad (85)$$

The coefficients are positive rational numbers.

Proof. Positivity follows from $\beta_n = h_n/h_{n-1}$, and rationality follows because Gram–Schmidt applied to the rational moment functional (77) uses only rational operations. Markov’s theorem for compactly supported positive measures identifies the continued fraction generated by (79) with the Cauchy transform and gives local uniform convergence off the support [7, 6]. $\square$

The all-degree rational coefficients, positivity, zero diagonal, determinant cross-ratio, orthogonality, and centered three-term recurrence are now formalized in `RvachevRationalJacobi.lean`. The generic polynomial-basis determinant transport and its ordinary-Legendre specialization are now formalized in `PolynomialMomentGramDeterminant.lean` and `FabiusLegendreHankelDeterminant.lean`. The second-kind-polynomial construction, identification of the finite or infinite continued fraction with R , and Markov convergence used in this theorem remain paper-level claims rather than results of those Lean modules.

Define the ordinary even-moment generating function

$$\mathcal{O}(t) = \sum_{n=0}^{\infty} \mu_{2n} t^n = \int_{-1}^1 \frac{\rho(x)}{1 - tx^2} \, dx, \quad |t| < 1. \quad (86)$$

Substituting $t = z^{-2}$ into (86) gives the contracted fraction.

Corollary 10.3 (Stieltjes fraction and product integral). *For $|t| < 1$,*

$$\mathcal{O}(t) = \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \frac{\beta_3 t}{1 - \ddots}}}} \quad (87)$$

$$\mathcal{O}(t) = \int_0^\infty e^{-u} \prod_{k=1}^\infty \frac{\sinh(\sqrt{t} u / 2^k)}{\sqrt{t} u / 2^k} du. \quad (88)$$

Proof. The continued fraction follows from $\mathcal{O}(z^{-2}) = zR(z)$. For the integral, use

$$\frac{1}{1 - tx^2} = \int_0^\infty e^{-u} \cosh(\sqrt{t} xu) du$$

and then average over X , obtaining $M(\sqrt{t} u)$. $\square$

For $s \geq 0$, $\mathcal{O}(-s)$ is completely monotone:

$$(-1)^m \frac{d^m}{ds^m} \mathcal{O}(-s) = m! \int_{-1}^1 \frac{x^{2m} \rho(x)}{(1 + sx^2)^{m+1}} dx > 0. \quad (89)$$

Thus the even moments form a Stieltjes moment sequence, and the fraction (87) belongs to the classical positive Stieltjes class.

10.4 Hankel determinants

Let

$$\Delta_n = \det[\mu_{i+j}]_{i,j=0}^{n-1}, \quad \Delta_0 = 1. \quad (90)$$

Then

$$h_n = \frac{\Delta_{n+1}}{\Delta_n}, \quad \beta_n = \frac{\Delta_{n+1} \Delta_{n-1}}{\Delta_n^2}. \quad (91)$$

Because odd moments vanish, each Δ_n factors after a parity permutation into an even-even and odd-odd moment determinant. This parity factorization is useful for exact arithmetic and should be exploited in future Lean formalization.

Remark 10.4 (Formal finite-algebra boundary). `FiniteMomentGram.lean` and `GramStieltjes.lean` provide a measure-free version of the finite determinant algebra: an arbitrary scalar sequence gives a moment functional, symmetric Hankel pairing, fixed-order nondegeneracy criterion, fraction-free adjugate polynomial, and—over a field with the leading minor nonzero—the unique normalized monic orthogonal polynomial with self-pairing Δ_{n+1}/Δ_n . The complementary up-measure modules `MomentHankelMatrix.lean`, `MomentHankelValues.lean`, and `OrthogonalPolynomial*.lean` prove strict positivity, identify the exact moments, construct the monic polynomials, prove parity and the symmetric three-term recurrence, and compute the first exact Jacobi data. The generic continuation `FiniteMomentJacobi.lean` defines the norm quotients, diagonal and subdiagonal Jacobi coefficients, proves the unconditional total-field identity

$$\beta_{n+1} = \frac{\Delta_{n+2} \Delta_n}{\Delta_{n+1}^2},$$

and, under the three consecutive nonzero-minor hypotheses, proves

$$XP_{n+1} = P_{n+2} + \alpha_{n+1}P_{n+1} + \beta_{n+1}P_n.$$

Its separate theorem `gramStieltjesPolynomial_three_term_zero` supplies the base equation $XP_0 = P_1 + \alpha_0P_0$ from the nonzero first minor. This complete finite recurrence layer uses no measure or positivity. Finally, `OrthogonalPolynomialGramBridge.lean` proves that the up-moment functional is integration against the canonical up measure and identifies the generic matrix, determinant, fraction-free polynomial, and monic polynomial exactly with `momentHankel`, `hankelDet`, `hankelOrthoPolynomial`, and `upOrthoPolynomial`; it also identifies the generic norm with `hankelRatio`, proves that symmetry kills the generic diagonal coefficient, and identifies the generic subdiagonal with the quotient of consecutive Hankel ratios. Root and quadrature theorems, second-kind polynomials, continued-fraction identification, and analytic convergence remain open.

The polynomial-basis transport leaf `PolynomialMomentGramDeterminant.lean` has exactly two public definitions—`polynomialCoefficientMatrix` and `polynomialMomentGramMatrix`—and seven public theorems: `polynomialCoefficientMatrix_apply`, `polynomialMomentGramMatrix_apply`, `polynomialMomentGramMatrix_eq_transpose_mul_hankel_mul`, `polynomialMomentGramMatrix_det_eq_coefficient_det_sq_mul`, `polynomialCoefficientMatrix_det_eq_prod_coeff`, `polynomialMomentGramMatrix_det_eq_prod_coeff_sq_mul`, and `gramStieltjesJacobiSubdiagonal_eq_polynomialMomentGramMatrix_det_ratio`. For a family whose k -th polynomial has degree at most k , the matrix congruence is over a commutative semiring and the determinant identities are over a commutative ring. The final field-level formula additionally assumes every diagonal coefficient is nonzero; it needs no Hankel nonvanishing because division is total. This is measure-free finite algebra and makes no positivity, orthogonality, root, quadrature, continued-fraction, or convergence claim.

The scalar-base-change leaf `GramStieltjesNaturality.lean` has zero definitions and exactly six public theorems: `momentPairing_map`, `map_gramStieltjesNumerator`, `map_gramStieltjesPolynomial`, `map_gramStieltjesNorm`, `map_gramStieltjesJacobiDiagonal`, and `map_gramStieltjesJacobiSubdiagonal`. The numerator result is valid at ring level; the normalized-polynomial and Jacobi-quotient results use fields. All six are finite algebra, with no measure, positivity, or convergence.

The rational specialization `RvachevRationalJacobi.lean` has exactly four public definitions—`rvachevHankelDetRat`, `rvachevOrthoPolynomialRat`, `rvachevOrthoNormRat`, and `rvachevJacobiSubdiagonalRat`—and thirteen public theorems: `upMoment_eq_rvachevRawMomentRat_cast`, `rvachevHankelDetRat_cast`, `rvachevHankelDetRat_pos`, `rvachevOrthoPolynomialRat_cast`, `rvachevOrthoPolynomialRat_isMonicOfDegree`, `momentPairing_rvachevOrthoPolynomialRat_eq_zero`, `rvachevOrthoNormRat_cast`, `rvachevOrthoNormRat_pos`, `gramStieltjesJacobiDiagonal_rvachevRawMomentRat_eq_zero`, `rvachevJacobiSubdiagonalRat_cast`, `rvachevJacobiSubdiagonalRat_pos`, `rvachevJacobiSubdiagonalRat_eq_det_ratio`, and `rvachevOrthoPolynomialRat_three_term`. This proves at every degree the exact rational Hankel determinant, every coefficient of the monic polynomial, the positive squared norm, zero diagonal, and positive subdiagonal, as well as the real-cast bridges, rational orthogonality, determinant cross-ratio, and centered recurrence. Its zero-based `rvachevJacobiSubdiagonalRat` at index n is the report’s conventional β_{n+1} . The definitions are noncomputable adjugate-based finite algebra, not an evaluator: no named theorem evaluates the report’s displayed H_4 or β_4 fraction. Nor does this tranche prove roots, Christoffel formulas, second-kind polynomials, Gaussian quadrature, continued-fraction/Padé identification or convergence, central products, or Szegő/Nevai/Rakhmanov asymptotics.

The Legendre specialization `FabiusLegendreHankelDeterminant.lean` has exactly two public definitions—`upLegendreGramMatrix` and `upLegendreGramDet`—and seven public theorems:

upLegendreGramMatrix_apply_eq_integral, upLegendreGramDet_eq_prod_leadingCoeff_sq_mul_hankelDet, upLegendreGramDet_zero, upLegendreGramDet_pos, coeff_legendrePolynomial_self_div_succ, gramStieltjesJacobiSubdiagonal_upMoment_eq_upLegendreGramDet_ratio, and rvachevJacobiSubdiagonalRat_cast_eq_upLegendreGramDet_ratio. The determinant transport and generic up-moment cross-ratio hold for every BoundedFabius; interpreting entries as integrals, proving strict positivity, and comparing with the exact rational subdiagonal additionally use IsFabius. Consecutive leading coefficients contribute $(n+1)/(2n+1)$, and zero-based index n denotes β_{n+1} . The module does not prove a Gaunt/Wigner-symbol expansion or entry rationality by that route, Christoffel reconstruction, roots, quadrature, an infinite Jacobi product/continued fraction, or asymptotics.

10.5 Exact coefficients and numerical evidence

The accompanying Python program computes the moments by (77) and then uses the exact symmetric Stieltjes procedure (79). Selected results are:

n	μ_{2n}	decimal
0	1	1
1	$\frac{1}{9}$	0.111111111111
2	$\frac{19}{675}$	0.0281481481481
3	$\frac{583}{59535}$	0.00979255899891
4	$\frac{132809}{32531625}$	0.00408245822334
5	$\frac{46840699}{24405225075}$	0.00191928977733
6	$\frac{4068990560161}{4133856862760625}$	0.000984308527181
7	$\frac{1204567303451311}{2232691548877164375}$	0.000539513532022
8	$\frac{4146897304424408411}{13301767332333178846875}$	0.00031175536309
9	$\frac{18814360006695807527868793}{100028040755473167511640090625}$	0.000188090857969
10	$\frac{21431473463327429953796293981397}{182171989134769427819794434994453125}$	0.000117644175513

n	β_n (exact when compact)	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.168435374150
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288
6	see data file	0.199759514028
7	see data file	0.205964295192
8	see data file	0.211147954951
9	see data file	0.215261988744
10	see data file	0.218406233337
11	see data file	0.221259971300
12	see data file	0.223692820062

The full numerators and denominators through $n = 40$ are in up_jacobi_coefficients.csv. Their size grows rapidly: by moderate order the reduced integers contain thousands of decimal digits. This arithmetic complexity is not a floating-point artifact.

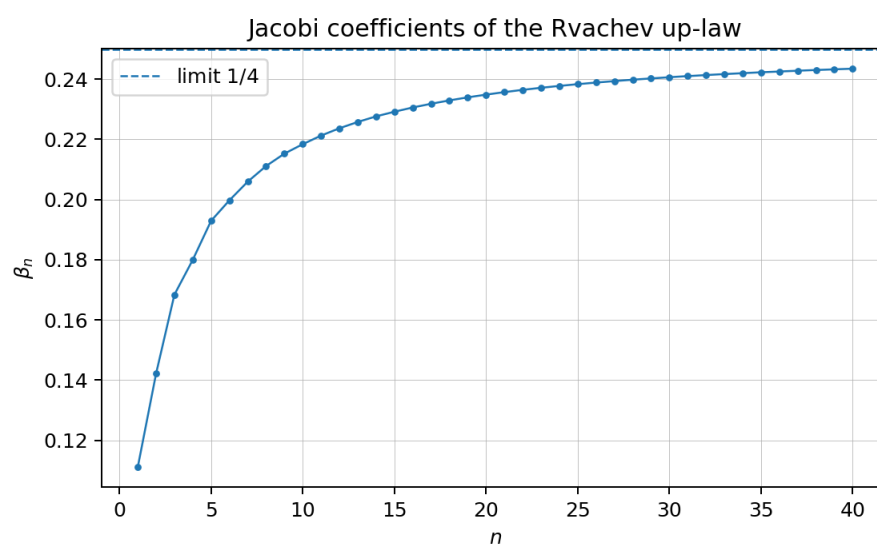


Figure 1: Exact Jacobi coefficients converted to decimals. The dashed line is the proved limit $1/4$.

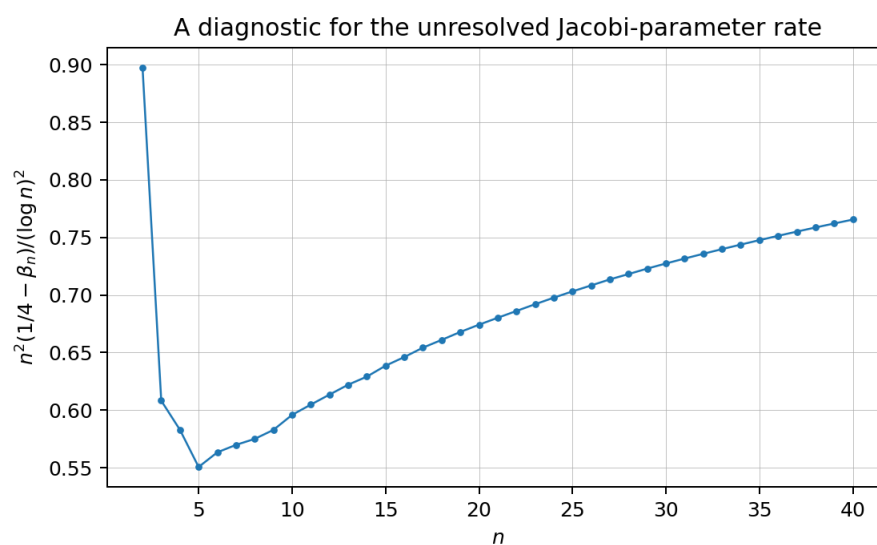


Figure 2: The diagnostic quantity $n^2(1/4 - \beta_n)/(\log n)^2$. The range $n \leq 40$ is evidence, not an asymptotic proof.

10.6 The limiting coefficient

Theorem 10.5 (Nevai-class limit). *The Jacobi coefficients satisfy*

$$\lim_{n \rightarrow \infty} \beta_n = \frac{1}{4}. \quad (92)$$

Proof. The measure $\rho(x) dx$ has essential support $[-1, 1]$, and its density is strictly positive for every $x \in (-1, 1)$. The Denisov–Rakhmanov theorem therefore places its orthonormal Jacobi matrix in the Nevai class of $[-1, 1]$ [3, 6]: the off-diagonal orthonormal coefficients tend to $1/2$, and the diagonal coefficients tend to zero. In the monic recurrence (79), β_n is the square of the off-diagonal orthonormal coefficient, hence tends to $1/4$. $\square$

10.7 Padé and Gaussian-quadrature representations

Let Q_{N-1} be the polynomial of the second kind

$$Q_{N-1}(z) = \int_{-1}^1 \frac{P_N(z) - P_N(x)}{z - x} \rho(x) dx. \quad (93)$$

The N -th Jacobi convergent is Q_{N-1}/P_N . If $x_{1,N}, \dots, x_{N,N}$ are the zeros of P_N , then

$$\frac{Q_{N-1}(z)}{P_N(z)} = \sum_{j=1}^N \frac{w_{j,N}}{z - x_{j,N}}, \quad w_{j,N} > 0, \quad \sum_j w_{j,N} = 1. \quad (94)$$

This is the Cauchy transform of the N -node Gaussian quadrature measure for ρ , and it matches moments through degree $2N - 1$. The exact error identity is

$$R(z) - \frac{Q_{N-1}(z)}{P_N(z)} = \frac{1}{P_N(z)^2} \int_{-1}^1 \frac{P_N(x)^2}{z - x} \rho(x) dx. \quad (95)$$

Hence every Gaussian resolvent approximant is a strict lower bound for $R(z)$ when $z > 1$, and a strict upper bound when $z < -1$.

The first nontrivial approximants are completely explicit:

$$R_1^G(z) = \frac{1}{z}, \quad (96)$$

$$R_2^G(z) = \frac{z}{z^2 - 1/9} = \frac{1}{2} \left(\frac{1}{z - 1/3} + \frac{1}{z + 1/3} \right), \quad (97)$$

$$R_3^G(z) = \frac{32/57}{z} + \frac{25/114}{z - \sqrt{19/75}} + \frac{25/114}{z + \sqrt{19/75}}. \quad (98)$$

Thus the continued fraction gives a hierarchy of positive finite atomic models for the up-law, complementary to the continuous finite-convolution models of Section 5.

11 Further combined representations

11.1 The Cauchy transform of the Fabius measure

Let

$$S(z) = \int_0^1 \frac{dF(x)}{z - x} = \mathbb{E} \frac{1}{z - Y}. \quad (99)$$

The affine relation (13) gives

$$\boxed{S(z) = 2R(2z - 1)}. \quad (100)$$

Consequently, every representation of R immediately yields one for S . For example, for $|2z - 1| > 1$,

$$S(z) = 2 \sum_{n=0}^{\infty} \frac{\mu_{2n}}{(2z - 1)^{2n+1}}, \quad (101)$$

and S has a Jacobi fraction obtained by the affine change of variable in (85).

11.2 A determinant form of the Fabius Laplace transform

Combining (48) with (74) gives

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right) - e^{-s}}{s}. \quad (102)$$

Similarly,

$$\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right)}{s}. \quad (103)$$

These are literal combinations of a compact-support transform with a trace-class spectral determinant.

11.3 A resolvent–product–continued-fraction triangle

For $|t| < 1$, the same analytic function $\mathcal{O}(t)$ has three structurally different representations:

$$\boxed{\begin{aligned} \mathcal{O}(t) &= \sum_{n \geq 0} \mu_{2n} t^n \\ &= \int_0^\infty e^{-u} \prod_{k \geq 1} \frac{\sinh(\sqrt{t} u / 2^k)}{\sqrt{t} u / 2^k} \, du \\ &= \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \ddots}}}. \end{aligned}} \quad (104)$$

The first line is arithmetic, the second is multiplicative and analytic, and the third is orthogonal-polynomial/geometric. Equality of the three creates a direct bridge among the Bernoulli–Bell moment formulas, the infinite hyperbolic product, and the exact rational Jacobi parameters.

12 Numerical experiments and reproducibility

12.1 Exact arithmetic

The script `rvachev_frontier_experiments.py` performs the following operations.

1. It computes $\mu_0, \mu_2, \dots$ exactly using (77).

2. It constructs the monic orthogonal polynomials using the symmetric three-term recurrence and exact rational inner products.
3. It writes all exact β_n through a user-selected order to CSV.
4. It checks (37) independently using a long moment expansion.
5. It compares the cotangent, tangent, and zero-set formulas for Φ'/Φ .
6. It generates Figures 1 and 2.

The default order is 40. On the reference environment the full run takes tens of seconds; the exact integers, rather than floating-point work, dominate at higher order.

12.2 Numerical verification of the resolvent equation

At 45-decimal mpmath precision, using moment expansions through orders 160 and 220, the computed values were:

z	$R'(z)$	$2(R(2z+1) - R(2z-1))$	relative discrepancy
1.25	-0.8312850091714830623446214	same	3.6×10^{-46}
1.75	-0.3679189599146538753803797	same	below printed precision
2.50	-0.1691589760955563001740583	same	4.4×10^{-47}

At $z = 1.25$, the change between the two truncation lengths was about 3.8×10^{-41} , so the identity error is below the visible truncation scale.

12.3 Numerical comparison of logarithmic derivatives

The cotangent and tangent formulas agreed to roughly 44 decimal places at the tested nonzero points. The zero-set formula, truncated at 10^5 , differed by 5.2×10^{-6} , 1.5×10^{-5} , and 2.9×10^{-5} at $z = 0.13, 0.37, 0.73$, respectively, consistent with its algebraic tail. The data file `numerical_checks.txt` records all values.

13 Conjectures and frontier problems

The following conjectures are deliberately separated from the proved results.

Conjecture 13.1 (Strict Jacobi monotonicity). The exact Jacobi coefficients are strictly increasing:

$$0 < \beta_1 < \beta_2 < \cdots < \frac{1}{4}. \quad (105)$$

Exact arithmetic verifies the inequalities through $n = 40$. The upper limit is proved by Theorem 10.5, but monotonicity does not follow from the Denisov–Rakhmanov theorem and appears to require a Fabius-specific argument.

Conjecture 13.2 (Logarithmically corrected recurrence rate). There exists a constant $C_{\text{up}} \in (0, \infty)$ such that

$$\frac{1}{4} - \beta_n \sim C_{\text{up}} \frac{(\log n)^2}{n^2}. \quad (106)$$

The scaled quantity in Figure 2 rises from approximately 0.674 at $n = 20$ to 0.766 at $n = 40$. This is too short a range to determine a constant or even to exclude a different slowly varying factor. The conjecture is motivated jointly by the data and by the squared-logarithmic endpoint scale of ρ . A Riemann–Hilbert or strong Szegő analysis adapted to the endpoint flatness is the natural next step.

Conjecture 13.3 (Dyadically regularized Mittag–Leffler expansion). Let $d_n = 1 + \nu_2(|n|)$, and let $\text{PP}_n(z)$ be the principal part of $1/\Phi(z)$ at the nonzero integer n . There exists a canonical dyadic grouping and a finite genus of subtraction polynomials such that

$$\frac{1}{\Phi(z)} = E(z) + \sum_{n \in \mathbb{Z} \setminus \{0\}}^{\text{dyadic}} (\text{PP}_n(z) - \text{canonical subtraction at } n), \quad (107)$$

where E is an explicitly controlled entire function, ideally constant or of very low order.

The local principal parts are computable from Theorem 8.2; the unresolved issue is a global summation scheme compatible with the unbounded dyadic multiplicities.

Problem 13.4 (Arithmetic of β_n). Determine sharp denominator-clearing factors for β_n , h_n , and the parity Hankel determinants. The first coefficients display repeated migration of prime factors between consecutive norms, suggesting that factored norm variables may be arithmetically simpler than reduced β_n themselves.

Problem 13.5 (Endpoint-sensitive Gaussian nodes). Determine the asymptotic distance of the extreme zero of P_n from 1, and compare it with the Chebyshev scale n^{-2} . The density is positive almost everywhere but flatter than every power at the endpoint, so the leading equilibrium distribution is classical while the first correction should retain the Lambert– W /log-square endpoint signature.

Problem 13.6 (A coupled resolvent hierarchy for exponent sequences). The repository’s Newton/Rvachev theory replaces the unit exponent sequence in the sinc product by general integer sequences. Construct the corresponding Cauchy transforms and derive coupled affine functional equations when the exponent sequence is closed under a Pascal or shift operator. The Fredholm model of Section 9 suggests that the Fourier side is easy; the real-space fixed-point law is the main obstacle.

Problem 13.7 (Fractional resolvent dynamics). Equation (37) gives every integer derivative as a finite Thue–Morse orbit. Seek a fractional-order interpolation

$$D_z^\alpha R(z) = \int_{\Gamma_\alpha} K_\alpha(j) R(2^\alpha z + \text{affine shift}(j)) \, dj$$

that reduces to (38) for $\alpha \in \mathbb{N}$. A semigroup built from the two affine branches $z \mapsto 2z \pm 1$ may provide the correct operator calculus.

Problem 13.8 (Formal verification roadmap). A Lean development can be staged as follows:

1. define the Cauchy transform of the canonical up probability measure and prove its density form, natural-domain holomorphy, first derivative, and affine bridge;
2. prove the logarithmic fixed-point identity first on $(1, \infty)$;
3. differentiate under the integral and formalize (37);
4. apply the generic open-domain theorem in `AffineDifferenceOrbit.lean` to obtain the Thue–Morse derivative tree once the preceding resolvent equation is available;
5. use the named comparison in `OrthogonalPolynomialGramBridge.lean` to identify the generic Gram–Stieltjes construction with the up-measure-specific monic polynomials;
6. construct second-kind polynomials and finite Jacobi convergents, then prove the Gaussian error identity;
7. postpone the Denisov–Rakhmanov limit, which depends on substantial external orthogonal-polynomial theory.

Item 1 is delivered by `CauchyTransform.lean`, now with the all-power affine bridge, named unit Stieltjes DDE, and both complex-slit adjacent-power recurrences; its reusable finite-measure and general geometric-law foundations are `MeasureCauchyTransform.lean` and `GeometricUniformCauchy.lean`. Item 2 is delivered exactly on the proposed first domain, real $z > 1$, by `StieltjesLogFixedPoint.lean`; complex logarithmic continuation remains open. `CauchyRenormalization.lean` delivers item 3 directly from the differential refinement equation and compact support, without depending on item 2, and instantiates the generic engine from `AffineDifferenceOrbit.lean` to deliver item 4.

The separate higher-kernel conversion stated in (39) remains open; the generic CDF/survival formulas at every positive kernel power do not by themselves prove that explicit Thue–Morse sum identity. The bounded-measure Laurent obligation is delivered generically by the one-definition, fifteen-theorem `MeasureCauchyMomentLaurent.lean` API under its `RCLike`, almost-everywhere relative-ball, and finite-mass hypotheses. The bridge declarations `ae_norm_sub_zero_le_one_rvachevMeasure` and `measureCauchyMoment_rvachevMeasure_zero` recover the existing up-specific $c = 0$, $R = 1$ Laurent results and their `upMoment` coefficients. This does not formalize the coefficient recursion or moment-determinacy argument. The up-specific Laurent, Hankel, and `OrthogonalPolynomialJacobi.lean` modules formalize the moment side and first exact Jacobi data, while the generic finite-algebra modules above formalize the determinant construction and complete finite three-term recurrence without positivity. Item 5 is now delivered by `OrthogonalPolynomialGramBridge.lean`. In addition, `PolynomialMomentGramDeterminant.lean`, `GramStieltjesNaturality.lean`, `RvachevRationalJacobi.lean`, and `FabiusLegendreHankelDeterminant.lean` deliver polynomial-basis determinant transport, scalar naturality, the all-degree centered rational determinants, polynomial coefficients, norms, zero diagonal, positive subdiagonals, determinant cross-ratio, orthogonality, recurrence, real-cast bridges, and the Legendre Gram-determinant specialization. The zero-based rational subdiagonal at index n is β_{n+1} ; this is exact finite algebra, not a native evaluator for the displayed H_4 or β_4 fractions. Item 6’s second-kind polynomials, complete finite convergents, and Gaussian error theorem and item 7’s asymptotics remain open.

14 Promising research program

14.1 Analytic program

The Cauchy transform should be treated as a primary Fabius–Rvachev object, on the same level as F , ρ , and Φ . Immediate goals are:

1. derive quantitative bounds for R near the cut from the sharp Fourier decay;
2. study the Hilbert transform (35), including its zeros and endpoint asymptotics;
3. use (37) to continue asymptotic expansions along dyadic affine rays;
4. compare finite-logarithmic and Jacobi rational approximants uniformly near the support.

14.2 Orthogonal-polynomial program

The exact J -fraction opens a separate research direction:

1. prove or disprove Conjecture 13.1;
2. determine strong asymptotics of P_n , h_n , and β_n ;
3. construct Fabius-specific Gauss–Kronrod or nested positive rules;

4. investigate whether the dyadic moment recurrence induces a nonlinear recurrence or integrable system for β_n ;
5. study the arithmetic Galois structure of Gaussian nodes and exact weights.

14.3 Operator and special-function program

The Fredholm determinant and Hurwitz-zeta hierarchy suggest:

1. regularized determinants and spectral shifts for deformed multiplicity sequences;
2. a Barnes-type compression of the determinant rather than only of the scalar product;
3. exact trace identities for logarithmic derivatives at rational and half-integer arguments;
4. meromorphic continuation of dyadically weighted spectral sums with local factors dictated by $\nu_2(n)$.

14.4 Inverse-function program

The pushforward formulas (58)–(61) show that many global integral questions about F^{-1} are easier than pointwise questions. Promising directions include:

1. exact orthogonal expansions of $I(u)$ using its known pushforward moments;
2. entropy and transport functionals of I , reduced to integrals against ρ ;
3. combining the global Cauchy transform with local Lambert- W endpoint inversions;
4. rational and Gaussian quadrature for inverse-function integrals without directly evaluating I .

15 Conclusion

The audited repository already contains most of the classical and modern representation theory one would first seek: convolution laws, sinc and cosine products, integer-zero products, moment and cumulant series, Thue–Morse splines, q-binomial dyadic values, Poisson and Legendre expansions, inverse asymptotics, and Fourier-decay envelopes. The most productive way to extend that corpus is therefore to introduce a new organizing transform rather than another isolated formula.

The Cauchy transform R accomplishes this. It joins the moment sequence to an analytic function, converts the dyadic law into a functional-differential equation, turns finite convolutions into logarithmic Thue–Morse sums, and generates a Jacobi continued fraction with exact rational coefficients. The Fourier image simultaneously acquires an operator determinant and a special-function logarithmic-derivative calculus. Closed one-sided transforms and inverse pushforward identities complete a network in which real-space, frequency-space, moment, orthogonal-polynomial, and operator representations are mutually convertible.

The proved limit $\beta_n \rightarrow 1/4$ anchors the new continued-fraction theory. The exact order-40 data, the monotonicity conjecture, the endpoint-sensitive rate problem, and the regularized Mittag–Leffler problem provide concrete frontier targets suitable for both analysis and formalization.

A Reproducibility instructions

From the directory containing this consolidated report, run

```
python3 assets/Fabius_Rvachev_Representation_Frontiers/
    rvachev_frontier_experiments.py \
    --order 40 --digits 45 --output-dir representation-frontiers-output
```

This regenerates:

- `up_even_moments.csv`;
- `up_jacobi_coefficients.csv`;
- `generated_tables.tex`;
- `numerical_checks.txt`;
- the two PDF/PNG figures.

The exact arithmetic uses Python’s `fractions.Fraction`. The numerical checks use `mpmath`; plotting uses `matplotlib`. The script contains detailed comments, normalization statements, argument validation, and exact orthogonality checks.

B Corpus audit ledger

The pinned 67-path inventory and the additional live draft families used in the audit are listed in the accompanying source fragment `corpus_inventory.tex`.

Pinned 67-source ledger

No.	Path relative to Analysis/FabiusFunction/docs
1	<code>FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>
2	<code>Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>
3	<code>Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex</code>
4	<code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>
5	<code>semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex</code>
6	<code>archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex</code>
7	<code>archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex</code>
8	<code>archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex</code>
9	<code>archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>
10	<code>archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex</code>
11	<code>archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>
12	<code>archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex</code>
13	<code>archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex</code>
14	<code>archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>

No.	Path relative to Analysis/FabiusFunction/docs
15	archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
16	archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
17	archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
18	archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
19	archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
20	archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
21	archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
22	archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
23	archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
24	archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
25	archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
26	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
27	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
28	archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
29	archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
30	archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
31	archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
32	archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
33	archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex
34	archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex
35	archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
36	archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
37	archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
38	semi-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex
39	semi-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex
40	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex
41	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex
42	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_New_Frontiers/Fabius_Rvachev_New_Frontiers.tex

No.	Path relative to Analysis/FabiusFunction/docs
43	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex
44	semi-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex
45	semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
46	semi-formalized-research-frontiers/drafts/fabius_frontier_new_results/fabius_frontier_new_results.tex
47	semi-formalized-research-frontiers/drafts/fabius_frontier_results_bundle/fabius_frontier_results.tex
48	semi-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.tex
49	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
50	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
51	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
52	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
53	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
54	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
55	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
56	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
57	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
58	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
59	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
60	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
61	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
62	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
63	papers/arXiv-1502.06738v2/Lacunary_products.tex
64	papers/arXiv-1702.05442v1/09-Function.tex
65	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66	papers/arXiv-2211.13570v1/document.tex
67	papers/ubiq15/ubiq15-corrected.tex

Live additions checked after the pinned ledger

- semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
- semi-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex

- `semi-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex`
- `semi-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/` (report source and auxiliaries)

Additional auxiliary TeX fragments inside these live report packages were checked through their parent report structure. The pinned ledger remains the stable reproducible count; the live tree changes during active research and therefore should not be assigned a permanent count without a fresh commit-pinned traversal.

C Selected exact data

The first eleven even moments and first twelve Jacobi coefficients are typeset in the main text. The CSV files contain the exact data through order 40. Selected late-order decimal values are:

n	β_n	$1/4 - \beta_n$	$n^2(1/4 - \beta_n)/(\log n)^2$
12	0.223692820061698	0.026307179938302	0.6135029154
15	0.229185734718218	0.020814265281782	0.6386018638
20	0.234873012373992	0.015126987626008	0.6742274757
25	0.238343193605764	0.011656806394236	0.7031551379
30	0.240650640426945	0.009349359573055	0.7273789024
35	0.242285352613889	0.007714647386111	0.7476320691
40	0.243489365492140	0.006510634507860	0.7655161747

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Part II

Representation Atlas and New Analytic Bridges for the Fabius and Rvachev Functions

Abstract

This report gives a source-audited atlas of integral, series, product, convolution, probabilistic, spectral, and transform representations of the bounded Fabius function F , Rvachev’s compactly supported even function up , the inverse Fabius function $Q = F^{-1}$, and the entire Fourier image $\Phi(z) = \widehat{up}(z)$. The baseline material is organized from the current formal exposition and the living non-formalized frontier documents in the ProveIt repository. Previously present results – including the sinc and cosine products, the arithmetic canonical product, Fredholm determinants, half-integer spectral series, q -binomial moment formulae, Thue–Morse splines, the generalized-gamma-convolution dual, and the logistic dual – are explicitly quarantined from novelty claims.

Five new-to-the-audited-corpus theorem packages are developed. First, a meromorphic *dyadic gamma function*

$$\Gamma_{dy}(z) = \prod_{h \geq 0} \Gamma(1 + 2^{-h}z)$$

factors the Rvachev transform as $\Phi(z) = (\Gamma_{dy}(z)\Gamma_{dy}(-z))^{-1}$, with functional equations, canonical products, digamma and heat-trace integrals, and a geometric- q extension. Second, the existing dyadic logistic series is identified in law with an explicit Gaussian variance mixture driven by the existing discrete Thorin random variable; this yields a convolution semigroup, a closed Levy density, complete monotonicity of the density as a function of x^2 , and dimension-free radial positive definiteness. Third, Jensen’s circular mean of $\log |\Phi|$ is evaluated exactly by binary digit sums and a finite factorial product. Fourth, the unbounded zero multiplicities prove that Φ is not D-finite, and neither are the dyadic gamma factors. Fifth, distributional and compact Fourier formulae for F are completed by an odd-harmonic sine series, while Q receives a new nonlinear Fourier transform and sine-series representation in terms of oscillatory integrals of F .

The formal source checkpoint 71ab6f6728fceb753c88d8b0573077a59acf2682 now covers the entire reciprocal product, its Mahler equation and zero orbit, the reflected sinc product and dyadic Rvachev bridge, and the exact dyadic zero and meromorphic pole orders. It does not cover the raw product of Gamma factors away from its poles, the arithmetic canonical regrouping, logarithmic or digamma expansions, Malmsten or heat-trace integrals, the displayed Fourier integrals, non-D-finiteness, the reciprocal-Gamma derivative coefficient used by the separate Thue–Morse tower, or the general integer-base collision order; those portions remain frontier claims despite occurring in the same theorem package below.

Current post-checkpoint overlay. The historical boundary above is retained verbatim for 71ab6f6728fceb753c88d8b0573077a59acf2682. At source commit 0ba35abd4, Reciprocal GammaJets.lean adds exactly five public declarations: `Fabius.deriv_Gamma_inv_neg_nat`, `Fabius.hasDerivAt_Gamma_inv_neg_nat`, `Fabius.hasDerivAt_Gamma_inv_zero`, `Fabius.analyticOrderAt_Gamma_inv_neg_nat`, and `Fabius.tendsto_Gamma_inv_div_add_nat`. They give, for every $r \in \mathbb{N}$, the first jet $(\Gamma^{-1})'(-r) = (-1)^r r!$, simple analytic order one, and the same punctured local coefficient. These are theorems about the entire reciprocal function; they do not assign a derivative to raw Gamma at a pole.

The same commit gives the first eight public declarations of `ThueMorseGammaTower.lean`: `Fabius.hasDerivAt_dirichletMellinContinuation_neg_nat`, `Fabius.deriv_dirichletMellinContinuation_neg_nat`, `Fabius.thueMorseGammaLog`, `Fabius.thueMorseGammaTower`, `Fabius.thueMorseGammaLog_eq_mellin`, `Fabius.thueMorseGammaLog_eq_integral`, `Fabius.thueMorseGammaLog_dyadic`, and `Fabius.thueMorseGammaTower_dyadic`. For every $r \in \mathbb{N}$ and $a > 0$, they identify the continuation first jet with $(-1)^r r!$ times the Mellin value, give its real-integral form, and prove the dyadic GammaLog and exponentiated tower laws. The GammaLog and tower definitions are total for every real a , but those identification theorems assume $a > 0$. GammaLog is a chosen

derivative coordinate, not a proved `Complex.log` identity. The current integrated tree adds the ninth public declaration `Fabijs.ofReal_exp_mplimit_eq_gammaTower_div`, the level-zero tower quotient for positive a, b . Accordingly the first-jet, Mellin/integral `GammaLog`, dyadic log/tower, and ratio layers are no longer frontiers. The parameter- a differential and iterated differential ladder remains open, as do the raw `Gamma tprod`, canonical regrouping, logarithmic/digamma, Gamma-product Malmsten/heat-trace integrals, D-finite, and general-base collision-order claims listed above.

A fully commented Python program independently checks the principal identities, produces four figures, and records truncation diagnostics. “Proved here” means proved in this report and not found by targeted collision searches in the audited repository; it is not a claim of priority over all external literature.

1 Scope, source audit, and status discipline

1.1 Repository boundary

The requested source boundary was the recursive collection of `*.tex` files below

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

The primary formal exposition was pinned to repository commit `0770442945d65bac5530c4f214c6c52c4cb9fbdc`, dated 27 August 2026. During the audit the branch changed: in particular, a support-free quantile substitution theorem entered the primary exposition. Consequently, quantile pushforward formulae that looked new against an earlier snapshot are classified here as repository consequences, not new results.

The recursive audit included the current formal exposition, the consolidated frontier notebook, the glossary and Lean walkthrough, the active frontier directories, the embedded literature sources, and the archive provenance map. The active frontier root contained seventeen named draft packages at the time of the audit. One of those packages records a snapshot with sixty-seven distinct TeX paths. Historical byte-identical copies were not counted as independent mathematical sources: the archive provenance file was used to map them to their living successors, after which the successor source was audited at formula level. This avoids manufacturing “novelty” merely by overlooking a renamed or moved draft.

1.2 Collision quarantine

The following structures were already present and are therefore baseline in this report:

- the infinite sinc product, weighted cosine product, and integer-zero canonical product;
- the arithmetic multiplicities $a_n = 1 + v_2(n)$, their Dirichlet series, counting law $\sum_{n \leq N} a_n = 2N - s_2(N)$, lobe signs, heat traces, spectral zeta functions, and Fredholm determinant;
- half-integer cosine expansions, spectral quadrature, sampling and alias formulae;
- moments and cumulants through Bell and Bernoulli numbers, q -binomial and multiple-zeta formulae, Appell sequences, Barnes-type prefixes, and Thue–Morse finite blocks;
- the discrete-Thorin generalized gamma convolution $\mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}$ and its complete-Bernstein/Stieltjes consequences;
- the dyadic logistic series whose characteristic function is the reciprocal moment generating function of the compact Rvachev law;
- exact negative-Laplace decompositions, periodic lower-Lambert endpoint asymptotics, inverse asymptotics, and the now-formal quantile substitution theorem.

Targeted searches did not find the phrases or exact structures “dyadic gamma”, “Gaussian variance mixture”, “Jensen circular mean”, “D-finite”, the principal-value transform of the global Fabius CDF, or the inverse sine coefficients derived in [Appendix 14](#). These results are marked **[proved here]**. Formulae obtained by combining known repository theorems without a substantively new bridge are marked **[derived synthesis]**.

1.3 Status legend and limits of the novelty statement

Status	Meaning
[repository]	Present in the formal exposition or an audited frontier document.
[derived synthesis]	A direct synthesis or reformulation of repository results; no novelty claim.
[proved here]	Proved in this report and not located by targeted collision searches in the audited corpus. This is a corpus-relative statement, not universal priority.
[conjecture]	A plausible but unproved statement, separated from theorems.

2 Normalization and the master probability diagram

2.1 The compact law and its CDF

Let $(V_j)_{j \geq 1}$ be independent uniform random variables on $[0, 1]$, let $U_j = 2V_j - 1$ be uniform on $[-1, 1]$, and define

$$X = \sum_{j \geq 1} 2^{-j} V_j, \quad Y = 2X - 1 = \sum_{j \geq 1} 2^{-j} U_j. \quad (108)$$

Then $X \in [0, 1]$, $Y \in [-1, 1]$, $X \stackrel{d}{=} 1 - X$, and $Y \stackrel{d}{=} -Y$. The bounded Fabius function and Rvachev density are

$$F(x) = \mathbb{P}(X \leq x), \quad \mathbb{P}(Y \in dy) = \text{up}(y) dy. \quad (109)$$

The normalizations used below are

$$\text{up}(y) = \begin{cases} F(y+1), & -1 \leq y \leq 0, \\ F(1-y) = 1 - F(y), & 0 \leq y \leq 1, \\ 0, & |y| > 1, \end{cases} \quad (110)$$

and

$$F'(x) = 2 \text{up}(2x - 1) \quad (0 < x < 1). \quad (111)$$

Thus up is even, nonnegative, C^∞ , supported on $[-1, 1]$, and has integral one. The density of X is $f_X(x) = 2 \text{up}(2x - 1)$ on $[0, 1]$.

2.2 Fixed-point integral equations

The random affine fixed point

$$Y \stackrel{d}{=} \frac{U + Y'}{2}, \quad (112)$$

where U is uniform on $[-1, 1]$ and Y' is an independent copy of Y , gives the exact refinement equation

$$\boxed{\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(t) dt.} \quad (113)$$

Differentiation gives

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1). \quad (114)$$

The bounded Fabius function is the fixed point of

$$F(x) = \begin{cases} \int_0^{2x} F(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} F(t) dt, & \frac{1}{2} \leq x \leq 1, \end{cases} \quad (115)$$

with constant continuation outside $[0, 1]$. Hence

$$F'(x) = \begin{cases} 2F(2x), & 0 < x < \frac{1}{2}, \\ 2F(2-2x), & \frac{1}{2} < x < 1. \end{cases} \quad (116)$$

These integral equations are often the most efficient starting point for exact quadrature, monotonicity, and contraction arguments.

2.3 The signed global extension and derivative tower

The repository's signed global extension $\mathcal{F}$ satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad (117)$$

so iteration gives the all-order identity

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x). \quad (118)$$

After the appropriate unit translation, this yields

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(x+1)). \quad (119)$$

Thus every derivative representation of $\mathcal{F}$ immediately becomes one for up, and vice versa. It is important not to apply (117) directly to the bounded CDF outside its valid normalization.

3 Infinite convolutions, finite splines, and head–tail representations

3.1 Box convolution

The density of $2^{-j}U_j$ is $2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}$. Therefore

$$\text{up} = \bigstar_{j=1}^{\infty} (2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}) . \quad (120)$$

This identity may be read in probability, in L^1 , or through uniform convergence of successive compact splines. It directly explains positivity, compact support, smoothness, and the sinc product of [Appendix 4](#).

Let

$$Y_N = \sum_{j=1}^N 2^{-j}U_j, \quad u_N = \bigstar_{j=1}^N (2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}) . \quad (121)$$

Then u_N is a piecewise polynomial of degree $N-1$ and

$$Y = Y_N + 2^{-N}Y', \quad \text{up} = u_N * (2^N \text{up}(2^N \cdot)) . \quad (122)$$

Equivalently,

$$\text{up}(x) = \mathbb{E} [2^N \text{up}(2^N(x - Y_N))] . \quad (123)$$

For the CDF,

$$F(x) = \mathbb{E} \left[F \left(2^N x - \sum_{j=1}^N 2^{N-j} V_j \right) \right], \quad (124)$$

where the inner F is the bounded, clamped CDF. Formula (124) is an exact N -dimensional cube integral and a useful recursive numerical scheme.

3.2 General truncated-power spline

For positive widths $a_1, \dots, a_N$, the density of $\sum_{j=1}^N a_j U_j$ is

$$b_N(x) = \frac{1}{(N-1)! 2^N \prod_{j=1}^N a_j} \sum_{\varepsilon \in \{\pm 1\}^N} \left(\prod_{j=1}^N \varepsilon_j \right) \left(x + \sum_{j=1}^N \varepsilon_j a_j \right)_+^{N-1}. \quad (125)$$

Its CDF is obtained by replacing $(N-1)!$ and the exponent $N-1$ by $N!$ and N . For $a_j = 2^{-j}$, binary indexing turns (125) into the exact Thue–Morse spline

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} \left(x + \frac{2^N - 1 - 2k}{2^N} \right)_+^{N-1}. \quad (126)$$

The corresponding finite CDF is

$$U_N(x) = \frac{2^{N(N-1)/2}}{N!} \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} \left(x + \frac{2^N - 1 - 2k}{2^N} \right)_+^N. \quad (127)$$

Prouhet cancellation makes the apparent polynomial tails cancel exactly. These formulae simultaneously represent the finite probability law, a compact spline, and a signed Thue–Morse power sum.

3.3 Hypercube and section representations

For bounded measurable g ,

$$\int_{-1}^1 g(y) \operatorname{up}(y) \, dy = \int_{[-1,1]^{\mathbb{N}}} g \left(\sum_{j \geq 1} 2^{-j} u_j \right) \bigotimes_{j \geq 1} \frac{du_j}{2}. \quad (128)$$

At finite depth this gives a coarea interpretation of u_N as the weighted $(N-1)$ -dimensional volume of the cube section $\sum_{j=1}^N 2^{-j} u_j = x$. The infinite-dimensional expression is often more useful than a closed formula: it converts integrals against up into product quadrature and supports concentration and coupling arguments.

4 Fourier-product atlas and spectral representations

4.1 Conventions and the three canonical products

We use

$$\widehat{g}(z) = \int_{\mathbb{R}} g(x) e^{-2\pi i x z} \, dx, \quad \operatorname{sinc} w = \frac{\sin w}{w}, \quad (129)$$

with the removable value at zero. Put

$$\Phi(z) = \widehat{\operatorname{up}}(z). \quad (130)$$

The repository establishes the entire products

$$\Phi(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right), \quad (131)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m, \quad (132)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{a_n}, \quad a_n = 1 + v_2(n). \quad (133)$$

They satisfy the Mahler-type renormalization

$$\Phi(z) = \operatorname{sinc}(\pi z) \Phi(z/2). \quad (134)$$

The zero at every nonzero integer n has exact order $a_{|n|}$.

4.2 Logarithmic Taylor series and partial fractions

For $|z| < 1$,

$$\log \Phi(z) = - \sum_{r \geq 1} \frac{\zeta(2r)}{r(1-4^{-r})} z^{2r}. \quad (135)$$

Logarithmic differentiation of the canonical product gives, away from the integer zeros,

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{n \geq 1} \frac{a_n}{n^2 - z^2}. \quad (136)$$

The arithmetic spectrum has Dirichlet generating function

$$\sum_{n \geq 1} \frac{a_n}{n^s} = \frac{\zeta(s)}{1-2^{-s}}, \quad \Re s > 1, \quad (137)$$

and exact counting function

$$A(N) = \sum_{n \leq N} a_n = \sum_{h \geq 0} \left\lfloor N/2^h \right\rfloor = 2N - s_2(N). \quad (138)$$

Thus the zero divisor simultaneously encodes the Riemann zeta function, dyadic valuation, and binary digit sum.

4.3 Linear and quadratic heat traces

Two Lambert/heat kernels occur naturally:

$$L_1(t) = \sum_{n \geq 1} a_n e^{-nt} = \sum_{h \geq 0} \frac{1}{e^{2^h t} - 1}, \quad (139)$$

$$K(t) = \sum_{n \geq 1} a_n e^{-n^2 t} = \sum_{h \geq 0} \sum_{m \geq 1} e^{-4^h m^2 t}. \quad (140)$$

Their Mellin transforms are

$$\int_0^\infty L_1(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(s)}{1-2^{-s}}, \quad \Re s > 1, \quad (141)$$

$$\int_0^\infty K(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(2s)}{1-4^{-s}}, \quad \Re s > \frac{1}{2}. \quad (142)$$

The consolidated frontier work develops theta completion and complex-dimension oscillations for K . The linear kernel L_1 will reappear in the dyadic-gamma integral and in the Levy density of the Gaussian-mixture dual.

4.4 Frullani–Lambert integral for the Fourier product

Summing the elementary Frullani identity over the zero divisor gives the mixed integral/product representation

$$\log \Phi(z) = -2 \int_0^\infty (\cosh(z t) - 1) L_1(t) \frac{dt}{t}, \quad |\Re z| < 1. \quad (143)$$

In particular, for real y ,

$$\log \Phi(iy) = 2 \int_0^\infty (1 - \cos(yt)) L_1(t) \frac{dt}{t}. \quad (144)$$

This writes the Wick-rotated product as the exponential of a positive Lambert-kernel energy. It is also the Levy–Khinchine exponent that underlies [Appendix 10](#).

4.5 Fredholm determinant and spectral zeta

Let D be the positive diagonal operator whose eigenvalue n occurs with multiplicity a_n . Since D^{-2} is trace class,

$$\Phi(z) = \det(I - z^2 D^{-2}). \quad (145)$$

The associated spectral zeta function is

$$\zeta_D(s) = \text{tr}(D^{-s}) = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1. \quad (146)$$

The determinant, zero multiplicities, heat trace, and arithmetic Dirichlet series are thus four views of the same spectrum.

5 Fourier inversion, sampling, and interval series

5.1 Whole-line inversion

Because $\text{up} \in C_c^\infty(\mathbb{R})$, its transform is rapidly decreasing on the real axis and

$$\text{up}(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (147)$$

$$= 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (148)$$

Every derivative may be obtained under the integral:

$$\text{up}^{(m)}(x) = \int_{\mathbb{R}} (2\pi i \xi)^m \Phi(\xi) e^{2\pi i x \xi} d\xi. \quad (149)$$

Combining this with (131), (133), or the dyadic-gamma factorization below gives distinct integral–product representations of the same bump.

5.2 Poisson partition and half-integer spectrum

The integer zeros imply the partition of unity

$$\sum_{k \in \mathbb{Z}} \text{up}(x - k) = 1. \quad (150)$$

On $[-1, 1]$, the half-integer samples give the rapidly convergent cosine expansion

$$\text{up}(t) = \frac{1}{2} + \sum_{k \geq 0} \Phi\left(k + \frac{1}{2}\right) \cos((2k + 1)\pi t). \quad (151)$$

The active half-integer spectral report develops weighted Prouhet identities, exact alias transforms, and spectral quadrature from this series.

5.3 Paley–Wiener sampling of the Fourier image

Since up is supported on $[-1, 1]$, Φ belongs to the Paley–Wiener space of bandwidth one. Shannon sampling at spacing $1/2$ gives

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k/2) \text{sinc}(\pi(2z - k)). \quad (152)$$

The rapid decay of the samples makes the series locally uniformly and absolutely convergent. Since $\Phi(m) = 0$ for every nonzero integer m ,

$$\Phi(z) = \text{sinc}(2\pi z) + \sum_{\substack{k \in \mathbb{Z} \\ k \text{ odd}}} \Phi(k/2) \text{sinc}(\pi(2z - k)). \quad (153)$$

Thus the entire transform is reconstructed from its half-integer arithmetic ray.

5.4 Fourier–Legendre representation

The compact interval also supports the uniformly convergent Legendre expansion

$$\text{up}(x) = \sum_{m \geq 0} \frac{2m + 1}{2} \left(\int_{-1}^1 \text{up}(t) P_m(t) dt \right) P_m(x). \quad (154)$$

Evenness removes the odd indices. The coefficient can be expressed either as a moment combination or by inserting (148) and using the Fourier–Bessel transform of P_m . This is a useful bridge between exact moments and orthogonal approximation.

6 Laplace, Mellin, moment, and q -series representations

6.1 Moment-generating and Laplace products

The random series gives the entire moment-generating functions

$$M_Y(t) = \mathbb{E}e^{tY} = \Phi\left(\frac{it}{2\pi}\right) = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j}, \quad (155)$$

$$M_X(t) = \mathbb{E}e^{tX} = \prod_{j \geq 1} \frac{e^{t/2^j} - 1}{t/2^j}, \quad (156)$$

$$L_X(s) = \mathbb{E}e^{-sX} = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (157)$$

For $c > 0$, the CDF has the Bromwich representation

$$F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{L_X(s)}{s} e^{sx} ds. \quad (158)$$

This contour formula is the natural entrance to the lower-Lambert saddle theory.

6.2 Mellin moments, including negative orders

For $\Re\alpha > 0$,

$$\mathbb{E}X^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty s^{\alpha-1} L_X(s) ds. \quad (159)$$

The endpoint flatness makes every negative moment finite. For $0 < \Re\alpha < 1$,

$$\mathbb{E}X^\alpha = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - L_X(s)) s^{-\alpha-1} ds. \quad (160)$$

For integer $m \geq 1$, integration by parts gives the survival representation

$$\mathbb{E}X^m = m \int_0^1 x^{m-1} (1 - F(x)) dx = m \int_0^1 x^{m-1} \text{up}(x) dx. \quad (161)$$

6.3 Cumulants, Bernoulli numbers, and Bell polynomials

From (135), the nonzero cumulants of Y are

$$\kappa_{2r}(Y) = \frac{B_{2r}}{2r(1-4^{-r})} = (-1)^{r+1} \frac{(2r)!\zeta(2r)}{r(1-4^{-r})(2\pi)^{2r}}, \quad \kappa_{2r+1}(Y) = 0. \quad (162)$$

Hence

$$\mathbb{E}Y^{2n} = B_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}), \quad (163)$$

where the B_{2n} on the right is the complete exponential Bell polynomial, not a Bernoulli number. The first values are

$$\mathbb{E}Y^2 = \frac{1}{9}, \quad \mathbb{E}Y^4 = \frac{19}{675}, \quad \mathbb{E}Y^6 = \frac{583}{59535}. \quad (164)$$

Moments of $X = (1 + Y)/2$ follow by binomial expansion.

6.4 Wick rotation, Newton recurrence, and q -MZV formula

Define

$$\mathcal{A}(w) = \Phi(i\sqrt{w}) = \prod_{n \geq 1} \left(1 + \frac{w}{n^2}\right)^{a_n} = \sum_{r \geq 0} C_r w^r. \quad (165)$$

Then

$$C_r = \frac{(2\pi)^{2r}}{(2r)!} \mathbb{E}Y^{2r}. \quad (166)$$

Writing

$$p_r = \sum_{n \geq 1} \frac{a_n}{n^{2r}} = \frac{\zeta(2r)}{1 - 4^{-r}}, \quad (167)$$

Newton's recurrence is

$$rC_r = \sum_{j=1}^r (-1)^{j+1} p_j C_{r-j}, \quad C_0 = 1. \quad (168)$$

At $q = 1/4$, the active frontier report further expands C_r by partitions. If $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash r$, $m_j(\lambda)$ is the multiplicity of part j , and

$$Z_\lambda = \sum_{\substack{n_1, \dots, n_\ell \geq 1 \\ n_i \neq n_j}} \frac{1}{n_1^{2\lambda_1} \dots n_\ell^{2\lambda_\ell}}, \quad (169)$$

then

$$C_r = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} \frac{q^{\lambda_i(\lambda_i-1)/2}}{(q; q)_{\lambda_i}}, \quad q = \frac{1}{4}. \quad (170)$$

Moreover Z_λ is a symmetrized multiple-zeta value. This formula keeps both the integer spectrum and its dyadic layer occupancy visible.

6.5 Appell and logistic representations already in the corpus

The Appell sequence associated with the compact Rvachev law is defined by

$$\frac{e^{xt}}{M_Y(t)} = \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}. \quad (171)$$

The audited Barnes/inverse draft introduces independent standard logistic variables Λ_j and

$$Z = \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}, \quad (172)$$

with

$$\mathbb{E} e^{itZ} = \frac{1}{M_Y(t)}, \quad A_n(x) = \mathbb{E}(x + iZ)^n. \quad (173)$$

This logistic dual is baseline. The new result in [Appendix 10](#) is its exact identification as a Gaussian variance mixture driven by the independently developed Thorin variable.

7 Thue–Morse, dyadic arithmetic, and finite product-to-sum bridges

Let $\varepsilon_k = (-1)^{s_2(k)}$. The finite Prouhet product is

$$\prod_{j=0}^{N-1} (1 - z^{2^j}) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k. \quad (174)$$

With $z = e^t$ this becomes

$$\sum_{k < 2^N} \varepsilon_k e^{kt} = \prod_{j=0}^{N-1} (1 - e^{2^j t}). \quad (175)$$

Coefficient comparison yields

$$\sum_{k < 2^N} \varepsilon_k k^m = 0 \quad (0 \leq m < N), \quad \sum_{k < 2^N} \varepsilon_k k^N = (-1)^N N! 2^{N(N-1)/2}. \quad (176)$$

These cancellations explain the compactness of the truncated-power spline (126); they also power the exact dyadic-value formulae, Gaussian-binomial interpolation, and Richardson extrapolants in the repository.

The finite Fourier product

$$\Phi_N(z) = \prod_{h=0}^{N-1} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) \quad (177)$$

can be converted into a signed exponential sum by expanding the sine factors. Conversely, Prouhet cancellations force high-order zeros in those finite sums. This is the basic product-to-sum bridge connecting Fourier analysis to exact binary arithmetic.

8 Endpoint Laplace phases, Lambert W , and inverse asymptotics

The active frontier synthesis writes

$$g(s) = \log L_X(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + R(\log_2 s) + E(s), \quad (178)$$

where R is real-analytic and one-periodic and

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (179)$$

The CDF integral (158) has saddle equation

$$x = -g'(s), \quad (180)$$

whose exact phase is solved by a lower branch of Lambert W , perturbed by the periodic function and its derivatives. The periodic term is essential: every nonzero Fourier mode is explicitly nonzero in the formal exposition, so replacing it by a constant loses the claimed sharp error scale.

The inverse $Q = F^{-1}$ consequently lives on the stretched-logarithmic scale

$$Q(y) \asymp \sqrt{\log(1/y)} \exp\left[-\sqrt{2 \log 2 \log(1/y)}\right] \quad (y \downarrow 0), \quad (181)$$

with a sharper periodic Lambert expansion in the audited inverse reports. Reflection gives $Q(1-y) = 1 - Q(y)$. These asymptotic representations complement, rather than replace, the exact Fourier and quantile identities developed below.

New theorem packages

9 The dyadic gamma factorization

9.1 Definition, convergence, and poles

Definition 9.1 (Dyadic gamma function [\[proved here\]](#)). For $z \in \mathbb{C}$, away from the negative integers, define

$$\Gamma_{\text{dy}}(z) = \prod_{h=0}^{\infty} \Gamma(1 + 2^{-h}z). \quad (182)$$

Theorem 9.2 (Meromorphy and Mahler equation [\[proved here\]](#)). The product (182) converges locally uniformly away from its poles and defines a meromorphic function satisfying

$$\Gamma_{\text{dy}}(z) = \Gamma(1 + z)\Gamma_{\text{dy}}(z/2), \quad \Gamma_{\text{dy}}(0) = 1. \quad (183)$$

Its pole at $-n$, $n \geq 1$, has exact order $a_n = 1 + v_2(n)$ and it has no other poles.

Proof. Uniformly on compact sets, $\log \Gamma(1 + w) = -\gamma w + \mathcal{O}(w^2)$ as $w \rightarrow 0$. The sums $\sum_h 2^{-h}$ and $\sum_h 4^{-h}$ converge, so the logarithms form a normally convergent series after finitely many factors are removed. Separating the factor $h = 0$ gives (183). At $z = -n$, the factor indexed by h has a simple gamma pole precisely when $n/2^h$ is a positive integer, namely for $0 \leq h \leq v_2(n)$. Their number is $1 + v_2(n)$. $\square$

Formal reciprocal version. In `GeometricReciprocalGamma.lean`, define $G_{\text{dy}}(z) = \text{Fabius.dyadicReciprocalGamma}(z) = \prod_{h \geq 0} \Gamma(1 + 2^{-h}z)^{-1}$ and define the Gamma-side representative $\text{Fabius.dyadicGamma} = G_{\text{dy}}^{-1}$. The reciprocal product is entire and normalized by `Fabius.dyadicReciprocalGamma_differentiable` and `Fabius.dyadicReciprocalGamma_zero`. Its Gamma-side inverse is meromorphic by `Fabius.dyadicGamma_meromorphic`, and the Mahler equation is the specialization of `Fabius.geometricGamma_mahler` at $q = 1/2$. In `DyadicGammaOrder.lean`, `Fabius.dyadicReciprocalGamma_eq_zero_iff` gives exactly the negative-integer zero set; `Fabius.analyticOrderAt_dyadicReciprocalGamma_neg_nat` gives zero order $1 + v_2(n)$, and `Fabius.meromorphicOrderAt_dyadicGamma_neg_nat` records the corresponding pole as meromorphic order $-(1 + v_2(n))$. Mathlib assigns `Complex.Gamma` value zero at a classical pole and also totalizes $0^{-1} = 0$, so the pole is encoded by this negative local order rather than an infinite point value. Neither module proves that the raw product in (182) converges locally uniformly off its poles or equals this meromorphic inverse there; that representation remains frontier material.

9.2 Reflection factorization of the Rvachev transform

Theorem 9.3 (Dyadic gamma factorization [\[proved here\]](#)). For every $z \in \mathbb{C}$,

$$\Phi(z) = \frac{1}{\Gamma_{\text{dy}}(z)\Gamma_{\text{dy}}(-z)}. \quad (184)$$

The equality is entire: at integer z , the poles of the gamma factors produce exactly the zeros of Φ with their arithmetic multiplicities.

Proof. Euler’s reflection formula in the shifted form, understood meromorphically and with the removable value supplied at $w = 0$,

$$\Gamma(1+w)\Gamma(1-w) = \frac{\pi w}{\sin(\pi w)} = \frac{1}{\operatorname{sinc}(\pi w)} \quad (185)$$

gives

$$\frac{1}{\Gamma(1+2^{-h}z)\Gamma(1-2^{-h}z)} = \operatorname{sinc}(\pi z/2^h).$$

Multiplication over $h \geq 0$ yields the sinc product (131). Normal convergence away from the poles and removable continuation at the integers complete the proof. $\square$

Formal version. The pointwise total identity actually used by Lean is

$$\Gamma(1+w)^{-1}\Gamma(1-w)^{-1} = \operatorname{sinc}(\pi w),$$

namely `Fabius.shiftedReciprocalGamma_mul_neg`; unlike the middle quotient in (185), it includes $w = 0$ without a removable-value convention. The entire dyadic identities (184) and (189) have formal analogues after replacing the raw $\Gamma_{\text{dy}}, G_{\text{dy}}$ pair by Lean’s totalized `Fabius.dyadicGamma, Fabius.dyadicReciprocalGamma` pair. Those analogues are respectively `Fabius.rvachevFourierProduct_eq_one_div_dyadicGamma_mul` and `Fabius.dyadicReciprocalGamma_mul_neg`; neither theorem identifies the raw Gamma product with the totalized inverse away from its poles.

Corollary 9.4 (Gamma integral representations of up and F [proved here]). *For real x ,*

$$\operatorname{up}(x) = \int_{\mathbb{R}} \frac{e^{2\pi i x \xi}}{\Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi, \quad (186)$$

and for real x ,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin(2\pi\xi(2x-1))}{\xi\Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi. \quad (187)$$

Both integrals are absolutely convergent after the usual removable interpretation at $\xi = 0$.

Frontier boundary. The reciprocal-Gamma factorization does not by itself provide an exact public Lean declaration for either integral above. Their convergence and their Gamma-form integrands therefore remain unformalized.

9.3 Canonical reciprocal product and power series

Let $G_{\text{dy}}(z) = 1/\Gamma_{\text{dy}}(z)$. Euler’s canonical product for $1/\Gamma$ and regrouping equal zeros give

$$G_{\text{dy}}(z) = e^{2\gamma z} \prod_{n \geq 1} \left[\left(1 + \frac{z}{n}\right) e^{-z/n} \right]^{a_n}. \quad (188)$$

Consequently

$$\Phi(z) = G_{\text{dy}}(z)G_{\text{dy}}(-z), \quad (189)$$

which splits the even zero divisor into a negative-integer factor and its reflection. Equation (189) has the formal analogue `Fabius.dyadicReciprocalGamma_mul_neg` after replacing raw G_{dy} by `Fabius.dyadicReciprocalGamma`; the arithmetic Weierstrass regrouping (188) is not formalized. For $|z| < 1$,

$$\log \Gamma_{\text{dy}}(z) = -2\gamma z + \sum_{m \geq 2} \frac{(-1)^m \zeta(m)}{m(1-2^{-m})} z^m. \quad (190)$$

The even part of (190) is exactly $-\frac{1}{2} \log \Phi(z)$, while the odd part is new information lost in the even Fourier product.

Frontier boundary. No public declaration at checkpoint 71ab6f6728fceb753c88d8b0573077a59acf2682 proves (188) or (190).

9.4 Dyadic digamma and Stieltjes resolvent

Define

$$\Psi_{\text{dy}}(z) = \frac{\Gamma'_{\text{dy}}(z)}{\Gamma_{\text{dy}}(z)}. \quad (191)$$

Termwise differentiation gives

$$\Psi_{\text{dy}}(z) = \sum_{h \geq 0} 2^{-h} \psi(1 + 2^{-h} z), \quad (192)$$

and the Fourier logarithmic derivative becomes

$$\boxed{\frac{\Phi'(z)}{\Phi(z)} = \Psi_{\text{dy}}(-z) - \Psi_{\text{dy}}(z) = -2z \sum_{n \geq 1} \frac{a_n}{n^2 - z^2}.} \quad (193)$$

Thus the arithmetic Stieltjes resolvent is the odd part of a dyadic digamma function. Differentiating repeatedly produces polygamma representations for every logarithmic jet of Φ .

Frontier boundary. The dyadic digamma, resolvent, and polygamma identities in this subsection are not consequences exposed by the reciprocal-product API and remain unformalized.

9.5 Malmsten and heat-trace integrals

For $\Re z > -1$, summing Malmsten's integral for $\log \Gamma(1 + z/2^h)$ gives

$$\log \Gamma_{\text{dy}}(z) = \int_0^\infty \left[2ze^{-t} - \frac{\sum_{h \geq 0} (1 - e^{-2^{-h}zt})}{e^t - 1} \right] \frac{dt}{t}. \quad (194)$$

The inner dyadic exponential sum has the entire expansion

$$\sum_{h \geq 0} (1 - e^{-2^{-h}w}) = \sum_{m \geq 1} \frac{(-1)^{m+1} w^m}{m!(1 - 2^{-m})}. \quad (195)$$

A second integral makes the integer spectrum explicit. From (188) and (139),

$$\boxed{\log \Gamma_{\text{dy}}(z) = -2\gamma z + \int_0^\infty (zt - (1 - e^{-zt})) L_1(t) \frac{dt}{t}, \quad \Re z > -1.} \quad (196)$$

This identity is a direct analytic bridge between the gamma factor, Lambert series, and the zero-divisor heat trace.

Frontier boundary. The Malmsten integral, dyadic exponential series, and heat-trace integral in this subsection have no exact public Lean counterparts.

9.6 Geometric- q extension

Theorem 9.5 (Geometric gamma factorization [\[proved here\]](#)). *Let $0 < q < 1$ and define*

$$\Gamma_{\langle q \rangle}(z) = \prod_{h \geq 0} \Gamma(1 + q^h z), \quad \Phi_q(z) = \prod_{h \geq 0} \operatorname{sinc}(\pi q^h z). \quad (197)$$

Then

$$\Gamma_{\langle q \rangle}(z) = \Gamma(1 + z) \Gamma_{\langle q \rangle}(qz), \quad (198)$$

$$\Phi_q(z) = \frac{1}{\Gamma_{\langle q \rangle}(z) \Gamma_{\langle q \rangle}(-z)}, \quad (199)$$

$$\log \Gamma_{\langle q \rangle}(z) = -\frac{\gamma z}{1 - q} + \sum_{m \geq 2} \frac{(-1)^m \zeta(m)}{m(1 - q^m)} z^m, \quad |z| < 1. \quad (200)$$

For $q = 1/b$ with integer $b \geq 2$, the pole at $-n$ has order $1 + \max\{h : b^h \mid n\}$.

Proof. The convergence argument is identical to the dyadic case, using $\sum q^h$ and $\sum q^{2^h}$. Reflection is applied factor by factor, and the power series follows by summing $\sum_h q^{hm} = (1 - q^m)^{-1}$. $\square$

Remark 9.6. $\Gamma_{\langle q \rangle}$ is not the classical Jackson q -gamma function. It is a geometric Mahler product of ordinary gamma functions. Its denominators $1 - q^m$ nevertheless align exactly with the q -series denominators already pervasive in the Fabius–Rvachev theory.

Formal reciprocal version and boundary. The formal theorem is stronger in its base hypothesis and narrower in analytic content: for every $q \in \mathbb{C}$ with $\|q\| < 1$, including $q = 0$, `Fabius.geometricReciprocalGamma_differentiable` proves that $G_q(z) = \prod_h \Gamma(1 + q^h z)^{-1}$ is entire, `Fabius.geometricReciprocalGamma_mahler` gives its reciprocal Mahler equation, `Fabius.geometricReciprocalGamma_eq_zero_iff` gives the exact affine zero orbit, `Fabius.geometricGamma_meromorphic` and `Fabius.geometricGamma_mahler` give the meromorphic inverse and Gamma-side Mahler equation, and `Fabius.geometricReciprocalGamma_mul_neg` gives the corresponding entire reciprocal analogue of (199). The raw Gamma product in (197) and the logarithmic series (200) are not formalized.

The general integer-base statement is now half formalized, and the halves are worth separating. Its zero set is, at every integer base: for $b \geq 2$,

$$G_{1/b}(z) = 0 \iff z \in \{-1, -2, -3, \dots\},$$

independently of b (`Fabius.geometricReciprocalGamma_inv_natCast_eq_zero_iff`, `ReciprocalIntegerGammaZeros.lean`). `GeometricReciprocalGamma.lean` states the zeros in the *affine* form $\exists n, m, 1 + q^n z = -m$, which is the shape that stays safe at $q = 0$ where no scale can be divided out; at reciprocal-integer base it collapses to the arithmetic one, since $(m + 1)b^n$ runs over the positive integers — surjectively already at $n = 0$.

The *order* is not, and is not a corollary of the set. At $b = 2$ it is obtained by transferring the integer-zero order of the Rvachev product through $G_{1/2}(z)G_{1/2}(-z) = \Phi(z)$, and the corpus has no Φ_b to transfer through at another base; a direct proof would have to split off the finitely many vanishing factors — the h with $b^h \mid n$ — and show the remainder is an analytic unit. Only the dyadic order has an exact counterpart, in `DyadicGammaOrder.lean`. The arithmetic side of the count is available at arbitrary base as `Fabius.weightedScaleMultiplicity`, so what is missing is the analytic splitting, not the combinatorics.

10 Gaussian variance mixtures and the logistic dual

10.1 The existing Thorin variable

The audited frontier corpus already constructs, for $\tau > 0$, a nonnegative random variable V_τ with

$$\mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}, \quad (201)$$

and grouped gamma representation

$$V_\tau \stackrel{d}{=} \sum_{n \geq 1} H_{n,\tau}, \quad H_{n,\tau} \sim \text{Gamma}(\tau a_n, \text{rate } n^2), \quad (202)$$

with independent summands. Its discrete Thorin measure is

$$U_\tau = \tau \sum_{n \geq 1} a_n \delta_{n^2}. \quad (203)$$

The construction and its complete-Bernstein consequences are [\[repository\]](#). We now connect it to the independently developed logistic dual.

10.2 Equality in law

Theorem 10.1 (Gaussian variance-mixture bridge [\[proved here\]](#)). *Let*

$$T_\tau = \frac{V_\tau}{4\pi^2}, \quad (204)$$

and let $N \sim \mathcal{N}(0, 1)$ be independent of T_τ . Define

$$Z_\tau = \sqrt{2T_\tau} N = \frac{\sqrt{V_\tau}}{\sqrt{2}\pi} N. \quad (205)$$

Then

$$\mathbb{E}e^{itZ_\tau} = \mathcal{A}\left(\frac{t^2}{4\pi^2}\right)^{-\tau}, \quad (206)$$

$$= \prod_{n \geq 1} \left(1 + \frac{t^2}{4\pi^2 n^2}\right)^{-\tau a_n}, \quad (207)$$

$$= \left[\prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} \right]^\tau, \quad (208)$$

$$= \left[\Gamma_{\text{dy}}\left(\frac{it}{2\pi}\right) \Gamma_{\text{dy}}\left(-\frac{it}{2\pi}\right) \right]^\tau. \quad (209)$$

For $\tau = 1$,

$$\boxed{\sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j} \stackrel{d}{=} \frac{\sqrt{V_1}}{\sqrt{2}\pi} N,} \quad (210)$$

where the Λ_j are independent standard logistic variables.

Proof. Conditioning on T_τ gives

$$\mathbb{E}(e^{it\sqrt{2T_\tau}N} \mid T_\tau) = e^{-t^2 T_\tau}.$$

Taking expectation and applying (201) at $w = t^2/(4\pi^2)$ proves (206); (207) follows from (165). Since $\mathcal{A}(t^2/(4\pi^2)) = M_Y(t)$, the reciprocal hyperbolic-sine product (208) follows from (155). Finally, (184) at $z = it/(2\pi)$ gives (209). At $\tau = 1$, the right-hand side is exactly the characteristic function of the repository's logistic series (172); uniqueness of characteristic functions proves (210). $\square$

Formalization status. The reciprocal-Gamma/Rvachev factor used in the last rewrite is formal, but the Gaussian variance-mixture theorem and its Gamma-form characteristic function do not have exact public Lean counterparts at the audited checkpoint.

The bridge is not a restatement of generalized gamma convolution or of the logistic series. It identifies two previously separate noncompact duals of the compact Rvachev law: one built additively from logistic coordinates and the other obtained by subordinating a Gaussian by the arithmetic Thorin variable.

10.3 Convolution semigroup, density, and moments

The family $(Z_\tau)_{\tau>0}$ is a symmetric convolution semigroup:

$$Z_{\tau_1} + Z'_{\tau_2} \stackrel{d}{=} Z_{\tau_1+\tau_2}. \quad (211)$$

Its density is the exact normal scale mixture

$$p_\tau(x) = \mathbb{E} \left[\frac{1}{\sqrt{4\pi T_\tau}} \exp \left(-\frac{x^2}{4T_\tau} \right) \right]. \quad (212)$$

The cumulants are

$$\kappa_{2r}(Z_\tau) = \tau \frac{(2r)!}{r(4\pi^2)^r} \frac{\zeta(2r)}{1-4^{-r}}, \quad \kappa_{2r+1}(Z_\tau) = 0. \quad (213)$$

In particular, $\text{Var } Z_\tau = \tau/9$. Conditional Gaussian moments yield

$$\mathbb{E} Z_\tau^{2m} = 2^m (2m-1)!! \mathbb{E} T_\tau^m, \quad (214)$$

where

$$\kappa_r(T_\tau) = \tau(r-1)! \frac{\zeta(2r)}{(4\pi^2)^r (1-4^{-r})}. \quad (215)$$

Thus (214) and complete Bell polynomials give a second moment calculus for the logistic/Appell dual.

10.4 Closed Levy density and self-decomposability

Theorem 10.2 (Levy density of the dual semigroup [\[proved here\]](#)). *The characteristic exponent of Z_τ is*

$$-\log \mathbb{E} e^{itZ_\tau} = 2\tau \int_0^\infty (1 - \cos(tx)) L_1(2\pi x) \frac{dx}{x}. \quad (216)$$

Consequently its symmetric Levy measure has density

$$\nu_\tau(x) = \frac{\tau}{|x|} L_1(2\pi|x|) = \frac{\tau}{|x|} \sum_{h \geq 0} \frac{1}{e^{2^{h+1}\pi|x|} - 1}, \quad x \neq 0. \quad (217)$$

Every Z_τ is self-decomposable.

Proof. Use

$$\log \left(1 + \frac{t^2}{\lambda^2} \right) = 2 \int_0^\infty (1 - \cos(tx)) e^{-\lambda x} \frac{dx}{x}$$

in (207), then sum over n . This proves (216) and (217). A symmetric infinitely divisible law with Levy density $k(|x|)/|x|$ is self-decomposable when k is decreasing. Here $k(x) = \tau L_1(2\pi x)$ is a positive sum of decreasing exponentials. $\square$

10.5 Complete monotonicity in the squared coordinate

Corollary 10.3 (Radial positivity in every dimension [\[proved here\]](#)). *For every $\tau > 0$, the function*

$$g_\tau(s) = p_\tau(\sqrt{s}), \quad s > 0, \quad (218)$$

is completely monotone:

$$(-1)^m g_\tau^{(m)}(s) \geq 0 \quad (m \geq 0, s > 0). \quad (219)$$

Consequently $x \mapsto p_\tau(\|x\|)$ is positive definite on $\mathbb{R}^d$ for every $d \geq 1$.

Proof. Formula (212) writes

$$g_\tau(s) = \mathbb{E} \left[(4\pi T_\tau)^{-1/2} e^{-s/(4T_\tau)} \right],$$

a positive mixture of exponentials in s . Bernstein's theorem gives complete monotonicity, and Schoenberg's theorem converts complete monotonicity of g_τ into radial positive definiteness of $g_\tau(\|x\|^2) = p_\tau(\|x\|)$ in every dimension. $\square$

11 Jensen means, factorial products, and binary digit sums

11.1 Exact circular mean

Define the Jensen mean

$$J(r) = \frac{1}{2\pi} \int_0^{2\pi} \log \left| \Phi(re^{i\theta}) \right| d\theta, \quad r > 0, \quad (220)$$

with the improper interpretation when the circle passes through an integer zero.

Theorem 11.1 (Arithmetic Jensen formula [\[proved here\]](#)). *Let $N = \lfloor r \rfloor$ and $N_j = \lfloor N/2^j \rfloor$. Then*

$$J(r) = 2 \sum_{n \leq N} a_n \log \frac{r}{n}, \quad (221)$$

$$= 2 \sum_{j \geq 0} \left[N_j \log \frac{r}{2^j} - \log \Gamma(N_j + 1) \right]. \quad (222)$$

The second sum is finite. Equivalently,

$$\exp \left(\frac{J(r)}{2} \right) = \prod_{n \leq N} \left(\frac{r}{n} \right)^{a_n} = \prod_{j \geq 0} \frac{(r/2^j)^{N_j}}{N_j!}. \quad (223)$$

Proof. Jensen's formula for the entire function Φ , normalized by $\Phi(0) = 1$, says that the mean is the sum of $\log(r/|\rho|)$ over zeros ρ in the disk, counted with multiplicity. The zeros occur at $\pm n$ with multiplicity a_n , giving (221). Expand $a_n = \sum_{j \geq 0} \mathbf{1}_{2^j | n}$, write $n = 2^j m$, and sum $\log(r/(2^j m))$ for $1 \leq m \leq N_j$. This proves (222) and (223). $\square$

11.2 Digit-sum derivative

Corollary 11.2 (Binary slope law [\[proved here\]](#)). For $N < r < N + 1$,

$$J'(r) = \frac{2A(N)}{r} = \frac{2(2N - s_2(N))}{r}. \quad (224)$$

Thus

$$\frac{rJ'(r)}{2} = 2 \lfloor r \rfloor - s_2(\lfloor r \rfloor) \quad (225)$$

recovers the binary digit sum directly from a circular mean of the Fourier image.

Remark 11.3. The zero-counting law $A(N) = 2N - s_2(N)$ was already in the repository. The new point is that it is the logarithmic radial derivative of a two-dimensional complex-analytic average, and that the full mean has the factorial product (223).

11.3 Asymptotic growth

Theorem 11.4 (Leading Jensen asymptotic [\[proved here\]](#)). As $r \rightarrow \infty$,

$$J(r) = 4r - \frac{(\log r)^2}{2 \log 2} + \mathcal{O}(\log r). \quad (226)$$

Proof. Let $N = \lfloor r \rfloor$. Partial summation applied to (221) gives

$$\frac{J(r)}{2} = A(N) \log(r/N) + \int_1^N \frac{A(\lfloor t \rfloor)}{t} dt.$$

The first term is $\mathcal{O}(1)$. Since $A(n) = 2n - s_2(n)$, the linear part contributes $2N + \mathcal{O}(\log N)$. The elementary summatory digit estimate

$$\sum_{n \leq x} s_2(n) = \frac{x \log x}{2 \log 2} + \mathcal{O}(x)$$

and another partial summation give

$$\int_1^N \frac{s_2(\lfloor t \rfloor)}{t} dt = \frac{(\log N)^2}{4 \log 2} + \mathcal{O}(\log N).$$

Multiply by two and replace N with r . □

11.4 Subunit circular energy

For $0 \leq r < 1$, Jensen's mean vanishes because the disk contains no zeros. The angular fluctuation nevertheless has an exact zeta-series energy. Put

$$c_m = \frac{\zeta(2m)}{m(1 - 4^{-m})}. \quad (227)$$

Then (135) and trigonometric orthogonality give

$$\frac{1}{2\pi} \int_0^{2\pi} \left(\log |\Phi(re^{i\theta})| \right)^2 d\theta = \frac{1}{2} \sum_{m \geq 1} c_m^2 r^{4m}. \quad (228)$$

More generally, for $0 \leq r, s < 1$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |\Phi(re^{i\theta})| \log |\Phi(se^{i\theta})| d\theta = \frac{1}{2} \sum_{m \geq 1} c_m^2 (rs)^{2m}. \quad (229)$$

This is a local angular covariance kernel for the Fourier determinant.

12 Non-holonomicity from the arithmetic zero divisor

Theorem 12.1 (The Rvachev transform is not D-finite [\[proved here\]](#)). *The entire function Φ does not satisfy any nontrivial linear differential equation*

$$p_R(z)y^{(R)} + p_{R-1}(z)y^{(R-1)} + \cdots + p_0(z)y = 0 \quad (230)$$

with polynomial coefficients $p_j \in \mathbb{C}[z]$. Equivalently, Φ is not holonomic over $\mathbb{C}(z)$.

Proof. Suppose (230) holds with $p_R \neq 0$. At any ordinary point z_0 with $p_R(z_0) \neq 0$, uniqueness for the associated analytic initial-value problem implies that a nonzero solution cannot have a zero of multiplicity R or larger: if the first R derivatives vanish, the solution is identically zero near z_0 .

The polynomial p_R has only finitely many zeros. But Φ has a zero at 2^k of multiplicity $k + 1$. Choose $k \geq R$ such that $p_R(2^k) \neq 0$. This gives an ordinary zero of multiplicity at least R , a contradiction. $\square$

Corollary 12.2 (Coefficient and algebraic consequences [\[proved here\]](#)). *The Maclaurin coefficient sequence of Φ is not P-recursive. In particular, Φ is not algebraic over $\mathbb{C}(z)$.*

Proof. A power series is D-finite if and only if its coefficient sequence is P-recursive. Every algebraic power series is D-finite. $\square$

Proposition 12.3 (The dyadic gamma factors are not D-finite [\[proved here\]](#)). *Neither Γ_{dy} nor $G_{\text{dy}} = 1/\Gamma_{\text{dy}}$ is D-finite.*

Proof. A D-finite germ has only finitely many possible finite singularities, whereas Γ_{dy} has poles at every negative integer. The reciprocal G_{dy} is entire but has a zero at -2^k of multiplicity $k + 1$, so the preceding multiplicity argument applies. $\square$

Remark 12.4. Non-D-finiteness is much weaker than differential transcendence. The latter is stated only as a conjecture in [Appendix 17](#).

Formalization status. Exact dyadic zero and pole orders are now available in `DyadicGammaOrder` .lean, but the repository still has no D-finite/P-recursive analytic ODE framework proving this proposition. It therefore remains a frontier result.

13 Fourier representations of the bounded Fabius function

13.1 The density transform

Let $f_X = F' = 2 \operatorname{up}(2x - 1)$ on $[0, 1]$ and zero outside. A change of variables gives

$$\widehat{f_X}(\xi) = e^{-\pi i \xi} \Phi(\xi/2). \quad (231)$$

Hence the compactly supported pieces

$$F_0(x) = F(x) \mathbf{1}_{[0,1]}(x), \quad S_0(x) = (1 - F(x)) \mathbf{1}_{[0,1]}(x) \quad (232)$$

have, for $\xi \neq 0$,

$$\widehat{F_0}(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (233)$$

$$\widehat{S_0}(\xi) = \frac{1 - e^{-\pi i \xi} \Phi(\xi/2)}{2\pi i \xi}. \quad (234)$$

Both values at zero are $1/2$. Inserting (131), (133), or (184) turns these into explicit product representations of the Fourier image of the bounded CDF.

13.2 The global CDF as a tempered distribution

Let H be the Heaviside distribution. Since $F = H * f_X$,

$$\widehat{H} = \frac{1}{2}\delta_0 + \frac{1}{2\pi i} \text{pv} \frac{1}{\xi} \quad (235)$$

gives the distributional formula

$$\widehat{F} = \frac{1}{2}\delta_0 + \frac{1}{2\pi i} \text{pv} \left(\frac{e^{-\pi i \xi} \Phi(\xi/2)}{\xi} \right). \quad (236)$$

This is the correct Fourier image of the globally bounded, nonintegrable CDF. Omitting the Dirac mass or principal value changes the distribution.

13.3 Sine integral and odd-harmonic series

Since $F(x) = \frac{1}{2} + \int_0^{2x-1} \text{up}(y) dy$, Fourier inversion yields

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(\xi) \frac{\sin(2\pi\xi(2x-1))}{\xi} d\xi. \quad (237)$$

This holds for all real x under the bounded-CDF convention. Integrating the interval cosine series (151) gives a second representation.

Theorem 13.1 (Odd-harmonic Fabius series [\[proved here\]](#)). For $0 \leq x \leq 1$,

$$F'(x) = 1 - 2 \sum_{k \geq 0} \Phi\left(k + \frac{1}{2}\right) \cos(2\pi(2k+1)x), \quad (238)$$

$$F(x) = x - \frac{1}{\pi} \sum_{k \geq 0} \frac{\Phi(k+1/2)}{2k+1} \sin(2\pi(2k+1)x). \quad (239)$$

Both series converge absolutely and uniformly together with every differentiated series allowed by the displayed factors.

Proof. Substitute $t = 2x - 1$ into (151) and multiply by two. Since

$$\cos((2k+1)\pi(2x-1)) = -\cos(2\pi(2k+1)x),$$

we obtain (238). Integrating from 0 to x and using $F(0) = 0$ gives (239). Rapid decay of the half-integer samples follows from $\text{up} \in C_c^\infty$. $\square$

13.4 Gamma versions

The new factorization makes the preceding formulae explicitly gamma-theoretic:

$$\widehat{f_X}(\xi) = \frac{e^{-\pi i \xi}}{\Gamma_{\text{dy}}(\xi/2)\Gamma_{\text{dy}}(-\xi/2)}, \quad (240)$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin(2\pi\xi(2x-1))}{\xi \Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi. \quad (241)$$

Thus ordinary gamma functions, assembled on a dyadic scale, provide direct integral and Fourier images for the Fabius CDF.

Frontier boundary. Although the common reciprocal-Gamma factor in these displays is formal, no public declaration states either Gamma-version Fourier formula above. They remain frontier claims together with the integral representations in (186)–(187).

14 Quantile and inverse-Fabius transform representations

14.1 Quantile pushforward and product transforms

Let $Q = F^{-1}$ be the clamped inverse. The current formal exposition proves the support-free quantile substitution theorem

$$\int_0^1 \varphi(Q(y)) \, dy = \mathbb{E}\varphi(X) \quad (242)$$

for a broad Banach-valued class of integrands. Therefore

$$\int_0^1 e^{-sQ(y)} \, dy = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad (243)$$

$$\int_0^1 e^{s(Q(y)-1/2)} \, dy = \prod_{j \geq 1} \frac{\sinh(s/2^{j+1})}{s/2^{j+1}}, \quad (244)$$

$$\int_0^1 e^{-2\pi i \xi Q(y)} \, dy = e^{-\pi i \xi} \Phi(\xi/2). \quad (245)$$

For $0 < \Re \alpha < 1$,

$$\int_0^1 Q(y)^\alpha \, dy = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty \left[1 - \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j} \right] t^{-\alpha-1} \, dt. \quad (246)$$

These are **[derived synthesis]** consequences of the now-formal quantile theorem.

14.2 A nonlinear Fourier transform of the inverse

Theorem 14.1 (Inverse-transform identity **[proved here]**). For $\eta \in \mathbb{C} \setminus \{0\}$,

$$\int_0^1 Q(y) e^{-2\pi i \eta y} \, dy = \int_0^1 x F'(x) e^{-2\pi i \eta F(x)} \, dx, \quad (247)$$

$$= \frac{\int_0^1 e^{-2\pi i \eta F(x)} \, dx - e^{-2\pi i \eta}}{2\pi i \eta}. \quad (248)$$

The value at $\eta = 0$ is $1/2$ by continuity.

Proof. Apply $y = F(x)$, $dy = F'(x) \, dx$, and $Q(F(x)) = x$ to obtain (247). Since $\frac{d}{d\eta} e^{-2\pi i \eta F(x)} = -2\pi i F'(x) e^{-2\pi i \eta F(x)}$, integration by parts gives (248). $\square$

14.3 Sine series of the inverse in terms of F

Theorem 14.2 (Nonlinear inverse-Fabius sine series [\[proved here\]](#)). *Put*

$$c_n = \frac{1}{\pi n} \int_0^1 \cos(2\pi n F(x)) \, dx, \quad n \geq 1. \quad (249)$$

Then

$$Q(y) = y + \sum_{n \geq 1} c_n \sin(2\pi n y), \quad 0 \leq y \leq 1. \quad (250)$$

The series converges in $L^2(0, 1)$ and pointwise at every $y \in [0, 1]$. Moreover $c_n = o(1/n)$.

Proof. The function $g(y) = Q(y) - y$ is continuous, vanishes at both endpoints, and is of bounded variation. The symmetry $Q(1 - y) = 1 - Q(y)$ gives $g(1 - y) = -g(y)$, so only the modes $\sin(2\pi n y)$ occur. Their coefficients are

$$2 \int_0^1 g(y) \sin(2\pi n y) \, dy.$$

For the Q term, use $y = F(x)$ and integrate by parts:

$$\begin{aligned} \int_0^1 Q(y) \sin(2\pi n y) \, dy &= \int_0^1 x F'(x) \sin(2\pi n F(x)) \, dx \\ &= -\frac{1}{2\pi n} + \frac{1}{2\pi n} \int_0^1 \cos(2\pi n F(x)) \, dx. \end{aligned}$$

Since $\int_0^1 y \sin(2\pi n y) \, dy = -1/(2\pi n)$, the difference is exactly (249). Dirichlet–Jordan convergence applies to the bounded-variation function g . Finally,

$$\int_0^1 \cos(2\pi n F(x)) \, dx = \int_0^1 \cos(2\pi n y) Q'(y) \, dy$$

with $Q' \in L^1(0, 1)$; the Riemann–Lebesgue lemma gives $c_n = o(1/n)$. $\square$

Remark 14.3. Formula (250) is not an asymptotic reversion. It is a global interval representation of the inverse in which every coefficient is a nonlinear oscillatory integral of the original CDF. It therefore complements the repository’s local Lambert–Bell inverse expansions.

15 Compact representation atlas

The following table is a lookup index rather than a proof substitute. It collects the main representations developed above and the principal established forms in the audited corpus. The domain column records the natural interpretation: pointwise, entire, distributional, probabilistic, or asymptotic.

Object	Representation	Domain	Status
F	$F(x) = \mathbb{P}(\sum_{j \geq 1} 2^{-j} V_j \leq x)$	all $x \in \mathbb{R}$	[repo]
up	$\mathbb{P}(Y \in dy) = \text{up}(y) \, dy, Y = \sum 2^{-j} U_j$	probability	[repo]
F, up	$F'(x) = 2 \text{up}(2x - 1)$	$0 < x < 1$	[repo]
up	$\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(t) \, dt$	pointwise	[repo]
up'	$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1)$	pointwise	[repo]
F	fixed integral equation (115)	$[0, 1]$	[repo]

Object	Representation	Domain	Status
$\mathcal{F}^{(m)}$	$2^{\binom{m+1}{2}} \mathcal{F}(2^m x)$	all orders	[repo]
up	$*_{j \geq 1} (2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]})$	L^1 /probability	[repo]
u_N	finite box convolution (121)	compact spline	[repo]
u_N	Thue–Morse truncated powers (126)	pointwise	[repo]
up	$u_N * (2^N \text{up}(2^N \cdot))$	exact head–tail	[repo]
F	expectation/cube recursion (124)	exact	[repo]
Φ	$\prod_{h \geq 0} \text{sinc}(\pi z / 2^h)$	entire	[repo]
Φ	$\prod_{m \geq 1} (\cos(\pi z / 2^m))^m$	entire	[repo]
Φ	$\prod_{n \geq 1} (1 - z^2 / n^2)^{1+v_2(n)}$	entire	[repo]
Φ	$[\Gamma_{\text{dy}}(z) \Gamma_{\text{dy}}(-z)]^{-1}$	entire	[new]
Φ	$\det(I - z^2 D^{-2})$	Fredholm	[repo]
$\log \Phi$	zeta series (135)	$ z < 1$	[repo]
Φ' / Φ	arithmetic resolvent (136)	off zeros	[repo]
$\log \Phi$	Frullani–Lambert integral (143)	$ \Re z < 1$	[derived]
up	Fourier inversion (147)	all real x	[repo]
up	cosine integral (148)	all real x	[repo]
up	half-integer cosine series (151)	$[-1, 1]$	[repo]
Φ	Shannon sampling (153)	all $z \in \mathbb{C}$	[repo]
up	$\sum_{k \in \mathbb{Z}} \text{up}(x - k) = 1$	all x	[repo]
up	Legendre series (154)	$[-1, 1]$	[repo]
M_Y	$\prod_{j \geq 1} \sinh(t/2^j) / (t/2^j)$	entire	[repo]
M_X	$\prod_{j \geq 1} (e^{t/2^j} - 1) / (t/2^j)$	entire	[repo]
L_X	$\prod_{j \geq 1} (1 - e^{-s/2^j}) / (s/2^j)$	entire in s	[repo]
F	Bromwich integral (158)	$c > 0$	[repo]
$\mathbb{E}X^{-\alpha}$	Mellin–Laplace integral (159)	$\Re \alpha > 0$	[derived]
$\mathbb{E}X^\alpha$	subtracted Mellin integral (160)	$0 < \Re \alpha < 1$	[derived]
$\mathbb{E}Y^{2n}$	Bernoulli cumulants and Bell polynomial (163)	exact	[repo]
C_r	Newton recurrence (168)	exact	[repo]
C_r	q -binomial/MZV partition sum (170)	$q = 1/4$	[repo]
F	density transform (231)	entire in ξ	[new]
F_0	compact transform (233)	$\xi \in \mathbb{R}$	[new]
F	tempered transform (236)	$\mathcal{S}'$	[new]
F	sine integral (237)	all real x	[new]
F	odd-harmonic sine series (239)	$[0, 1]$	[new]
Γ_{dy}	gamma product (182)	meromorphic	[new]
$1/\Gamma_{\text{dy}}$	arithmetic canonical product (188)	entire	[new]
Γ_{dy}	heat-trace integral (196)	$\Re z > -1$	[new]
V_τ	gamma convolution (202)	probability	[repo]
Z	dyadic logistic sum (172)	probability	[repo]
Z_τ	Gaussian variance mixture (205)	probability	[new]
Z_τ	gamma/sinc characteristic products (208)–(209)	real t	[new]
Z_τ	normal-mixture density (212)	real x	[new]
Z_τ	Levy density (217)	$x \neq 0$	[new]
$J(r)$	zero sum and factorial product (221)–(223)	$r > 0$	[new]
$J'(r)$	$(4 \lfloor r \rfloor - 2s_2(\lfloor r \rfloor)) / r$	noninteger r	[new]
$J(r)$	$4r - (\log r)^2 / (2 \log 2) + \mathcal{O}(\log r)$	$r \rightarrow \infty$	[new]

Object	Representation	Domain	Status
Q	quantile pushforward (242)	measurable tests	[repo]
Q	Laplace/product transform (243)	complex s	[derived]
Q	nonlinear Fourier transform (248)	$\eta \neq 0$	[new]
Q	sine series (250)	$[0, 1]$	[new]
F, Q	periodic lower-Lambert endpoint expansions	endpoint	[repo]

The tags in this table record the audited-corpus novelty boundary, not Lean status. At formal checkpoint 71ab6f6728fceb753c88d8b0573077a59acf2682, the reflected reciprocal-Gamma row for Φ , the entire reciprocal product, its Mahler and zero-orbit laws, and the exact dyadic zero/pole orders are formal. The raw meromorphic Gamma product in (182), the arithmetic canonical row (188), and the heat-trace row remain frontier material, as do the Gamma-form probability and Fourier consequences unless named separately in the primary coverage map.

16 Numerical experiments and reproducibility

The package contains `numerical_experiments.py`, written to be run without network access. It uses double precision, a fixed seed 20260827, vectorized dyadic products, and explicit tail diagnostics. The program creates the figures in this section and writes a plain-text audit trail to `figures/numerical_results.txt`.

16.1 Dyadic gamma factorization

Three complex test points were evaluated with sixty product layers. The absolute errors in (184) were between 1.8×10^{-15} and 3.6×10^{-15} ; the Mahler functional equation (183) was verified at comparable accuracy. The finite-layer identity is exact mathematically, so the plateau in Fig. 3 is floating-point accumulation rather than truncation error. The reciprocal identity and the totalized Gamma-denominator form are now proved independently in `GeometricReciprocalGamma.lean`; this figure remains a reproducibility check, not the formal justification.

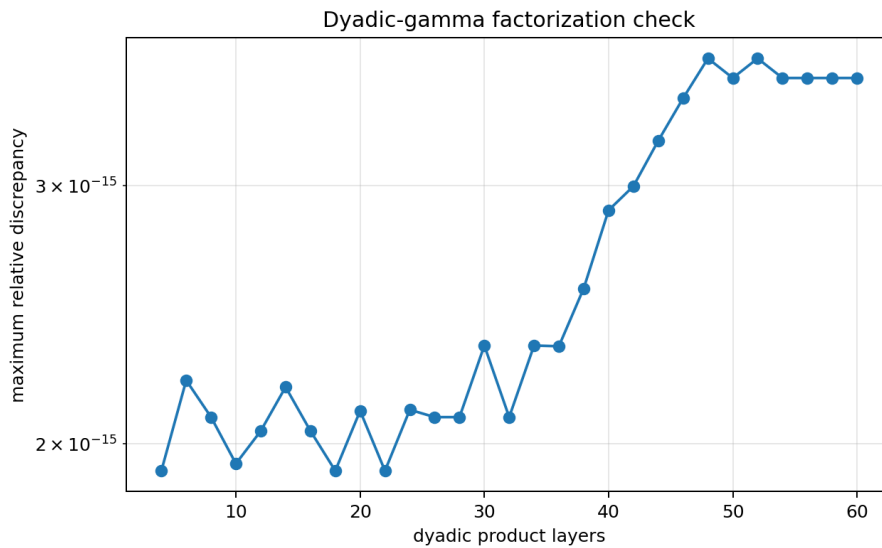


Figure 3: Finite-layer check of the dyadic-gamma reflection factorization.

16.2 Jensen mean

Angular trapezoidal quadrature with 8192 nodes was compared with the exact zero-divisor formula on 76 radii. The maximum absolute discrepancy was 7.11×10^{-15} . Representative values are

r	angular quadrature	exact formula	error
0.6	-3.6082×10^{-16}	0	-3.61×10^{-16}
1.4	0.6729444732424255	0.6729444732424258	-3.33×10^{-16}
3.7	5.496849257625430	5.496849257625430	0
10.3	26.15154490220755	26.15154490220754	3.55×10^{-15}

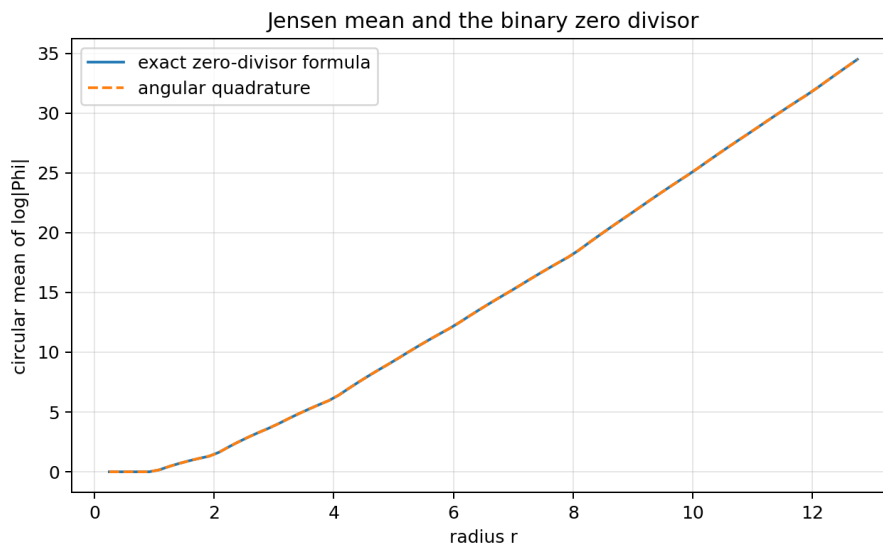


Figure 4: Jensen’s angular mean versus the exact factorial/zero-divisor expression.

16.3 Logistic series versus Gaussian mixture

The simulation used 100,000 samples. The logistic series retained sixteen dyadic layers; its omitted standard deviation was 5.09×10^{-6} . The gamma mixture retained the first 250 spectral sites and added the exact mean of the positive tail. The tail standard deviation was 2.07×10^{-4} , negligible compared with the exact standard deviation $1/3$ of Z_1 . The two-sample Kolmogorov–Smirnov statistic was 0.00347 with p-value 0.5825. Sample variances were 0.1114725 and 0.1110627, compared with the exact $1/9 = 0.111111 \dots$

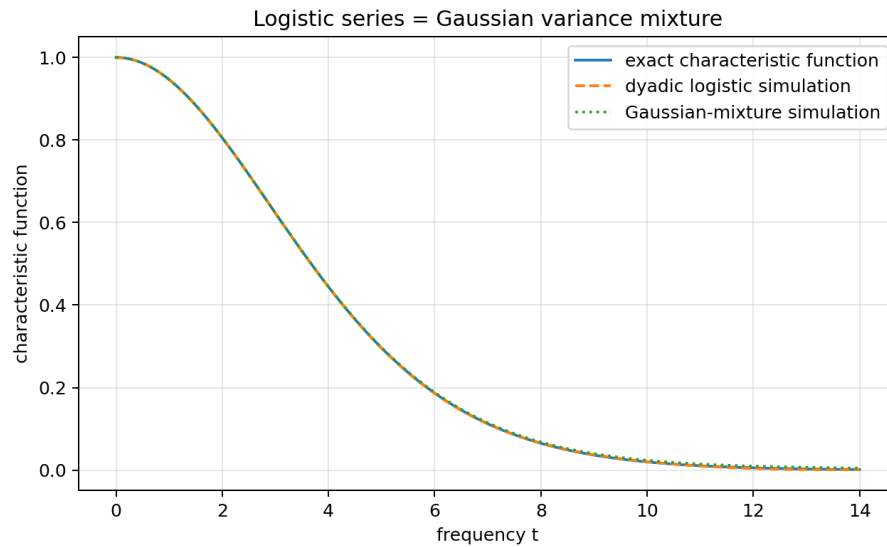


Figure 5: The exact characteristic function and two independent simulations of the law in (210).

16.4 Fabius sine series

The series (239) was compared with the independent Fourier integral (237). At $x = 0.25$, for example,

$$F(0.25) = 0.06944444445136150 \quad (\text{integral}), \quad 0.06944444444444434 \quad (128 \text{ harmonics}).$$

The discrepancy is governed by the numerical quadrature floor after roughly thirty-two harmonics. The rapid early convergence is visible in Fig. 6.

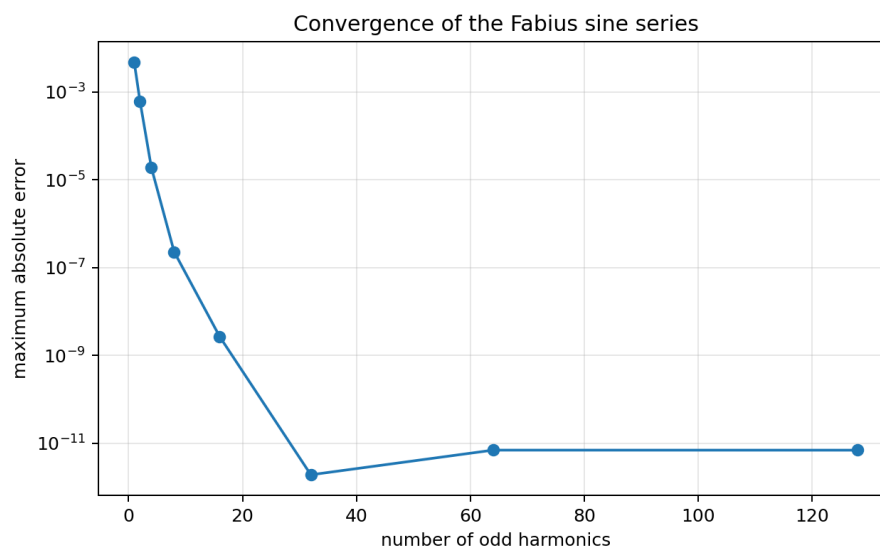


Figure 6: Maximum error of the odd-harmonic Fabius series against Fourier inversion on ten sample points.

16.5 Reproduction command

From the directory containing this consolidated report, run

```
python3 assets/fabius_rvachev_representation_frontier/numerical_experiments.py
```

The script creates no hidden cache and writes all outputs beneath `assets/fabius_rvachev_representation_frontier/figures/`. The source comments explain the product truncations, gamma-tail correction, and quadrature splitting at integer zeros.

17 Conjectures and testable frontier statements

Conjecture 17.1 (Differential transcendence of the Fourier determinant **[conjecture]**). The functions Φ , G_{dy} , and Γ_{dy} are differentially transcendental over $\mathbb{C}(z)$: none satisfies a nonzero algebraic differential equation with coefficients in $\mathbb{C}(z)$.

The non-D-finite theorem is evidence but does not approach the full differential-algebraic claim. A promising route is to combine the Mahler equation (183) with differential Galois theory for mixed differential–difference systems and the nonperiodic pole multiplicities.

Conjecture 17.2 (Full Delange expansion of the Jensen mean **[conjecture]**). There exist explicit continuous one-periodic functions P_1, P_0 such that

$$J(r) = 4r - \frac{(\log r)^2}{2 \log 2} + (\log r)P_1(\log_2 r) + P_0(\log_2 r) + o(1), \quad (251)$$

away from a controlled boundary layer around the integers. The functions should be computable from the Delange Fourier series for the binary digit sum and from the fractional part $r - \lfloor r \rfloor$.

Conjecture 17.3 (Asymptotics of inverse Fourier coefficients **[conjecture]**). Let c_n be defined by (249). Then there is a bounded, nonconstant, phase-dependent function P such that

$$\pi n c_n = Q(1/n) (P(\log_2 \lambda_n) + o(1)), \quad (252)$$

where λ_n is the exact lower-Lambert saddle phase for $F(1/n)$. In particular, $\log |n c_n|$ should have leading scale $-\sqrt{2 \log 2 \log n}$ with the same periodic modulation that governs the inverse endpoint.

This statement is testable by computing the oscillatory integral in (249) with the repository’s certified endpoint saddle machinery. It would turn the local inverse asymptotic into a global spectral law for Q .

Conjecture 17.4 (Hyperbolic complete monotonicity of the mixture density **[conjecture]**). For every $\tau > 0$, the function $s \mapsto p_\tau(\sqrt{s})$ is hyperbolically completely monotone, not merely completely monotone. Equivalently, the normal variance mixture may lie in the Bondesson class with a stronger multiplicative total-positivity structure inherited from the discrete Thorin measure.

Conjecture 17.5 (Arithmetic special values of the dyadic gamma function **[conjecture]**). At rational arguments with odd denominator, suitable products of $\Gamma_{\text{dy}}(r)$, powers of 2π , and classical Barnes multiple-gamma constants reduce to a finite family of dyadic L -values and logarithmic q -products. The functional equation (183) should produce explicit distribution relations indexed by the binary orbit of r .

Conjecture 17.6 (Optimal rational approximation from the Stieltjes resolvent **[conjecture]**). Diagonal Padé approximants obtained from $\Psi_{\text{dy}}(-z) - \Psi_{\text{dy}}(z)$ provide alternating, globally sharp rational bounds for $\Phi(iy)$ and for the characteristic function of Z_τ on $y > 0$, with error constants controlled explicitly by the next unused spectral atom.

18 Promising research directions

18.1 Formalization roadmap

The shortest Lean route to the new results is modular.

1. *Reciprocal tranche delivered.* `ScaledInfiniteProducts.lean`, `GeometricReciprocalGamma.lean`, and `DyadicGammaOrder.lean` construct the entire reciprocal product, prove its generic Mahler/zero/reflection laws, identify its dyadic reflection with Φ , and give exact dyadic zero and pole orders. Still open here is the distinct raw-product claim: construct $\prod_h \Gamma(1 + 2^{-h}z)$ locally uniformly away from its poles and identify it there with the formal meromorphic inverse.
2. *First-jet/tower follow-up delivered.* `ReciprocalGammaJets.lean` proves the exact first jet, simple analytic order, and punctured local coefficient of reciprocal Gamma at every nonpositive integer. `ThueMorseGammaTower.lean` transfers those jets to every trivial zero of the entire Dirichlet–Mellin continuation and proves the Mellin/integral `GammaLog`, dyadic log/tower, and level-zero master-product ratio laws. Its definitions are total in the real parameter, while the theorems use positive parameters; the remaining parameter-differential ladder requires differentiation under the Mellin integral.
3. Reuse the formal zero-multiplicity theorem to prove non-D-finiteness in a small analytic-ODE module.
4. Build Z_τ by conditioning a standard Gaussian on the already developed V_τ , then identify characteristic functions.
5. Formalize Jensen’s formula for the canonical product, followed by the finite factorial regrouping. The digit-sum derivative then becomes arithmetic simplification.
6. Derive the compact and distributional Fourier transforms of F from the existing transform of up and a formal Heaviside distribution library.
7. Use the support-free quantile theorem and bounded-variation Fourier convergence to formalize (248) and (250).

18.2 Spectral geometry and complex analysis

The exact Jensen mean suggests treating Φ as an arithmetic spectral determinant in the complex plane, not only on the real axis. Natural next objects include logarithmic capacity, angular large deviations of $\log |\Phi|$, Nevanlinna characteristics refined by binary digit sums, and de Branges spaces generated by the reflected factors $G_{\text{dy}}(z)$ and $G_{\text{dy}}(-z)$.

18.3 Probability and semigroups

The family Z_τ gives a canonical subordinate-Brownian semigroup attached to the Rvachev zero divisor. Its transition density, fractional generator, entropy production, and small-time scaling should be expressible through the two heat kernels L_1 and K . The explicit Levy density (217) makes simulation, self-decomposability, and potential theory accessible without reconstructing the gamma series term by term.

18.4 Base- b and geometric- q families

The entire reciprocal form of the geometric factorization (199) is delivered for every complex q with $\|q\| < 1$, in parallel with the repository’s base-generic Gaussian-binomial interpolation. The

remaining comparison concerns raw Gamma products and exact analytic collision orders: for integer b , the claimed order is a whole-base valuation, whereas for nonintegral q^{-1} the pole orbits are typically noncoincident. Only the dyadic order is currently formal. Comparing the remaining regimes may separate which phenomena are genuinely dyadic from those caused only by geometric scaling.

18.5 Inverse spectral analysis

The coefficients (249) provide a concrete meeting point for global Fourier analysis and endpoint Lambert asymptotics. Certified steepest-descent evaluation of these coefficients could reveal periodic modulation, sign patterns, and optimal approximation rates for the inverse. A parallel study of $\int_0^1 e^{-2\pi i \eta F(x)} dx$ would produce a nonlinear Fourier theory of the CDF itself.

18.6 Computer-assisted certification

The existing Python checks are exploratory. A rigorous package should combine interval arithmetic for gamma products, exact rational moment recurrences, certified quadrature split at the integer zero set, and explicit tail bounds from the rapid Fourier decay. Such a tool could test the conjectural Jensen fluctuation and inverse-coefficient phases at hundreds of dyadic scales without confusing truncation error with genuine periodic oscillation.

19 Conclusion

The Fabius–Rvachev system is unusually rich because the same object can be read as a smooth compact bump, a CDF, an infinite uniform convolution, a Thue–Morse spline limit, an entire sinc determinant, an arithmetic spectrum, a q -series, and a Lambert-saddle problem. The source audit shows that many attractive formulae already coexist in the repository; treating them as separate discoveries would obscure rather than advance the subject.

The new results in this report add bridges rather than isolated identities. The dyadic gamma function splits the even Fourier determinant into reflected arithmetic gamma factors. The Gaussian-mixture theorem joins the logistic/Appell dual to the Thorin/gamma dual and exposes a new Levy semigroup. Jensen’s formula turns binary digit sums into a radial complex-analytic observable. Unbounded multiplicities settle non-holonomicity in one stroke. Finally, the bounded CDF and its inverse acquire transform formulae that are global, explicit, and compatible with the existing endpoint theory. Each bridge supplies both a theorem and a research program, with a comparatively short path to formal verification.

The reciprocal-product core of the first bridge is now formal at checkpoint 71ab6f6728fceb753c88d8b0573077a59acf2682: it includes generic complex contractions, Mahler and zero-orbit laws, reflection, the dyadic Rvachev identity, and exact dyadic zero/pole orders. The raw Gamma-product realization and the canonical, logarithmic, digamma, integral, heat-trace, D-finite, derivative-tower, and general-base extensions retain their frontier status.

The current overlay sharpens two of those historical boundaries: commit 0ba35abd4 supplies all reciprocal-Gamma first jets and the first eight continuation GammaLog/tower declarations, and the current integrated tree adds the level-zero master-product ratio. It also closes the zero set of $G_{1/b}$ at every integer base $b \geq 2$ in `ReciprocalIntegerGammaZeros.lean`. Thus the Mellin/integral GammaLog and dyadic log/tower layers are formal at every natural level. Only the parameter-differential and iterated differential ladder remains frontier within that tower; the raw Gamma product and the canonical, logarithmic/digamma, Malmsten/heat-trace, and D-finite extensions remain frontier, as does the collision *order* at general integer base.

A Audited living document families

The active draft root at the audited snapshot contained the following document families. Their exact names are retained to make future collision searches reproducible.

Draft family	Principal audited themes
Fabius_Arithmetic_Rays_Frontier_Report	arithmetic rays, exact values, spectral sampling
Fabius_Dyadic_Self_Sampling_Frontier_Package	dyadic self-sampling and reconstruction
Fabius_Half_Integer_Spectral_Frontier_Report	half-integer cosine spectrum, Prouhet identities, alias transforms
Fabius_Inverse_Frontier_Report_Source_and_PDF	inverse germs, Catalan and Lambert–Bell asymptotics
Fabius_Newton_Rvachev_Frontier_Report	Newton recurrences, exponent sequences, exact FFT checks
Fabius_Rvachev_Frontier_Report-2	arithmetic spectrum, zero multiplicities, heat trace, lobe signs
Fabius_Rvachev_Frontier_Report	tail quadrature, exact dyadic error laws, q -Richardson acceleration
Fabius_Rvachev_New_Frontiers	q -MZV moments, Thorin dual, complete Bernstein theory, theta completion
Fabius_Rvachev_Thue_Morse_Frontier_Results	finite blocks, deconvolution, zeta–Lambert tails
Spectral_Arithmetic_Pascal_Rvachev_Hierarchy	spectral arithmetic and Pascal-type hierarchies
Thue_Morse_Formula_Atlas	broad Thue–Morse product, series, automata, and arithmetic atlas
fabius_frontier_dyadic_inverse_barnes_report	dyadic inverse, Barnes prefixes, Appell and logistic dual
fabius_frontier_new_results	logarithmic q -Richardson and Lambert phase locking
fabius_frontier_results	q -Richardson sinc products and acceleration
fabius_frontier_results_bundle	Fredholm determinant, spectral zeta, zero jets, q -Euler determinant
fabius_frontier_spectral_endpoint_report_bundle	endpoint spectral products, Mellin and saddle analysis
rvachev_up_fourier_decay	successive Fourier-decay audits and certified bounds

The source audit also included the primary formal article, the consolidated non-formalized-research-frontiers.tex, glossary, Lean walkthrough, non-elementarity document, embedded papers, and the archive provenance map. The archive map was essential for identifying superseded or byte-identical historical variants without treating them as independent sources.

B Numerical package contents

The reproducibility assets retained for this part live under assets/fabius_rvachev_representation_frontier/:

- numerical_experiments.py, with detailed comments and deterministic seed;
- figures/numerical_results.txt;
- the four PNG figures embedded in the report;
- SOURCE_AUDIT.md, a concise audit and novelty-collision record;

- README.md, reproduction instructions and file descriptions.

C Selected derivations in one line

For convenient reuse, here are the central bridges in compressed form:

$$\begin{array}{ll}
 \Gamma(1+w)^{-1}\Gamma(1-w)^{-1} = \text{sinc}(\pi w) & \implies \Phi(z) = G_{\text{dy}}(z)G_{\text{dy}}(-z) \\
 & \quad = 1/[\Gamma_{\text{dy}}(z)\Gamma_{\text{dy}}(-z)], \\
 \mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}, \quad \mathbb{E}(e^{it\sqrt{2T}N} \mid T) = e^{-t^2T} & \implies Z_1 \stackrel{\text{d}}{=} \sqrt{2T_1}N, \\
 \text{zeros}(\Phi) = \{\pm n \text{ with multiplicity } a_n\} & \implies J(r) = 2 \sum_{n \leq r} a_n \log(r/n), \\
 a_{2k} = k+1 \rightarrow \infty & \implies \Phi \text{ is not D-finite,} \\
 F'(x) = 2 \text{up}(2x-1), \quad \widehat{\text{up}} = \Phi & \implies \widehat{F}'(\xi) = e^{-\pi i \xi} \Phi(\xi/2), \\
 y = F(x), \quad Q(F(x)) = x & \implies \widehat{Q}(\eta) = \frac{\int_0^1 e^{-2\pi i \eta F(x)} dx - e^{-2\pi i \eta}}{2\pi i \eta}.
 \end{array}$$

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Part III

Dyadic Multiresolution and Product-Series Representations in the Fabius–Rvachev System

Abstract

The existing ProveIt corpus develops the Fabius function, its inverse, Rvachev’s compactly supported up-function, the Thue–Morse sequence, exact dyadic values, q -binomial formulas, infinite sinc products, Bell and Bernoulli polynomial machinery, and phase-resolved Lambert– W endpoint asymptotics in considerable depth. This report first organizes those results into a representation atlas and then develops a complementary multiresolution layer that was not located in the audited LaTeX corpus.

Let $p = F'$ and let $\Delta_{N,k} = F((k+1)2^{-N}) - F(k2^{-N})$ be the exact dyadic cell masses. The same rational projective array produces four different finite models. It produces a fixed-cell uniform mixture for p and up, an exact rational unnormalized Haar expansion of p , a rational Faber–Schauder expansion of F , and the finite Walsh–Hadamard transform of p . Their Fourier images yield locally uniform limits of rational exponential polynomials multiplied by sinc windows. The Walsh transform of every basis function is itself a finite product of sines and cosines. Consequently the infinite sinc product admits a new series whose summands are finite trigonometric products and whose coefficients are exact Walsh transforms of dyadic Fabius masses.

A Walsh mode indexed by m has exactly $\text{wt}_2(m)$ vanishing algebraic moments. Writing $\sigma(m) = \sum_j j b_j$ for the positions of the nonzero binary digits, its first nonzero moment is

$$\int_0^1 x^{\text{wt}_2(m)} w_m(x) dx = (-1)^{\text{wt}_2(m)} \text{wt}_2(m)! 2^{-2 \text{wt}_2(m) - \sigma(m)}.$$

Summing over a fixed binary weight produces the Gaussian binomial coefficient $\begin{bmatrix} N \\ r \end{bmatrix}_{1/2}$. Higher Walsh moments are encoded by a finite Bell-polynomial expression whose logarithmic coefficients are explicit Bernoulli-number sums. This gives a direct bridge among Walsh cancellation, Thue–Morse signs, q -binomial coefficients, and the Bell–Bernoulli structure of the Fabius moments.

A second positive approximation is obtained by differentiating the Bernstein polynomial of F : p is the uniform limit of finite mixtures of beta densities with rational weights $\Delta_{N,k}$. Its moment-generating and Fourier transforms are therefore limits of positive rational mixtures of confluent hypergeometric functions. A dual construction uses dyadic values of the inverse $G = F^{-1}$: equal probability cells with variable spatial widths give a finite sinc-mixture formula for $\widehat{\text{up}}$, as well as polynomial limits for all moments.

Finally, the exact Haar detail energy E_N is rational and has an all-orders continuum expansion

$$E_N \sim \frac{h^2}{16} \|p'\|_2^2 - \frac{h^4}{384} \|p''\|_2^2 + \frac{h^6}{20480} \|p'''\|_2^2 - \cdots, \quad h = 2^{-N}.$$

The complete coefficient sequence is generated by sinc^4 , hence by complete Bell polynomials in Bernoulli numbers. The beyond-all-orders aliasing remainder provides a new place where the known log-periodic Fourier decay should reappear. The report distinguishes repository input, results proved here, numerical checks, and conjectures; “new” always means new relative to the audited repository corpus, not an assertion of worldwide priority.

1 Audit protocol, scope, and novelty boundary

1.1 Pinned corpus and cumulative audit

The documentation tree

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>

was audited recursively. The Git tree contained 89 files with extension `.tex` at the source-level inventory used in the audit. During this investigation the branch advanced; the report is pinned to commit `924652f4a9a656fd957d2936b2f053747c6f6c47`. The five-commit delta from the preceding pin modified the primary exposition, its coverage ledger, and an integral-transform report, but added no new TeX path. The delta was checked before the novelty boundary below was frozen.

The audit is cumulative rather than a naive count of duplicated revisions. Earlier source-level passes covered the complete path inventory; the present pass reread the canonical primary exposition, current audit ledgers, active frontier syntheses, and every recent report capable of overlapping the proposed results. Historical copies under `archive/` were normalized against their live successors. Imported papers were used as literature baselines, not counted as independent repository discoveries.

What the current formal source already contains

The current primary exposition and Lean modules include the probability law of weighted uniform series, finite atomic and spline approximants, exact dyadic values, quantile transport, sinc and cosine products, Fourier inversion and Poisson summation, moment recurrences, a Fourier–Legendre expansion, normalized integer Volterra calculus, inverse-Fabius infrastructure, and the proved Lambert-phase endpoint machinery. These are inputs below. Repackaging one of those statements with different notation is not counted as a new result.

1.2 Explicit exclusion map

The following recent manuscripts were treated as an exclusion map, not as a source of claims to relabel.

Already developed in the corpus	Excluded from novelty claims here
Primary Fabius–Rvachev synthesis	Function triad, probability construction, refinement laws, exact dyadic arithmetic, sinc product, Fourier inversion, moments, Fourier–Legendre series, and formal Lambert asymptotics.
Integral and transform reports	Weighted antiderivatives, inverse-pair area identities, Mellin/Newton continuation, negative moments, fractional primitives, order statistics, Stieltjes/Cauchy transforms, autocorrelation, and Gini-type integrals.
Spectral determinant and Barnes reports	Zero multiplicities, spectral zeta, Fredholm determinants, heat traces, Barnes deconvolution, reciprocal-sinc poles, and coefficient-threshold asymptotics.
Confluent digital extrapolation report	Reciprocal-moment Appell systems, Strang–Fix/cardinal reproduction, q -binomial scale closure, confluent Richardson filters, and Lambert-phase tomography.

Already developed in the corpus	Excluded from novelty claims here
Gamma-duality/total-positivity report	Laguerre–Pólya and PF_∞ structure, generalized gamma-convolution duality, Thorin measure, theta completion, mod-Gamma endpoint tilting, and large-order transfer.
Fourier-decay campaign	Exact zero orders, Thue–Morse lobe signs, sharp global envelopes, log-periodic fluctuations, and phase-sensitive real-axis decay.

Targeted searches were then made for the exact structures used below: Faber–Schauder expansions of F or G , Haar expansions from dyadic Fabius masses, Walsh–Hadamard expansions of p , finite-product Fourier transforms of Walsh modes, Bernstein beta-mixtures, and the sinc^4 Haar-energy expansion. No corresponding package was located in the audited TeX corpus. General Haar, Walsh, Schauder, and Bernstein theory is classical; the frontier content is the exact specialization and synthesis for the Fabius–Rvachev system.

1.3 Status ledger

ID	Statement	Status
A	Projective rational dyadic mass array and fixed-cell uniform-mixture representations, including locally uniform entire transform limits.	proved here
B	Rational unnormalized Haar and Faber–Schauder representations, with an orthonormal Haar/Fourier product expansion.	proved here
C	Walsh–Hadamard representation, reflection selection rule, and a series for $\widehat{\text{up}}$ whose terms are finite trigonometric products.	proved here
D	Exact Hamming-weight cancellation order, first nonzero Walsh moment, and $q = 1/2$ Gaussian-binomial aggregation.	proved here
E	Bell–Bernoulli formula for every higher Walsh moment.	proved here
F	Positive beta-density mixture and hypergeometric transform limits.	proved here
G	Equal-weight inverse-quantile chord mixtures and their finite sinc-product transform formula.	proved here
H	All-orders Haar-energy expansion generated by sinc^4 , with a superalgebraically small alias remainder.	proved here
I	Lambert-phase asymptotics for the boundary Schauder coefficients of $G = F^{-1}$ and boundary domination of the global interpolation error.	conjectural
J	A log-periodic transseries for the beyond-all-orders Haar alias remainder.	conjectural

2 Normalization and the function triad

2.1 Bounded Fabius function, density, and Rvachev function

Throughout, $F : \mathbb{R} \rightarrow [0, 1]$ is the bounded Fabius distribution function,

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

and up is Rvachev's even C^∞ function supported on $[-1, 1]$. On their common fundamental intervals,

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (253)$$

$$F'(x) = 2 \text{up}(2x - 1) =: p(x), \quad 0 < x < 1, \quad (254)$$

$$F(x) + F(1 - x) = 1, \quad 0 \leq x \leq 1, \quad (255)$$

$$\text{up}(x) = 1 - F(x) = F(1 - x), \quad 0 \leq x \leq 1. \quad (256)$$

The density p , extended by zero outside $[0, 1]$, is a compactly supported C^∞ function satisfying

$$p(x) = p(1 - x), \quad \int_0^1 p(x) \, dx = 1. \quad (257)$$

The ordinary inverse of $F|_{[0,1]}$ is denoted

$$G = F^{-1} : [0, 1] \longrightarrow [0, 1], \quad G(1 - u) = 1 - G(u). \quad (258)$$

The signed global extension $\mathcal{F}$, important in the repository's repeated-integration theory, is not needed for the multiresolution proofs; it appears only in the baseline atlas.

2.2 Fourier convention

For an integrable function f we use angular frequency,

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-itx} \, dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t) e^{itx} \, dt. \quad (259)$$

Set

$$\Phi(t) = \widehat{\text{up}}(t), \quad M(z) = \int_{-1}^1 \text{up}(x) e^{zx} \, dx = \Phi(iz). \quad (260)$$

The compact-interval Fourier image of the bounded Fabius function is

$$\mathcal{F}_F(t) = \int_0^1 F(x) e^{-itx} \, dx. \quad (261)$$

This notation avoids the meaningless full-line Fourier transform of a function that equals one on $[1, \infty)$.

2.3 Probabilistic model

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$ and let $V_j = (U_j + 1)/2$, uniform on $[0, 1]$. Then

$$Y = \sum_{j \geq 1} 2^{-j} U_j, \quad X = \frac{Y + 1}{2} = \sum_{j \geq 1} 2^{-j} V_j. \quad (262)$$

The law of Y has density up, and F is the distribution function of X . Consequently

$$X \stackrel{d}{=} G(U), \quad Y \stackrel{d}{=} 2G(U) - 1, \quad U \sim \text{Unif}[0, 1]. \quad (263)$$

3 A representation atlas from the existing theory

This section collects a broad common baseline. Most formulas are already present, sometimes in stronger form, in the audited repository. They are included so that the genuinely new representations later can be compared against one normalization without forcing the reader to move among several manuscripts.

3.1 Infinite convolution, cube integrals, and finite heads

Let ν_a denote the uniform probability density on $[-a, a]$. Independence in (262) gives the infinite convolution

$$\text{up} = \nu_{1/2} * \nu_{1/4} * \nu_{1/8} * \cdots. \quad (264)$$

In distributional notation this is the infinite-cube integral

$$\text{up}(x) = \int_{[-1,1]^{\mathbb{N}}} \delta\left(x - \sum_{j \geq 1} 2^{-j} u_j\right) \prod_{j \geq 1} \frac{du_j}{2}, \quad (265)$$

and

$$F(x) = \int_{[0,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum_{j \geq 1} 2^{-j} v_j \leq x\}} \prod_{j \geq 1} dv_j. \quad (266)$$

For the finite head

$$Y_N = \sum_{j=1}^N 2^{-j} U_j, \quad A_N = \sum_{j=1}^N 2^{-j} = 1 - 2^{-N},$$

write B_N for its density. Inclusion–exclusion for a box spline gives

$$B_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x + A_N - 2 \sum_{j=1}^N \varepsilon_j 2^{-j} \right)_+^{N-1}. \quad (267)$$

The signs $(-1)^{|\varepsilon|}$ are the finite Thue–Morse signs. Likewise, for $X_N = \sum_{j=1}^N 2^{-j} V_j$,

$$\mathbb{P}(X_N \leq x) = \frac{2^{N(N+1)/2}}{N!} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=1}^N \varepsilon_j 2^{-j} \right)_+^N. \quad (268)$$

Taking $N \rightarrow \infty$ gives $B_N \rightarrow \text{up}$ and the finite CDFs tend to F .

The tail is an independent scaled copy:

$$X \stackrel{d}{=} X_N + 2^{-N} X', \quad Y \stackrel{d}{=} Y_N + 2^{-N} Y'. \quad (269)$$

Hence the exact finite-scale identities

$$F(x) = \mathbb{E} F(2^N(x - X_N)), \quad (270)$$

$$\text{up}(x) = 2^N \int_{\mathbb{R}} B_N(y) \text{up}(2^N(x - y)) \, dy. \quad (271)$$

These formulas combine a finite Thue–Morse spline with a shrunken copy of the same limiting function.

3.2 Sinc, hyperbolic-sinc, and renormalization products

The Fourier and moment-generating products are

$$\Phi(t) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}, \quad (272)$$

$$M(z) = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (273)$$

They converge locally uniformly on $\mathbb{C}$, and

$$\Phi(2t) = \operatorname{sinc}(t)\Phi(t), \quad M(2z) = \frac{\sinh z}{z}M(z). \quad (274)$$

Iteration at any finite depth gives

$$\Phi(t) = \Phi(t/2^N) \prod_{j=1}^N \operatorname{sinc}(t/2^j), \quad (275)$$

which is the transform counterpart of (271).

Euler's product for sine yields two canonical products:

$$\Phi(t) = \prod_{m=1}^{\infty} \left(\frac{t^2}{4\pi^2 m^2}; \frac{1}{4} \right)_{\infty}, \quad (276)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{t^2}{4\pi^2 n^2} \right)^{1+v_2(n)}. \quad (277)$$

Thus $2\pi n$ is a zero of multiplicity $1 + v_2(n)$. For $|t| < 2\pi$,

$$\log \Phi(t) = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)} t^{2r}. \quad (278)$$

The Wick-rotated determinant

$$\mathcal{A}(s) = \Phi(i\sqrt{s}) = \prod_{n=1}^{\infty} \left(1 + \frac{s}{4\pi^2 n^2} \right)^{1+v_2(n)} \quad (279)$$

underlies the spectral-zeta, total-positivity, and generalized-gamma-convolution results already developed in the corpus.

3.3 Moments, cumulants, Bell polynomials, and Bernoulli numbers

Write

$$\mu_n = \int_{-1}^1 x^n \operatorname{up}(x) \, dx = \mathbb{E}Y^n. \quad (280)$$

Odd moments vanish. The even cumulants are

$$\kappa_{2r}(Y) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (281)$$

Therefore

$$\mu_n = Y_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (282)$$

where Y_n is the complete exponential Bell polynomial, and

$$\Phi(t) = \sum_{r=0}^{\infty} \frac{(-1)^r \mu_{2r}}{(2r)!} t^{2r}. \quad (283)$$

For $m_n = \mathbb{E}X^n$ the smoothing equation $X \stackrel{d}{=} (V + X')/2$ gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (284)$$

All m_n are rational, and

$$\mu_{2r} = \frac{2m_{2r+1}}{2r + 1}. \quad (285)$$

3.4 Fourier inversion, twisted Poisson summation, and cosine series

Since Φ is rapidly decreasing on the real axis,

$$\text{up}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t) e^{itx} dt = \frac{1}{\pi} \int_0^\infty \Phi(t) \cos(tx) dt. \quad (286)$$

The distribution function of the centered law gives, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(t) \frac{\sin((2x-1)t)}{t} dt. \quad (287)$$

Combining this with $F(x) = \text{up}(x-1)$ produces the integral identity

$$\int_0^\infty \Phi(t) \cos((1-x)t) dt = \frac{\pi}{2} + \int_0^\infty \Phi(t) \frac{\sin((2x-1)t)}{t} dt. \quad (288)$$

The partition of unity $\sum_{k \in \mathbb{Z}} \text{up}(x-k) = 1$ kills all nonzero integer Fourier modes. Twisting by $(-1)^k$ leaves the half-integer modes:

$$\sum_{k \in \mathbb{Z}} (-1)^k \text{up}(x-k) = \sum_{m \in \mathbb{Z}} \Phi((2m+1)\pi) e^{i(2m+1)\pi x}. \quad (289)$$

For $x \in [0, 1]$ the left side is $1 - 2F(x)$. With

$$A_m = \Phi((2m+1)\pi), \quad (290)$$

we obtain the rapidly convergent series

$$\text{up}(x) = \frac{1}{2} + \sum_{m=0}^\infty A_m \cos((2m+1)\pi x), \quad |x| \leq 1, \quad (291)$$

$$F(x) = \frac{1}{2} - \sum_{m=0}^\infty A_m \cos((2m+1)\pi x), \quad 0 \leq x \leq 1, \quad (292)$$

$$p(x) = \sum_{m=0}^\infty A_m (2m+1)\pi \sin((2m+1)\pi x). \quad (293)$$

Every derivative may be taken termwise.

3.5 Fourier images of p , F , and $1 - F$

The affine relation between p and up gives

$$\widehat{p}(t) = e^{-it/2} \Phi(t/2). \quad (294)$$

Integration by parts then yields the exact compact-interval transforms

$$\mathcal{F}_F(t) = \frac{e^{-it/2} \Phi(t/2) - e^{-it}}{it}, \quad (295)$$

$$\int_0^1 (1 - F(x)) e^{-itx} dx = \frac{1 - e^{-it/2} \Phi(t/2)}{it}. \quad (296)$$

The singularities at $t = 0$ are removable. After Wick rotation,

$$\int_0^1 e^{zx} F(x) dx = \frac{e^z - e^{z/2} M(z/2)}{z}, \quad (297)$$

$$\int_0^1 e^{zx} (1 - F(x)) dx = \frac{e^{z/2} M(z/2) - 1}{z}. \quad (298)$$

3.6 Quantile, Stieltjes, and correlation representations

Quantile transport (263) gives

$$\Phi(t) = \int_0^1 \exp(-it(2G(u) - 1)) \, du, \quad (299)$$

$$M(z) = \int_0^1 \exp(z(2G(u) - 1)) \, du, \quad (300)$$

$$\mu_n = \int_0^1 (2G(u) - 1)^n \, du. \quad (301)$$

For $z \notin [-1, 1]$, the Cauchy transform is

$$\mathcal{S}(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} \, dx = \int_0^1 \frac{du}{z - (2G(u) - 1)}. \quad (302)$$

If $\Im z > 0$,

$$\mathcal{S}(z) = -i \int_0^\infty e^{itz} \Phi(t) \, dt. \quad (303)$$

The autocorrelation and convolution powers are

$$\int_{\mathbb{R}} \text{up}(x) \text{up}(x + y) \, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t)^2 e^{ity} \, dt, \quad (304)$$

$$\text{up}^{*r}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t)^r e^{itx} \, dt, \quad r \in \mathbb{N}. \quad (305)$$

3.7 Legendre and Chebyshev coefficient integrals

The repository contains a Fourier–Legendre expansion. Fourier inversion gives a useful transform-side form of its coefficients. If

$$\text{up}(x) = \sum_{n \geq 0} \lambda_n P_n(x), \quad \lambda_n = \frac{2n+1}{2} \int_{-1}^1 \text{up}(x) P_n(x) \, dx, \quad (306)$$

then odd coefficients vanish and, with j_n the spherical Bessel function,

$$\lambda_{2r} = \frac{4r+1}{\pi} (-1)^r \int_0^\infty \Phi(t) j_{2r}(t) \, dt. \quad (307)$$

Indeed, $\int_{-1}^1 e^{itx} P_n(x) \, dx = 2i^n j_n(t)$.

A parallel Chebyshev representation, apparently not stated in the audited corpus, is

$$\text{up}(x) = \frac{a_0}{2} + \sum_{n \geq 1} a_n T_n(x), \quad a_n = \frac{2}{\pi} \int_0^\pi \text{up}(\cos \theta) \cos(n\theta) \, d\theta. \quad (308)$$

The odd coefficients vanish and

$$a_{2r} = \frac{2(-1)^r}{\pi} \int_0^\infty \Phi(t) J_{2r}(t) \, dt. \quad (309)$$

The function $\theta \mapsto \text{up}(\cos \theta)$ is smooth and periodic, so the Chebyshev coefficients decrease faster than every power.

4 The projective dyadic mass array

4.1 Definition and exact algebra

Put

$$n_N = 2^N, \quad h_N = 2^{-N}, \quad I_{N,k} = [kh_N, (k+1)h_N), \quad (310)$$

with the last interval closed at 1, and define

$$\Delta_{N,k} = F((k+1)h_N) - F(kh_N), \quad 0 \leq k < n_N. \quad (311)$$

These are the probabilities that X lies in the dyadic cells.

Theorem 4.1 (Projective rational mass array). *For every $N \geq 0$, put*

$$\Delta_N = (\Delta_{N,0}, \dots, \Delta_{N,n_N-1}).$$

This vector has the following properties:

$$\Delta_{N,k} \in \mathbb{Q}_{>0}, \quad (312)$$

$$\sum_{k=0}^{n_N-1} \Delta_{N,k} = 1, \quad (313)$$

$$\Delta_{N,n_N-1-k} = \Delta_{N,k}, \quad (314)$$

$$\Delta_{N,k} = \Delta_{N+1,2k} + \Delta_{N+1,2k+1}. \quad (315)$$

Consequently the mass polynomial

$$D_N(z) = \sum_{k=0}^{n_N-1} \Delta_{N,k} z^k \quad (316)$$

has rational coefficients, satisfies $D_N(1) = 1$, and is palindromic:

$$D_N(z) = z^{n_N-1} D_N(z^{-1}). \quad (317)$$

Proof. Rationality is the exact dyadic-value theorem from the repository. Positivity comes from strict increase of F on $(0, 1)$. The sum telescopes. Reflection (314) follows from $F(1-x) = 1 - F(x)$, and (315) is additivity after bisecting a cell. The polynomial statements are immediate. $\square$

The array is therefore a rational projective system of probability vectors. It is useful to regard it as the common discrete skeleton behind all subsequent representations. No approximation has yet been made.

4.2 Fixed-cell uniform mixtures

Define the dyadic conditional-expectation density

$$p_N(x) = n_N \Delta_{N,k}, \quad x \in I_{N,k}. \quad (318)$$

Equivalently,

$$p_N(x) dx = \sum_{k=0}^{n_N-1} \Delta_{N,k} \text{Unif}(I_{N,k}). \quad (319)$$

This is a finite mixture of equal-width uniform laws with variable rational weights. On the centered up scale, put

$$u_{N,k} = \frac{2k+1}{n_N} - 1. \quad (320)$$

The corresponding approximation is

$$u_{p_N}(u) = \frac{1}{2} p_N \left(\frac{u+1}{2} \right) = \frac{n_N}{2} \Delta_{N,k}, \quad |u - u_{N,k}| < \frac{1}{n_N}. \quad (321)$$

Theorem 4.2 (Martingale and uniform convergence). *Let $\mathcal{D}_N$ be the dyadic sigma-algebra on $[0, 1]$. Then*

$$p_N = \mathbb{E}[p \mid \mathcal{D}_N]. \quad (322)$$

Moreover,

$$\|p_N - p\|_\infty \longrightarrow 0, \quad \|p_N - p\|_{L^q(0,1)} \longrightarrow 0 \quad (0 < q < \infty), \quad (323)$$

and $u_{p_N} \rightarrow u_p$ in the corresponding centered norms. If $p \in C^1$, then

$$\|p_N - p\|_\infty \leq h_N \|p'\|_\infty. \quad (324)$$

Proof. The average of p on $I_{N,k}$ is $h_N^{-1} \int_{I_{N,k}} p = n_N \Delta_{N,k}$, proving (322). Uniform continuity gives uniform convergence of cell averages; the displayed bound follows from the mean-value theorem. Finite-measure L^q convergence follows from uniform convergence. $\square$

Remark 4.3. Although p_N is only piecewise constant, all its coefficients are exact. It is not the finite-head spline B_N of (267): B_N is obtained by truncating the random series, whereas p_N is obtained by conditioning the limiting density on a spatial dyadic partition. The two approximations therefore probe different structures.

4.3 Rational exponential-polynomial limits for the transforms

The transform of (319) can be written directly in terms of the mass polynomial.

Theorem 4.4 (Primal dyadic transform representations). *For every $z \in \mathbb{C}$,*

$$\widehat{p}_N(z) = \frac{n_N(1 - e^{-iz/n_N})}{iz} D_N(e^{-iz/n_N}). \quad (325)$$

Consequently

$$\Phi_N^{\text{cell}}(t) := e^{it} \widehat{p}_N(2t) \quad (326)$$

$$= \text{sinc}(t/n_N) \sum_{k=0}^{n_N-1} \Delta_{N,k} \exp \left[-it \left(\frac{2k+1}{n_N} - 1 \right) \right]. \quad (327)$$

The limits

$$\widehat{p}(z) = \lim_{N \rightarrow \infty} \frac{n_N(1 - e^{-iz/n_N})}{iz} D_N(e^{-iz/n_N}), \quad (328)$$

$$\Phi(t) = \lim_{N \rightarrow \infty} \text{sinc}(t/n_N) \sum_{k=0}^{n_N-1} \Delta_{N,k} e^{-itu_{N,k}} \quad (329)$$

are locally uniform on $\mathbb{C}$. Thus the infinite sinc product is a locally uniform limit of rational exponential polynomials times one elementary sinc window.

Proof. Integrating p_N cell by cell gives

$$\int_{kh_N}^{(k+1)h_N} n_N \Delta_{N,k} e^{-izx} dx = \Delta_{N,k} \frac{n_N(1 - e^{-iz/n_N})}{iz} e^{-izk/n_N},$$

which sums to (325). The affine change from p to up gives (327). If $K \subset \mathbb{C}$ is compact, then

$$\sup_{z \in K} |\widehat{p}_N(z) - \widehat{p}(z)| \leq e^{\sup_{z \in K} |\Im z|} \|p_N - p\|_1,$$

so convergence is locally uniform. The same argument applies after the affine change of variables. $\square$

Deleting the sinc window gives the midpoint-atomic representation

$$\Phi(t) = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} e^{-itu_{N,k}}. \quad (330)$$

The cell version is usually more accurate because it integrates the test function across each cell rather than concentrating the mass at its midpoint.

After Wick rotation one obtains the moment-generating limit

$$M(z) = \lim_{N \rightarrow \infty} \frac{\sinh(z/n_N)}{z/n_N} \sum_{k=0}^{n_N-1} \Delta_{N,k} \exp \left[z \left(\frac{2k+1}{n_N} - 1 \right) \right]. \quad (331)$$

Equations (329) and (331) are the same finite mixture on the imaginary and real axes.

4.4 A second-order quadrature principle

The conditional-expectation property gives better weak accuracy than the pointwise bound (324) suggests.

Proposition 4.5 (Cell-average quadrature). *If $p, g \in C^1[0, 1]$, then*

$$\left| \int_0^1 g(x)(p(x) - p_N(x)) dx \right| \leq h_N^2 \|g'\|_\infty \|p'\|_\infty. \quad (332)$$

In particular, on each compact set $|z| \leq R$,

$$\widehat{p}_N(z) - \widehat{p}(z) = \mathcal{O}_R(4^{-N}), \quad \Phi_N^{\text{cell}}(z) - \Phi(z) = \mathcal{O}_R(4^{-N}). \quad (333)$$

Proof. On a cell I choose any $x_I \in I$. Since $\int_I (p - p_N) = 0$,

$$\int_I g(p - p_N) = \int_I (g - g(x_I))(p - p_N).$$

Now $\text{osc}_I g \leq h_N \|g'\|_\infty$ and $\int_I |p - p_N| \leq h_N \text{osc}_I p \leq h_N^2 \|p'\|_\infty$. Summing $n_N = h_N^{-1}$ cells proves (332). For exponential test functions, their derivatives are uniformly bounded on compact subsets of $\mathbb{C}$. $\square$

The 4^{-N} behavior is visible in the numerical experiment of [Appendix 13](#). The atomic midpoint formula is also second order for a smooth density and smooth test function, but the sinc-window formula has a clearer probabilistic interpretation: it is exactly the transform of the conditional-expectation density.

4.5 Moments, L^q norms, and discrete entropies

For $q > 0$, uniform convergence gives

$$\int_0^1 p(x)^q dx = \lim_{N \rightarrow \infty} n_N^{q-1} \sum_{k=0}^{n_N-1} \Delta_{N,k}^q. \quad (334)$$

For $q = 2$ this is an exact rational approximation to the collision integral,

$$\int_{-1}^1 \text{up}(x)^2 dx = \frac{1}{2} \lim_{N \rightarrow \infty} n_N \sum_{k=0}^{n_N-1} \Delta_{N,k}^2. \quad (335)$$

Differentiating (334) at $q = 1$ gives the Shannon asymptotic

$$-\sum_{k=0}^{n_N-1} \Delta_{N,k} \log \Delta_{N,k} = \log n_N - \int_0^1 p(x) \log p(x) dx + o(1). \quad (336)$$

Here $p \log p$ extends continuously by zero at the endpoints. Thus the continuous entropy of the Fabius density is the constant term in the entropy of its exact rational dyadic mass vectors.

For ordinary moments, integrating a monomial over each uniform cell gives

$$m_r = \int_0^1 x^r p(x) dx = \lim_{N \rightarrow \infty} \frac{1}{(r+1)n_N^r} \sum_{k=0}^{n_N-1} \Delta_{N,k}((k+1)^{r+1} - k^{r+1}). \quad (337)$$

Every finite approximant is rational.

5 Haar and Faber–Schauder representations

5.1 Rational unnormalized Haar increments

Let

$$\eta_{n,k} = \mathbf{1}_{I_{n+1,2k}} - \mathbf{1}_{I_{n+1,2k+1}}, \quad 0 \leq k < 2^n, \quad (338)$$

and let $h_{n,k} = 2^{n/2} \eta_{n,k}$ be the orthonormal Haar functions. Define

$$\gamma_{n,k} = 2^n (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) \in \mathbb{Q}, \quad (339)$$

$$c_{n,k} = \langle p, h_{n,k} \rangle = 2^{n/2} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = 2^{-n/2} \gamma_{n,k}. \quad (340)$$

Thus the coefficients of the unnormalized Haar representation are rational, while the orthonormal coefficients lie in $2^{n/2} \mathbb{Q}$.

Theorem 5.1 (Exact Haar expansion). *For every $N \geq 1$,*

$$p_N(x) = 1 + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} \gamma_{n,k} \eta_{n,k}(x) = 1 + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} c_{n,k} h_{n,k}(x). \quad (341)$$

The power-of-two partial sums converge uniformly and in every finite L^q :

$$p(x) = 1 + \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} \gamma_{n,k} \eta_{n,k}(x). \quad (342)$$

Reflection gives

$$\gamma_{n,2^n-1-k} = -\gamma_{n,k}, \quad c_{n,2^n-1-k} = -c_{n,k}. \quad (343)$$

Proof. The difference $p_{n+1} - p_n$ is constant with opposite signs on the two children of each parent cell. On the left child it equals $2^n(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = \gamma_{n,k}$, so $p_{n+1} - p_n = \sum_k \gamma_{n,k} \eta_{n,k}$. Telescoping from $p_0 \equiv 1$ proves (341). Convergence is [Theorem 4.2](#); antisymmetry follows from (314). $\square$

The coefficients can also be read directly from three dyadic values:

$$c_{n,k} = 2^{n/2} \left[2F\left(\frac{2k+1}{2^{n+1}}\right) - F\left(\frac{k}{2^n}\right) - F\left(\frac{k+1}{2^n}\right) \right]. \quad (344)$$

This is a discrete second derivative of F at a dyadic midpoint.

5.2 Fourier image of the Haar expansion

A direct integration gives

$$\hat{h}_{n,k}(z) = 2^{n/2} e^{-izk/2^n} \frac{(1 - e^{-iz/2^{n+1}})^2}{iz}. \quad (345)$$

Therefore:

Theorem 5.2 (Haar product-series transform). *Locally uniformly on $\mathbb{C}$,*

$$\hat{p}(z) = \frac{1 - e^{-iz}}{iz} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k} 2^{n/2} e^{-izk/2^n} \frac{(1 - e^{-iz/2^{n+1}})^2}{iz}. \quad (346)$$

Equivalently,

$$\Phi(t) = \text{sinc}(t) + e^{it} \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k} 2^{n/2} e^{-2itk/2^n} \frac{(1 - e^{-2it/2^{n+1}})^2}{2it}. \quad (347)$$

Proof. Transform the finite identity (341) and let $N \rightarrow \infty$. Local uniform convergence follows from uniform L^1 convergence of p_N to p . Absolute convergence by scale also follows directly: smoothness gives $c_{n,k} = \mathcal{O}(2^{-3n/2})$ uniformly in k , while $\hat{h}_{n,k}(z) = \mathcal{O}_K(2^{-3n/2})$ on a compact K ; summing 2^n locations leaves a convergent $\mathcal{O}_K(4^{-n})$ scale contribution. $\square$

The factor $(1 - e^{-iz/2^{n+1}})^2$ is the frequency signature of one vanishing moment. In [Appendix 6](#) it will be replaced by a product with as many sine factors as there are nonzero binary digits in the Walsh index.

5.3 Faber–Schauder series of the Fabius function

Let $\psi_{n,k}$ be the peak-one tent supported on $I_{n,k}$, with apex at

$$m_{n,k} = \frac{2k+1}{2^{n+1}}. \quad (348)$$

Define

$$d_{n,k} = F(m_{n,k}) - \frac{1}{2}F(k2^{-n}) - \frac{1}{2}F((k+1)2^{-n}) \quad (349)$$

$$= \frac{1}{2}(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = 2^{-n/2-1} c_{n,k} \in \mathbb{Q}. \quad (350)$$

Theorem 5.3 (Rational Faber–Schauder representation). *The piecewise-linear interpolant $L_N F$ through the dyadic values $F(k2^{-N})$ is*

$$L_N F(x) = x + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} d_{n,k} \psi_{n,k}(x). \quad (351)$$

It converges uniformly to F , hence

$$F(x) = x + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} d_{n,k} \psi_{n,k}(x), \quad (352)$$

where the series is understood by complete dyadic levels. The coefficients are rational and obey

$$d_{n,2^n-1-k} = -d_{n,k}. \quad (353)$$

Moreover,

$$\|F - L_N F\|_{\infty} \leq \frac{\|F''\|_{\infty}}{8} 4^{-N}. \quad (354)$$

Proof. The classical Faber–Schauder construction says that the coefficient added at a new midpoint is its value minus the chord average, giving (349). Equation (350) follows by expanding the two child masses. Uniform convergence follows from continuity; the displayed error is the standard linear-interpolation bound on intervals of length 2^{-N} . Symmetry follows from $F(1-x) = 1-F(x)$. $\square$

Because F is convex on $[0, 1/2]$ and concave on $[1/2, 1]$, one also has

$$d_{n,k} < 0 \quad (0 \leq k < 2^{n-1}), \quad d_{n,k} > 0 \quad (2^{n-1} \leq k < 2^n), \quad (355)$$

with the $n = 0$ coefficient zero by symmetry.

5.4 A sinc-squared series for the Fourier image of F

The transform of a peak-one tent of width 2^{-n} is

$$\int_0^1 \psi_{n,k}(x) e^{-izx} dx = \frac{2^{-n}}{2} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (356)$$

Since

$$\int_0^1 x e^{-izx} dx = \frac{e^{-iz}(1+iz) - 1}{z^2}, \quad (357)$$

we obtain a representation of the Fourier image of the Fabius function itself.

Corollary 5.4 (Schauder–sinc representation of $\mathcal{F}_F$). *Locally uniformly on $\mathbb{C}$,*

$$\mathcal{F}_F(z) = \frac{e^{-iz}(1+iz) - 1}{z^2} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} d_{n,k} 2^{-n-1} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (358)$$

All coefficients $d_{n,k}$ are rational. Equation (358) is identical, after summation, to the compact product formula (295).

The identity equates a single infinite sinc product to a doubly indexed sum of translated sinc^2 atoms. It is a time–frequency multiresolution counterpart of the canonical zero product (277).

6 Walsh–Hadamard and Thue–Morse product representations

6.1 Paley Walsh functions and the finite Hadamard transform

Let

$$r_j(x) = (-1)^{\lfloor 2^{j+1}x \rfloor}, \quad j \geq 0, \quad (359)$$

away from dyadic endpoints. If

$$m = \sum_{j \geq 0} b_j 2^j, \quad b_j \in \{0, 1\}, \quad (360)$$

then the Paley Walsh function is

$$w_m(x) = \prod_{j \geq 0} r_j(x)^{b_j}. \quad (361)$$

Write

$$r(m) = \text{wt}_2(m) = \sum_j b_j, \quad \sigma(m) = \sum_j j b_j. \quad (362)$$

The top mode w_{2^N-1} on a depth- N partition is the signed Thue–Morse step function.

Define the Walsh coefficient of the Fabius density by

$$a_m = \int_0^1 p(x) w_m(x) dx. \quad (363)$$

If $m < n_N$, then w_m is constant on every depth- N cell. With

$$\varepsilon_{m,N}(k) = w_m\left(\frac{k + 1/2}{n_N}\right) \in \{\pm 1\}, \quad (364)$$

we have the exact finite formula

$$a_m = \sum_{k=0}^{n_N-1} \varepsilon_{m,N}(k) \Delta_{N,k}. \quad (365)$$

Thus every a_m is rational.

Let $H_N = (\varepsilon_{m,N}(k))_{0 \leq m, k < n_N}$. In vector notation,

$$a^{(N)} = H_N \Delta_N, \quad \Delta_N = n_N^{-1} H_N a^{(N)}, \quad H_N H_N^\top = n_N I. \quad (366)$$

This is an exact rational Walsh–Hadamard transform of the dyadic mass vector.

Theorem 6.1 (Walsh expansion and finite Parseval). *For every N ,*

$$p_N(x) = \sum_{m=0}^{n_N-1} a_m w_m(x). \quad (367)$$

Consequently the power-of-two Walsh partial sums converge uniformly:

$$p(x) = \lim_{N \rightarrow \infty} \sum_{m=0}^{n_N-1} a_m w_m(x). \quad (368)$$

At each finite level,

$$\sum_{m=0}^{n_N-1} a_m^2 = n_N \sum_{k=0}^{n_N-1} \Delta_{N,k}^2 = \|p_N\|_2^2. \quad (369)$$

Hence

$$\|p\|_2^2 = \sum_{m=0}^{\infty} a_m^2, \quad \|\text{up}\|_2^2 = \frac{1}{2} \sum_{m=0}^{\infty} a_m^2. \quad (370)$$

Proof. On a depth- N cell, (367) is exactly the inverse Hadamard formula in (366). Uniform convergence is Theorem 4.2. Parseval follows from orthonormality of the Walsh functions or directly from $H_N H_N^\top = n_N I$. $\square$

6.2 Reflection selection rules

Binary reflection complements every Rademacher sign, modulo a null set:

$$w_m(1-x) = (-1)^{r(m)} w_m(x). \quad (371)$$

Since $p(1-x) = p(x)$, this gives a sharp parity filter.

Proposition 6.2 (Hamming-parity selection). *The density coefficients satisfy*

$$a_m = 0 \quad \text{whenever } r(m) \text{ is odd.} \quad (372)$$

If

$$b_m = \int_0^1 \left(F(x) - \frac{1}{2} \right) w_m(x) dx, \quad (373)$$

then

$$b_m = 0 \quad \text{whenever } r(m) \text{ is even.} \quad (374)$$

Proof. Substitute $x \mapsto 1-x$. For a_m the integral is multiplied by $(-1)^{r(m)}$; for b_m , the antisymmetry $F(1-x) - 1/2 = -(F(x) - 1/2)$ contributes one additional minus sign. $\square$

Thus the Walsh support of the density occupies only even Hamming layers, while the centered distribution function occupies only odd layers. This is a digital analogue of the disappearance of odd ordinary Fourier and polynomial moments under reflection symmetry.

6.3 Finite-product Fourier transform of a Walsh mode

Set

$$W_m(z) = \int_0^1 w_m(x) e^{-izx} dx. \quad (375)$$

For $m > 0$, let

$$N(m) = 1 + \lfloor \log_2 m \rfloor. \quad (376)$$

Theorem 6.3 (Walsh finite-product transform). *For $m = 0$,*

$$W_0(z) = e^{-iz/2} \frac{2 \sin(z/2)}{z}. \quad (377)$$

For $m > 0$, with $N = N(m)$ and $r = r(m)$,

$$W_m(z) = \frac{1 - e^{-iz/2^N}}{iz} \prod_{j=0}^{N-1} \left(1 + (-1)^{b_j} e^{-iz/2^{j+1}} \right) \quad (378)$$

$$= e^{-iz/2} \frac{2^{N+1} i^r}{z} \sin\left(\frac{z}{2^{N+1}}\right) \prod_{b_j=0} \cos\left(\frac{z}{2^{j+2}}\right) \prod_{b_j=1} \sin\left(\frac{z}{2^{j+2}}\right). \quad (379)$$

Both right sides have removable singularities and define the same entire function.

Proof. At depth N , integrate the constant value of w_m on each cell. The common cell integral is $(1 - e^{-iz/2^N})/(iz)$, and the signed geometric sum factorizes according to the binary digits of the cell index:

$$\sum_{k=0}^{2^N-1} \varepsilon_{m,N}(k) q^k = \prod_{j=0}^{N-1} \left(1 + (-1)^{b_j} q^{2^{N-1-j}} \right), \quad q = e^{-iz/2^N}.$$

This proves (378). Pairing exponentials into sines and cosines gives (379). $\square$

6.4 A series of finite products for the infinite sinc product

Transforming the Walsh partial sums gives a representation of Φ that is qualitatively different from the canonical product: the coefficients encode Fabius arithmetic, while each basis transform is a finite product.

Theorem 6.4 (Walsh product-series for Φ). *Locally uniformly on $\mathbb{C}$,*

$$\Phi(t) = e^{it} \sum_{m=0}^{\infty} a_m W_m(2t), \quad (380)$$

where the sum is taken through complete blocks $m < 2^N$. More explicitly,

$$\begin{aligned} \Phi(t) = \operatorname{sinc}(t) + \sum_{\substack{m \geq 1 \\ r(m) \text{ even}}} a_m \frac{2^{N(m)} (-1)^{r(m)/2}}{t} \sin\left(\frac{t}{2^{N(m)}}\right) \\ \times \prod_{b_j=0} \cos\left(\frac{t}{2^{j+1}}\right) \prod_{b_j=1} \sin\left(\frac{t}{2^{j+1}}\right). \end{aligned} \quad (381)$$

Each coefficient a_m is rational and is given exactly by (365).

Proof. Since $p_N \rightarrow p$ in L^1 and all functions are supported in $[0, 1]$, their Fourier transforms converge locally uniformly. Substitute the finite Walsh expansion (367) and use $\Phi(t) = e^{it} \widehat{p}(2t)$. Equation (381) follows from Theorem 6.3 and the parity filter (372). $\square$

The corresponding hyperbolic product-series is obtained by $t \mapsto iz$:

$$M(z) = e^{-z} \sum_{m=0}^{\infty} a_m W_m(2iz), \quad (382)$$

so each term becomes a finite product of sinh and cosh factors. Equations (272) and (381) therefore give a product-versus-product-series duality for the same entire function.

6.5 Exact Hamming-weight cancellation

The number of sine factors in (379) is not cosmetic: it is the exact number of vanishing algebraic moments.

Theorem 6.5 (Walsh cancellation order and first moment). *For $m > 0$, put $r = r(m)$ and $\sigma = \sigma(m)$. Then*

$$\int_0^1 x^q w_m(x) dx = 0, \quad 0 \leq q < r, \quad (383)$$

and

$$\int_0^1 x^r w_m(x) dx = (-1)^r r! 2^{-2r-\sigma}. \quad (384)$$

Thus W_m has a zero of exact order $r(m)$ at the origin.

Proof. Extract the small factors from (379). One obtains

$$W_m(z) = i^r 2^{-2r-\sigma} z^r R_m(z), \quad R_m(0) = 1, \quad (385)$$

where R_m is entire. On the other hand,

$$W_m(z) = \sum_{q=0}^{\infty} \frac{(-iz)^q}{q!} \int_0^1 x^q w_m(x) dx. \quad (386)$$

Comparing the first nonzero Taylor coefficient proves both statements. $\square$

For the top Thue–Morse mode, $m = 2^N - 1$, one has $r = N$ and $\sigma = N(N-1)/2$. Hence

$$\int_0^1 x^q w_{2^N-1}(x) dx = 0 \quad (q < N), \quad \int_0^1 x^N w_{2^N-1}(x) dx = (-1)^N N! 2^{-N(N+3)/2}. \quad (387)$$

This is the continuous, normalized Prouhet cancellation attached to the Thue–Morse sign pattern.

6.6 The Gaussian-binomial cancellation spectrum

The first nonzero moments aggregate exactly by Hamming weight.

Theorem 6.6 ($q = 1/2$ Gaussian-binomial aggregation). *For $0 \leq r \leq N$,*

$$\sum_{\substack{0 \leq m < 2^N \\ r(m)=r}} 2^{-\sigma(m)} = 2^{-r(r-1)/2} \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2}. \quad (388)$$

Consequently

$$\sum_{\substack{0 \leq m < 2^N \\ r(m)=r}} \int_0^1 x^r w_m(x) dx = (-1)^r r! 2^{-2r-r(r-1)/2} \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2}. \quad (389)$$

Proof. The finite q -binomial theorem gives

$$\prod_{j=0}^{N-1} (1 + y 2^{-j}) = \sum_{r=0}^N 2^{-r(r-1)/2} \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2} y^r.$$

The coefficient of y^r on the left is the sum of $2^{-\sigma(m)}$ over all subsets of r bit positions. Combine with (384). $\square$

The appearance of $q = 1/2$, rather than the $q = 1/4$ common in the even-moment and scale-energy theory, has a simple origin: it records bit *positions*, not squared dyadic lengths. Squaring amplitudes or passing to even cumulants naturally changes $1/2$ to $1/4$.

6.7 All higher Walsh moments: a Bell–Bernoulli formula

The product formula encodes every moment, not only the first nonzero one. Define the even entire factor

$$E_m(z) = \operatorname{sinc}\left(\frac{z}{2^{N+1}}\right) \prod_{b_j=0} \cos\left(\frac{z}{2^{j+2}}\right) \prod_{b_j=1} \operatorname{sinc}\left(\frac{z}{2^{j+2}}\right), \quad (390)$$

where $N = N(m)$. Then

$$R_m(z) = e^{-iz/2} E_m(z) \quad (391)$$

in (385). Write

$$E_m(z) = \sum_{j=0}^{\infty} e_{m,j} z^{2j}. \quad (392)$$

For $q \geq 1$, set

$$\lambda_{2q}(m) = -\frac{2^{2q-1}|B_{2q}|}{q} \left[2^{-2q(N+1)} + \sum_{b_j=1} 2^{-2q(j+2)} + (2^{2q} - 1) \sum_{b_j=0} 2^{-2q(j+2)} \right]. \quad (393)$$

The classical expansions of log sinc and log cos show that

$$\log E_m(z) = \sum_{q=1}^{\infty} \lambda_{2q}(m) \frac{z^{2q}}{(2q)!}. \quad (394)$$

Therefore

$$e_{m,j} = \frac{1}{(2j)!} Y_{2j}(0, \lambda_2(m), 0, \lambda_4(m), \dots, \lambda_{2j}(m)). \quad (395)$$

Theorem 6.7 (Bell–Bernoulli formula for Walsh moments). *Let $m > 0$, $r = r(m)$, and $\ell \geq 0$. Then*

$$\int_0^1 x^{r+\ell} w_m(x) dx = (-1)^r \frac{(r+\ell)!}{2^{2r+\sigma(m)}} \sum_{j=0}^{\lfloor \ell/2 \rfloor} \frac{(-1)^j e_{m,j}}{2^{\ell-2j}(\ell-2j)!}, \quad (396)$$

with $e_{m,j}$ given explicitly by (395) and (393).

Proof. Use $W_m(z) = i^r 2^{-2r-\sigma} z^r e^{-iz/2} E_m(z)$, expand the exponential and the even series (392), and compare with (386). Formula (395) is the standard exponential Bell-polynomial coefficient formula applied to (394). $\square$

This theorem places the Bernoulli numbers in the *basis moments*, while the rational coefficients a_m carry the Fabius-specific arithmetic. The ordinary Fabius moments can therefore be reconstructed as

$$m_q = \lim_{N \rightarrow \infty} \sum_{m=0}^{2^N-1} a_m \int_0^1 x^q w_m(x) dx. \quad (397)$$

Only modes with even $r(m) \leq q$ contribute. For example, since $m_2 = 5/18$,

$$\sum_{\substack{m \geq 1 \\ r(m)=2}} 2^{-\sigma(m)} a_m = -\frac{4}{9}. \quad (398)$$

Indeed, the constant Walsh mode contributes $\int_0^1 x^2 dx = 1/3$, and (384) contributes $2^{-3-\sigma(m)} a_m$ from every weight-two mode.

Remark 6.8. Equation (397) is naturally ordered by dyadic blocks, not by arbitrary rearrangement. Blockwise convergence follows immediately from uniform convergence of the Walsh conditional expectations. Establishing absolute convergence after regrouping by Hamming weight is a separate and interesting coefficient-decay problem.

7 Bernstein polynomials and positive beta-mixture limits

7.1 Rational Bernstein approximants

For $n = n_N = 2^N$, define the Bernstein polynomial

$$\mathcal{B}_N F(x) = \sum_{k=0}^n F(k/n) \binom{n}{k} x^k (1-x)^{n-k}. \quad (399)$$

Every coefficient in the Bernstein basis is a rational dyadic Fabius value. The power-basis form is also rational:

$$\mathcal{B}_N F(x) = \sum_{j=0}^n \binom{n}{j} \left(\sum_{k=0}^j (-1)^{j-k} \binom{j}{k} F(k/n) \right) x^j. \quad (400)$$

Thus

$$F(x) = \lim_{N \rightarrow \infty} \mathcal{B}_N F(x) \quad (401)$$

uniformly, as a limit of explicit rational polynomials.

The dyadic Faber interpolants of [Appendix 5](#) converge at order 4^{-N} because they exploit the C^2 smoothness on a mesh of width 2^{-N} . The Bernstein approximants use a binomial cloud of width $n^{-1/2}$ and have the classical expansion

$$\mathcal{B}_N F(x) - F(x) = \frac{x(1-x)}{2n} F''(x) + \mathcal{O}(n^{-2}) \quad (402)$$

under the available smoothness. Their advantage is positivity and the probabilistic mixture structure of the derivative.

7.2 Derivative as a beta-density mixture

Differentiating (399) gives

$$p_N^B(x) := (\mathcal{B}_N F)'(x) \quad (403)$$

$$= n \sum_{k=0}^{n-1} \Delta_{N,k} \binom{n-1}{k} x^k (1-x)^{n-1-k} \quad (404)$$

$$= \sum_{k=0}^{n-1} \Delta_{N,k} \frac{x^k (1-x)^{n-1-k}}{B(k+1, n-k)}. \quad (405)$$

Theorem 7.1 (Positive rational beta-mixture representation). *The functions p_N^B are probability densities and are finite mixtures of $\text{Beta}(k+1, n-k)$ densities with the exact rational weights $\Delta_{N,k}$. Moreover,*

$$p_N^B \longrightarrow p \quad \text{uniformly on } [0, 1]. \quad (406)$$

Consequently p is a uniform limit of positive polynomial densities whose mixture weights are exactly the dyadic Fabius masses.

Proof. Formula (405) proves positivity and unit mass. The classical derivative theorem for Bernstein polynomials states that $(\mathcal{B}_n f)' \rightarrow f'$ uniformly when $f \in C^1[0, 1]$. $\square$

This representation is structurally different from the uniform-cell mixture:

model	component family	weights
p_N	disjoint equal-width uniforms	$\Delta_{N,k}$
p_N^B	overlapping beta densities	$\Delta_{N,k}$

The same rational vector is therefore a mixing law for both a local and a global positive basis.

7.3 Hypergeometric transform representations

The moment-generating function of a $\text{Beta}(a, b)$ variable is ${}_1F_1(a; a + b; z)$. Hence:

Theorem 7.2 (Confluent-hypergeometric mixture limits). *Locally uniformly on $\mathbb{C}$,*

$$\int_0^1 p(x)e^{zx} dx = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; z), \quad (407)$$

$$\Phi(t) = e^{it} \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; -2it), \quad (408)$$

$$M(z) = e^{-z} \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; 2z). \quad (409)$$

Thus the infinite sinc and hyperbolic-sinc products are limits of positive rational mixtures of confluent hypergeometric functions.

Proof. Integrate (405) against e^{zx} and use uniform convergence. The affine transform relations $\Phi(t) = e^{it} \hat{p}(2t)$ and $M(z) = e^{-z} \int_0^1 p(x)e^{2zx} dx$ give the remaining formulas. $\square$

The beta moments give another rational moment limit:

$$m_r = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} \frac{(k+1)_r}{(n_N+1)_r}, \quad r \in \mathbb{N}, \quad (410)$$

where $(a)_r$ is the rising factorial. Comparison of (410) with the cell formula (337) gives two positive rational quadrature schemes for the same exact Fabius moments.

8 Dual quantile cells and equal-weight sinc mixtures

8.1 The inverse dyadic width array

The primal mass array fixes spatial widths and varies probabilities. The inverse construction fixes probabilities and varies spatial widths. Put

$$g_{N,k} = G(k/n_N), \quad q_{N,k} = g_{N,k+1} - g_{N,k} > 0, \quad 0 \leq k < n_N. \quad (411)$$

Then

$$\sum_{k=0}^{n_N-1} q_{N,k} = 1, \quad (412)$$

$$q_{N,n_N-1-k} = q_{N,k}, \quad (413)$$

$$q_{N,k} = q_{N+1,2k} + q_{N+1,2k+1}. \quad (414)$$

Unlike $\Delta_{N,k}$, these widths are not expected to be rational in general. They are exact inverse-Fabius data.

Let $G_N = L_N G$ be the piecewise-linear interpolant through the points $(k/n_N, g_{N,k})$. Pushing uniform measure forward by G_N gives the density

$$p_N^Q(x) = \frac{1}{n_N q_{N,k}}, \quad g_{N,k} < x < g_{N,k+1}. \quad (415)$$

Each interval has probability $1/n_N$. Thus p_N^Q is a finite mixture of variable-width uniforms with *equal* weights.

Primal–dual cell dictionary

	spatial interval	probability weight
primal Fabius cells	fixed width $1/n_N$	variable $\Delta_{N,k}$
dual quantile cells	variable width $q_{N,k}$	fixed $1/n_N$

The two projective arrays are interchanged by F and G : the first samples increments of F on a uniform spatial grid, the second samples increments of G on a uniform probability grid.

8.2 Exact chord integration and the Fourier image

On the k th probability cell, G_N moves linearly from $g_{N,k}$ to $g_{N,k+1}$. Integrating an exponential along that chord gives a sinc factor.

Theorem 8.1 (Inverse-quantile chord representation). *Locally uniformly on $\mathbb{C}$,*

$$\Phi(t) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{-it(g_{N,k} + g_{N,k+1} - 1)} \operatorname{sinc}(t q_{N,k}), \quad (416)$$

$$M(z) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{z(g_{N,k} + g_{N,k+1} - 1)} \frac{\sinh(z q_{N,k})}{z q_{N,k}}. \quad (417)$$

Equivalently, Φ is the locally uniform limit of the Fourier transforms of finite equal-weight uniform mixtures on the centered intervals

$$[2g_{N,k} - 1, 2g_{N,k+1} - 1]. \quad (418)$$

Proof. Uniform continuity of G gives $G_N \rightarrow G$ uniformly. Therefore

$$\int_0^1 e^{-it(2G_N(u)-1)} du \longrightarrow \int_0^1 e^{-it(2G(u)-1)} du = \Phi(t)$$

locally uniformly in t . On a probability cell of length $1/n_N$, direct integration of the affine phase gives

$$\frac{1}{n_N} e^{-it(g_{N,k} + g_{N,k+1} - 1)} \operatorname{sinc}(t q_{N,k}).$$

Summing proves (416); Wick rotation gives (417). $\square$

Removing the chord average gives the simpler midpoint Riemann representation

$$\Phi(t) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} \exp \left[-it \left(2G \left(\frac{k + 1/2}{n_N} \right) - 1 \right) \right]. \quad (419)$$

The chord formula is preferable when inverse values at endpoints are already available, because it integrates the linear interpolant exactly.

8.3 Moment and Laplace limits from inverse dyadic values

For the uncentered law, exact integration of powers along each chord yields:

Corollary 8.2 (Inverse-dyadic moment polygons). *For every integer $r \geq 0$,*

$$m_r = \lim_{N \rightarrow \infty} \frac{1}{n_N(r+1)} \sum_{k=0}^{n_N-1} \frac{g_{N,k+1}^{r+1} - g_{N,k}^{r+1}}{g_{N,k+1} - g_{N,k}} \quad (420)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{n_N(r+1)} \sum_{k=0}^{n_N-1} \sum_{j=0}^r g_{N,k}^j g_{N,k+1}^{r-j}. \quad (421)$$

For arbitrary $z \in \mathbb{C}$,

$$\int_0^1 p(x) e^{zx} dx = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{z(g_{N,k} + g_{N,k+1})/2} \operatorname{sinhc}\left(\frac{z q_{N,k}}{2}\right), \quad (422)$$

where $\operatorname{sinhc} w = \sinh(w)/w$.

The first line is a limit of complete homogeneous symmetric polynomials in adjacent inverse-dyadic values. It complements the repository's local inverse series and exact inverse values at special dyadic levels: local inverse arithmetic enters a global moment formula through a simple polygonal limit.

8.4 Dual L^q spectra and entropy constants

Because $q_{N,k} = \int_{k/n_N}^{(k+1)/n_N} G'(u) du$, the slope $n_N q_{N,k}$ is the dyadic conditional expectation of G' . This gives a discrete counterpart of the inverse-derivative norm threshold.

Theorem 8.3 (Primal–dual Rényi spectra). *For $s > 0$,*

$$n_N^{s-1} \sum_k \Delta_{N,k}^s \longrightarrow \int_0^1 p(x)^s dx. \quad (423)$$

For $0 < s < 1$,

$$n_N^{s-1} \sum_k q_{N,k}^s \longrightarrow \int_0^1 G'(u)^s du = \int_0^1 p(x)^{1-s} dx. \quad (424)$$

At $s = 1$ both normalized sums equal one. For $s > 1$, the dual normalized sum tends to $+\infty$, consistently with $G' \notin L^s(0, 1)$.

Proof. The primal statement is (334). On each probability cell, $n_N q_{N,k} = \mathbb{E}[G' \mid \mathcal{D}_N]$. The dyadic martingale converges to G' in L^1 . For $0 < s < 1$, the inequality $|a^s - b^s| \leq |a - b|^s$ and Hölder's inequality give convergence of the s th powers in L^1 . The change of variables $u = F(x)$ gives

$$\int_0^1 G'(u)^s du = \int_0^1 p(x)^{1-s} dx.$$

For $s > 1$, conditional-expectation L^s norms increase to the infinite L^s norm of G' . $\square$

The one-sided derivatives at $s = 1$ give entropy constants. In addition to (336), the endpoint asymptotics imply $\log p \in L^1(0, 1)$, and

$$- \sum_{k=0}^{n_N-1} q_{N,k} \log q_{N,k} = \log n_N + \int_0^1 \log p(x) dx + o(1). \quad (425)$$

Thus the two exact partitions retain different continuous entropies:

$$H(\Delta_N) - \log n_N \longrightarrow - \int p \log p, \quad (426)$$

$$H(q_N) - \log n_N \longrightarrow \int \log p. \quad (427)$$

The second constant is sensitive to endpoint flatness, whereas the first is weighted by the probability density itself.

9 Haar scale energies and the sinc^4 continuum expansion

9.1 Three exact forms of the energy

Define the detail energy

$$E_n = \|p_{n+1} - p_n\|_2^2 = \sum_{k=0}^{2^n-1} c_{n,k}^2. \quad (428)$$

Using (340),

$$E_n = 2^n \sum_{k=0}^{2^n-1} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1})^2 \in \mathbb{Q}. \quad (429)$$

Thus a sum of squares of generally quadratic Haar coefficients is rational. Orthogonality gives

$$\|p_N\|_2^2 = 1 + \sum_{n=0}^{N-1} E_n, \quad \|p\|_2^2 = 1 + \sum_{n=0}^{\infty} E_n. \quad (430)$$

The Haar and Walsh filtrations are the same nested dyadic subspaces. Hence:

Proposition 9.1 (Haar–Walsh shell identity). *For every $n \geq 0$,*

$$E_n = \sum_{m=2^n}^{2^{n+1}-1} a_m^2. \quad (431)$$

Only even-Hamming-weight indices contribute to the right side.

Proof. Both sides are the squared norm of the orthogonal projection of p onto $V_{n+1} \ominus V_n$, where V_n is the space of functions constant on depth- n dyadic cells. The Haar basis organizes this complement by location; the Walsh basis organizes it by sequency. $\square$

Thus the same rational E_n is simultaneously a local curvature statistic of F , a sibling-mass contrast, and a Walsh spectral-shell energy.

9.2 Local Taylor expansion of a Haar coefficient

Let $h = 2^{-n}$ and let $m_{n,k} = (k + 1/2)h$ be the center of a parent cell. Flatness of p at the endpoints permits uniform Taylor expansion to arbitrary order. Direct integration of odd Taylor terms gives

$$c_{n,k} \sim - \sum_{r=0}^{\infty} \frac{h^{2r+3/2}}{2^{2r+1}(2r+2)!} p^{(2r+1)}(m_{n,k}). \quad (432)$$

The first terms are

$$c_{n,k} = -h^{3/2} \left[\frac{p'(m_{n,k})}{4} + \frac{h^2 p'''(m_{n,k})}{192} + \frac{h^4 p^{(5)}(m_{n,k})}{23040} + \mathcal{O}(h^6) \right]. \quad (433)$$

This explains the scale $E_n \asymp 4^{-n}$ despite the infinite smoothness of p : the Haar wavelet has only one vanishing moment. Higher-order multiwavelets would expose the superalgebraic smoothness more efficiently.

9.3 A frequency multiplier and a superalgebraic alias remainder

Let

$$K_h(x) = h^{-1/2} (\mathbf{1}_{[-h/2,0]}(x) - \mathbf{1}_{[0,h/2]}(x)). \quad (434)$$

Its squared Fourier modulus is

$$|\widehat{K}_h(\xi)|^2 = \frac{16}{h\xi^2} \sin^4\left(\frac{h\xi}{4}\right). \quad (435)$$

The coefficients $c_{n,k}$ are samples of a translate of $p * K_h$ at the cell centers. Poisson summation for the squared samples separates a zero alias from all nonzero aliases.

Theorem 9.2 (sinc^4 multiplier formula). *For $h = 2^{-n}$,*

$$E_n = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{p}(\xi)|^2 \frac{h^2 \xi^2}{16} \text{sinc}^4\left(\frac{h\xi}{4}\right) d\xi + R(h), \quad (436)$$

where for every $M > 0$,

$$R(h) = \mathcal{O}_M(h^M) \quad (h \downarrow 0). \quad (437)$$

Proof. Apply Poisson summation to the samples of $|p * K_h|^2$ on the shifted lattice $h(\mathbb{Z} + 1/2)$. The zero Fourier mode is

$$\frac{1}{h} \|p * K_h\|_2^2 = \frac{1}{2\pi h} \int |\widehat{p}(\xi)|^2 |\widehat{K}_h(\xi)|^2 d\xi,$$

which simplifies to the integral in (436). Every nonzero alias contains a factor $\widehat{p}(\xi + 2\pi\ell/h)$ or its conjugate. Because the zero extension of p is $C_c^\infty(\mathbb{R})$, $\widehat{p}$ is Schwartz; the nonzero aliases are therefore smaller than every power of h after summation over $\ell \neq 0$. $\square$

9.4 All orders and Bell–Bernoulli coefficients

Write

$$\text{sinc}^4 z = \sum_{j=0}^{\infty} \alpha_j z^{2j}, \quad \alpha_0 = 1, \quad \alpha_1 = -\frac{2}{3}, \quad \alpha_2 = \frac{1}{5}, \quad \alpha_3 = -\frac{34}{945}, \dots \quad (438)$$

Expanding the multiplier in (436) and using Parseval gives:

Theorem 9.3 (All-orders Haar-energy expansion). *As $n \rightarrow \infty$, with $h = 2^{-n}$,*

$$E_n \sim \sum_{j=0}^{\infty} \frac{\alpha_j}{16^{j+1}} h^{2j+2} \|p^{(j+1)}\|_2^2. \quad (439)$$

In particular,

$$E_n \sim \frac{h^2}{16} \|p'\|_2^2 - \frac{h^4}{384} \|p''\|_2^2 + \frac{h^6}{20480} \|p'''\|_2^2 - \frac{17h^8}{30965760} \|p^{(4)}\|_2^2 + \dots \quad (440)$$

Proof. Taylor-expand $\text{sinc}^4(h\xi/4)$ to an arbitrary fixed order in (436). The Schwartz decay of $\widehat{p}$ justifies termwise integration and bounds the remainder. Parseval identifies $(2\pi)^{-1} \int \xi^{2j+2} |\widehat{p}(\xi)|^2 d\xi$ with $\|p^{(j+1)}\|_2^2$. The alias term is smaller than every algebraic order. $\square$

The coefficients themselves are Bell–Bernoulli quantities. Since

$$\log \text{sinc}^4 z = - \sum_{r=1}^{\infty} \frac{2^{2r+1} |B_{2r}|}{r(2r)!} z^{2r}, \quad (441)$$

put

$$\beta_{2r} = - \frac{2^{2r+1} |B_{2r}|}{r}, \quad \beta_{2r+1} = 0. \quad (442)$$

Then

$$\alpha_j = \frac{1}{(2j)!} Y_{2j}(0, \beta_2, 0, \beta_4, \dots, \beta_{2j}). \quad (443)$$

Thus the same Bell–Bernoulli mechanism that governs the moment expansion of Φ also governs the continuum expansion of its dyadic Haar energies, but through the fourth power of a single-scale sinc filter.

9.5 Derivative norms from half-integer samples of the sinc product

The cosine series (291) gives exact rapidly convergent sums for every derivative norm:

$$\|p^{(r)}\|_2^2 = 2^{2r+1} \sum_{m=0}^{\infty} A_m^2 ((2m+1)\pi)^{2r}. \quad (444)$$

Substitution into (439) expresses the rational scale energies asymptotically through discrete samples of the infinite sinc product at the odd half-lattice. The first numerical values are reported in [Appendix 13](#).

9.6 A meromorphic generating function for the rational energies

The complete scale expansion has a strong analytic consequence. Define

$$\mathcal{H}(z) = \sum_{n=0}^{\infty} E_n z^n. \quad (445)$$

Put

$$C_j = \frac{\alpha_j}{16^{j+1}} \|p^{(j+1)}\|_2^2. \quad (446)$$

Then (439) says, at every fixed depth J ,

$$E_n = \sum_{j=0}^J C_j 4^{-(j+1)n} + \mathcal{O}_J(4^{-(J+2)n}). \quad (447)$$

Theorem 9.4 (Meromorphic Haar-energy generating function). *The power series $\mathcal{H}$ has radius of convergence 4 and admits a meromorphic continuation to the whole complex plane. Its only possible finite poles are simple poles at*

$$z = 4, 16, 64, \dots, 4^{j+1}, \dots \quad (448)$$

Near $z = 4^{j+1}$ its principal part is

$$\frac{C_j}{1 - z/4^{j+1}}, \quad (449)$$

and hence the residue is

$$\operatorname{Res}_{z=4^{j+1}} \mathcal{H}(z) = -\frac{\alpha_j}{4^{j+1}} \|p^{(j+1)}\|_2^2. \quad (450)$$

In particular,

$$\operatorname{Res}_{z=4} \mathcal{H}(z) = -\frac{1}{4} \|p'\|_2^2. \quad (451)$$

Proof. The positive leading asymptotic $E_n \sim C_0/4^n$, with $C_0 = \|p'\|_2^2/16 > 0$, gives radius 4. For a fixed J , subtract from $\mathcal{H}$ the finite rational function

$$\sum_{j=0}^J \frac{C_j}{1 - z/4^{j+1}}.$$

By (447), the remaining coefficient sequence is $\mathcal{O}_J(4^{-(J+2)n})$, so its power series is holomorphic in $|z| < 4^{J+2}$. Letting J grow gives compatible continuations to every finite disc. The principal parts and residues are immediate. $\square$

This is a new $q = 1/4$ spectral object attached to the dyadic multiresolution, but unlike the zero determinant already present in the corpus its coefficients are rational sibling-mass energies. The pole at 4^{j+1} directly records the Sobolev norm of derivative order $j + 1$, multiplied by the Bell–Bernoulli coefficient α_j .

9.7 Beyond all orders: a phase-locked alias frontier

The remainder $R(2^{-n})$ is superalgebraic but not expected to be a featureless error term. Its Poisson representation samples cross-products of $\widehat{p}$ at frequencies separated by $2\pi 2^n \ell$. The corpus proves that Φ has a log-square envelope with a nonconstant periodic phase in $\log_2 |t|$. Along a dyadic scale sequence, the phase of each fixed alias ℓ is locked at the fractional part of $\log_2(2\pi|\ell|)$.

Conjecture 9.5 (Phase-locked Haar alias transseries). After subtraction of any fixed number of terms in (439), the first nonzero Poisson aliases admit a transseries whose exponential scale is determined by the proved real-axis decay of $\Phi(2\pi 2^n \ell)$ and whose amplitudes depend on the corresponding locked Lambert phase. In particular, the optimally truncated remainder should be asymptotic to a finite sum of $\ell = \pm 1$ alias contributions except at isolated cancellation phases.

A proof would require inserting the phase-resolved Fourier asymptotic into the exact Poisson cross-correlation integral, not merely into a pointwise sample. This is a concrete meeting point between the repository’s Fourier-decay campaign and the new multiresolution energy theory.

10 Multiresolution of the inverse Fabius function

10.1 Faber–Schauder series and reflection

The inverse G is continuous and strictly increasing, so it has the universal Faber–Schauder expansion

$$G(u) = u + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} e_{n,k} \psi_{n,k}(u), \quad (452)$$

where

$$e_{n,k} = G\left(\frac{2k+1}{2^{n+1}}\right) - \frac{1}{2}G\left(\frac{k}{2^n}\right) - \frac{1}{2}G\left(\frac{k+1}{2^n}\right). \quad (453)$$

The sum is taken through complete dyadic levels and converges uniformly. From $G(1-u) = 1-G(u)$,

$$e_{n,2^n-1-k} = -e_{n,k}. \quad (454)$$

Since F is convex on the left half and concave on the right half, G is concave on $[0, 1/2]$ and convex on $[1/2, 1]$; hence

$$e_{n,k} > 0 \quad (k < 2^{n-1}), \quad e_{n,k} < 0 \quad (k \geq 2^{n-1}). \quad (455)$$

The coefficients are generally not expected to be rational. They package the inverse-dyadic values studied elsewhere in the corpus into a global basis.

10.2 Fourier and Walsh images of the inverse

Transforming the tents gives, levelwise and locally uniformly,

$$\int_0^1 G(u) e^{-iz u} du = \frac{e^{-iz}(1+iz) - 1}{z^2} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} e_{n,k} 2^{-n-1} e^{-iz m_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (456)$$

This is the inverse analogue of (358). It is particularly well suited to low-frequency moments, because derivatives at $z = 0$ produce finite polynomial combinations of inverse-dyadic coefficients at each level.

The centered inverse $G(u) - 1/2$ is antisymmetric under $u \mapsto 1-u$. Therefore its Walsh coefficients

$$g_m = \int_0^1 \left(G(u) - \frac{1}{2}\right) w_m(u) du \quad (457)$$

vanish for even Hamming weight. The inverse and the density thus have complementary Walsh parity:

$$a_m \neq 0 \Rightarrow r(m) \text{ even}, \quad g_m \neq 0 \Rightarrow r(m) \text{ odd}. \quad (458)$$

The finite-product transform (379) applies unchanged to the Walsh expansion of $G - 1/2$.

10.3 Boundary coefficients and slow variation

The leftmost coefficient is

$$e_{n,0} = G(2^{-n-1}) - \frac{1}{2}G(2^{-n}). \quad (459)$$

The phase-resolved Lambert endpoint theory in the repository implies the leading inverse scale

$$\log \frac{1}{G(y)} \sim \sqrt{2 \log 2 \log \frac{1}{y}} \quad (y \downarrow 0), \quad (460)$$

and, more strongly, the fixed-ratio slow-variation law

$$\frac{G(cy)}{G(y)} \longrightarrow 1 \quad (y \downarrow 0), \quad c > 0. \quad (461)$$

Applying $c = 1/2$ to (459) gives a rigorous first-order consequence.

Proposition 10.1 (Boundary Schauder scale). *Assuming the proved phase-resolved inverse endpoint expansion from the primary corpus,*

$$e_{n,0} \sim \frac{1}{2}G(2^{-n}) = 2^{-\sqrt{2n}+o(\sqrt{n})}. \quad (462)$$

Thus the inverse Schauder coefficients at the boundary decay faster than every power of n but slower than every geometric sequence in the level n .

The crude leading model $G(2^{-n}) \approx 2^{-\sqrt{2n}}$ predicts one further term:

$$\frac{e_{n,0}}{G(2^{-n})} \approx \frac{1}{2} - \frac{\log 2}{\sqrt{2n}}. \quad (463)$$

The actual correction must include the lower-Lambert logarithmic terms and the nonconstant phase already known for the endpoint problem.

Conjecture 10.2 (Phase-resolved boundary coefficient). There is an explicit expansion

$$\frac{e_{n,0}}{G(2^{-n})} = \frac{1}{2} - \frac{\log 2}{\sqrt{2n}} + \frac{P_1(\log n, \Theta_n)}{n} + \frac{P_2(\log n, \Theta_n)}{n^{3/2}} + \dots, \quad (464)$$

where Θ_n is the exact lower-Lambert phase inherited from the endpoint expansion and each P_j is explicitly computable from its Bell-polynomial coefficient calculus.

10.4 Boundary domination of inverse interpolation

Let $L_N G$ denote the polygon through the inverse dyadic values $G(k2^{-N})$. Uniform continuity gives $L_N G \rightarrow G$, but the usual C^2 error bound fails because G' diverges at both endpoints. Concavity on the left shows that the interpolation error there is positive, and the first cell already has midpoint error $e_{N,0}$.

Conjecture 10.3 (Boundary-dominated inverse interpolation). The global polygonal error is asymptotically attained in the two boundary cells:

$$\|G - L_N G\|_\infty \sim e_{N,0} \sim \frac{1}{2} G(2^{-N}). \quad (465)$$

In particular,

$$\log_2 \|G - L_N G\|_\infty = -\sqrt{2N} + o(\sqrt{N}). \quad (466)$$

A proof should combine monotonicity of the inverse curvature on successive endpoint cells with the phase-resolved ratio asymptotics $G(2^{-N-j})/G(2^{-N})$. Even a two-sided constant-factor result would provide a sharp complexity theorem for inverse-Fabius interpolation.

11 Comparison of the representation families

The formulas developed above are not redundant. They expose different parts of the same function and have different numerical and formal strengths.

representation	finite data		building blocks	principal strength
Infinite convolution	dyadic scales	uniform	compact uniforms	Probability, support, smoothing equations, finite head–tail factorization.
Sinc product	no sampled values		$\text{sinc}(t/2^j)$	Zeros, entire-function structure, Fourier decay, canonical products.
Half-integer cosine	$\Phi((2m+1)\pi)$		cosines	Fast point evaluation, derivatives, Sobolev norms, twisted Poisson summation.

representation	finite data	building blocks	principal strength
Dyadic cell mixture	rational $\Delta_{N,k}$	equal-width uni- forms	Positive local model, 4^{-N} transform quadrature, exact mass polynomial.
Haar	rational $\gamma_{n,k}$	local sign steps	Curvature by scale and location, exact rational energies, meromorphic energy generating function.
Faber–Schauder	rational $d_{n,k}$	tents and sinc^2 atoms	Uniform piecewise-linear approximation of F and a Fourier series of translated sinc^2 functions.
Walsh–Hadamard	rational a_m	digital characters	Exact Hamming cancellation, Thue–Morse top mode, finite trigonometric-product transforms.
Bernstein/beta	rational $\Delta_{N,k}$	beta densities, ${}_1F_1$	Positive polynomial approximation and hypergeometric transform mixtures.
Quantile chords	inverse values $g_{N,k}$	variable-width uni- forms	Equal probability cells, direct use of inverse Fabius values, finite sinc mixtures.
Legendre– Chebyshev	Bessel integrals of Φ	orthogonal polynomials	Global spectral approximation and transform-to-polynomial bridges.

11.1 A commuting diagram of finite transforms

At level N , one rational vector Δ_N enters three exact linear maps:

$$\Delta_N \xrightarrow{H_N} a^{(N)}, \quad \Delta_N \xrightarrow{\text{sibling differences}} \gamma_n, \quad \Delta_N \xrightarrow{\text{beta kernel}} p_N^B. \quad (467)$$

The same vector also determines the mass polynomial D_N . The resulting functions all integrate to one and converge to p , but their Fourier images factor differently:

$$\widehat{p}_N(z) = \text{exponential polynomial} \times \text{sinc window}, \quad (468)$$

$$\widehat{p}_N(z) = \sum_{m < 2^N} a_m \times \text{finite sine-cosine product}, \quad (469)$$

$$\widehat{p}_N^B(z) = \sum_k \Delta_{N,k} \times {}_1F_1(k+1; 2^N+1; -iz). \quad (470)$$

This makes it possible to transfer a question among a local partition, a digital frequency basis, and a positive global polynomial basis without changing the exact arithmetic input.

11.2 Computational observations

For direct evaluation of $\Phi(t)$ at moderate t , the canonical sinc product remains the simplest representation. The new formulas are useful for other purposes:

- the cell and Haar formulas expose exact rationality and give certified dyadic approximation errors;

- the Walsh formula isolates cancellation order before numerical evaluation, avoiding subtraction of low moments;
- the beta mixture preserves positivity and is convenient for rigorous interval bounds on the real Laplace axis;
- the quantile chord formula is naturally adapted to a table of inverse values and does not require differentiating G ;
- the energy expansion converts fine-scale rational data into derivative norms and vice versa.

12 Further deductions and frontier directions

12.1 Exact arithmetic of multiresolution coefficients

The formulas

$$\begin{aligned}\gamma_{n,k} &= 2^n (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}), \\ d_{n,k} &= \frac{1}{2} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}), \\ a_m &= \sum_k \varepsilon_{m,N}(k) \Delta_{N,k}\end{aligned}$$

reduce the arithmetic of three basis systems to exact dyadic Fabius values. The existing two-adic theory should therefore induce valuation laws for all three coefficient arrays.

Open problem 12.1 (Two-adic multiresolution spectrum). Determine sharp formulas or lower envelopes for $v_2(\gamma_{n,k})$, $v_2(d_{n,k})$, and $v_2(a_m)$ in terms of binary carries, Hamming weight, and the known valuations of dyadic Fabius values. In particular, classify the indices for which sibling subtraction creates additional two-adic cancellation.

A promising approach is to compute the exact mass vectors by the repository's binary reduction algorithm and apply the fast Walsh–Hadamard transform over $\mathbb{Q}_2$, retaining valuations before rational reconstruction.

12.2 Mass-polynomial renormalization

The palindromic rational polynomials D_N are projectively compatible, but (315) alone does not close them under a scalar renormalization. The functional equation of F supplies additional relations among sibling masses.

Open problem 12.2 (Closed recursion for D_N). Find a matrix or q -difference recursion that computes D_{N+1} from a small number of transforms of D_N and finitely many boundary values, preferably in a form compatible with the existing q -binomial dyadic formulas. Such a recursion would turn (329) into an autonomous sequence of rational entire approximants.

A two-component recursion separating the left and right halves is more plausible than a scalar recursion, because $F'(x) = 2F(2x)$ on the left couples a cell mass to an integral of F , not merely to another mass.

12.3 Hamming layers and Walsh pressure

Let

$$\mathcal{W}_{N,r} = \sum_{\substack{m < 2^N \\ r(m)=r}} a_m^2. \quad (471)$$

The exact parity filter removes odd r , and [Theorem 6.6](#) controls the corresponding basis moments. The coefficient energy itself is a new digital spectrum.

Conjecture 12.3 (Walsh Hamming-layer pressure). There exists a limiting pressure function $P(\theta)$ for the bivariate sums

$$\sum_{m < 2^N} |a_m|^2 e^{\theta r(m)}, \quad (472)$$

possibly after subtraction of a deterministic quadratic scale term. Its Legendre transform should describe the distribution of Walsh energy across Hamming layers and should be related, but not identical, to the multifractal pressure already found for Fourier lobes.

The exact shell identity [\(431\)](#) supplies the marginal in binary length; the new problem is to resolve each shell by Hamming weight.

12.4 Higher-order rational multiwavelets

Haar has only one vanishing moment, explaining the geometric energy scale. Alpert-type multiwavelets on dyadic cells have any prescribed number r of vanishing moments. Their coefficients are finite linear combinations of integrals $\int_I x^j p(x) dx$, which the repository's antiderivative and exact dyadic machinery can evaluate rationally.

Open problem 12.4 (Fabius–Alpert hierarchy). Construct an exact rational multiwavelet hierarchy for p and derive its Fourier product-series. For order r , the detail energy should start at $h^{2r} \|p^{(r)}\|_2^2$ with a multiplier determined by the corresponding finite filter. Determine the Bell–Bernoulli coefficients and the meromorphic pole lattice of its scale-energy generating function.

This would replace the single sinc^4 multiplier by squared higher-order finite-difference symbols and produce a family of rational spectral objects.

12.5 Beta mixtures and confluent dyadic extrapolation

The beta mixture converges at the binomial scale $n^{-1} = 2^{-N}$, slower than the cell transform's 4^{-N} rate. Its complete Voronovskaya expansion is in integer powers of 2^{-N} , so the repository's confluent geometric filters can be applied without abandoning positivity at the base level.

Open problem 12.5 (Accelerated hypergeometric mixtures). Derive the full operator expansion of $\sum_k \Delta_{N,k} F_1(k+1; 2^N+1; z)$ on compact z -sets and construct $q = 1/2$ extrapolation filters that cancel the first r Bernstein corrections. Quantify how acceleration affects positivity, conditioning, and zero locations.

12.6 Quantile-width extremes and Lambert phases

The largest quantile width is the boundary width $q_{N,0} = G(2^{-N})$, up to symmetry. Hence the extremes of the projective array q_N are governed by the same Lambert phase as the endpoint inverse. The dual Rényi spectrum diverges for $s > 1$ because these boundary widths are much larger than 2^{-N} .

Conjecture 12.6 (Extreme-width partition function). For $s > 1$,

$$\sum_{k=0}^{2^N-1} q_{N,k}^s \sim 2G(2^{-N})^s \mathcal{C}_s(\Theta_N), \quad (473)$$

where $\mathcal{C}_s$ is a positive phase-dependent correction obtained by summing a finite boundary layer $G((j+1)2^{-N}) - G(j2^{-N})$ and Θ_N is the lower-Lambert phase.

This would turn inverse endpoint asymptotics into a discrete partition function and provide a phase-resolved refinement of the L^s threshold for G' .

12.7 Base- b and nonuniform scale systems

The multiresolution arguments do not fundamentally require base two. A base- b weighted-uniform series would produce b -adic mass arrays, b -ary Haar systems, and characters of $(\mathbb{Z}/b\mathbb{Z})^N$. The Walsh moment cancellation would be replaced by digit-character cancellation, while q -binomial aggregates would naturally involve $q = 1/b$.

Open problem 12.7 (Base- b cancellation spectrum). For the geometric scale b^{-j} , derive the analogue of [Theorems 6.5](#) and [6.6](#), identify the first nonzero moments of b -ary characters, and determine which Gaussian or multinomial q -coefficients aggregate the digit-position factors.

For nongeometric weights the finite character product still exists, but the clean q -binomial aggregation is replaced by an elementary symmetric function of the scales.

12.8 Formalization route

Most new statements separate into finite algebra plus standard convergence:

1. define dyadic mass vectors and prove projective consistency;
2. formalize the finite Hadamard matrix and the identity $H_N H_N^T = 2^N I$;
3. identify Walsh partial sums with dyadic conditional expectations;
4. prove the finite product [\(378\)](#) by induction on bit length;
5. extract vanishing moments from the order of zero at the origin;
6. use the finite q -binomial theorem for [\(388\)](#);
7. obtain beta mixtures by differentiating the already elementary Bernstein polynomial;
8. formalize quantile chord integration as an affine integral on each cell.

The analytic bottleneck is the all-orders alias estimate, which requires a Schwartz-space Poisson-summation API. The finite identities and all rationality claims should be substantially easier to integrate into Lean.

13 Numerical and exact-symbolic verification

The accompanying script `numerical_experiments.py` is not used as a substitute for proof. It serves four narrower purposes:

1. independently check the Fourier conventions and finite-product signs;
2. verify the exact Walsh moment and q -binomial identities over a large finite range using rational arithmetic;
3. compare four finite transform representations against the canonical sinc product;
4. test the constants in the Haar-energy expansion.

The default run uses 55 decimal digits, 160 sinc factors, and 200 half-integer cosine modes. It completes in a few seconds in the supplied environment and writes both a plain-text ledger and two figures.

13.1 Exact checks

SymPy rational arithmetic checked all vanishing moments and first nonzero moments for every nonzero Walsh index of binary length at most seven:

$$1016 \text{ exact moment identities checked, } \quad 0 \text{ failures.} \quad (474)$$

It also checked all 35 pairs (N, r) with $1 \leq N \leq 7$ in (388), again with zero failures. Selected exact moments are

m	binary m	$r(m)$	$\int_0^1 x^{r(m)} w_m(x) dx$
1	1	1	$-1/4$
2	10	1	$-1/8$
3	11	2	$1/16$
5	101	2	$1/32$
6	110	2	$1/64$
7	111	3	$-3/256$
15	1111	4	$3/2048$
31	11111	5	$-15/131072$

At the complex test point $z = 3.7 + 0.4i$, the largest discrepancy between the cell-by-cell integral of W_m and the finite product (379), for $0 \leq m \leq 15$, was 3.75×10^{-56} . At dyadic level seven, the largest numerically computed coefficient with odd Hamming weight was 1.73×10^{-54} , while finite Hadamard Parseval had zero residual at the working precision.

As a normalization check, the cosine evaluator gives

$$F(1/4) = 0.06944444444444445713\dots, \quad \frac{5}{72} = 0.06944444444444444\dots;$$

the 1.27×10^{-16} difference is the deliberately finite half-integer Fourier tail, not a failure of the exact dyadic value.

13.2 Transform convergence

The test frequencies were $t = 0.7, 3.1, 7.3, 11.2$. The table reports the maximum absolute error relative to $\Phi(t) = \prod_{j \geq 1} \text{sinc}(t/2^j)$.

N	cell/Walsh	midpoint atomic	beta mixture	Walsh product series
2	$9.720 \cdot 10^{-2}$	$8.750 \cdot 10^{-1}$	$3.312 \cdot 10^{-1}$	$9.720 \cdot 10^{-2}$
3	$2.780 \cdot 10^{-2}$	$1.416 \cdot 10^{-2}$	$1.985 \cdot 10^{-1}$	$2.780 \cdot 10^{-2}$
4	$7.012 \cdot 10^{-3}$	$3.517 \cdot 10^{-3}$	$1.098 \cdot 10^{-1}$	$7.012 \cdot 10^{-3}$
5	$1.760 \cdot 10^{-3}$	$8.805 \cdot 10^{-4}$	$5.796 \cdot 10^{-2}$	$1.760 \cdot 10^{-3}$
6	$4.403 \cdot 10^{-4}$	$2.202 \cdot 10^{-4}$	$2.980 \cdot 10^{-2}$	$4.403 \cdot 10^{-4}$
7	$1.101 \cdot 10^{-4}$	$5.505 \cdot 10^{-5}$	$1.512 \cdot 10^{-2}$	$1.101 \cdot 10^{-4}$

The equality of the cell and Walsh errors is exact, not accidental: the power-of-two Walsh partial sum is precisely p_N , whose transform is the cell mixture. The observed factor near four per level agrees with (333). The Bernstein/beta approximation displays its slower factor near two per level.

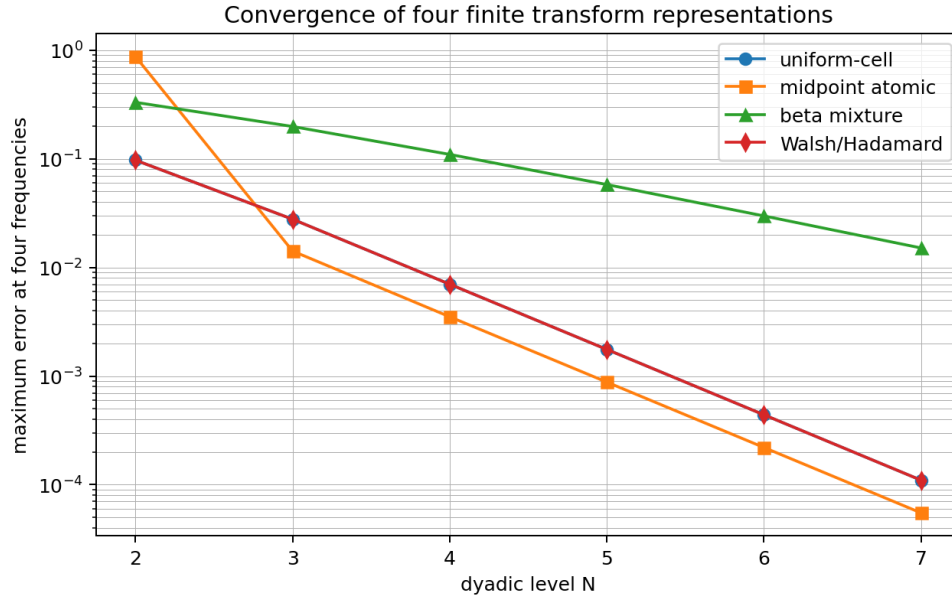


Figure 7: Convergence of the fixed-cell, midpoint-atomic, beta-mixture, and Walsh–Hadamard transform representations at four sample frequencies.

13.3 Haar energy asymptotics

The half-integer Fourier samples give

$$\|p'\|_2^2 = 25.8839531509750580138\dots, \quad (475)$$

$$\|p''\|_2^2 = 1656.57300166240371284\dots, \quad (476)$$

$$\|p'''\|_2^2 = 424082.688425575063737\dots \quad (477)$$

The exact rational energies and the three-term approximation from (440) behave as follows.

n	E_n	$16 \cdot 4^n E_n$	relative error of three terms
2	$6.2500000 \cdot 10^{-2}$	16.0000000000	$4.290 \cdot 10^{-1}$
3	$2.5292927 \cdot 10^{-2}$	25.8999572626	$3.914 \cdot 10^{-2}$
4	$6.2471084 \cdot 10^{-3}$	25.5881558642	$1.220 \cdot 10^{-3}$
5	$1.5757619 \cdot 10^{-3}$	25.8172829143	$1.627 \cdot 10^{-5}$
6	$3.9470093 \cdot 10^{-4}$	25.8671202723	$4.231 \cdot 10^{-8}$
7	$9.8723379 \cdot 10^{-5}$	25.8797415033	$4.139 \cdot 10^{-11}$

The scaled energy converges to $\|p'\|_2^2$, while the three-term expansion quickly becomes much more accurate than the displayed leading convergence.

13.4 Reproducibility files

The archive contains:

file	purpose
numerical_experiments.py	fully commented exact and numerical checks
numerical_results.txt	complete deterministic output ledger
multiresolution_convergence.pdf/.png	transform convergence figure
haar_energy_asymptotics.pdf/.png	scale-energy figure

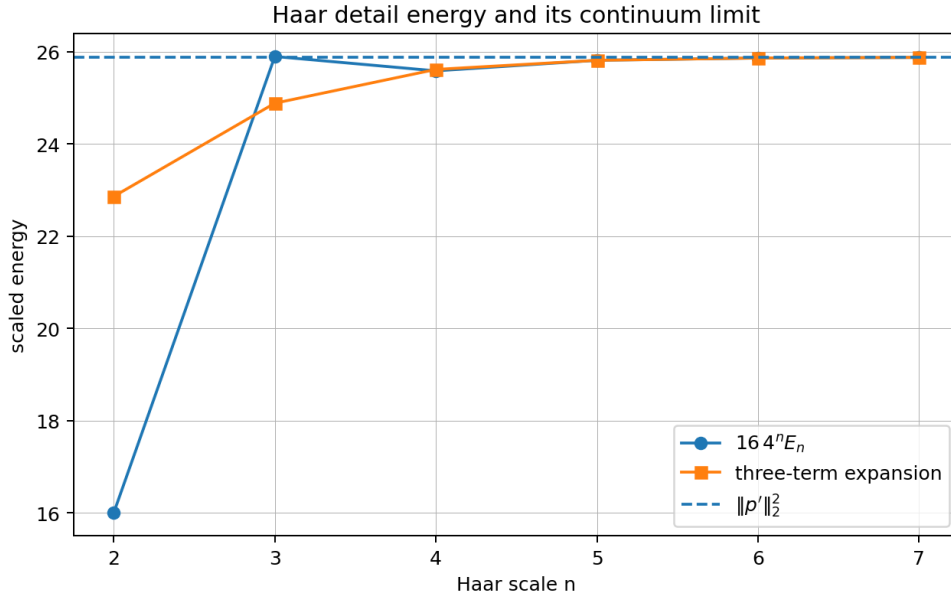


Figure 8: The scaled exact Haar energy, its three-term continuum expansion, and the limiting derivative norm.

The script contains independent direct and product implementations of the Walsh transform, rather than comparing a formula with a rearrangement of itself.

14 Conclusions

The central organizing object of this report is the exact dyadic mass array $\Delta_{N,k}$. It simultaneously yields:

1. a positive uniform-cell approximation and a rational exponential-polynomial transform limit;
2. rational unnormalized Haar coefficients and rational Faber–Schauder coefficients;
3. exact Walsh–Hadamard coefficients, with a finite sine–cosine product for every basis transform;
4. a positive beta-density mixture and a confluent-hypergeometric transform limit;
5. exact rational scale energies whose generating function has poles at the geometric lattice $4, 16, 64, \dots$

The dual inverse array $q_{N,k}$ exchanges fixed widths and variable weights for fixed weights and variable widths. Its chord integrals give finite sinc mixtures for Φ and complete homogeneous polynomial limits for the Fabius moments.

The strongest new structural bridge is

$$\begin{aligned} \text{binary Hamming weight} &\longleftrightarrow \text{Walsh vanishing moments} \\ &\longleftrightarrow \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2} \longleftrightarrow \text{Bell–Bernoulli higher moments.} \end{aligned}$$

It complements the repository’s existing $q = 1/4$ moment and scale structures. The strongest analytic bridge is

$$\begin{aligned} \text{rational Haar detail energies} &\longleftrightarrow \text{sinc}^4 \text{ multiplier} \\ &\longleftrightarrow \|p^{(r)}\|_2^2 \longleftrightarrow \text{meromorphic poles at } 4^r. \end{aligned}$$

Both bridges are finite and exact before limits are taken, making them good candidates for formalization.

The most promising next steps are the two-adic classification of the coefficient arrays, a higher-order rational multiwavelet hierarchy, phase-resolved asymptotics of the inverse boundary coefficients, and an exact analysis of the first Poisson aliases beyond the sinc^4 energy expansion.

A Compact formula index

object	equation	representation type
up	(264)	infinite convolution of dyadic uniforms
up	(265)	infinite-dimensional delta integral
up	(267)	limit of finite Thue–Morse splines
up	(271)	finite spline convolved with a scaled copy
Φ	(272)	infinite sinc product
Φ	(276)	q -Pochhammer product at $q = 1/4$
Φ	(277)	canonical zero product with $1 + v_2(n)$ multiplicity
$\log \Phi$	(278)	zeta power series
up	(286)	Fourier cosine integral
F	(287)	sine-over-frequency integral
F	(292)	half-integer cosine series
p	(293)	half-integer sine series
$\mathcal{F}_F$	(295)	compact transform through Φ
Φ	(299)	inverse-Fabius quantile integral
up	(307)	Fourier–Legendre/Bessel coefficient integral
up	(309)	Chebyshev/Bessel coefficient integral
p	(342)	rational unnormalized Haar series
F	(352)	rational Faber–Schauder series
$\mathcal{F}_F$	(358)	translated sinc^2 product-series
p	(368)	rational Walsh–Hadamard series
Φ	(381)	series of finite sine–cosine products
Walsh moments	(396)	Bell–Bernoulli finite-product coefficients
F	(401)	limit of rational Bernstein polynomials
p	(405)	positive rational beta mixture
Φ	(408)	confluent-hypergeometric mixture limit
Φ	(329)	fixed-width rational sinc mixture
Φ	(416)	equal-weight variable-width sinc mixture
Fabius moments	(337)	rational fixed-cell polynomial limit
Fabius moments	(410)	rational beta-moment limit
Fabius moments	(421)	inverse-dyadic polygon limit
E_n	(436)	sinc^4 Fourier multiplier plus aliases
E_n	(439)	Bell–Bernoulli Sobolev expansion

object	equation	representation type
$\mathcal{H}$	(445)	meromorphic rational-energy generating series
G	(452)	inverse Faber–Schauder series
$\widehat{G}$	(456)	inverse translated-sinc ² series

B Proof details for two auxiliary identities

B.1 Derivation of the local Haar expansion

Let m be the center of a parent interval of length h . Then

$$\begin{aligned}
 c_{n,k} &= h^{-1/2} \left(\int_{m-h/2}^m p(x) \, dx - \int_m^{m+h/2} p(x) \, dx \right) \\
 &= -h^{-1/2} \sum_{r \geq 0} \frac{2p^{(2r+1)}(m)}{(2r+1)!} \frac{(h/2)^{2r+2}}{2r+2} \\
 &= - \sum_{r \geq 0} \frac{h^{2r+3/2}}{2^{2r+1}(2r+2)!} p^{(2r+1)}(m),
 \end{aligned}$$

which is (432). Squaring and summing midpoint samples recovers the first coefficients in (440); the multiplier proof packages all integrations by parts into one sinc⁴ symbol.

B.2 Absolute convergence of the first-moment Hamming layers

Letting $N \rightarrow \infty$ in (388) gives

$$\sum_{\substack{m \geq 0 \\ r(m)=r}} 2^{-\sigma(m)} = \frac{2^{-r(r-1)/2}}{(1/2; 1/2)_r}. \quad (478)$$

Since $|a_m| \leq \int_0^1 p = 1$, the weight- r contribution formed from the first nonzero Walsh moments is absolutely summable. In particular, the sum rule (398) is independent of the blockwise ordering.

C Reproducibility ledger

Dyadic Multiresolution and Product-Series Representations
Numerical and exact-symbolic verification

=====

```

mpmath decimal digits : 55
sinc-product factors : 160
half-integer Fourier modes : 200

```

1. Normalizations

```

F(1/4) = 0.06944444444444445713227823382417
Expected F(1/4) = 0.069444444444444444444444444444444444
absolute residual = 1.2687834e-16
sum of level-7 masses = 1.0

```

2. Exact Walsh moment and q-binomial checks

Walsh moment identities : 1016 checked, 0 failed
 q-binomial aggregates : 35 checked, 0 failed

Selected first nonzero Walsh moments:

m binary r integral $x^r w_m(x) dx$
 1 1 1 -1/4
 2 10 1 -1/8
 3 11 2 1/16
 5 101 2 1/32
 6 110 2 1/64
 7 111 3 -3/256
 15 1111 4 3/2048
 23 10111 4 3/4096
 31 11111 5 -15/131072

3. Walsh finite-product transform

max residual m=0,...,15 at z=3.7+0.4i: 3.745842e-56

4. Symmetry selection rule for Walsh coefficients

max $|a_m|$, odd Hamming weight (should be 0): 1.720445e-54
 max $|a_m|$, nonzero even weight : 0.722222222222221715
 Hadamard Parseval residual, level 7 : 0.0

5. Transform convergence

level max uniform-cell error max atomic error max beta-mixture error max Walsh error
 2 0.09719740288 0.8749804357 0.3311617664 0.09719740288
 3 0.0278015789 0.01416477865 0.1984606384 0.0278015789
 4 0.007011640078 0.003516689896 0.1097810782 0.007011640078
 5 0.001759556884 0.0008804669754 0.05795858183 0.001759556884
 6 0.0004403020799 0.0002201940863 0.02980477899 0.0004403020799
 7 0.0001101013485 5.505336511e-5 0.01511621869 0.0001101013485

6. Haar energy and all-orders continuum expansion

$||p'||_2^2 = 25.8839531509750580137998$
 $||p''||_2^2 = 1656.57300166240371284162$
 $||p'''||_2^2 = 424082.688425575063736863$
 $n E_n 16^{4^n} E_n \text{ rel.error(3-term)}$
 2 0.0625 16.000000000000006 0.42900991
 3 0.025292927014271 25.899957262613 0.039135771
 4 0.0062471083652827 25.58815586419799 0.0012204175
 5 0.0015757618966264 25.81728291432679 1.6265272e-5
 6 0.00039470093189003 25.86712027234482 4.2313174e-8
 7 9.872337914768e-5 25.87974150328945 4.1389921e-11

7. Low-level rational-structure samples (high-precision decimals)

level 1: Haar $c_{\{n,k\}}$ = -0.510688230857, 0.510688230857
 Schauder $d_{\{n,k\}}$ = -0.180555555556, 0.180555555556
 level 2: Haar $c_{\{n,k\}}$ = -0.125, -0.125, 0.125, 0.125
 Schauder $d_{\{n,k\}}$ = -0.03125, -0.03125, 0.03125, 0.03125
 level 3: Haar $c_{\{n,k\}}$ = -0.00943081845124, -0.0789575291971, -0.0789575291971, -0.00943081845124,
 0.00943081845124, 0.0789575291971, 0.0789575291971, 0.00943081845124
 Schauder $d_{\{n,k\}}$ = -0.00166714891975, -0.0139578510802, -0.0139578510802, -0.00166714891975,
 0.00166714891975, 0.0139578510802, 0.0139578510802, 0.00166714891975
 level 4: Haar $c_{\{n,k\}}$ = -0.000271267361112, -0.00808376736111, -0.0231662326389, -0.0309787326389,
 -0.0309787326389, -0.0231662326389, -0.00808376736111, -0.00027126736111, 0.00027126736111,
 0.00808376736111, 0.0231662326389, 0.0309787326389, 0.0309787326389, 0.0231662326389,
 0.00808376736111, 0.000271267361112
 Schauder $d_{\{n,k\}}$ = -3.3908420139e-5, -0.00101047092014, -0.00289577907986, -0.00387234157986,
 -0.00387234157986, -0.00289577907986, -0.00101047092014, -3.39084201388e-5, 3.39084201388e-5,
 0.00101047092014, 0.00289577907986, 0.00387234157986, 0.00387234157986,
 0.00289577907986, 0.00101047092014, 3.3908420139e-5

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Part IV

Integral, Series, Product, and Operator Representations for the Fabius–Rvachev System

Abstract

The Fabius function, Rvachev’s compactly supported up-function, their signed Thue–Morse extension, and their Fourier images admit an unusually dense web of representations: self-differential and refinement equations, random uniform series, infinite convolutions, finite box splines, signed truncated-power sums, dyadic sinc and cosine products, canonical products indexed by the 2-adic valuation, Fourier and Poisson inversion formulas, cosine and sampling series, moment and cumulant expansions, exact dyadic composition and q -binomial formulas, Laplace products, Mellin transforms, and Lambert- W_{-1} endpoint coordinates. This report first organizes those formulas into a normalization-safe atlas after a recursive audit of the repository’s living references, frozen historical lineages, active frontier drafts, and transcribed papers. Results already present in the audited tree are explicitly marked as such rather than relabeled as discoveries.

The main new contribution relative to the pinned corpus is a specified operator model. Let $G = (-d^2/dx^2)_D^{-1}$ be the Dirichlet Green operator on $L^2(0, 1)$, and put

$$B = \bigoplus_{n \geq 1} 4^{-n} G.$$

Then B is positive trace class and the angular Fourier transform of up is the Fredholm determinant

$$\varphi(t) = \det(I - t^2 B).$$

Equivalently, B^{-1} has eigenvalues $(2\pi m)^2$ with multiplicity $1 + \nu_2(m)$. This specified operator upgrades the scalar “spectral determinant” of the existing corpus to a concrete Dirichlet-Laplacian spectrum. It yields new resolvent and heat-kernel representations, Brownian-bridge quadratic functionals, complete-Bernstein structure on the imaginary axis, and a Mellin–Barnes determinant formula for the endpoint Laplace transform.

A second new bridge identifies normalized even moments with exterior-power traces,

$$\frac{\mathbb{E} Z^{2k}}{(2k)!} = \mathrm{Tr}(\wedge^k B),$$

where Z has density up. Combining this with the exact reciprocal-dyadic identity gives the positive spectral formula

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\mathrm{Tr}(\wedge^k B)}{(n - 2k)!}.$$

Newton identities convert it into finite combinations of $\zeta(2j)/(\pi^{2j}(4^j - 1))$, hence Bernoulli numbers. Moreover the sequence $\{\mathbb{E} Z^{2k}/(2k)!\}_{k \geq 0}$ is a Pólya-frequency sequence of infinite order; therefore every Toeplitz minor is nonnegative and strict Newton–Turán inequalities follow. High-precision numerical experiments verify the product/trace identity, the exact dyadic formula, the inequalities, and the Mellin–Barnes integral. The report ends with carefully separated conjectures and a research program for Fredholm minors, operator deformations, inverse-Fabius saddles, derivative-zero statistics, and formalization.

1 Scope, audit boundary, and proof-status discipline

1.1 What was audited

The requested source boundary is the recursive `*.tex` tree below `Analysis/FabiusFunction/docs`. The repository’s own most recent audit draft pins commit `32d6d36c51d803289e6d6a0dc0c37753766eba47` and records 67 distinct LaTeX paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (including twelve Fourier-decay dossiers), and seven transcribed or corrected papers [12]. The present study used those same lineages and compared the living primary exposition, the consolidated frontier volume, the active draft families, the archive provenance listings, and the paper-source folders. Historical revisions were treated as mathematical lineages rather than as independent claims.

The most important distinction is between two layers.

- (i) The living primary exposition records definitions and results aligned with Lean declarations: the bounded distribution function F , the compact bump up, the signed global extension, probability models, Fourier products and inversion, exact dyadic arithmetic, and the proved part of the endpoint theory.
- (ii) The non-formalized frontier volume and its descendant reports contain developed but not yet fully formalized mathematics: all-order endpoint expansions, periodic Mellin corrections, inverse asymptotics, sharp sinc-product decay, zero spectra and scalar spectral-determinant formulas, arithmetic rays, half-integer sampling, Pascal-weighted hierarchies, q -Richardson schemes, and arbitrary exponent-sequence generalizations.

Meaning of “new”. Throughout this report, “new” means absent from the pinned repository corpus after path, phrase, and formula comparison, and derived or proved in this manuscript. It is not a claim of exhaustive worldwide priority. Classical ingredients are cited, and open statements are labeled as conjectures.

1.2 Prior art inside the repository

The following are treated as **corpus-known** and are used as inputs rather than claimed as discoveries:

- the probability-series model, convolution structure, refinement equations, and finite Thue–Morse box-spline approximants;
- the dyadic sinc product, repeated cosine product, integer zero set, multiplicity $1 + \nu_2(m)$, canonical product, zero-count defect, scalar spectral zeta, scalar heat trace, local zero jets, and q -deformations;
- Fourier inversion, Poisson summation, half-integer cosine/sampling formulas, moment, cumulant, Bell-polynomial, and Bernoulli-polynomial identities;
- exact dyadic values, ordered-composition and path sums, determinant recurrences, half-base q -Pochhammer and Gaussian-binomial formulas, denominator results, and interpolation/Richardson transforms;
- endpoint Laplace products, Mellin harmonic sums, nonconstant logarithmic-periodic corrections, the lower Lambert coordinate, all-order asymptotics, and the inverse Fabius expansion;
- Thue–Morse Mahler products, Prouhet cancellation, rarefied sums, and automatic sign laws;

- arbitrary nonnegative exponent sequences and the Pascal-weighted hierarchy as scalar products and probability laws.

The phrase audit found no counterpart in the principal/current frontier sources for a specified Fredholm operator built from the Dirichlet Green kernel, Brownian-bridge quadratic functionals, exterior-power traces, Pólya-frequency/Toeplitz total positivity of normalized even moments, or the exact exterior-power formula for $F(2^{-n})$. One audited hierarchy report explicitly warns that its word “spectral” refers to the zero divisor and not to the spectrum of a specified operator; that gap is the starting point of [Appendix 4](#).

2 Normalization dictionary and the probability model

Normalization errors are particularly easy here because three Fourier conventions and three closely related functions occur in the literature. We fix all conventions at the outset.

2.1 The three real functions

Let $F: \mathbb{R} \rightarrow [0, 1]$ be the Fabius distribution function, normalized by

$$F(x) = 0 \quad (x \leq 0), \quad F(1) = 1,$$

with

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad F(1-x) = 1 - F(x) \quad (0 \leq x \leq 1). \quad (479)$$

Rvachev’s compact bump is

$$\text{up}(x) = \begin{cases} F(1 - |x|), & |x| \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (480)$$

Consequently

$$F(x) = \text{up}(x-1) \quad (0 \leq x \leq 1), \quad F'(x) = 2 \text{up}(2x-1). \quad (481)$$

The signed global extension is

$$\mathcal{F}(x) = \sum_{m \geq 0} (-1)^{s_2(m)} \text{up}(x - 2m - 1), \quad (482)$$

where the supports are disjoint apart from endpoints. It satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (483)$$

These identities are **corpus-known**; see the classical constructions in [\[5, 1, 13\]](#).

2.2 Random uniform series

Let U_n be independent $\text{Unif}[0, 1]$ variables and let $V_n = 2U_n - 1$, so $V_n \sim \text{Unif}[-1, 1]$. Define

$$X = \sum_{n \geq 1} 2^{-n} U_n, \quad Z = 2X - 1 = \sum_{n \geq 1} 2^{-n} V_n. \quad (484)$$

Then

$$F(x) = \mathbb{P}(X \leq x), \quad \text{up}(x) \, dx = \mathbb{P}(Z \in \, dx). \quad (485)$$

Thus F is a CDF while up is the density of the centered and rescaled law. This classical probability representation goes back to the distribution-function viewpoint of Jessen–Wintner and Fabius [8, 5]. It simultaneously explains positivity, monotonicity, symmetry, support, convolution formulas, and positive definiteness of the Fourier transform. The variance is

$$\mathbb{E}Z^2 = \frac{1}{3} \sum_{n \geq 1} 4^{-n} = \frac{1}{9}. \quad (486)$$

2.3 Fourier conventions

We use the angular transform

$$\varphi(t) = \int_{-1}^1 \text{up}(x) e^{-itx} dx = \mathbb{E}e^{-itZ}, \quad (487)$$

and the cycles-per-unit transform

$$\Phi(z) = \int_{-1}^1 \text{up}(x) e^{-2\pi izx} dx = \varphi(2\pi z). \quad (488)$$

We write $\text{sinc } t = \sin t/t$ in angular formulas and explicitly write $\sin(\pi z)/(\pi z)$ in normalized formulas.

3 A representation atlas from the audited corpus

This section is a compact, normalization-consistent synthesis of the principal representations already developed across the source tree. Later sections attach new operator meanings and consequences to this atlas.

3.1 Refinement and functional-differential forms

The bump satisfies the two-scale equation

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1), \quad (489)$$

with $\text{up} \geq 0$, support $[-1, 1]$, and integral one. Integrating or differentiating (489) produces a hierarchy of shifted/scaled formulas; folding by (480) recovers (479). On the signed extension, (483) removes the two shifts at the price of the Thue–Morse sign pattern.

A Volterra form useful for repeated integration is

$$F(x) = \int_0^{2x} F(t) dt \quad (x \geq 0), \quad (490)$$

where the globally signed continuation is understood once $2x > 1$. Iteration gives simplex integrals and triangular recurrences for dyadic samples.

3.2 Infinite convolution and finite box splines

Let

$$b_a(x) = \frac{1}{2a} \mathbf{1}_{[-a, a]}(x).$$

Then

$$\text{up} = b_{1/2} * b_{1/4} * b_{1/8} * \cdots \quad (491)$$

with convergence in every standard sense appropriate to the smooth limiting density. For the finite sum $Z_N = \sum_{n=1}^N 2^{-n} V_n$, write u_N for its density and $a_N = 1 - 2^{-N}$. Inclusion–exclusion gives the finite box-spline formula

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} (-1)^{s_2(j)} \left(x + a_N - \frac{j}{2^{N-1}} \right)_+^{N-1}. \quad (492)$$

This makes the Thue–Morse sequence visible before the smooth limit is taken. It is also a practical representation for derivatives of order below N , because each finite approximant is piecewise polynomial.

3.3 Dyadic sinc and cosine products

Independence in (484) yields

$$\varphi(t) = \prod_{n=1}^{\infty} \text{sinc}\left(\frac{t}{2^n}\right), \quad \Phi(z) = \prod_{h=0}^{\infty} \frac{\sin(\pi z/2^h)}{\pi z/2^h}. \quad (493)$$

Using $\text{sinc } x = \prod_{j \geq 1} \cos(x/2^j)$ and counting pairs of scales gives

$$\varphi(t) = \prod_{r=1}^{\infty} \cos\left(\frac{t}{2^{r+1}}\right)^r, \quad \Phi(z) = \prod_{r=1}^{\infty} \cos\left(\frac{\pi z}{2^r}\right)^r. \quad (494)$$

The scale shift immediately gives the Mahler/refinement identities

$$\varphi(2t) = \text{sinc}(t)\varphi(t), \quad \Phi(2z) = \frac{\sin(2\pi z)}{2\pi z} \Phi(z). \quad (495)$$

3.4 Collapsed canonical product and 2-adic multiplicity

Euler’s product for sine collapses all scales onto the positive integers:

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \quad (496)$$

Indeed, the factor at scale h vanishes at an integer m precisely when $2^h \mid m$, so

$$\text{ord}_{z=\pm m} \Phi = 1 + \nu_2(m). \quad (497)$$

The associated Dirichlet series and exact counting function are

$$\sum_{m \geq 1} \frac{1 + \nu_2(m)}{m^s} = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1, \quad (498)$$

$$\sum_{m=1}^M (1 + \nu_2(m)) = 2M - s_2(M). \quad (499)$$

All of (496)–(499) are **corpus-known**; the new point in [Appendix 4](#) is that the same multiplicity becomes the actual degeneracy of a specified self-adjoint operator.

3.5 Logarithmic, zeta, and Bernoulli series

Inside the first zero, $|t| < 2\pi$, logarithmic expansion of (496) gives

$$\log \varphi(t) = - \sum_{j \geq 1} \frac{t^{2j}}{j} \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (500)$$

Equivalently, for the moment generating function $M_Z(s) = \varphi(is)$,

$$\log M_Z(s) = \sum_{j \geq 1} (-1)^{j+1} \frac{s^{2j}}{j} \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (501)$$

Therefore the even cumulants are

$$\kappa_{2j}(Z) = \frac{2^{2j-1} B_{2j}}{j(2^{2j} - 1)}, \quad \kappa_{2j+1}(Z) = 0. \quad (502)$$

Complete Bell polynomials then give

$$\mathbb{E} Z^{2n} = \mathcal{B}_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}). \quad (503)$$

The first values are

$$\mathbb{E} Z^2 = \frac{1}{9}, \quad \mathbb{E} Z^4 = \frac{19}{675}, \quad \mathbb{E} Z^6 = \frac{583}{59535}, \quad \mathbb{E} Z^8 = \frac{132809}{32531625}. \quad (504)$$

3.6 Fourier inversion, cosine series, and sampling

Because $\text{up} \in C_c^\infty(\mathbb{R})$, its transform is rapidly decreasing on the real axis, and Fourier inversion is absolutely convergent:

$$\text{up}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} \varphi(t) dt = \frac{1}{\pi} \int_0^{\infty} \cos(tx) \varphi(t) dt. \quad (505)$$

For $0 < x < 1$, the symmetric CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin((2x-1)t)}{t} \varphi(t) dt. \quad (506)$$

The period-two cosine expansion of the compact bump is

$$\text{up}(x) = \frac{1}{2} + \sum_{k \geq 0} \Phi\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi x), \quad |x| \leq 1. \quad (507)$$

Integer Fourier coefficients vanish because of (497), leaving only half-integers. Paley–Wiener sampling similarly yields

$$\Phi(z) = \frac{\sin(2\pi z)}{2\pi z} + \sum_{k \in \mathbb{Z}} \Phi\left(\frac{2k+1}{2}\right) \frac{\sin \pi(2z - 2k - 1)}{\pi(2z - 2k - 1)}, \quad (508)$$

with the standard L^2 and locally uniform interpretations. Half-integer quadrature and arithmetic-ray refinements are developed extensively in the audited frontier reports.

3.7 Endpoint Laplace products

The moment generating function of X is

$$A(z) = \mathbb{E}e^{zX} = \prod_{n \geq 1} \frac{e^{z/2^n} - 1}{z/2^n} = e^{z/2} \varphi(iz/2). \quad (509)$$

The negative Laplace transform

$$L(s) = \mathbb{E}e^{-sX} = A(-s) = \prod_{n \geq 1} \frac{1 - e^{-s/2^n}}{s/2^n} = e^{-s/2} \varphi(is/2) \quad (510)$$

obeys

$$L(2s) = \frac{1 - e^{-s}}{s} L(s). \quad (511)$$

Writing $\Lambda(s) = \log L(s)$, the corpus derives the harmonic-sum Mellin transform

$$\int_0^\infty \Lambda(s) s^{w-1} ds = \frac{2^w}{1 - 2^w} [-\Gamma(w) \zeta(1+w)], \quad -1 < \Re w < 0, \quad (512)$$

and, after Mellin inversion, the logarithmic-periodic decomposition

$$\Lambda(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + P(\log_2 s) + E(s), \quad (513)$$

where P is nonconstant and one-periodic and E is rapidly small. This theory is **corpus-known**; [Appendix 10](#) derives a Fredholm/Mellin–Barnes representation whose poles reproduce the same decomposition.

3.8 Exact reciprocal-dyadic values

Let $d_n = \mathbb{E}X^n$. A central exact identity in the corpus, closely related to the arithmetic formulas of Arias de Reyna [2], is

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!} = 2^{-\binom{n}{2}} [z^n] A(z). \quad (514)$$

Among the many equivalent forms are the positive ordered-composition sum

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{(r_1, \dots, r_\ell) \models n} \prod_{j=1}^{\ell} \frac{1}{(2^{r_1 + \dots + r_j} - 1)(r_j + 1)!}, \quad (515)$$

Bell-polynomial and lower-Hessenberg determinant forms, and a finite half-base q -binomial formula of the shape

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2} (1/2; 1/2)_n}. \quad (516)$$

The numerator $\mathcal{N}_n$ is an explicit alternating Gaussian-binomial sum with Thue–Morse inner sums. The new formula in [Appendix 7](#) replaces these by a finite positive sum of exterior-power traces of one operator.

3.9 Endpoint and inverse asymptotics

For $L_2 = \log 2$, the lower-Lambert coordinate

$$\lambda_n = -\frac{1}{L_2} W_{-1}(-L_2 2^{-n}), \quad \lambda_n - \log_2 \lambda_n = n, \quad (517)$$

organizes the sharp reciprocal-power asymptotics. The audited frontier volume gives

$$\log F(2^{-n}) = -\frac{L_2}{2} \lambda_n^2 + \left(1 + \frac{L_2}{2}\right) \lambda_n - \frac{1}{2} \log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}), \quad (518)$$

where Ψ is smooth, nonconstant, mean-zero, and one-periodic, with explicit Gamma–zeta Fourier coefficients. All-order corrections use Bell polynomials. The same saddle machinery is inverted to describe $F^{-1}(y)$ as $y \downarrow 0$. These results are **corpus-known** and provide the asymptotic benchmark for the new operator formulas below.

4 A specified Dirichlet operator and a genuine Fredholm determinant

Status. The scalar sinc products, canonical product, zero multiplicities, and scalar spectral zeta in this section are **corpus-known**. The specified Hilbert space, Dirichlet Green operator, Fredholm determinant, unitary renormalization, and probabilistic operator consequences are **proved here** relative to the pinned corpus.

4.1 The Green block

Let

$$D = -\frac{d^2}{dx^2}, \quad \mathcal{D}(D) = H^2(0, 1) \cap H_0^1(0, 1), \quad (519)$$

be the positive Dirichlet Laplacian on $L^2(0, 1)$. Its normalized eigenfunctions and eigenvalues are

$$e_k(x) = \sqrt{2} \sin(k\pi x), \quad D e_k = \pi^2 k^2 e_k, \quad k \geq 1. \quad (520)$$

The inverse $G = D^{-1}$ is the positive trace-class integral operator

$$(Gf)(x) = \int_0^1 g(x, y) f(y) dy, \quad g(x, y) = \min(x, y) - xy, \quad (521)$$

with

$$G e_k = \frac{1}{\pi^2 k^2} e_k, \quad \text{Tr } G^s = \frac{\zeta(2s)}{\pi^{2s}} \quad (\Re s > \tfrac{1}{2}). \quad (522)$$

The same kernel is the covariance kernel of a standard Brownian bridge; that fact will be used in [Appendix 6](#).

4.2 The dyadic direct sum

Set

$$\mathcal{H} = \bigoplus_{n=1}^{\infty} L^2(0, 1), \quad B = \bigoplus_{n=1}^{\infty} 4^{-n} G. \quad (523)$$

Since

$$\mathrm{Tr} B = \sum_{n \geq 1} 4^{-n} \mathrm{Tr} G = \frac{1}{3} \frac{\zeta(2)}{\pi^2} = \frac{1}{18}, \quad (524)$$

B is positive trace class. Its nonzero eigenvalues, initially indexed by $(n, k) \in \mathbb{N}^2$, are

$$\lambda_{n,k} = \frac{4^{-n}}{\pi^2 k^2}. \quad (525)$$

Theorem 4.1 (Specified Fredholm realization). *For every $t \in \mathbb{C}$,*

$$\varphi(t) = \det_{\mathcal{H}}(I - t^2 B). \quad (526)$$

Equivalently,

$$\Phi(z) = \det_{\mathcal{H}}(I - 4\pi^2 z^2 B). \quad (527)$$

Both are ordinary trace-class Fredholm determinants, not zeta-regularized products.

Proof. The Fredholm determinant converges locally uniformly because B is trace class. Using (525) and Euler's sine product,

$$\begin{aligned} \det(I - t^2 B) &= \prod_{n \geq 1} \prod_{k \geq 1} \left(1 - \frac{t^2}{\pi^2 4^n k^2} \right) \\ &= \prod_{n \geq 1} \frac{\sin(t/2^n)}{t/2^n} = \varphi(t). \end{aligned}$$

The normalized statement follows from $\Phi(z) = \varphi(2\pi z)$. $\square$

This result gives an operator interpretation to every coefficient in the logarithmic series. For $|t| < 2\pi$,

$$\log \varphi(t) = \mathrm{Tr} \log(I - t^2 B) = - \sum_{j \geq 1} \frac{t^{2j}}{j} \mathrm{Tr} B^j, \quad (528)$$

where

$$p_j := \mathrm{Tr} B^j = \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (529)$$

Thus (500) is literally a trace expansion.

4.3 A self-adjoint operator whose degeneracies are 2-adic

Let

$$\mathcal{L} = B^{-1} = \bigoplus_{n \geq 1} 4^n D \quad (530)$$

with its natural direct-sum domain. This is positive self-adjoint with compact resolvent. Reindexing $m = 2^{n-1}k$ in (525) gives

$$\mathrm{Spec}(\mathcal{L}) = \{(2\pi m)^2 : m \geq 1\}, \quad \mathrm{mult}_{(2\pi m)^2} \mathcal{L} = 1 + \nu_2(m). \quad (531)$$

Indeed, $m = 2^{n-1}k$ is possible for exactly $n = 1, \dots, 1 + \nu_2(m)$. Therefore

$$\varphi(t) = \det(I - t^2 \mathcal{L}^{-1}) = \prod_{m \geq 1} \left(1 - \frac{t^2}{(2\pi m)^2}\right)^{1+\nu_2(m)}. \quad (532)$$

This proves that the 2-adic zero multiplicity is an actual spectral degeneracy, not merely a multiplicity in a formal zero divisor.

The spectral zeta function of $\mathcal{L}$ is

$$\zeta_{\mathcal{L}}(s) = \text{Tr } \mathcal{L}^{-s} = \frac{\zeta(2s)}{\pi^{2s}(4^s - 1)}, \quad \Re s > \frac{1}{2}, \quad (533)$$

which extends meromorphically through the right-hand side. Its poles at $s = \pi i k / \log 2$, including the collision at $s = 0$, are the operator source of the logarithmic-periodic oscillations encountered in endpoint asymptotics.

4.4 Exact renormalization by removing one block

The dyadic direct sum satisfies the unitary equivalences

$$4B \simeq G \oplus B, \quad \mathcal{L} \simeq 4D \oplus 4\mathcal{L}. \quad (534)$$

Taking determinants in the first identity gives

$$\begin{aligned} \det(I - 4t^2 B) &= \det(I - t^2 G) \det(I - t^2 B), \\ \varphi(2t) &= \text{sinc}(t) \varphi(t), \end{aligned} \quad (535)$$

so the familiar Fourier scaling law is an exact block-decimation identity. On the imaginary axis, if

$$D_B(q) = \det(I + qB), \quad q \in \mathbb{C}, \quad (536)$$

then

$$D_B(4q) = \frac{\sinh \sqrt{q}}{\sqrt{q}} D_B(q). \quad (537)$$

Equations (511) and (537) are the same renormalization after

$$L(s) = e^{-s/2} D_B(s^2/4). \quad (538)$$

4.5 Geometric-scale and exponent-sequence operators

The construction is not confined to the classical scale. Let $0 < q < 1$, $c > 0$, and let $a_n \in \mathbb{Z}_{\geq 0}$ satisfy

$$\sum_{n \geq 0} a_n c^2 q^{2n} < \infty. \quad (539)$$

On a direct sum with a_n copies of $L^2(0, 1)$ at scale n , put

$$B_{a,q,c} = \bigoplus_{n \geq 0} (c^2 q^{2n} G)^{\oplus a_n}. \quad (540)$$

Then

$$\det(I - t^2 B_{a,q,c}) = \prod_{n \geq 0} \text{sinc}(c q^n t)^{a_n}, \quad (541)$$

while

$$\mathrm{Tr} B_{a,q,c}^s = \frac{\zeta(2s)}{\pi^{2s}} c^{2s} \sum_{n \geq 0} a_n q^{2ns}. \quad (542)$$

For $a_n \equiv 1$, this becomes

$$\mathrm{Tr} B_{q,c}^s = \frac{c^{2s} \zeta(2s)}{\pi^{2s} (1 - q^{2s})}. \quad (543)$$

The classical Fabius–Rvachev operator is $q = c = 1/2$. Thus the exponent-sequence Dirichlet series in the audited hierarchy is realized as an honest operator trace. In particular, scalar “zero zeta” identities become spectral zeta identities whenever the exponents are nonnegative integers.

5 Resolvent, heat, theta, and Stieltjes representations

5.1 The logarithmic derivative as a resolvent trace

For t away from the zeros of φ , differentiating the Fredholm determinant gives

$$\frac{\varphi'(t)}{\varphi(t)} = -2t \mathrm{Tr} [B(I - t^2 B)^{-1}]. \quad (544)$$

Using either spectral indexing yields two complementary partial-fraction forms:

$$\frac{\varphi'(t)}{\varphi(t)} = -2t \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 - t^2}, \quad (545)$$

$$= \sum_{n \geq 1} \left[2^{-n} \cot\left(\frac{t}{2^n}\right) - \frac{1}{t} \right]. \quad (546)$$

Although each bracket in (546) contains $-1/t$, the bracket as a whole is $\mathcal{O}(4^{-n})$ on compact sets away from poles, so the series is locally normally convergent. Equating (545) and (546) gives a 2-adic cotangent identity:

$$\sum_{n \geq 1} \left[2^{-n} \cot\left(\frac{t}{2^n}\right) - \frac{1}{t} \right] = -2t \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 - t^2}. \quad (547)$$

This is simultaneously a scale expansion and a spectral resolvent expansion.

For the positive determinant $D_B(q)$,

$$\frac{d}{dq} \log D_B(q) = \mathrm{Tr} [B(I + qB)^{-1}] = \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 + q}. \quad (548)$$

Thus the logarithmic derivative is the Stieltjes transform of the discrete spectral measure

$$\rho = \sum_{m \geq 1} (1 + \nu_2(m)) \delta_{(2\pi m)^2}. \quad (549)$$

5.2 Heat trace and dyadic theta series

Define

$$H(x) = \mathrm{Tr} e^{-x\mathcal{L}} = \sum_{n \geq 1} \sum_{k \geq 1} e^{-\pi^2 4^n k^2 x} = \sum_{m \geq 1} (1 + \nu_2(m)) e^{-(2\pi m)^2 x}. \quad (550)$$

In terms of Jacobi's theta function $\vartheta_3(0, q) = 1 + 2 \sum_{k \geq 1} q^{k^2}$,

$$H(x) = \frac{1}{2} \sum_{n \geq 1} \left[\vartheta_3(0, e^{-\pi^2 4^n x}) - 1 \right]. \quad (551)$$

Removing the first block gives the exact heat renormalization

$$H(x/4) = \sum_{k \geq 1} e^{-\pi^2 k^2 x} + H(x). \quad (552)$$

The Mellin transform is

$$\int_0^\infty x^{s-1} H(x) dx = \Gamma(s) \frac{\zeta(2s)}{\pi^{2s}(4^s - 1)}, \quad \Re s > \frac{1}{2}. \quad (553)$$

This is a theta-integral representation of the spectral zeta and a direct route from Poisson summation to the endpoint logarithmic period.

5.3 Exponentials of heat integrals

The elementary identity

$$\log(1 + q\lambda) = \int_0^\infty \frac{1 - e^{-qx}}{x} e^{-x/\lambda} dx, \quad q \geq 0, \quad (554)$$

can be summed over the spectrum because B is trace class.

Theorem 5.1 (Heat-kernel determinant representations). *For $q \geq 0$,*

$$\log D_B(q) = \int_0^\infty \frac{1 - e^{-qx}}{x} H(x) dx. \quad (555)$$

For $|t| < 2\pi$,

$$\log \varphi(t) = - \int_0^\infty \frac{e^{t^2 x} - 1}{x} H(x) dx. \quad (556)$$

The latter identity extends by analytic continuation from the disk bounded by the first zero.

Proof. Apply (554) to each eigenvalue λ_j and use monotone convergence for $q \geq 0$. Since $\sum_j e^{-x/\lambda_j} = H(x)$, this proves (555). Replacing q by $-t^2$ is justified when $t^2 < \lambda_1^{-1} = 4\pi^2$, and gives (556). $\square$

Combining (555) and (551) produces the explicit product–integral–theta representation

$$\varphi(i\sqrt{q}) = \exp \left\{ \frac{1}{2} \sum_{n \geq 1} \int_0^\infty \frac{1 - e^{-qx}}{x} [\vartheta_3(0, e^{-\pi^2 4^n x}) - 1] dx \right\}. \quad (557)$$

5.4 Complete Bernstein structure

Let $f(q) = \log D_B(q)$ for $q \geq 0$. From (548),

$$f'(q) = \int_{(0, \infty)} \frac{\rho(dy)}{q + y} \quad (558)$$

with ρ from (549). Hence f' is a Stieltjes function, and f is a complete Bernstein function in the sense of [14]. Equivalently, its Lévy–Khintchine representation is exactly (555), whose Lévy density is

$$\frac{H(x)}{x} = \frac{1}{x} \sum_{m \geq 1} (1 + \nu_2(m)) e^{-(2\pi m)^2 x}. \quad (559)$$

In particular,

$$(-1)^r f^{(r+1)}(q) = r! \sum_j \frac{\lambda_j^{r+1}}{(1 + q\lambda_j)^{r+1}} > 0, \quad r \geq 0. \quad (560)$$

This supplies an infinite family of monotonicity and convexity inequalities for the imaginary-axis Fourier transform.

6 Probability laws generated by the operator

The Dirichlet block has an immediate probabilistic meaning that is not visible in the scalar canonical product. This section turns the determinant and its reciprocal powers into Laplace transforms of explicit random series.

6.1 Brownian-bridge quadratic functionals

Let β be a standard Brownian bridge on $[0, 1]$. Its Karhunen–Loève expansion is [9]

$$\beta(x) = \sum_{k \geq 1} \frac{\xi_k}{\pi k} \sqrt{2} \sin(\pi k x), \quad \xi_k \stackrel{\text{iid}}{\sim} N(0, 1), \quad (561)$$

and hence

$$\int_0^1 \beta(x)^2 dx = \sum_{k \geq 1} \frac{\xi_k^2}{\pi^2 k^2} \quad \text{almost surely and in } L^1. \quad (562)$$

Take independent bridges β_n , $n \geq 1$, and define

$$Q = \sum_{n \geq 1} 4^{-n} \int_0^1 \beta_n(x)^2 dx. \quad (563)$$

The series converges almost surely because $\mathbb{E}Q = \text{Tr } B = 1/18$.

Theorem 6.1 (Brownian-bridge determinant law). *For every $q \geq 0$,*

$$\mathbb{E} \exp\left(-\frac{q}{2} Q\right) = D_B(q)^{-1/2} = \varphi(i\sqrt{q})^{-1/2}. \quad (564)$$

Equivalently, after grouping equal eigenvalues,

$$\frac{Q}{2} \stackrel{d}{=} \sum_{m \geq 1} \Gamma_m, \quad \Gamma_m \sim \text{Gamma}\left(\frac{1 + \nu_2(m)}{2}, \frac{1}{(2\pi m)^2}\right), \quad (565)$$

where the second parameter is the scale and the variables are independent.

Proof. Insert (562) into (563). The coefficients of the independent squares $\xi_{n,k}^2$ are exactly the eigenvalues $\lambda_{n,k}$ of B . For a standard normal variable, $\mathbb{E} \exp(-q\lambda\xi^2/2) = (1 + q\lambda)^{-1/2}$. Multiplication over the spectrum proves (564). Reindexing by $m = 2^{n-1}k$ collects $1 + \nu_2(m)$ chi-square degrees of freedom at scale $(2\pi m)^{-2}$, which is (565). $\square$

This result places the Rvachev Fourier image beside classical quadratic-functionals such as the Cramér–von Mises statistic, but with an infinite dyadic tower of independent bridges. In particular, every fractional reciprocal power $D_B(q)^{-\alpha}$, $\alpha > 0$, is completely monotone: replace each gamma shape in (565) by $\alpha(1 + \nu_2(m))$.

6.2 An exponential series and a generalized gamma convolution

Let $E_{n,k}$ be independent unit exponential random variables and set

$$S = \sum_{n,k \geq 1} \lambda_{n,k} E_{n,k}. \quad (566)$$

Again $\mathbb{E}S = \text{Tr } B < \infty$, so the series converges almost surely.

Proposition 6.2 (Reciprocal determinant as a Laplace transform). *For $q \geq 0$,*

$$\boxed{\mathbb{E}e^{-qS} = D_B(q)^{-1}.} \quad (567)$$

Moreover,

$$S \stackrel{d}{=} \sum_{m \geq 1} Y_m, \quad Y_m \sim \text{Gamma}\left(1 + \nu_2(m), \frac{1}{(2\pi m)^2}\right) \quad (568)$$

independently.

The cumulants of S are therefore

$$\kappa_j(S) = (j-1)! \text{Tr } B^j = (j-1)! \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}, \quad j \geq 1. \quad (569)$$

If $h_j(B)$ denotes the complete homogeneous symmetric function of degree j in the eigenvalues of B , then

$$\frac{1}{D_B(-q)} = \sum_{j \geq 0} h_j(B) q^j, \quad \mathbb{E}S^j = j! h_j(B). \quad (570)$$

Thus the determinant simultaneously packages two dual symmetric-function families: its coefficients are exterior powers, while the reciprocal coefficients are symmetric powers.

6.3 A Thorin and heat-measure description

Equation (567) shows that S is a generalized gamma convolution with discrete Thorin measure

$$U = \sum_{m \geq 1} (1 + \nu_2(m)) \delta_{(2\pi m)^2}. \quad (571)$$

Indeed,

$$-\log \mathbb{E}e^{-qS} = \int_{(0,\infty)} \log\left(1 + \frac{q}{y}\right) U(dy), \quad (572)$$

which is the same formula as (555). The Lévy density $H(x)/x$ is therefore not an auxiliary analytic device: it is the Lévy density of an explicit infinitely divisible probability law. This creates a three-way dictionary

dyadic sinc product $\longleftrightarrow$ Dirichlet spectrum $\longleftrightarrow$ generalized gamma convolution.

7 Exterior powers, exact moments, and reciprocal-dyadic values

7.1 Exterior powers are normalized even moments

For a positive trace-class operator, the Fredholm determinant has the absolutely convergent trace-ideal expansion [16, 7]

$$D_B(q) = \det(I + qB) = \sum_{k=0}^{\infty} e_k(B) q^k, \quad e_k(B) = \text{Tr}(\wedge^k B). \quad (573)$$

On the other hand, by the probability model,

$$D_B(s^2) = \varphi(is) = \mathbb{E} e^{sZ} = \sum_{k=0}^{\infty} \frac{\mathbb{E} Z^{2k}}{(2k)!} s^{2k}. \quad (574)$$

Comparison proves the following exact identification.

Theorem 7.1 (Moment–exterior-power identity). *For every $k \geq 0$,*

$$e_k(B) = \text{Tr}(\wedge^k B) = \frac{\mathbb{E} Z^{2k}}{(2k)!}. \quad (575)$$

In particular all exterior-power traces are positive rational numbers.

The rationality follows from the power traces $p_j = \text{Tr } B^j$, since Euler’s evaluation of $\zeta(2j)$ gives

$$p_j = \frac{(-1)^{j+1} 2^{2j-1} B_{2j}}{(2j)!(4^j - 1)} \in \mathbb{Q}_{>0}. \quad (576)$$

Newton’s identities become

$$n e_n = \sum_{j=1}^n (-1)^{j-1} p_j e_{n-j}, \quad e_0 = 1, \quad (577)$$

and the determinant form is

$$e_n = \frac{1}{n!} \det \begin{pmatrix} p_1 & 1 & 0 & \cdots & 0 \\ p_2 & p_1 & 2 & \cdots & 0 \\ p_3 & p_2 & p_1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & n-1 \\ p_n & p_{n-1} & \cdots & p_2 & p_1 \end{pmatrix}. \quad (578)$$

Equivalently, the partition expansion is

$$e_n = \sum_{m_1+2m_2+\cdots+nm_n=n} \prod_{j=1}^n \frac{1}{m_j!} \left(\frac{(-1)^{j-1} p_j}{j} \right)^{m_j}. \quad (579)$$

Although signs occur in (579), positivity is transparent from the spectral sum

$$e_n = \sum_{j_1 < \cdots < j_n} \lambda_{j_1} \cdots \lambda_{j_n}. \quad (580)$$

The first values are

$$e_0 = 1, \quad e_1 = \frac{1}{18}, \quad e_2 = \frac{19}{16200}, \quad e_3 = \frac{583}{42865200}, \quad e_4 = \frac{132809}{1311675120000}. \quad (581)$$

7.2 A positive spectral formula for $F(2^{-n})$

The audited corpus supplies (514). The centered variable gives

$$\mathbb{E}X^n = 2^{-n}\mathbb{E}(1+Z)^n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} \mathbb{E}Z^{2k}. \quad (582)$$

Substituting (575) into (514) proves the promised exact representation.

Theorem 7.2 (Exterior-power formula for reciprocal dyadic values). *For every integer $n \geq 0$,*

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\text{Tr}(\wedge^k B)}{(n-2k)!}. \quad (583)$$

Equivalently,

$$2^{n(n+1)/2} F(2^{-n}) = [z^n] e^z D_B(z^2). \quad (584)$$

The formula is a finite positive sum even though each exterior trace is itself an infinite multiple series. Combining (583) with (579) gives a finite Bernoulli–partition expression:

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{(n-2k)!} \sum_{m_1+2m_2+\dots+km_k=k} \prod_{j=1}^k \frac{1}{m_j!} \left[\frac{(-1)^{j-1}}{j} \frac{\zeta(2j)}{\pi^{2j}(4^j-1)} \right]^{m_j}. \quad (585)$$

A purely positive multiple-zeta form follows from (580):

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{(n-2k)!} \sum_{j_1 < \dots < j_k} \prod_{r=1}^k \lambda_{j_r}. \quad (586)$$

Here the single index j may be replaced by either the pair (a, b) with $\lambda_{a,b} = 4^{-a}/(\pi^2 b^2)$, or the collapsed integer index with multiplicity $1 + \nu_2(m)$.

Table 10: Exact values produced by (583).

n	2^{-n}	$F(2^{-n})$
0	1	1
1	1/2	1/2
2	1/4	5/72
3	1/8	1/288
4	1/16	143/2073600
5	1/32	19/33177600
6	1/64	1153/561842749440
7	1/128	583/179789679820800
8	1/256	1616353/704200217922109440000

7.3 A contour coefficient representation

Cauchy’s formula applied to (584) yields, for any positively oriented circle γ around the origin,

$$F(2^{-n}) = \frac{2^{-n(n+1)/2}}{2\pi i} \int_{\gamma} \frac{e^z \det(I + z^2 B)}{z^{n+1}} dz. \quad (587)$$

This representation invites saddle-point analysis directly on the Fredholm determinant. Its stationary equation is the resolvent equation

$$1 + 2z \operatorname{Tr}[B(I + z^2 B)^{-1}] - \frac{n+1}{z} = 0, \quad (588)$$

which is an operator counterpart of the lower-Lambert saddle used in the corpus.

8 Pólya-frequency structure and real-zero consequences

8.1 Toeplitz total positivity

The product

$$D_B(q) = \prod_{j \geq 1} (1 + \lambda_j q), \quad \lambda_j > 0, \quad \sum_j \lambda_j < \infty, \quad (589)$$

places its coefficient sequence in the classical Aissen–Schoenberg–Whitney–Edrei class [3, 4, 15].

Theorem 8.1 (Pólya-frequency theorem for normalized even moments). *The sequence*

$$a_n = e_n(B) = \frac{\mathbb{E} Z^{2n}}{(2n)!}, \quad n \geq 0, \quad (590)$$

is a Pólya-frequency sequence of infinite order. Equivalently, every finite minor of the Toeplitz matrix

$$T(a) = (a_{j-i})_{i,j \geq 0}, \quad a_r = 0 \ (r < 0), \quad (591)$$

is nonnegative.

Proof. Every finite product $\prod_{j=1}^N (1 + \lambda_j q)$ has only real nonpositive zeros, so its coefficient sequence is PF_∞ . Since $\sum_j \lambda_j < \infty$, the finite products converge locally uniformly to D_B , and every fixed Toeplitz minor converges to the corresponding minor of $T(a)$. Total positivity is closed under this limit. $\square$

The dual Jacobi–Trudi identity gives more information. If λ and μ are partitions with $\mu \subseteq \lambda$, then an appropriate Toeplitz minor is the skew Schur function

$$\det(e_{\lambda'_i - \mu'_j - i + j}(B)) = s_{\lambda/\mu}(\lambda_1, \lambda_2, \dots) \geq 0. \quad (592)$$

Through the Frobenius character formula, every such minor is an explicit finite combination of the power traces (529), and hence of Bernoulli numbers. This extends the single sequence of moments to an infinite hierarchy of spectral inequalities.

8.2 Strict Newton–Turán inequalities

Newton’s inequalities for entire functions of Laguerre–Pólya type imply strong log-concavity of the factorially normalized coefficients.

Corollary 8.2 (Strict Newton inequalities). *For every $n \geq 1$,*

$$e_n(B)^2 > \frac{n+1}{n} e_{n-1}(B) e_{n+1}(B). \quad (593)$$

Equivalently, the moment sequence satisfies

$$\left(\frac{\mathbb{E} Z^{2n}}{(2n)!} \right)^2 > \frac{n+1}{n} \frac{\mathbb{E} Z^{2n-2}}{(2n-2)!} \frac{\mathbb{E} Z^{2n+2}}{(2n+2)!}. \quad (594)$$

Equivalently, after normalizing by $D_B(1)$, the coefficients are the law of a countable Bernoulli sum and are ultra-log-concave [11]. Strictness follows because D_B has infinitely many nonzero roots; the equality case of the limiting Newton inequality is the zero-free exponential e^{cq} , not the determinant (589). Numerically, $e_1^2/(e_0 e_2) = 50/19 \approx 2.63158$, while the lower bound is 2.

8.3 Laguerre–Pólya class and derivative zeros

The finite determinant truncations are polynomials with real zeros $\pm \lambda_j^{-1/2}$. They converge locally uniformly to φ , so

$$\varphi \in \mathcal{LP}, \quad (595)$$

where $\mathcal{LP}$ is the Laguerre–Pólya class. Therefore every derivative $\varphi^{(r)}$ has only real zeros. At $t = 2\pi m$, differentiation removes one unit of the known multiplicity until it is exhausted; all remaining derivative zeros are constrained to the real gaps between the spectral points. This elementary but strong consequence is useful for validating numerical Fourier calculations and motivates [conjecture 12.1](#) below.

The power traces also give Euler–Rayleigh bounds for the first positive zero 2π : for every $j \geq 1$,

$$p_j^{-1/(2j)} < 2\pi < \sqrt{\frac{p_j}{p_{j+1}}}, \quad p_j = \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (596)$$

Although the zero is already known exactly, these inequalities illustrate how the Bernoulli trace data alone recover the spectral edge.

9 Determinant inversion formulas for up and F

Substitution of the Fredholm determinant into classical inversion formulas produces representations that involve no unnamed Fourier transform.

9.1 Real Fourier integrals

For every $x \in \mathbb{R}$,

$$\text{up}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} \det(I - t^2 B) dt = \frac{1}{\pi} \int_0^{\infty} \cos(tx) \det(I - t^2 B) dt. \quad (597)$$

For $0 < x < 1$,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin((2x-1)t)}{t} \det(I - t^2 B) dt. \quad (598)$$

The integrands are Schwartz on the real axis, despite the determinant's exponential type in nonreal directions. Differentiating (598) recovers $F'(x) = 2 \text{up}(2x - 1)$.

9.2 Bromwich–Fredholm inversion

Since $L(s) = \int_{[0,1]} e^{-sx} dF(x)$, integration by parts gives

$$\int_0^1 e^{-sx} F(x) dx = \frac{L(s) - e^{-s}}{s}. \quad (599)$$

It is often cleaner to invert the measure directly. At every $0 < x < 1$ and any $c > 0$, the standard Laplace–Stieltjes inversion formula gives

$$F(x) = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{e^{sx}}{s} L(s) ds. \quad (600)$$

Using (538),

$$F(x) = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{e^{s(x-1/2)}}{s} \det\left(I + \frac{s^2}{4} B\right) ds. \quad (601)$$

Similarly, at interior points,

$$2 \operatorname{up}(2x - 1) = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} e^{s(x-1/2)} \det\left(I + \frac{s^2}{4} B\right) ds. \quad (602)$$

These contour formulas are particularly suited to saddle deformation, because their phase derivative is the explicit resolvent trace (548).

9.3 A determinant series on compact subdisks

Within $|t| < 2\pi$, exponentiating the trace series gives

$$\det(I - t^2 B) = \exp \left[- \sum_{j \geq 1} \frac{p_j}{j} t^{2j} \right] = \sum_{n \geq 0} (-1)^n e_n(B) t^{2n}. \quad (603)$$

Substituting the first exponential in (598) is useful after splitting the Fourier integral at any radius below 2π ; substituting the second gives a moment/exterior-power expansion of the low-frequency contribution. This is not a global termwise Fourier inversion—the Taylor series is local—but it provides rigorous error-controlled hybrid quadrature when paired with the superpolynomial decay estimates from the audited Fourier dossiers.

10 A Mellin–Barnes Fredholm representation of the endpoint transform

The corpus’s Mellin harmonic-sum analysis and the new determinant model meet in a particularly direct contour integral. The bridge is the classical Mellin–Barnes identity [6]:

$$\log(1+x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{u \sin \pi u} x^u du, \quad x > 0, \quad 0 < c < 1. \quad (604)$$

Theorem 10.1 (Endpoint Mellin–Barnes determinant formula). *Let $s > 0$ and $1/2 < c < 1$. Then*

$$\log L(s) = -\frac{s}{2} + \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{u \sin \pi u} \left(\frac{s}{2\pi}\right)^{2u} \frac{\zeta(2u)}{4^u - 1} du. \quad (605)$$

The integral is absolutely convergent on the vertical line.

Proof. By (538),

$$\log L(s) = -\frac{s}{2} + \sum_j \log \left(1 + \frac{s^2}{4} \lambda_j \right).$$

Apply (604) term by term. The choice $c > 1/2$ ensures $\sum_j \lambda_j^c = \text{Tr } B^c < \infty$, so Fubini's theorem is applicable. The spectral trace (533) then gives

$$\left(\frac{s^2}{4}\right)^u \text{Tr } B^u = \left(\frac{s}{2\pi}\right)^{2u} \frac{\zeta(2u)}{4^u - 1}.$$

The factor $1/\sin \pi u$ supplies exponential decay on vertical lines, while the remaining factors have at most polynomial growth. $\square$

10.1 Pole geometry and the periodic endpoint term

The integrand in (605) has three kinds of relevant singularities.

(i) At $u = 1/2$, the pole of $\zeta(2u)$ has residue $s/2$, exactly cancelling the explicit term $-s/2$.

(ii) At

$$\tau_k = \frac{\pi i k}{\log 2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad (606)$$

the zeros of $4^u - 1$ produce the logarithmic-periodic Fourier modes.

(iii) At $u = 0$, the collision of the pole of $1/(4^u - 1)$ with the double pole of $\pi/(u \sin \pi u)$ produces a triple pole and hence the quadratic polynomial in $\log s$.

Put $L_2 = \log 2$, $a = \log(2\pi)$, and

$$C_0 = \frac{a^2/2 + \zeta''(0) - \pi^2/24}{L_2} - \frac{L_2}{12}. \quad (607)$$

A Laurent expansion at zero gives the residue

$$-\frac{(\log s)^2}{2L_2} + \frac{1}{2} \log s + C_0. \quad (608)$$

For $k \neq 0$, define

$$c_k = \frac{\pi \zeta(2\tau_k)}{2L_2 \tau_k \sin(\pi \tau_k)}. \quad (609)$$

Then $c_{-k} = \overline{c_k}$, and the exponential size of $\sin(\pi \tau_k)$ implies rapid convergence of the real periodic function

$$P_{\text{MB}}(y) = \sum_{k \neq 0} c_k e^{2\pi i k y}. \quad (610)$$

Shifting the contour across the imaginary-axis poles yields, for every $A > 0$,

$$\log L(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + C_0 + P_{\text{MB}}\left(\log_2 \frac{s}{2\pi}\right) + \mathcal{O}_A(s^{-A}). \quad (611)$$

The negative integers cause no additional residues: the poles of $1/\sin \pi u$ are cancelled by the trivial zeros $\zeta(2u) = 0$. Formula (611) is the Fredholm-residue version of the corpus-known endpoint periodicity. Standard functional-equation manipulations convert (609) to the Gamma-zeta coefficients used in the harmonic-sum derivation.

Remark 10.2 (What the contour shift proves). The contour may be moved to an arbitrary fixed left half-plane, giving a remainder smaller than every fixed negative power of s . The numerical data and the original dyadic product indicate a much sharper exponentially small remainder. A complete uniform transseries statement is isolated as [conjecture 12.5](#) rather than being assumed here.

10.2 A second Mellin representation through the heat trace

Combining (555) with (538) gives

$$\log L(s) = -\frac{s}{2} + \int_0^\infty \frac{1 - e^{-s^2 x/4}}{x} H(x) dx. \quad (612)$$

Inserting (551) produces the mixed theta–Laplace integral

$$\log L(s) = -\frac{s}{2} + \frac{1}{2} \sum_{n \geq 1} \int_0^\infty \frac{1 - e^{-s^2 x/4}}{x} \left[\vartheta_3(0, e^{-\pi^2 4^n x}) - 1 \right] dx. \quad (613)$$

Taking the Mellin transform in $s^2/4$ recovers (605). Thus the endpoint oscillation can be viewed in three equivalent ways: poles of the scale-harmonic Mellin transform, complex dimensions of the Dirichlet tower, or complex poles of the Fredholm spectral zeta.

11 Numerical experiments and reproducibility

The archive contains `numerical_experiments.py`, written with detailed comments and using `mpmath`, `sympy`, and `matplotlib`. It regenerates every CSV table and figure in this section. All transcendental checks use 50 decimal digits; the exact moment and dyadic calculations use rational arithmetic.

11.1 Fourier product versus trace exponential

For $|t| < 2\pi$, the code compares

$$\prod_{n=1}^{120} \operatorname{sinc}(t/2^n) \quad \text{with} \quad \exp\left(-\sum_{j=1}^{180} \frac{p_j t^{2j}}{j}\right).$$

Representative values are shown in Table 11.

Table 11: Dyadic product and trace-series agreement.

t	dyadic sinc product	trace exponential	absolute error
0.25	0.9965323558631412	0.9965323558631412	1.6×10^{-50}
0.75	0.9691186832108649	0.9691186832108649	1.6×10^{-50}
1.50	0.8807851433857338	0.8807851433857338	9.4×10^{-51}
2.75	0.6413734517278066	0.6413734517278066	2.3×10^{-50}
4.00	0.3617631389237471	0.3617631389237471	2.7×10^{-51}
5.25	0.1258123621455857	0.1258123621455857	1.3×10^{-31}

The slower convergence near $t = 5.25$ reflects the radius 2π set by the first spectral zero.

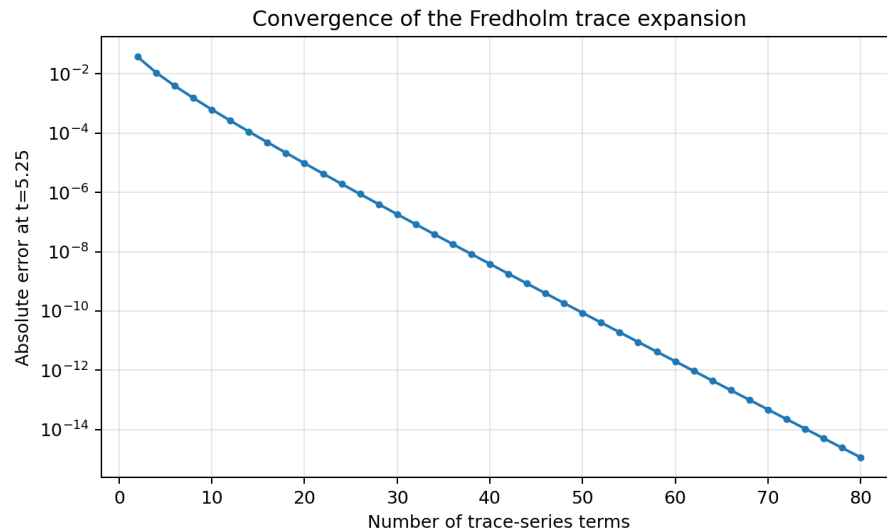


Figure 9: Convergence of the trace expansion at $t = 5.25$.

11.2 Exact exterior powers and Turán ratios

Newton's identities were evaluated exactly through e_{18} , and (583) was checked through $n = 15$. The generated files `exterior_coefficients.csv` and `dyadic_exterior_values.csv` contain all fractions. The strict inequality ratio

$$R_n = \frac{e_n^2}{e_{n-1}e_{n+1}} \quad (614)$$

was compared with $(n+1)/n$ for $1 \leq n \leq 14$. The gaps remain positive and decrease smoothly, as shown in Fig. 10.

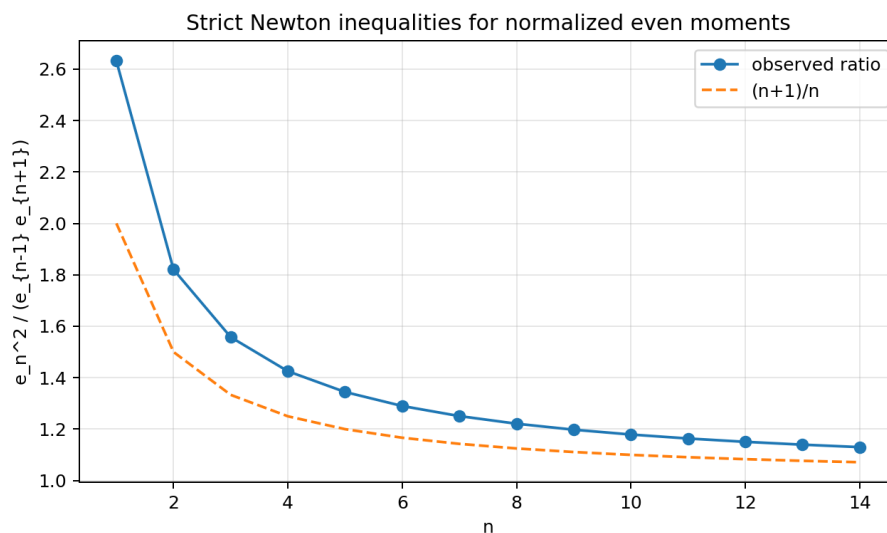


Figure 10: Observed Newton–Turán ratios and the sharp limiting lower bound.

11.3 Mellin–Barnes validation

The vertical integral in (605) was computed on $\Re u = 3/4$, using conjugation symmetry and a cutoff $|\Im u| = 35$. Direct evaluation of the endpoint product and the contour integral agree as follows.

Table 12: Numerical check of the Mellin–Barnes formula.

s	$\log L(s)$, product	$\log L(s)$, MB	absolute error
0.5	-0.2465292231716876	-0.2465292231716876	1.0×10^{-50}
2	-0.9448093175660127	-0.9448093175660127	1.2×10^{-50}
10	-3.782217787719464	-3.782217787719464	3.2×10^{-50}
50	-10.19248359859169	-10.19248359859169	1.7×10^{-49}

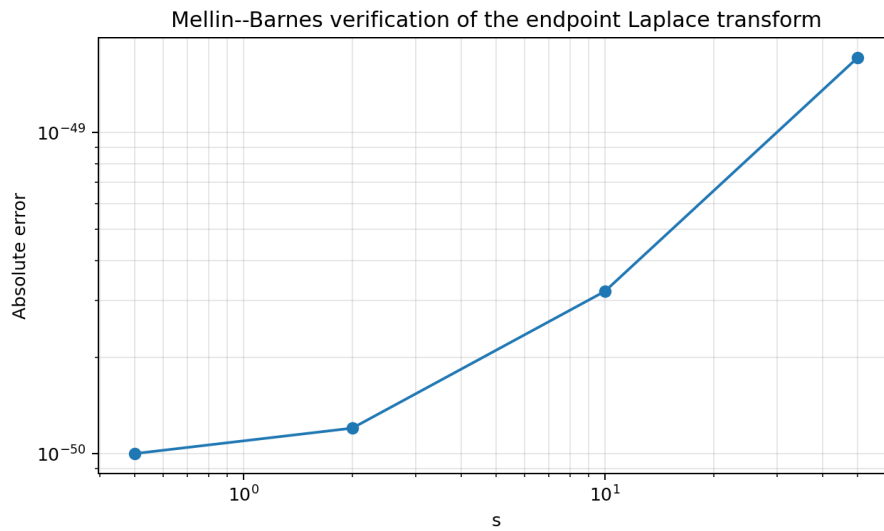


Figure 11: Absolute errors in the Mellin–Barnes check at 50-digit precision.

The near-machine-precision agreement is not evidence by itself, but it is a useful independent check of constants, scaling by 2π , and the cancellation of the $u = 1/2$ residue.

12 Further proved deductions and frontier conjectures

12.1 A unique critical point in every spectral gap

The logarithmic derivative can be written symmetrically as

$$\frac{\varphi'(t)}{\varphi(t)} = \sum_{m \geq 1} (1 + \nu_2(m)) \left(\frac{1}{t - 2\pi m} + \frac{1}{t + 2\pi m} \right). \quad (615)$$

On each interval $(2\pi m, 2\pi(m+1))$, its derivative is strictly negative and its limits are $+\infty$ and $-\infty$. Hence there is exactly one new zero of φ' in every positive spectral gap, in addition to the inherited zeros at $2\pi m$ of multiplicity $\nu_2(m)$. Denote this gap zero by c_m ; it is the unique solution of

$$\sum_{j \geq 1} \frac{1 + \nu_2(j)}{c_m^2 - (2\pi j)^2} = 0, \quad 2\pi m < c_m < 2\pi(m+1). \quad (616)$$

This resolvent characterization appears to be a useful starting point for the finer question below.

Conjecture 12.1 (2-adic modulation of derivative-zero locations). After subtracting a smooth expansion in descending powers of m , the normalized gap position

$$\theta_m = \frac{c_m - 2\pi m}{2\pi} \in (0, 1) \quad (617)$$

possesses a bounded nonconstant correction determined by the local 2-adic valuation pattern near m . More precisely, finite windows of $(\nu_2(m - r), \dots, \nu_2(m + r))$ should determine successively accurate approximations to θ_m .

12.2 Fredholm minors for pointwise values

The determinant inversion formulas represent $\text{up}(x)$ and $F(x)$ by integrals of a scalar determinant. A more local operator formula would be stronger.

Conjecture 12.2 (A pointwise Fredholm-minor model). There exists a natural trace-class integral operator K_x , obtained from the self-similar Dirichlet tower by a boundary evaluation or rank-one coupling, such that $\text{up}(x)$ on $(-1, 1)$ is a Fredholm minor or a ratio of Fredholm determinants of K_x . The family can be chosen so that its Schur complement encodes the refinement equation (489).

A promising construction is to replace the disconnected direct sum by a metric tree whose edge lengths decrease geometrically and to impose Kirchhoff conditions at the scale vertices. The direct-sum spectrum should then appear as the decoupled limit, while suitable boundary Green functions may recover pointwise values.

12.3 Operator saddle theory for the inverse Fabius function

The coefficient saddle (588) and the Bromwich phase derivative both involve

$$R(q) = \text{Tr}[B(I + qB)^{-1}] = \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 + q}. \quad (618)$$

Its complete monotonicity gives robust uniqueness and error bounds for positive saddles. This suggests an operator derivation of the lower-Lambert coordinate without first expanding the scalar product.

Conjecture 12.3 (Resolvent-controlled inverse expansion). The all-order endpoint expansion of $F^{-1}(y)$ can be organized entirely in terms of $R(q)$ and its derivatives at the unique positive saddle. After the leading Lambert inversion, every coefficient is a universal Bell polynomial in the spectral traces $\text{Tr}[B^j(I + qB)^{-j}]$, with the logarithmic-periodic component inherited from the poles of $\zeta_{\mathcal{L}}$.

12.4 Beyond the classical dyadic tower

The family (540) turns exponent-sequence products into operators, but several arithmetic questions remain.

Conjecture 12.4 (Schur positivity and arithmetic denominators). For the classical spectrum, suitably scaled Schur values $s_\lambda(\lambda_1, \lambda_2, \dots)$ have denominator bounds controlled by the hook lengths of λ and the factors $2^{2j} - 1$ occurring in (576). These bounds should refine the known denominator theorems for $F(2^{-n})$ when $\lambda = (1^k)$, and should extend to Pascal-weighted exponent sequences.

For a geometric ratio $q \neq 1/2$, the spectral zeta denominator $1 - q^{2s}$ moves the complex dimensions to an arithmetic progression with period $|\log q|$. This makes the operator family a controlled laboratory for separating features caused by compact convolution from those caused specifically by base two.

12.5 Exponentially small endpoint structure

Conjecture 12.5 (Resurgent endpoint remainder). After subtracting the quadratic logarithmic polynomial and the complete periodic Fourier series in (611), the remainder admits an exponentially small transseries whose actions are generated by the dyadic semigroup $\{j2^r : j, r \in \mathbb{N}\}$. Its Borel singularities are determined by the factors $1 - e^{-s/2^n}$ in (510), and optimal truncation of the shifted Mellin–Barnes contour recovers that transseries.

12.6 Formalization targets

The most accessible Lean targets are algebraic/operator statements that do not require the full endpoint saddle machinery:

1. construction of finite block operators $B_N = \bigoplus_{n=1}^N 4^{-n}G$ and their determinant products;
2. passage to the trace-class limit and proof of (526);
3. Newton identities and the finite exact formula (583);
4. finite total positivity of the coefficient sequences, followed by stabilization of each fixed minor;
5. the resolvent identity (547) on compact subsets away from poles.

These steps would turn a substantial part of the new frontier package into a formal extension of the living repository.

13 Conclusion

The Fabius–Rvachev system has no shortage of formulas; the challenge is to organize them without confusing normalization, proof status, or historical priority. The corpus-known representation web begins with a random uniform series and reaches Thue–Morse splines, sinc and cosine products, 2-adic canonical products, Fourier and sampling integrals, exact dyadic q -arithmetic, Mellin harmonic sums, and Lambert-periodic endpoint asymptotics.

The specified operator

$$B = \bigoplus_{n \geq 1} 4^{-n} (-d^2/dx^2)_D^{-1}$$

adds a unifying layer. Its Fredholm determinant is the Fourier image; its inverse has 2-adic spectral degeneracies; its heat trace gives theta integrals; its resolvent gives partial fractions and saddle equations; its exterior powers are normalized even moments; its symmetric powers are moments of an exponential random series; and its gamma/Brownian laws explain complete monotonicity. The exact formula (583) connects reciprocal-dyadic Fabius values directly to this operator, while Pólya-frequency theory yields a new hierarchy of total-positivity and Turán inequalities. Finally, the Mellin–Barnes formula (605) identifies the endpoint logarithmic period with the complex dimensions of the same Dirichlet tower.

These links do not replace the existing q -binomial, Thue–Morse, Bell, Bernoulli, or Lambert descriptions. They explain why those descriptions coexist and supply new routes between them. The most promising next step is to develop Fredholm minors and operator saddles far enough to produce pointwise and inverse-function formulas, then to formalize the finite-block approximations and their trace-class limit.

A Master table of representations

Table 13: A compact map of the principal representations.

Object	type of representation	status	reference
F	CDF of $X = \sum 2^{-n} U_n$	corpus-known	(485)
up	density of $Z = \sum 2^{-n} V_n$	corpus-known	(485)
up	infinite convolution of centered boxes	corpus-known	(491)
u_N	finite Thue–Morse truncated-power spline	corpus-known	(492)
$\mathcal{F}$	signed translate series and global self-differential equation	corpus-known	(482), (483)
φ	dyadic sinc product	corpus-known	(493)
φ	multiplicity-weighted cosine product	corpus-known	(494)
Φ	2-adic canonical product	corpus-known	(496)
φ	Fredholm determinant of a specified trace-class operator	proved here	(526)
φ	trace exponential involving $\zeta(2j)$	operator form new	(528)
φ'/φ	resolvent, valuation partial fractions, and cotangent scales	operator bridge new	(547)
$\varphi(i\sqrt{q})$	heat-kernel exponential and theta integral	proved here	(557)
$D_B^{-1/2}$	Brownian-bridge quadratic-functional Laplace transform	proved here	(564)
D_B^{-1}	exponential/gamma random-series Laplace transform	proved here	(567), (568)
$e_n(B)$	exterior-power trace and normalized even moment	proved here	(575)
$F(2^{-n})$	positive finite exterior-power sum	proved here	(583)
$F(2^{-n})$	Cauchy integral of a Fredholm determinant	proved here	(587)
$\{e_n\}$	PF_∞ , Toeplitz minors, Schur functions	proved here	(591), (592)
up	cosine Fourier determinant integral	operator form new	(597)
F	sine Fourier determinant integral	operator form new	(598)
F	Bromwich–Fredholm contour integral	operator form new	(601)
L	one-sided dyadic product	corpus-known	(510)
L	heat/theta integral	proved here	(612), (613)
L	Mellin–Barnes spectral-zeta integral	proved here	(605)
$\log L$	quadratic logarithm plus periodic residue series	new derivation of known structure	(611)
F^{-1}	lower-Lambert and periodic saddle expansion	corpus-known	(517), (518)

B Proof details for the exterior-power dyadic formula

For completeness, the coefficient calculation behind [Theorem 7.2](#) can be written in one line. From $X = (1 + Z)/2$ and symmetry,

$$\begin{aligned}
 [z^n]A(z) &= \frac{\mathbb{E}X^n}{n!} = \frac{1}{2^n n!} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} \mathbb{E}Z^{2k} \\
 &= \frac{1}{2^n} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{e_k(B)}{(n-2k)!}.
 \end{aligned}$$

Multiplication by $2^{-\binom{n}{2}}$ proves (583). Equivalently,

$$A(2z) = e^z \varphi(iz) = e^z D_B(z^2), \quad (619)$$

so coefficient extraction immediately gives (584).

The first three nontrivial cases illustrate the mechanism:

$$F(1/4) = 2^{-3} \left(\frac{1}{2!} + e_1 \right) = \frac{1}{8} \left(\frac{1}{2} + \frac{1}{18} \right) = \frac{5}{72}, \quad (620)$$

$$F(1/8) = 2^{-6} \left(\frac{1}{3!} + e_1 \right) = \frac{1}{64} \left(\frac{1}{6} + \frac{1}{18} \right) = \frac{1}{288}, \quad (621)$$

$$F(1/16) = 2^{-10} \left(\frac{1}{4!} + \frac{e_1}{2!} + e_2 \right) = \frac{143}{2073600}. \quad (622)$$

C Recursive corpus map and nonduplication checklist

The repository tree is unusually rich in revisions and parallel frontier reports. The audit was therefore conducted by lineage and theorem content rather than by comparing filenames alone. The following map records the groups that informed this report.

Corpus group	material checked and its role
Living exposition	Definitions aligned with formal declarations; probability law; Fourier product and inversion; moments; exact dyadic arithmetic; source-of-truth normalizations.
Mathematical glossary	Notation reconciliation among F , up, signed extensions, Thue–Morse data, q -products, Bell and Bernoulli polynomials, and Lambert coordinates.
Non-elementarity and Lean walkthroughs	Logical status, proof dependencies, and the boundary between formalized and research-only statements.
Consolidated frontier volume	Endpoint Mellin analysis; logarithmic periodicity; inverse Fabius asymptotics; Fourier decay; local zero jets; arithmetic rays; half-integer sampling; integral and interpolation directions.
Exponent-sequence/Newton hierarchy	Collapsed multiplicities, zero zeta, scalar spectral determinant, heat trace, arbitrary exponent sequences, Pascal weighting, and explicit warning that no operator spectrum was specified.
Fourier-decay dossier family	Sharp real-axis bounds, dyadic scale selection, stationary-scale envelopes, numerical validation, and nonduplication constraints for new Fourier formulas.
Archived revisions	Historical development of exact dyadic, q -binomial, Richardson, endpoint, and Thue–Morse results; used to prevent renaming an older result as new.
Transcribed papers	Classical definitions, probability constructions, arithmetic of dyadic values, and comparison with the independent literature.

The specific nonduplication checks most relevant to the new results were searches for operator-defining terms (Fredholm, trace class, Dirichlet Laplacian, Green operator), probability terms (Brownian bridge, gamma convolution), symmetric-function terms (exterior power, Schur, Pólya frequency, total positivity), and the exact spectral formula (583). The scalar zero-spectrum material was treated as prior art throughout.

D Numerical archive contents

The ZIP archive contains the following reproducibility files.

`numerical_experiments.py`

High-precision product, trace, exact rational, Turán, and Mellin–Barnes checks. The script is self-contained apart from the standard Python packages listed in its header.

`fourier_product_trace_checks.csv`

Numerical comparison of (493) and (528).

`exterior_coefficients.csv`

Exact $e_n(B)$ and $\mathbb{E}Z^{2n}$ through $n = 18$.

`dyadic_exterior_values.csv`

Exact values from (583) through $n = 15$.

`turan_inequalities.csv`

Ratios in (593) through $n = 14$.

`mellin_barnes_checks.csv`

Direct-product and vertical-contour values for (605).

`*.png`

The three figures embedded in Appendix 11.

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Part V

Polyphase, Operator, and Jump-Measure Representations

Abstract

The recursive LaTeX corpus under `Analysis/FabiusFunction/docs` already contains a large fraction of the modern analytic theory of the Fabius function, its inverse, Rvachev’s compactly supported up-function, the Thue–Morse sequence, dyadic and Gaussian q -binomial arithmetic, sinc products, spectral determinants, Bell–Bernoulli moment algebra, and periodic Lambert– W endpoint asymptotics. The purpose of this report is therefore twofold: first, to organize the existing material as a single representation atlas; second, to add a complementary layer that was not located in the audited corpus.

The new organizing object is the sparse-scale Fourier product

$$\Psi_r(z) = \prod_{m \geq 0} \operatorname{sinc}_\pi\left(\frac{z}{2^m}\right), \quad r \in \mathbb{N}_{>0}.$$

Partitioning the scales of the ordinary Rvachev product modulo r gives the exact polyphase factorization

$$\widehat{\operatorname{up}}(z) = \prod_{a=0}^{r-1} \Psi_r(z/2^a).$$

If g_r is the probability density with Fourier image Ψ_r , then

$$\operatorname{up} = g_r * (2g_r(2 \cdot)) * \cdots * (2^{r-1}g_r(2^{r-1} \cdot)).$$

This one identity generates cosine, Gamma, canonical-zero, logarithmic-zeta, hyperbolic-product, cumulant, Bell-polynomial, spectral-zeta, and integral representations. The component zero multiplicity is $1 + \lfloor \nu_2(n)/r \rfloor$, and its spectral zeta is $\zeta(s)/(1 - 2^{-rs})$. Mellin analysis shows an exact roots-of-unity cancellation: the r -periodic pole lattice of a sparse component recombines so that only the ordinary dyadic harmonics survive.

The probability factorization yields a new family of finite-dimensional representations. With $w_r = (1, 2^{-1}, \dots, 2^{-(r-1)})$ and independent coordinates having density g_r , the Fabius function is both a sublevel-set volume and a parallel hyperplane section:

$$F(x) = \int_{w_r \cdot y \leq 2x-1} \prod_{a=0}^{r-1} g_r(y_a) \, dy = \frac{1}{\|w_r\|_2} \int_{w_r \cdot y = x-1} \prod_{a=0}^{r-1} g_r(y_a) \, d\sigma(y).$$

The equality of a moving volume and a differently placed section is the multidimensional form of the self-CDF identity $F(x) = \text{up}(x-1)$.

A second new package extracts the Thue–Morse sequence from analytic data. Finite sparse box splines $g_{r,N}$ have knots $x_{r,N,k} = -A_{r,N} + \sum_j b_j(k)2^{-rj}$, and their highest classical derivative has jumps

$$\varepsilon_k = (-1)^{s_2(k)} = 2^{-r\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}.$$

Equivalently, the complete signed block is the inverse Fourier transform of a deconvolution quotient:

$$\sum_{k < 2^N} \varepsilon_k \delta_{x_{r,N,k}} = 2^{-r\binom{N}{2}} D^N \mathcal{F}^{-1} \left[\frac{\Psi_r(z)}{\Psi_r(z/2^{rN})} \right].$$

Taking the positive and negative Jordan parts gives equally explicit formulas for the unsigned sequence. Pairing this measure with high powers gives exact Bell–Bernoulli formulas for all surviving Thue–Morse power sums and new limit representations for every moment of g_r and up .

Finally, convolution is recast as an infinite product of averaging operators,

$$\text{up} = \exp \left(\sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!(1-4^{-m})} D^{2m} \right) \delta_0,$$

with a rigorous polynomial interpretation and a strong-operator interpretation through uniform averaging. Commented Python code checks the polyphase products, Fourier deconvolution, exact power sums, and cumulant recombination. Throughout, “new” means new relative to the pinned repository snapshot; no claim of worldwide bibliographic priority is made without a separate exhaustive literature search.

1 Scope, corpus audit, and novelty discipline

1.1 Pinned source boundary

The branch moved during the audit, so the mathematical source boundary is the exact commit printed on the title page. The recursive documentation tree was classified into five strata:

1. the primary theorem-backed exposition and mathematical glossary;
2. the living non-formalized frontier synthesis;
3. active thematic reports on Thue–Morse, spectra, arithmetic rays, exponent sequences, transforms, inverse sampling, multiresolution, and Fourier decay;
4. frozen historical lineages under `archive/`;
5. imported or corrected primary-paper sources under `papers/`.

Every TeX path was included in the recursive inventory. Superseded or nearly identical archive revisions were treated through their provenance lineages rather than counted as independent mathematical discoveries. The canonical current descendants were used as the source of record, while targeted full-corpus searches were made for every proposed new structure.

The most relevant exclusion sources are the primary Fabius–Rvachev synthesis [10], the unified Thue–Morse atlas [11], the finite-block calculus [12], the representation-frontier report [13], the multiresolution report [14], the inverse/sampling synthesis [15], the exponent-sequence synthesis [16], and the spectral-arithmetic synthesis [17]. The imported literature was also checked, notably Fabius’s original probabilistic example, Rvachev’s compact support theory, Arias de Reyna’s analytic and arithmetic papers, and the lacunary-product literature [2, 5, 3, 4, 9].

1.2 Status convention

The report uses four status tags.

Tag	Meaning
K	Explicit in the audited repository or classical source; retained to make the atlas usable.
D	A short transform, specialization, or recombination of known material; proved for completeness but not promoted as a substantive novelty.
N	Apparently absent from the pinned TeX corpus; a proof is supplied here.
C	Conjectural or proposed research direction; numerical evidence is identified separately.

1.3 What is deliberately not claimed as new

The following substantial structures already occur in the corpus and are therefore used only as baseline:

- the probability law of a geometrically weighted uniform series, finite atomic and spline approximants, the Fabius/up fold, and quantile transport;
- the sinc and triangular cosine products, Fourier inversion, critical Poisson cosine series, partition of unity, and exact integer-zero jets;
- exact dyadic values, Gaussian q -binomial formulae, Thue–Morse products, Prouhet cancellation, Mahler equations, Dirichlet–Mellin theory, and the large signed/unsigned formula atlas;

- sharp Fourier decay, log-periodic fluctuations, arithmetic rays, half-integer spectra, cyclotomic traces, spectral determinants, heat traces, and Barnes/Gamma dualities;
- Cauchy–Stieltjes resolvents, logarithmic finite-level resolvents, Jacobi and Stieltjes continued fractions, and Fredholm determinant models;
- Haar, Faber–Schauder, Walsh–Hadamard, beta-mixture, sinc-mixture, Fourier–Legendre, and inverse self-sampling representations;
- integer and fractional integration packages, inverse endpoint reversion, Bell-polynomial coefficient calculus, and periodic lower-Lambert asymptotics.

Novelty boundary in one sentence

The new contribution is not another isolated formula for the same sinc product. It is a modular-scale factorization that simultaneously produces a sparse Rvachev family, a Mellin alias-cancellation law, multidimensional volume–section identities, and analytic jump formulas for the signed and unsigned Thue–Morse sequences.

2 Normalization and structural dictionary

2.1 The probability law, the bump, and the bounded function

Let $(U_n)_{n \geq 0}$ be independent random variables uniformly distributed on $[-1, 1]$, and set

$$X = \sum_{n=0}^{\infty} 2^{-n-1} U_n. \quad (623)$$

The series converges absolutely almost surely, $X \in [-1, 1]$, and its density is Rvachev’s even bump

$$u(x) := \text{up}(x), \quad \int_{-1}^1 u(x) \, dx = 1, \quad u(0) = 1. \quad (624)$$

Let

$$Y = \frac{X + 1}{2} \in [0, 1]. \quad (625)$$

The bounded Fabius function is the distribution function of Y :

$$F(x) = \mathbb{P}(Y \leq x) = \mathbb{P}(X \leq 2x - 1), \quad 0 \leq x \leq 1. \quad (626)$$

The special self-similarity of the law gives the stronger pointwise fold

$$F(x) = u(x - 1), \quad F(1 - x) = u(x), \quad 0 \leq x \leq 1. \quad (627)$$

Thus the same function value is at once a probability of a half-line and a density value at a different point.

The inverse Fabius function is denoted

$$G = F^{-1} : (0, 1) \longrightarrow (0, 1). \quad (628)$$

Endpoint totalizations can be added when needed; no ambiguity below depends on them.

2.2 Fourier and sinc conventions

The Fourier transform is

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i z x} \, dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(z) e^{2\pi i z x} \, dz. \quad (629)$$

Write

$$\operatorname{sinc}_{\pi}(z) = \frac{\sin(\pi z)}{\pi z}, \quad \operatorname{sinc}_{\pi}(0) = 1, \quad \Phi(z) = \widehat{u}(z). \quad (630)$$

Then

$$\Phi(z) = \prod_{n=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{z}{2^n}\right), \quad \Phi(z) = \operatorname{sinc}_{\pi}(z) \Phi(z/2). \quad (631)$$

The moment-generating function is

$$M(s) = \mathbb{E} e^{sX} = \Phi\left(\frac{is}{2\pi}\right) = \prod_{n=0}^{\infty} \frac{\sinh(s/2^{n+1})}{s/2^{n+1}}. \quad (632)$$

2.3 Thue–Morse normalizations

For $k \geq 0$, write

$$\tau(k) = s_2(k) \bmod 2 \in \{0, 1\}, \quad \varepsilon_k = (-1)^{s_2(k)} = 1 - 2\tau(k). \quad (633)$$

The signed normalization is natural for products, differences, and atomic measures. The unsigned normalization is recovered by

$$\tau(k) = \frac{1 - \varepsilon_k}{2}. \quad (634)$$

3 A compact representation atlas

This section records the main representation families already available in the corpus, plus immediate transforms needed later. It is intentionally compact: the detailed novelty begins in [Appendix 4](#). The tags refer to the convention of [Appendix 1](#).

3.1 Probability, infinite-cube, and distribution representations

From (623), for every bounded Borel function h ,

$$\int_{-1}^1 h(x) u(x) \, dx = \mathbb{E} h\left(\sum_{n \geq 0} 2^{-n-1} U_n\right), \quad (635)$$

$$F(x) = \int_{[-1,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum 2^{-n-1} v_n \leq 2x-1\}} \prod_{n \geq 0} \frac{dv_n}{2}. \quad (636)$$

These are [K](#). They immediately imply the quantile-transport formula

$$\int_0^1 h(G(y)) \, dy = \int_0^1 h(x) \, dF(x) = \mathbb{E} h(Y), \quad (637)$$

again [K](#).

3.2 Differential and refinement representations

The bump satisfies

$$u'(x) = 2(u(2x+1) - u(2x-1)), \quad (638)$$

and F satisfies

$$F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2), \quad F(1-x) = 1 - F(x). \quad (639)$$

Together with (627), these give two useful inverse-derivative forms:

$$G'(y) = \frac{1}{F'(G(y))} = \frac{1}{u'(G(y)-1)} = \frac{1}{2u(2G(y)-1)}. \quad (640)$$

All are [K](#).

3.3 Fourier inversion and CDF inversion

Rapid decay of Φ gives

$$u(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z x} dz = 2 \int_0^\infty \Phi(z) \cos(2\pi z x) dz, \quad (641)$$

$$F(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z(x-1)} dz, \quad 0 \leq x \leq 1. \quad (642)$$

Applying the Gil–Pelaez inversion formula to X gives the companion CDF form

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(z) \frac{\sin(2\pi z(2x-1))}{z} dz. \quad (643)$$

The equality of (642) and (643) is a Fourier form of the self-CDF identity. Status: [K/D](#).

3.4 Laplace, Bromwich, and Fourier images of the bounded function

The Laplace–Stieltjes transform of F is

$$\int_0^1 e^{sx} dF(x) = \mathbb{E}e^{sY} = e^{s/2} M(s/2). \quad (644)$$

Integration by parts therefore yields, for $s \neq 0$,

$$\int_0^1 e^{sx} F(x) dx = \frac{e^s - e^{s/2} M(s/2)}{s}, \quad (645)$$

with limiting value $1/2$ at $s = 0$. Setting $s = -2\pi i \xi$ gives the exact Fourier image

$$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (646)$$

again continued by $1/2$ at the origin. Conversely, for $c > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^{s(x-1/2)}}{s} M(s/2) ds. \quad (647)$$

These formulas are [K](#) or immediate one-line transforms of known identities.

3.5 Product, canonical-zero, and Gamma representations

The audited corpus contains the following equivalent products:

$$\Phi(z) = \prod_{n=0}^{\infty} \operatorname{sinc}_{\pi}(z/2^n), \quad (648)$$

$$= \prod_{m=1}^{\infty} \cos(\pi z/2^m)^m, \quad (649)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}, \quad (650)$$

$$= \prod_{n=0}^{\infty} \frac{1}{\Gamma(1+z/2^n)\Gamma(1-z/2^n)}. \quad (651)$$

Thus the nonzero integer n is a zero of order $1 + \nu_2(n)$. The local logarithmic series is

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1-4^{-m})} z^{2m}, \quad |z| < 1, \quad (652)$$

and the meromorphic logarithmic derivative is

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^2 - z^2}. \quad (653)$$

Status: K.

3.6 Moments, cumulants, and Bell–Bernoulli series

Writing

$$\log M(s) = \sum_{m \geq 1} \kappa_{2m} \frac{s^{2m}}{(2m)!}, \quad (654)$$

comparison with (632) gives

$$\kappa_{2m} = \frac{B_{2m}}{2m(1-4^{-m})}, \quad \kappa_{2m+1} = 0. \quad (655)$$

Hence

$$\mu_n := \int x^n u(x) \, dx = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (656)$$

where $\mathcal{B}_n$ is the complete exponential Bell polynomial. In particular

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \kappa_4 = -\frac{2}{225}. \quad (657)$$

Status: K.

3.7 Finite splines, Poisson series, and multiresolution forms

The corpus supplies finite uniform-convolution splines whose coefficients are Thue–Morse signs, critical Poisson cosine series on $[-1, 1]$, exact partition of unity, Fourier–Legendre expansions, and

rational Haar, Walsh, and Faber–Schauder systems. Representative formulas are

$$u(x) = \frac{1}{2} + \sum_{k \geq 0} \Phi(k + \frac{1}{2}) \cos((2k + 1)\pi x), \quad (658)$$

$$\sum_{j \in \mathbb{Z}} u(x - j) = 1, \quad (659)$$

with the convergence and endpoint conventions established in the primary source. The multiresolution report expresses the same function as limits of exact rational step mixtures, Haar details, Walsh finite products, Bernstein beta mixtures, and inverse-quantile sinc mixtures [14]. Status: K.

3.8 Resolvent and determinant forms

Let

$$R(z) = \int_{-1}^1 \frac{u(x)}{z - x} dx, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (660)$$

Then

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}} \quad (661)$$

for $|z| > 1$, and the representation-frontier report derives a logarithmic fixed point, a Thue–Morse derivative tree, finite logarithmic resolvents, and the Jacobi continued fraction of R [13]. Likewise (650) is the Fredholm determinant

$$\Phi(z) = \det(I - z^2 T), \quad (662)$$

where T is diagonal with eigenvalue n^{-2} repeated $1 + \nu_2(n)$ times. Status: K.

Transition

The formulas above are a map, not the claimed contribution. The remainder of the report introduces a decomposition by congruence classes of the scale index. It simultaneously refines the Fourier product, the probability law, the zero divisor, the Mellin pole lattice, the finite splines, and the Thue–Morse coefficient measure.

4 Polyphase factorization by scale classes

4.1 Sparse-scale products and laws

Fix an integer $r \geq 1$ and put

$$q_r = 2^{-r}, \quad \Psi_r(z) = \prod_{m=0}^{\infty} \text{sinc}_{\pi}(q_r^m z). \quad (663)$$

Let Z_r be the random series

$$Z_r = \sum_{m=0}^{\infty} \frac{q_r^m}{2} U_m, \quad (664)$$

where the U_m are independent uniform variables on $[-1, 1]$. Its support radius is

$$A_r = \frac{1}{2(1 - q_r)}, \quad (665)$$

and its density is denoted g_r . Then

$$\widehat{g}_r(z) = \Psi_r(z), \quad \text{supp } g_r = [-A_r, A_r]. \quad (666)$$

The generic geometrically weighted uniform law is part of the repository baseline. What is new here is its use as a polyphase factor of the ordinary Rvachev law.

Theorem 4.1 (Polyphase product and convolution factorization). *For every integer $r \geq 1$,*

$$\Phi(z) = \prod_{a=0}^{r-1} \Psi_r(z/2^a). \quad (667)$$

Equivalently, if

$$g_{r,a}(x) = 2^a g_r(2^a x) \quad (668)$$

is the density of $2^{-a}Z_r$, then

$$u = g_{r,0} * g_{r,1} * \cdots * g_{r,r-1}. \quad (669)$$

In random-variable form,

$$X \stackrel{d}{=} \sum_{a=0}^{r-1} 2^{-a} Z_r^{(a)}, \quad (670)$$

where the $Z_r^{(a)}$ are independent copies of Z_r .

Proof. Every nonnegative integer has a unique decomposition $n = a + rm$ with $0 \leq a < r$ and $m \geq 0$. Therefore

$$\prod_{n \geq 0} \text{sinc}_\pi(z/2^n) = \prod_{a=0}^{r-1} \prod_{m \geq 0} \text{sinc}_\pi\left(\frac{z}{2^{a+rm}}\right) = \prod_{a=0}^{r-1} \Psi_r(z/2^a).$$

The products converge locally uniformly, so regrouping is legitimate. The Fourier transform of $g_{r,a}$ is $\Psi_r(z/2^a)$; multiplication of the transforms gives (669). The random-series identity is the same partition of the summands of (623). $\square$

Remark 4.2. The support radii add exactly:

$$\sum_{a=0}^{r-1} 2^{-a} A_r = \frac{1}{2(1-2^{-r})} \sum_{a=0}^{r-1} 2^{-a} = 1. \quad (671)$$

Thus no slack is introduced by the factorization.

Example 4.3 (The even–odd scale split). For $r = 2$,

$$\Phi(z) = \Psi_2(z)\Psi_2(z/2), \quad u = g_2 * (2g_2(2 \cdot)), \quad (672)$$

where g_2 is supported on $[-2/3, 2/3]$. One factor contains scales $1, 1/4, 1/16, \dots$ in the full widths; the other contains the intervening scales after one dyadic contraction.

4.2 A sparse Rvachev differential equation

The product (663) satisfies

$$\Psi_r(z) = \text{sinc}_\pi(z) \Psi_r(q_r z). \quad (673)$$

The random fixed point is

$$Z_r \stackrel{d}{=} \frac{U}{2} + q_r Z'_r, \quad (674)$$

with independent $U \sim \text{Unif}[-1, 1]$ and $Z'_r \stackrel{d}{=} Z_r$.

Proposition 4.4 (Sparse-scale Rvachev equation). *The density g_r obeys*

$$g_r(x) = \int_{2^r x - 2^{r-1}}^{2^r x + 2^{r-1}} g_r(t) dt, \quad (675)$$

$$g'_r(x) = 2^r (g_r(2^r x + 2^{r-1}) - g_r(2^r x - 2^{r-1})). \quad (676)$$

For $r = 1$ this is the ordinary Rvachev equation.

Proof. Condition on $U/2$ in (674). The density of $q_r Z'_r$ is $q_r^{-1} g_r(\cdot/q_r)$, hence

$$g_r(x) = \int_{-1/2}^{1/2} q_r^{-1} g_r\left(\frac{x-y}{q_r}\right) dy.$$

The substitution $t = (x - y)/q_r$ gives (675); differentiation gives (676). $\square$

4.3 Cosine, Gamma, canonical-zero, and logarithmic forms

The sparse product has a complete set of analogues of the ordinary Rvachev representations.

Theorem 4.5 (Equivalent products for the sparse component). *For every $r \geq 1$,*

$$\Psi_r(z) = \prod_{m=0}^{\infty} \text{sinc}_\pi(z/2^{rm}), \quad (677)$$

$$= \prod_{n=1}^{\infty} \cos(\pi z/2^n)^{1+\lfloor (n-1)/r \rfloor}, \quad (678)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\lfloor \nu_2(n)/r \rfloor}, \quad (679)$$

$$= \prod_{m=0}^{\infty} \frac{1}{\Gamma(1 + z/2^{rm})\Gamma(1 - z/2^{rm})}. \quad (680)$$

For $|z| < 1$,

$$\log \Psi_r(z) = - \sum_{j=1}^{\infty} \frac{\zeta(2j)}{j(1 - 2^{-2rj})} z^{2j}. \quad (681)$$

Away from its integer zeros,

$$\frac{\Psi'_r(z)}{\Psi_r(z)} = -2z \sum_{n=1}^{\infty} \frac{1 + \lfloor \nu_2(n)/r \rfloor}{n^2 - z^2}. \quad (682)$$

Proof. The sinc and Gamma forms follow from $\text{sinc}_\pi w = (\Gamma(1+w)\Gamma(1-w))^{-1}$. Expanding each sinc by $\text{sinc}_\pi w = \prod_{j \geq 1} \cos(\pi w/2^j)$, a fixed cosine at scale 2^{-n} occurs once for each $m \geq 0$ with $rm < n$, giving $1 + \lfloor (n-1)/r \rfloor$ copies. Expanding instead by Euler's sine product, a zero at a nonzero integer n receives one copy for each m with $rm \leq \nu_2(n)$, hence multiplicity $1 + \lfloor \nu_2(n)/r \rfloor$. The logarithmic series follows from $\log \text{sinc}_\pi z = -\sum_{j \geq 1} \zeta(2j)z^{2j}/j$ and a geometric sum. Logarithmic differentiation of the canonical product gives (682). $\square$

Corollary 4.6 (Exact splitting of the integer zero divisor). *Let $v = \nu_2(n)$. The zero order $1 + v$ of Φ at $n \neq 0$ decomposes as*

$$1 + v = \sum_{a=0}^{r-1} \mathbf{1}_{a \leq v} \left(1 + \left\lfloor \frac{v-a}{r} \right\rfloor \right). \quad (683)$$

The a th summand is the order contributed by $\Psi_r(z/2^a)$.

Proof. The right side counts pairs (a, m) satisfying $0 \leq a < r$, $m \geq 0$, and $a + rm \leq v$. Uniqueness of Euclidean division identifies these pairs with the integers $0, 1, \dots, v$. $\square$

4.4 Spectral zeta and determinant splitting

Let

$$m_r(n) = 1 + \left\lfloor \frac{\nu_2(n)}{r} \right\rfloor. \quad (684)$$

The zero divisor of Ψ_r has spectral zeta

$$\mathcal{Z}_r(s) = \sum_{n \geq 1} \frac{m_r(n)}{n^s}. \quad (685)$$

Proposition 4.7 (Sparse spectral zeta). *For $\Re s > 1$,*

$$\boxed{\mathcal{Z}_r(s) = \frac{\zeta(s)}{1 - 2^{-rs}}}. \quad (686)$$

Moreover the scaled factors satisfy the exact recombination

$$\sum_{a=0}^{r-1} 2^{-as} \mathcal{Z}_r(s) = \frac{\zeta(s)}{1 - 2^{-s}}, \quad (687)$$

which is the spectral zeta of Φ .

Proof. Each factor $\text{sinc}_\pi(z/2^{rm})$ contributes one zero at every nonzero multiple of 2^{rm} . Therefore

$$\mathcal{Z}_r(s) = \sum_{m \geq 0} \sum_{k \geq 1} (2^{rm}k)^{-s} = \zeta(s) \sum_{m \geq 0} 2^{-rms}.$$

Scaling the zeros by 2^a multiplies their zeta contribution by 2^{-as} . The finite geometric sum cancels $1 - 2^{-rs}$. $\square$

In particular,

$$\Psi_r(z) = \det(I - z^2 T_r), \quad (688)$$

where T_r has eigenvalue n^{-2} repeated $m_r(n)$ times. The ordinary Rvachev determinant is the direct-sum product of the r scaled component operators.

4.5 Hyperbolic products, cumulants, and Bell polynomials

The component moment-generating function is

$$M_r(s) = \mathbb{E}e^{sZ_r} = \Psi_r\left(\frac{is}{2\pi}\right) = \prod_{m=0}^{\infty} \frac{\sinh(s/2^{rm+1})}{s/2^{rm+1}}. \quad (689)$$

Theorem 4.8 (Sparse cumulants and exact recombination). *The component cumulants are*

$$\kappa_{2j}^{(r)} = \frac{B_{2j}}{2j(1 - 2^{-2rj})}, \quad \kappa_{2j+1}^{(r)} = 0. \quad (690)$$

The $2j$ th cumulant of the polyphase sum (670) is

$$\sum_{a=0}^{r-1} 2^{-2aj} \kappa_{2j}^{(r)} = \frac{B_{2j}}{2j(1 - 2^{-2j})} = \kappa_{2j}. \quad (691)$$

Consequently

$$\mu_n^{(r)} := \mathbb{E}Z_r^n = \mathcal{B}_n(\kappa_1^{(r)}, \dots, \kappa_n^{(r)}). \quad (692)$$

Proof. Use

$$\log \frac{\sinh x}{x} = \sum_{j \geq 1} \frac{2^{2j-1} B_{2j}}{j(2j)!} x^{2j}$$

in (689) and sum the resulting geometric progression. The scaling law for cumulants and the finite geometric sum give (691). The Bell-polynomial relation between moments and cumulants is standard. $\square$

The first component values are

$$\text{Var } Z_r = \frac{1}{12(1 - 2^{-2r})}, \quad \kappa_4^{(r)} = -\frac{1}{120(1 - 2^{-4r})}. \quad (693)$$

For $r = 2$, for example,

$$\kappa_2^{(2)} = \frac{4}{45}, \quad \kappa_4^{(2)} = -\frac{32}{3825}. \quad (694)$$

The weighted sum of the two dilated cumulants recovers $1/9$ and $-2/225$.

Contribution I: one object, many representations

The sparse product Ψ_r is simultaneously a Fourier image, a Gamma product, a cosine product with staircase exponents, a canonical product with quotient valuation multiplicities, a Fredholm determinant, a hyperbolic moment product, and the transform of a sparse Rvachev density. Its r dyadic dilates multiply or convolve to the classical object. This modular-scale package was not found as such in the pinned corpus.

5 Mellin alias cancellation and log-periodic phases

The polyphase identity does more than regroup convergent products. It explains how the denser dyadic pole lattice is synthesized from a sparse one.

5.1 A general harmonic-sum lemma

Let $h : (0, \infty) \rightarrow \mathbb{C}$ be such that its Mellin transform

$$H(s) = \int_0^\infty h(t)t^{s-1} dt \quad (695)$$

exists in a vertical strip with $\Re s < 0$. Define

$$L_r(t) = \sum_{m=0}^{\infty} h(2^{-rm}t). \quad (696)$$

When necessary, polynomial or logarithmic principal parts are subtracted before applying this statement, exactly as in standard Mellin analysis of harmonic sums [6].

Theorem 5.1 (Polyphase Mellin cancellation). *In every common strip of absolute convergence,*

$$\mathcal{M}[L_r](s) = \frac{H(s)}{1 - 2^{rs}}, \quad (697)$$

$$L_1(t) = \sum_{a=0}^{r-1} L_r(2^{-a}t), \quad (698)$$

$$\mathcal{M}[L_1](s) = \frac{H(s)}{1 - 2^s}. \quad (699)$$

At a pole

$$s_k = \frac{2\pi i k}{r \log 2} \quad (700)$$

created by $1 - 2^{rs}$, the scale-sum multiplier is

$$\sum_{a=0}^{r-1} 2^{as_k} = \sum_{a=0}^{r-1} e^{2\pi i k a/r} = \begin{cases} r, & r \mid k, \\ 0, & r \nmid k. \end{cases} \quad (701)$$

Thus recombination cancels every sparse harmonic except every r th one.

Proof. For one term, $t = 2^{rm}u$ gives

$$\int_0^\infty h(2^{-rm}t)t^{s-1} dt = 2^{rms}H(s).$$

Summation yields (697). Euclidean division of the scale index gives (698); the Mellin transform of $L_r(2^{-a}t)$ is $2^{as}\mathcal{M}[L_r](s)$. The finite geometric numerator cancels $1 - 2^{rs}$ and gives (699). Evaluating the numerator on the pole lattice gives the roots-of-unity sum. $\square$

5.2 Application to logarithmic Fourier and Laplace products

Take, after the usual local regularization,

$$h(t) = \log \operatorname{sinc}_\pi(t) \quad \text{or} \quad h(t) = \log \frac{\sinh(t/2)}{t/2}. \quad (702)$$

Then L_r is respectively $\log \Psi_r$ or $\log M_r$. The component has a logarithmic-scale pole lattice with fundamental period $r \log 2$; the product of its r dyadic dilates retains only the harmonics compatible with period $\log 2$. This is the Mellin-side analogue of (683) and (691).

Corollary 5.2 (Periodic coefficient filter). *Suppose the regularized asymptotic expansion of a sparse component contains a Fourier series in the logarithmic phase*

$$P_r(\theta) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k \theta / r}. \quad (703)$$

Then the corresponding coefficient after polyphase recombination is obtained by the exact projector

$$P_1(\theta) = \sum_{a=0}^{r-1} P_r(\theta - a) = r \sum_{\ell \in \mathbb{Z}} c_{r\ell} e^{2\pi i \ell \theta}. \quad (704)$$

Remark 5.3. The existing Fourier-decay campaign identifies nontrivial orbit-dependent and periodic behavior of the full product. The theorem above does not replace that analysis. It factors its periodic data into a sparse phase and an exact cyclic projection, suggesting a new way to compute or certify the fluctuation by working at the longer period $r \log 2$ and then filtering.

Contribution II: Mellin polyphase projector

Partitioning the product by scale residues induces a literal roots-of-unity filter on Mellin residues. Zero multiplicities, cumulants, and log-periodic asymptotic harmonics are three manifestations of the same finite cyclic projection.

6 Multidimensional volume and section representations

The polyphase random decomposition turns the one-dimensional self-CDF identity into a relation between two different geometric operations on the same product density: integration over a half-space and integration over a parallel hyperplane.

6.1 Radon notation and the product density

Let

$$w_r = (1, 2^{-1}, \dots, 2^{-(r-1)}) \in \mathbb{R}^r, \quad p_r(y) = \prod_{a=0}^{r-1} g_r(y_a). \quad (705)$$

For an integrable function p on $\mathbb{R}^r$, define its cumulative and ordinary Radon transforms in direction $w \neq 0$ by

$$\mathcal{C}_w p(t) = \int_{w \cdot y \leq t} p(y) \, dy, \quad (706)$$

$$\mathcal{R}_w p(t) = \frac{1}{\|w\|_2} \int_{w \cdot y = t} p(y) \, d\sigma_{r-1}(y). \quad (707)$$

The coarea formula gives $(\mathcal{C}_w p)' = \mathcal{R}_w p$ at every continuity point. For the smooth compactly supported density p_r , this holds everywhere.

Theorem 6.1 (Polyphase Radon representation). *For every $r \geq 1$ and every $t \in \mathbb{R}$,*

$$u(t) = \mathcal{R}_{w_r} p_r(t), \quad \mathbb{P}(X \leq t) = \mathcal{C}_{w_r} p_r(t). \quad (708)$$

Consequently, for $0 \leq x \leq 1$,

$$F(x) = \mathcal{C}_{w_r} p_r(2x - 1) = \mathcal{R}_{w_r} p_r(x - 1). \quad (709)$$

a relation between a half-space volume at level $2x - 1$ and a parallel section at level $x - 1$.

Proof. By [Theorem 4.1](#), the linear form $w_r \cdot (Z_r^{(0)}, \dots, Z_r^{(r-1)})$ has the same law as X . Its distribution function is therefore the cumulative Radon transform in [\(708\)](#). The coarea formula identifies its density with the ordinary Radon transform. Substitute the defining relation $F(x) = \mathbb{P}(X \leq 2x - 1)$ and the self-CDF identity $F(x) = u(x - 1)$. $\square$

Remark 6.2. At $r = 1$, [\(709\)](#) is exactly the one-dimensional identity

$$\int_{-\infty}^{2x-1} u(t) \, dt = u(x - 1).$$

For $r > 1$, the same identity becomes genuinely multidimensional: the left side is an r -dimensional sublevel volume, while the right side is a weighted $(r - 1)$ -dimensional slice.

6.2 Coordinate integral forms

Let

$$H_r(t) = \int_{-\infty}^t g_r(v) \, dv \quad (710)$$

be the CDF of the sparse component. Eliminating the last coordinate in the half-space and in the hyperplane gives two explicit $(r - 1)$ -fold integrals.

Corollary 6.3 (Volume–section integral pair). *For $r \geq 2$ and $0 \leq x \leq 1$,*

$$F(x) = \int_{\mathbb{R}^{r-1}} \left(\prod_{a=0}^{r-2} g_r(y_a) \right) H_r \left(2^{r-1} \left(2x - 1 - \sum_{a=0}^{r-2} 2^{-a} y_a \right) \right) \, dy, \quad (711)$$

$$F(x) = 2^{r-1} \int_{\mathbb{R}^{r-1}} \left(\prod_{a=0}^{r-2} g_r(y_a) \right) g_r \left(2^{r-1} \left(x - 1 - \sum_{a=0}^{r-2} 2^{-a} y_a \right) \right) \, dy. \quad (712)$$

Both integrals are effectively over a compact box because g_r is compactly supported.

Proof. In the volume formula, condition on $y_0, \dots, y_{r-2}$ and solve the inequality for y_{r-1} . In the section formula, solve $w_r \cdot y = x - 1$ for the last coordinate. The Jacobian is 2^{r-1} . $\square$

For the first nontrivial split $r = 2$, writing H_2 for the CDF of g_2 ,

$$F(x) = \int_{\mathbb{R}} g_2(y) H_2(4x - 2 - 2y) \, dy, \quad (713)$$

$$= 2 \int_{\mathbb{R}} g_2(y) g_2(2x - 2 - 2y) \, dy. \quad (714)$$

Thus an integral involving the CDF of g_2 at one affine scale equals an integral involving its density at another.

6.3 A geometric representation of the inverse Fabius function

The preceding theorem also gives an implicit representation of $G = F^{-1}$ that uses only one fixed product density and moving parallel hyperplanes.

Corollary 6.4 (Two-level geometry of the quantile). *For $0 < y < 1$, the unique $x = G(y)$ is characterized by*

$$\boxed{\mathcal{C}_{w_r p_r}(2x - 1) = \mathcal{R}_{w_r p_r}(x - 1) = y.} \quad (715)$$

Its derivative is the reciprocal of a third section:

$$G'(y) = \frac{1}{2\mathcal{R}_{w_r}p_r(2G(y) - 1)}. \quad (716)$$

Equivalently, the same product density has section mass y at level $G(y) - 1$, while the section at level $2G(y) - 1$ controls the quantile elasticity.

Proof. Equation (715) is (709) with $x = G(y)$. Since $F'(x) = 2u(2x - 1)$, the inverse-function theorem and (708) give (716). $\square$

The quantile-transport transform becomes, for every integrable test function h ,

$$\int_0^1 h(G(y)) \, dy = \int_{\mathbb{R}^r} h\left(\frac{1 + w_r \cdot z}{2}\right) p_r(z) \, dz. \quad (717)$$

In particular,

$$\int_0^1 e^{sG(y)} \, dy = e^{s/2} \prod_{a=0}^{r-1} M_r(s/2^{a+1}), \quad (718)$$

which reduces, by cumulant recombination, to $e^{s/2}M(s/2)$.

6.4 Polyphase transforms of the bounded Fabius function

Substitution of (667) into the transform formulas of Appendix 3 gives product–integral representations involving only the sparse component:

$$F(x) = 2 \int_0^\infty \left(\prod_{a=0}^{r-1} \Psi_r(z/2^a) \right) \cos(2\pi z(x - 1)) \, dz, \quad (719)$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \left(\prod_{a=0}^{r-1} \Psi_r(z/2^a) \right) \frac{\sin(2\pi z(2x - 1))}{z} \, dz, \quad (720)$$

$$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = \frac{e^{-\pi i \xi} \prod_{a=0}^{r-1} \Psi_r(\xi/2^{a+1}) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (721)$$

These are [D](#), but they make the new factorization directly usable for evaluating F rather than only u .

6.5 Nested polyphase factorizations

The scale-class decomposition is coherent under further subdivision.

Theorem 6.5 (Divisibility-lattice factorization). *For positive integers r and s ,*

$$\Psi_r(z) = \prod_{a=0}^{s-1} \Psi_{rs}(z/2^{ra}), \quad (722)$$

$$g_r = \bigstar_{a=0}^{s-1} (2^{ra} g_{rs}(2^{ra} \cdot)). \quad (723)$$

The factorization is associative: subdividing first into classes modulo r and then each class modulo s gives exactly the same factors as a direct split modulo rs .

Proof. Write the sparse scale index as $m = a + s\ell$ and regroup the locally uniformly convergent product. Fourier inversion gives the convolution formula. $\square$

On spectral zeta functions the same coherence is the finite geometric identity

$$\frac{\zeta(z)}{1 - 2^{-rz}} = \sum_{a=0}^{s-1} 2^{-raz} \frac{\zeta(z)}{1 - 2^{-rsz}}. \quad (724)$$

Thus the family $(\Psi_r)_{r \geq 1}$ forms a compatible factor system over the divisibility poset.

Contribution III: a volume–section identity

Every modular-scale factorization of the Fourier product yields an r -dimensional product density whose cumulative Radon transform at $2x - 1$ and ordinary Radon transform at $x - 1$ are the same Fabius value. The inverse function is the unique position where these two moving geometric quantities coincide at a prescribed level.

7 Finite sparse splines and Thue–Morse jump measures

Fix $r \geq 1$, put $q = 2^{-r}$, and let $N \geq 1$. Define the finite random sum

$$Z_{r,N} = \sum_{j=0}^{N-1} \frac{q^j}{2} U_j, \quad (725)$$

with density $g_{r,N}$. Its support radius and subset-sum knots are

$$A_{r,N} = \frac{1 - q^N}{2(1 - q)}, \quad (726)$$

$$\lambda_{r,N}(k) = \sum_{j=0}^{N-1} b_j(k) q^j, \quad (727)$$

$$x_{r,N,k} = -A_{r,N} + \lambda_{r,N}(k), \quad 0 \leq k < 2^N. \quad (728)$$

Here $b_j(k)$ is the j th binary digit of k .

7.1 Truncated-power and Fourier-quotient formulas

Theorem 7.1 (Finite sparse box spline). *The density $g_{r,N}$ is the degree- $(N - 1)$ compact spline*

$$g_{r,N}(x) = \frac{q^{-\binom{N}{2}}}{(N - 1)!} \sum_{k=0}^{2^N-1} \varepsilon_k (x - x_{r,N,k})_+^{N-1}. \quad (729)$$

Its Fourier transform has the equivalent finite-product and deconvolution forms

$$\widehat{g_{r,N}}(z) = \prod_{j=0}^{N-1} \operatorname{sinc}_{\pi}(q^j z) = \frac{\Psi_r(z)}{\Psi_r(q^N z)}. \quad (730)$$

Moreover,

$$g_r = g_{r,N} * (q^{-N} g_r(\cdot/q^N)), \quad (731)$$

so the omitted tail is exactly a q^N -dilate of the full sparse law.

Proof. The density of the j th summand is $q^{-j} \mathbf{1}_{[-q^j/2, q^j/2]}$. The standard inclusion–exclusion formula for a convolution of nonuniform boxes gives (729); the product of all widths is $q^{\binom{N}{2}}$ and the left endpoints produce the knots (728). Multiplying the individual Fourier images gives the first part of (730), while splitting the infinite product at N gives the quotient. The random-series tail is $q^N Z'_r$ with $Z'_r \stackrel{d}{=} Z_r$, proving (731). $\square$

7.2 The derivative-jump formula

The N th distributional derivative of a normalized truncated power is a Dirac mass. Applying this to (729) exposes the entire Thue–Morse block as the singular derivative of one positive spline.

Theorem 7.2 (Thue–Morse jump measure). *In the space of compactly supported distributions,*

$$D^N g_{r,N} = q^{-\binom{N}{2}} \sum_{k=0}^{2^N-1} \varepsilon_k \delta_{x_{r,N,k}}. \quad (732)$$

Consequently the piecewise constant derivative $D^{N-1} g_{r,N}$ has jump

$$\varepsilon_k = q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}. \quad (733)$$

The unsigned bit is therefore

$$\tau(k) = \frac{1}{2} \left(1 - q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}} \right). \quad (734)$$

These formulas hold for every N large enough to contain the binary expansion of k .

Proof. Distributionally, $D^N (x - a)_+^{N-1} = (N-1)! \delta_a$. This gives (732). The mass of $D^N g_{r,N}$ at a knot is exactly the classical jump of $D^{N-1} g_{r,N}$ there, proving (733); (734) follows from $\tau = (1 - \varepsilon)/2$. $\square$

Corollary 7.3 (Smooth integral extraction). *Let φ_k be any C^N function satisfying $\varphi_k(x_{r,N,\ell}) = \delta_{k\ell}$. Then*

$$\varepsilon_k = q^{\binom{N}{2}} \langle D^N g_{r,N}, \varphi_k \rangle \quad (735)$$

$$= (-1)^N q^{\binom{N}{2}} \int_{\mathbb{R}} g_{r,N}(x) \varphi_k^{(N)}(x) \, dx. \quad (736)$$

One may take φ_k to be the Lagrange cardinal polynomial on the knot set. Thus each Thue–Morse sign has an exact integral representation against a positive spline.

7.3 Reading the sequence from left to right

Let $\text{rev}_N(j)$ denote reversal of the N binary digits of j . Since $q \leq 1/2$, each weight q^j exceeds the sum of all later finite weights, except for a strict finite remainder at $q = 1/2$. The spatial order of the knots is therefore lexicographic in $(b_0, b_1, \dots, b_{N-1})$.

Proposition 7.4 (Bit reversal and spatial order). *If the knots are listed increasingly as $x_{r,N}^\uparrow(0) < \dots < x_{r,N}^\uparrow(2^N - 1)$, then*

$$x_{r,N}^\uparrow(j) = x_{r,N,\text{rev}_N(j)}. \quad (737)$$

Because binary weight is invariant under reversal,

$$q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N}^\uparrow(j)} = \varepsilon_{\text{rev}_N(j)} = \varepsilon_j. \quad (738)$$

Hence scanning the jumps from the left endpoint to the right endpoint spells the ordinary length- 2^N signed Thue–Morse word in its usual order; replacing $+1, -1$ by $0, 1$ gives the unsigned word.

Proof. The first differing digit in the order $b_0, b_1, \dots$ determines the larger subset sum because $q^j > \sum_{\ell=j+1}^{N-1} q^\ell$ for $q < 1/2$, and the same inequality remains strict for the finite sum when $q = 1/2$. Lexicographic order in low-to-high binary digits is ordinary integer order after bit reversal. Finally, $s_2(\text{rev}_N(j)) = s_2(j)$. $\square$

7.4 Atomic convolution, Fourier, and Laplace products

Define the signed and unsigned knot combs

$$\Theta_{r,N} = \sum_{k < 2^N} \varepsilon_k \delta_{x_{r,N,k}}, \quad (739)$$

$$\Lambda_{r,N} = \sum_{k < 2^N} \delta_{x_{r,N,k}}. \quad (740)$$

They factor as convolutions of two-point measures:

$$\Theta_{r,N} = \bigstar_{j=0}^{N-1} (\delta_{-q^j/2} - \delta_{q^j/2}), \quad (741)$$

$$\Lambda_{r,N} = \bigstar_{j=0}^{N-1} (\delta_{-q^j/2} + \delta_{q^j/2}). \quad (742)$$

Their Fourier transforms are the finite lacunary products

$$\widehat{\Theta}_{r,N}(z) = (2i)^N \prod_{j=0}^{N-1} \sin(\pi q^j z) \quad (743)$$

$$= (2\pi i z)^N q^{\binom{N}{2}} \widehat{g_{r,N}}(z), \quad (744)$$

$$\widehat{\Lambda}_{r,N}(z) = 2^N \prod_{j=0}^{N-1} \cos(\pi q^j z). \quad (745)$$

Equivalently,

$$\Theta_{r,N} = q^{\binom{N}{2}} D^N \mathcal{F}^{-1} \left[\frac{\Psi_r(z)}{\Psi_r(q^N z)} \right]. \quad (746)$$

This is the announced Fourier deconvolution representation of the signed sequence.

The bilateral Laplace transforms are similarly

$$\int e^{tx} d\Theta_{r,N}(x) = (-2)^N \prod_{j=0}^{N-1} \sinh(tq^j/2), \quad (747)$$

$$\int e^{tx} d\Lambda_{r,N}(x) = 2^N \prod_{j=0}^{N-1} \cosh(tq^j/2). \quad (748)$$

More generally, the Hamming-weight enumerator is the master product

$$\sum_{k < 2^N} y^{s_2(k)} e^{tx_{r,N,k}} = e^{-tA_{r,N}} \prod_{j=0}^{N-1} (1 + ye^{tq^j}). \quad (749)$$

The choices $y = -1$ and $y = 1$ recover (747) and (748).

7.5 Unsigned evil and odious measures

Let

$$E_{r,N} = \sum_{\tau(k)=0} \delta_{x_{r,N,k}}, \quad (750)$$

$$O_{r,N} = \sum_{\tau(k)=1} \delta_{x_{r,N,k}}. \quad (751)$$

Then the Jordan decomposition is exact:

$$E_{r,N} = \frac{\Lambda_{r,N} + \Theta_{r,N}}{2}, \quad O_{r,N} = \frac{\Lambda_{r,N} - \Theta_{r,N}}{2}. \quad (752)$$

Accordingly,

$$\widehat{E}_{r,N}(z) = 2^{N-1} \left(\prod_{j < N} \cos(\pi q^j z) + i^N \prod_{j < N} \sin(\pi q^j z) \right), \quad (753)$$

$$\widehat{O}_{r,N}(z) = 2^{N-1} \left(\prod_{j < N} \cos(\pi q^j z) - i^N \prod_{j < N} \sin(\pi q^j z) \right). \quad (754)$$

These are product representations for the unsigned Thue–Morse partition on the sparse knot set. Individual membership is given by the jump formula (734); aggregate transforms are given by (753)–(754).

Contribution IV: Thue–Morse as analytic singular data

The signed and unsigned Thue–Morse sequences can be recovered pointwise from jumps of the highest classical derivative of a positive compact spline, measure-theoretically from its next distributional derivative, integrally by testing that derivative against cardinal functions, and spectrally from a quotient of two sparse infinite sinc products. In spatial order no external binary labels are needed: the jumps already occur in Thue–Morse order.

8 Bell–Bernoulli power sums on sparse knots

The jump measure immediately converts the moments of a positive box spline into high-degree signed power sums. This extends ordinary Prouhet cancellation past its first nonzero degree and gives every surviving sum in closed Bell form.

8.1 Prouhet cancellation and all surviving powers

Let

$$S_{r,N}(d) = \sum_{k=0}^{2^N-1} \varepsilon_k x_{r,N,k}^d. \quad (755)$$

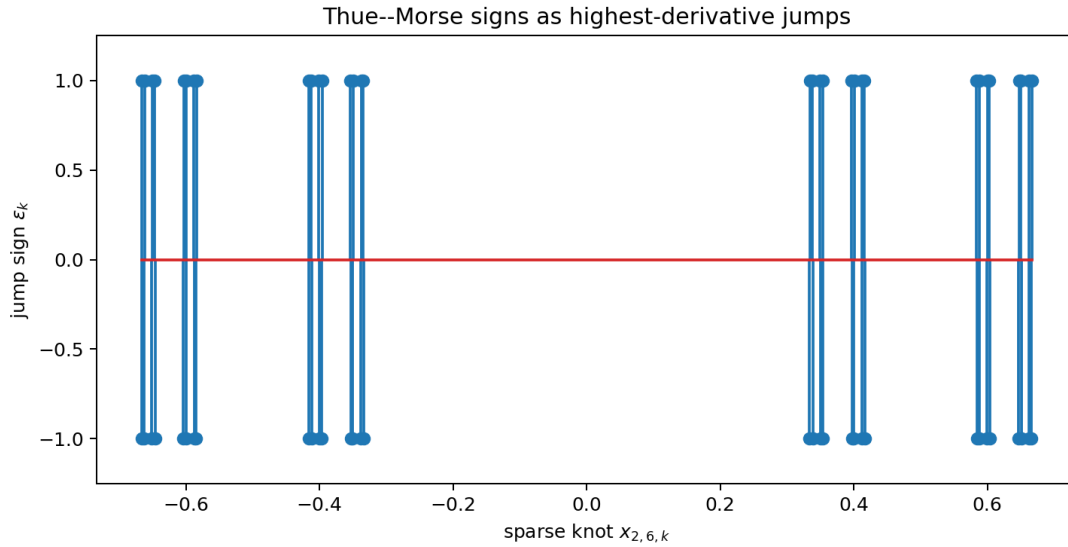


Figure 12: The distribution $q^{\binom{N}{2}} D^N g_{r,N}$ for a finite sparse spline. Reading its signed atoms from left to right gives the ordinary Thue–Morse word because the spatial bit-reversal leaves binary parity unchanged. The plotted example is generated by the accompanying script.

Theorem 8.1 (Moment–power-sum duality). *For $0 \leq d < N$,*

$$S_{r,N}(d) = 0. \quad (756)$$

For every $m \geq 0$,

$$S_{r,N}(N+m) = (-1)^N q^{\binom{N}{2}} \frac{(N+m)!}{m!} \mu_m^{(r,N)}, \quad (757)$$

where

$$\mu_m^{(r,N)} = \int_{\mathbb{R}} x^m g_{r,N}(x) \, dx. \quad (758)$$

Equivalently,

$$\mu_m^{(r,N)} = (-1)^N q^{-\binom{N}{2}} \frac{m!}{(N+m)!} \sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+m}. \quad (759)$$

Odd m give zero by symmetry.

Proof. Pair (732) with the polynomial x^d . If $d < N$, its N th derivative vanishes. If $d = N + m$, distributional integration by parts gives

$$q^{-\binom{N}{2}} S_{r,N}(N+m) = (-1)^N \frac{(N+m)!}{m!} \int x^m g_{r,N}(x) \, dx.$$

Rearrange. □

The first two nontrivial cases are

$$\sum_{k < 2^N} \varepsilon_k x_{r,N,k}^N = (-1)^N N! q^{\binom{N}{2}}, \quad (760)$$

$$\sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+2} = (-1)^N q^{\binom{N}{2}} \frac{(N+2)!}{24} \frac{1 - q^{2N}}{1 - q^2}. \quad (761)$$

Thus the full hierarchy refines the usual statement that the first surviving Prouhet power occurs in degree N .

8.2 Finite cumulants and complete Bell form

The moment-generating function of $Z_{r,N}$ is

$$M_{r,N}(s) = \prod_{j=0}^{N-1} \frac{\sinh(sq^j/2)}{sq^j/2}. \quad (762)$$

Its cumulants are finite geometric truncations of the sparse cumulants.

Proposition 8.2 (Finite sparse cumulants). *For $\ell \geq 1$,*

$$\kappa_{2\ell}^{(r,N)} = \frac{B_{2\ell}}{2\ell} \frac{1 - q^{2\ell N}}{1 - q^{2\ell}}, \quad \kappa_{2\ell+1}^{(r,N)} = 0. \quad (763)$$

Hence

$$\mu_m^{(r,N)} = \mathcal{B}_m(0, \kappa_2^{(r,N)}, 0, \kappa_4^{(r,N)}, \dots). \quad (764)$$

Combining with (757) gives an explicit Bell–Bernoulli formula for every signed power sum $S_{r,N}(N+m)$.

In particular, for $\ell \geq 0$,

$$S_{r,N}(N+2\ell) = (-1)^N q^{\binom{N}{2}} \frac{(N+2\ell)!}{(2\ell)!} \mathcal{B}_{2\ell}(0, \kappa_2^{(r,N)}, 0, \dots, \kappa_{2\ell}^{(r,N)}). \quad (765)$$

This is a finite product–series–Bell representation in which every parameter is rational for $q = 2^{-r}$.

8.3 Unsigned power sums

The total knot comb $\Lambda_{r,N}$ is the unnormalized law of the centered Bernoulli sum

$$W_{r,N} = \sum_{j=0}^{N-1} q^j (B_j - \tfrac{1}{2}), \quad \mathbb{P}(B_j = 0) = \mathbb{P}(B_j = 1) = \tfrac{1}{2}. \quad (766)$$

Its even cumulants are

$$\lambda_{2\ell}^{(r,N)} = \frac{(2^{2\ell} - 1)B_{2\ell}}{2\ell} \frac{1 - q^{2\ell N}}{1 - q^{2\ell}}, \quad (767)$$

with odd cumulants zero. Therefore

$$T_{r,N}(d) := \sum_{k < 2^N} x_{r,N,k}^d = 2^N \mathcal{B}_d(0, \lambda_2^{(r,N)}, 0, \lambda_4^{(r,N)}, \dots). \quad (768)$$

If

$$T_{r,N}^{\text{evil}}(d) = \sum_{\tau(k)=0} x_{r,N,k}^d, \quad (769)$$

$$T_{r,N}^{\text{odious}}(d) = \sum_{\tau(k)=1} x_{r,N,k}^d, \quad (770)$$

then

$$T_{r,N}^{\text{evil}}(d) = \frac{T_{r,N}(d) + S_{r,N}(d)}{2}, \quad (771)$$

$$T_{r,N}^{\text{odious}}(d) = \frac{T_{r,N}(d) - S_{r,N}(d)}{2}. \quad (772)$$

For $d < N$ the two values coincide; for $d = N + m$ their exact difference is (757). Thus the unsigned partition inherits a complete Bell–Bernoulli moment calculus, not only the usual equality up to degree $N - 1$.

8.4 Infinite moment limits

Since $Z_{r,N} \rightarrow Z_r$ almost surely and uniformly in the underlying cube, all moments converge. Therefore (759) gives

$$\mu_m^{(r)} = \lim_{N \rightarrow \infty} (-1)^N q^{-\binom{N}{2}} \frac{m!}{(N+m)!} \sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+m}. \quad (773)$$

At $r = 1$, these are the moments of Rvachev's up-function:

$$\int_{-1}^1 x^m u(x) \, dx = \lim_{N \rightarrow \infty} (-1)^N 2^{\binom{N}{2}} \frac{m!}{(N+m)!} \sum_{k < 2^N} \varepsilon_k x_{1,N,k}^{N+m}. \quad (774)$$

The enormous factor $q^{-\binom{N}{2}}$ exactly renormalizes the Prouhet cancellation.

Moments of the bounded Fabius law and powers of the inverse follow from $Y = (X + 1)/2$:

$$\int_0^1 G(y)^m \, dy = \mathbb{E}Y^m = 2^{-m} \sum_{j=0}^m \binom{m}{j} \mu_j, \quad (775)$$

$$\int_0^1 x^p F(x) \, dx = \frac{1 - \mathbb{E}Y^{p+1}}{p+1}, \quad \Re p > -1. \quad (776)$$

Substitution of (774) therefore yields explicit Thue–Morse limit representations for both integrals.

8.5 A Gaussian q -binomial side identity

The same binary digits carry a second geometric statistic $\sigma(k) = \sum_j j b_j(k)$. Its joint Hamming-weight enumerator is

$$\sum_{k < 2^N} y^{s_2(k)} q^{\sigma(k)} = \prod_{j=0}^{N-1} (1 + yq^j) = \sum_{h=0}^N y^h q^{\binom{h}{2}} \begin{bmatrix} N \\ h \end{bmatrix}_q. \quad (777)$$

This familiar Gaussian-binomial identity is not the same as the knot transform (749): one weights a selected digit by q^j , the other places it at the geometric coordinate q^j . Their coexistence is useful in symbolic calculations because the first organizes selected scale indices, while the second organizes the actual spline knots.

Contribution V: all Prouhet survivors

The finite sparse spline does more than explain cancellation below degree N . Its moments give every surviving signed power sum in degree $N + m$, and finite uniform cumulants turn the answer into a closed Bell polynomial in Bernoulli numbers and geometric factors. Adding the cosh-product moments of the unsigned knot comb gives separate exact formulas for the evil and odious classes.

9 Infinite products of averaging operators

The Fourier product can be read literally as a product of physical-space averaging operators. This provides a representation which is simultaneously probabilistic, convolutional, and an infinite-order differential calculus.

9.1 Uniform averaging and strong product limits

For $a > 0$, define

$$(\mathcal{U}_a f)(x) = \frac{1}{2a} \int_{-a}^a f(x-t) \, dt. \quad (778)$$

Its Fourier multiplier and formal differential symbol are

$$\widehat{\mathcal{U}_a f}(z) = \text{sinc}_\pi(2az) \widehat{f}(z), \quad \mathcal{U}_a = \frac{\sinh(aD)}{aD}. \quad (779)$$

The second equality is exact on polynomials and on every class on which the translation exponential may be integrated termwise.

Put

$$\mathcal{G}_{r,N} = \prod_{j=0}^{N-1} \mathcal{U}_{q^j/2}, \quad \mathcal{G}_r = \prod_{j=0}^{\infty} \mathcal{U}_{q^j/2}. \quad (780)$$

The factors commute.

Theorem 9.1 (Averaging-operator representation). *For every bounded uniformly continuous f ,*

$$(\mathcal{G}_{r,N} f)(x) = (g_{r,N} * f)(x) = \mathbb{E} f(x - Z_{r,N}), \quad (781)$$

$$(\mathcal{G}_r f)(x) = (g_r * f)(x) = \mathbb{E} f(x - Z_r), \quad (782)$$

and $\mathcal{G}_{r,N} f \rightarrow \mathcal{G}_r f$ uniformly. The same convergence is strong on $L^p(\mathbb{R})$ for $1 \leq p < \infty$. Distributionally,

$$g_{r,N} = \mathcal{G}_{r,N} \delta_0, \quad g_r = \mathcal{G}_r \delta_0. \quad (783)$$

For $r = 1$ this gives

$$u = \left(\prod_{j=0}^{\infty} \mathcal{U}_{2^{-j-1}} \right) \delta_0. \quad (784)$$

Proof. Each operator convolves with the density of a centered uniform random variable of half-width $q^j/2$. Finite products therefore give the law of $Z_{r,N}$. The tail satisfies $|Z_r - Z_{r,N}| \leq A_r q^N$ pointwise on the underlying product space, which proves uniform convergence for uniformly continuous f . Strong L^p convergence follows from continuity of translations in L^p and dominated averaging. Applying the convolution operators to δ_0 gives the densities. $\square$

9.2 Bernoulli logarithm of the operator product

The classical expansion

$$\log \frac{\sinh z}{z} = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m} \quad (785)$$

shows that the logarithm of the sparse averaging product is controlled by the same Bernoulli cumulants as the random law.

Define

$$c_{r,m} = \frac{B_{2m}}{2m(2m)!(1 - q^{2m})}. \quad (786)$$

Theorem 9.2 (Infinite-order differential representation). *As a formal differential operator, and exactly on every polynomial,*

$$\mathcal{G}_r = \exp \left(\sum_{m=1}^{\infty} c_{r,m} D^{2m} \right), \quad (787)$$

$$\mathcal{G}_{r,N} = \exp \left(\sum_{m=1}^{\infty} c_{r,m} (1 - q^{2mN}) D^{2m} \right). \quad (788)$$

If f has Fourier support in a compact subset of $(-1, 1)$, then (787) is also an exact Fourier-multiplier identity. For the classical up-function,

$$u = \exp \left(\frac{D^2}{18} - \frac{D^4}{2700} + \frac{D^6}{178605} - \frac{D^8}{9639000} + \cdots \right) \delta_0, \quad (789)$$

where the equality is understood through the exact averaging product and, termwise, through its formal or band-limited expansion.

Proof. From (779) and (785),

$$\log \mathcal{U}_{q^j/2} = \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} q^{2mj} D^{2m}.$$

Summing over j gives (787); truncating the geometric sum gives (788). On a polynomial only finitely many powers of D survive. On Fourier support $|z| < 1$, substitution $D \mapsto 2\pi iz$ gives precisely the convergent series (681). The displayed coefficients in (789) are $c_{1,m}$. $\square$

Using Euler's formula for $\zeta(2m)$, the same operator can be written

$$\mathcal{G}_r = \exp \left(\sum_{m=1}^{\infty} \frac{(-1)^{m+1} \zeta(2m)}{m(1 - q^{2m})(2\pi)^{2m}} D^{2m} \right). \quad (790)$$

Thus the Bernoulli, zeta, cumulant, and Fourier-logarithm representations are literally the same coefficients in four coordinate systems.

9.3 Exact tail operators and series-product combinations

Let $\mathcal{T}_{r,N}$ be convolution by the scaled tail density $q^{-N} g_r(\cdot/q^N)$. Then

$$\mathcal{G}_r = \mathcal{T}_{r,N} \mathcal{G}_{r,N}, \quad \widehat{\mathcal{T}_{r,N} f}(z) = \Psi_r(q^N z) \widehat{f}(z). \quad (791)$$

On polynomials and in the low-frequency calculus,

$$\mathcal{T}_{r,N} = \exp \left(\sum_{m=1}^{\infty} c_{r,m} q^{2mN} D^{2m} \right). \quad (792)$$

Consequently, for every polynomial p ,

$$\int p(x - y) g_r(y) \, dy = \exp \left(\sum_{m \geq 1} c_{r,m} D^{2m} \right) p(x), \quad (793)$$

and

$$\int p(x-y)q^{-N}g_r(y/q^N) dy = \exp\left(\sum_{m \geq 1} c_{r,m} q^{2mN} D^{2m}\right) p(x). \quad (794)$$

Expanding the last exponential gives the exact finite identity

$$\mathbb{E}p(x - q^N Z_r) = \sum_{m=0}^{\deg p} \frac{(-q^N)^m \mu_m^{(r)}}{m!} p^{(m)}(x), \quad (795)$$

whose odd terms vanish. This is a rigorous series–integral representation, not an asymptotic expansion, whenever the test function is a polynomial.

For analytic or smooth functions the same coefficients organize the familiar small-tail expansion. The first two even orders are

$$\mathcal{T}_{r,N} = I + c_{r,1} q^{2N} D^2 + \left(c_{r,2} + \frac{c_{r,1}^2}{2}\right) q^{4N} D^4 + \mathcal{O}_{\text{formal}}(q^{6N} D^6). \quad (796)$$

The coefficient of $q^{2mN} D^{2m}$ is equivalently $\mu_{2m}^{(r)}/(2m)!$, by the Bell relation between cumulants and moments.

9.4 Reciprocal and deconvolution operators

Since $\widehat{g_{r,N}} = \Psi_r/\Psi_r(q^N \cdot)$, the finite spline is a formal deconvolution of the smooth limit:

$$g_{r,N} = \exp\left(-\sum_{m \geq 1} c_{r,m} q^{2mN} D^{2m}\right) g_r. \quad (797)$$

This identity is exact as a germ of Fourier multipliers at the origin and gives all algebraic orders in q^{2N} . Its global use requires the zero-aware factorization (730), because the reciprocal of $\Psi_r(q^N z)$ has poles which are cancelled by zeros of $\Psi_r(z)$. That distinction is important: the product quotient is globally exact, while the exponential reciprocal is a local or asymptotic coordinate chart.

The first logarithmic coefficients, generated exactly by the accompanying script, are shown in Table 15.

$2m$	$c_{1,m}$	$c_{2,m}$	$c_{3,m}$
2	1/18	2/45	8/189
4	-1/2700	-4/11475	-64/184275
6	1/178605	64/11609325	4096/743175405
8	-1/9639000	-32/309652875	-8192/79272340875
10	1/478533825	1024/490497170625	1048576/502269581253825

Table 15: Coefficients of D^{2m} in $\log \mathcal{G}_r$. The complete exact table through order 16 is included as `operator_coefficients.csv`.

Contribution VI: an operator logarithm

The infinite sinc product is the Fourier symbol of an infinite product of uniform averaging operators. Its logarithm is an even differential operator whose coefficients are simultaneously sparse cumulants, Bernoulli numbers, and zeta values. On polynomials the exponential representation is exact; on band-limited functions it is an exact multiplier; globally the averaging product supplies the canonical interpretation.

10 Twisted Poisson and zero-slope representations

The critical half-integer cosine series is one member of a continuous family obtained by twisting Poisson summation. This family also yields a representation of F through the derivatives of Φ at its integer zeros.

10.1 A one-parameter Zak family

For real α , define

$$Z_u(x, \alpha) = \sum_{n \in \mathbb{Z}} u(x + n) e^{-2\pi i n \alpha}. \quad (798)$$

Rapid decay of Φ and compact support of u justify Poisson summation absolutely in the forms used below.

Theorem 10.1 (Twisted spectral representation of F). *For $0 \leq x \leq 1$,*

$$Z_u(x, \alpha) = \sum_{k \in \mathbb{Z}} \Phi(k + \alpha) e^{2\pi i (k + \alpha)x} \quad (799)$$

and also

$$Z_u(x, \alpha) = 1 + (e^{2\pi i \alpha} - 1)F(x). \quad (800)$$

Hence, whenever $\alpha \notin \mathbb{Z}$,

$$F(x) = \frac{\sum_{k \in \mathbb{Z}} \Phi(k + \alpha) e^{2\pi i (k + \alpha)x} - 1}{e^{2\pi i \alpha} - 1}. \quad (801)$$

Using polyphase factors, each coefficient may be replaced by $\prod_{a=0}^{r-1} \Psi_r((k + \alpha)/2^a)$.

Proof. The modulated Poisson formula gives (799). On the interval $0 < x < 1$, only the translates $n = 0$ and $n = -1$ meet the support of u , so

$$Z_u(x, \alpha) = u(x) + e^{2\pi i \alpha} u(x - 1) = 1 - F(x) + e^{2\pi i \alpha} F(x).$$

Continuity supplies the endpoint values. □

At $\alpha = 1/2$, (801) becomes the real cosine series

$$F(x) = \frac{1}{2} - \sum_{k=0}^{\infty} \Phi(k + \frac{1}{2}) \cos((2k + 1)\pi x), \quad (802)$$

which is the shifted form of (658). Rational values of α give cyclotomic spectral expansions; irrational values give a continuous family of quasiperiodic representations.

10.2 Differentiating through the untwisted limit

Both numerator and denominator of (801) vanish at $\alpha = 0$. Differentiation gives a new zero-jet representation.

Corollary 10.2 (Integer-zero slope series). *For $0 \leq x \leq 1$,*

$$F(x) = x + \frac{1}{2\pi i} \sum_{k \in \mathbb{Z} \setminus \{0\}} \Phi'(k) e^{2\pi i k x} \quad (803)$$

$$= x + \frac{1}{\pi} \sum_{k=1}^{\infty} \Phi'(k) \sin(2\pi k x) \quad (804)$$

$$= x - \frac{1}{\pi} \sum_{m=0}^{\infty} \frac{\Phi(m + \frac{1}{2})}{2m+1} \sin(2\pi(2m+1)x). \quad (805)$$

Only odd positive integers contribute in the middle series.

Proof. Differentiate (799) at $\alpha = 0$ and use (800). Since $\Phi(0) = 1$, $\Phi'(0) = 0$, and $\Phi(k) = 0$ for nonzero integers, division by $2\pi i$ gives (803). Pair k with $-k$ to obtain the real sine series. If k is even, the zero has order at least two, so $\Phi'(k) = 0$. For odd $k = 2m + 1$, the first sinc factor has a simple zero and

$$\Phi'(2m+1) = -\frac{\Phi(m + \frac{1}{2})}{2m+1}, \quad (806)$$

which gives (805). $\square$

The transform formula (646) also yields, for real $\xi \neq 0$,

$$\int_0^1 F(x) \cos(2\pi \xi x) \, dx = \frac{\sin(2\pi \xi) - \Phi(\xi/2) \sin(\pi \xi)}{2\pi \xi}, \quad (807)$$

$$\int_0^1 F(x) \sin(2\pi \xi x) \, dx = \frac{\Phi(\xi/2) \cos(\pi \xi) - \cos(2\pi \xi)}{2\pi \xi}. \quad (808)$$

At integer frequencies the cosine coefficients vanish, while the sine coefficients are explicit samples of Φ .

Contribution VII: a continuous spectral family

The usual half-integer cosine expansion sits inside a twisted Poisson family parameterized by α . Letting the twist tend to zero converts spectral values into spectral slopes and yields a sine series for F supported exactly on the simple odd zeros of the Rvachev Fourier product.

11 Sparse lobe signs and base- 2^r digit sums

The staircase zero multiplicities of Ψ_r generate a natural automatic sign sequence. Polyphase recombination then factors the ordinary Thue–Morse sign into r such sparse signs.

Let $s_b(n)$ denote the sum of the base- b digits of n .

Theorem 11.1 (Lobe signs of a sparse sinc product). *For $n \geq 0$ and any $x \in (n, n+1)$,*

$$\operatorname{sgn} \Psi_r(x) = (-1)^{\sum_{j=1}^n m_r(j)} \quad (809)$$

$$= (-1)^{n + \sum_{\ell \geq 1} \lfloor n/2^{r\ell} \rfloor} \quad (810)$$

$$= \boxed{(-1)^{s_{2^r}(n)}}. \quad (811)$$

Thus the signs of the consecutive real lobes of Ψ_r form the parity-of- digit-sum sequence in base 2^r .

Proof. The sign starts positive on $(0, 1)$ and flips once for each zero multiplicity crossed, giving (809). Since $m_r(j) = 1 + \lfloor \nu_2(j)/r \rfloor$,

$$\sum_{j=1}^n \left\lfloor \frac{\nu_2(j)}{r} \right\rfloor = \sum_{\ell \geq 1} \#\{j \leq n : 2^{r\ell} \mid j\} = \sum_{\ell \geq 1} \left\lfloor \frac{n}{2^{r\ell}} \right\rfloor.$$

For $b = 2^r$, the base- b Legendre identity is

$$\sum_{\ell \geq 1} \left\lfloor \frac{n}{b^\ell} \right\rfloor = \frac{n - s_b(n)}{b - 1}.$$

Because $b - 1$ is odd and b is even, $n + (n - s_b(n))/(b - 1)$ has the same parity as $s_b(n)$. $\square$

At $r = 1$, (811) is the familiar formula $\text{sgn } \Phi(x) = \varepsilon_n$ on the n th lobe. For $r > 1$ it supplies a companion automatic sequence attached to the sparse Rvachev density.

Corollary 11.2 (Polyphase factorization of the Thue–Morse sign). *For every $r \geq 1$ and $n \geq 0$,*

$$\varepsilon_n = \prod_{a=0}^{r-1} (-1)^{s_{2^r}(\lfloor n/2^a \rfloor)}. \quad (812)$$

Equivalently,

$$\tau(n) \equiv \sum_{a=0}^{r-1} s_{2^r}(\lfloor n/2^a \rfloor) \pmod{2}. \quad (813)$$

The same representation is analytic:

$$\varepsilon_n = \prod_{a=0}^{r-1} \text{sgn } \Psi_r\left(\frac{n + 1/2}{2^a}\right). \quad (814)$$

Proof. Take signs in the polyphase product $\Phi(n + 1/2) = \prod_a \Psi_r((n + 1/2)/2^a)$. The argument of the a th factor lies in the lobe indexed by $\lfloor n/2^a \rfloor$. Apply Theorem 11.1 and use $\text{sgn } \Phi(n + 1/2) = \varepsilon_n$. The unsigned parity form is the exponent modulo two. $\square$

An entirely arithmetical proof follows by summing the floor exponents:

$$\sum_{a=0}^{r-1} \sum_{m \geq 0} \left\lfloor \frac{n}{2^{a+rm}} \right\rfloor = \sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor = 2n - s_2(n), \quad (815)$$

whose parity is $s_2(n)$. Formula (812) therefore expresses binary digit parity as a cyclic product of overlapping base- 2^r digit parities.

Corollary 11.3 (Thue–Morse from odd zero slopes). *For every $n \geq 0$,*

$$\varepsilon_n = -\text{sgn } \Phi'(2n + 1), \quad \tau(n) = \frac{1 + \text{sgn } \Phi'(2n + 1)}{2}. \quad (816)$$

Proof. Equation (806) says $\Phi'(2n + 1) = -\Phi(n + 1/2)/(2n + 1)$. The lobe sign of the half-integer value is ε_n . $\square$

Contribution VIII: a new digital factorization

The sparse component does not merely split amplitudes. Its lobe signs are base- 2^r digit-parity characters, and their r shifted polyphase copies multiply to the binary Thue–Morse character. This gives both a new purely digital formula and a product of analytic sign samples. Odd zero slopes of the full Fourier image give a second pointwise analytic representation.

12 Representation concordance

This section gathers the formulas into a compact lookup index. A reader interested in a particular mode—probability, transform, product, series, operator, geometry, or combinatorics—can use the rightmost column to locate the hypotheses and proof. The list includes both corpus baseline and the new polyphase layer; it is not a novelty list.

12.1 Rvachev’s up-function and its Fourier image

Viewpoint	Representation	Status
Random series	$X = \sum_{n \geq 0} 2^{-n-1} U_n$, with u the density of X .	K
Infinite cube	$u(x) = \int_{[-1,1]^{\mathbb{N}}} \delta(x - \sum 2^{-n-1} v_n) \prod dv_n / 2$ in the distributional sense.	K
Finite convolution	$u = \lim_{N \rightarrow \infty} g_{1,N}$, where $g_{1,N}$ is an explicit nonuniform box spline.	K
Self-CDF	$u(x) = \mathbb{P}(X \leq 2x + 1)$ for $-1 \leq x \leq 0$, equivalently $u(x) = F(x + 1)$.	K
Delay equation	$u'(x) = 2(u(2x + 1) - u(2x - 1))$.	K
Derivative cascade	$u^{(N)}(x) = 2^{N(N+1)/2} \sum_{k < 2^N} \varepsilon_k u(2^N x + 2^N - 1 - 2k)$.	K
Fourier inversion	$u(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z x} dz = 2 \int_0^\infty \Phi(z) \cos(2\pi z x) dz$.	K
Characteristic expectation	$\Phi(z) = \mathbb{E} e^{-2\pi i z X}$.	K
Sinc product	$\Phi(z) = \prod_{n \geq 0} \text{sinc}_\pi(z/2^n)$.	K
Triangular cosine product	$\Phi(z) = \prod_{m \geq 1} \cos(\pi z/2^m)^m$.	K
Canonical zero product	$\Phi(z) = \prod_{n \geq 1} (1 - z^2/n^2)^{1+\nu_2(n)}$.	K
Gamma product	$\Phi(z) = \prod_{n \geq 0} [\Gamma(1 + z/2^n) \Gamma(1 - z/2^n)]^{-1}$.	K
Logarithmic zeta series	$\log \Phi(z) = - \sum_{m \geq 1} \zeta(2m) z^{2m} / [m(1 - 4^{-m})]$, $ z < 1$.	K
Cotangent logarithmic derivative	$\Phi'/\Phi = \sum_{j \geq 0} [(\pi/2^j) \cot(\pi z/2^j) - 1/z]$.	D
Tangent logarithmic derivative	$\Phi'/\Phi = -\pi \sum_{m \geq 1} m 2^{-m} \tan(\pi z/2^m)$.	D
Partial fractions	$\Phi'/\Phi = -2z \sum_{n \geq 1} (1 + \nu_2(n)) / (n^2 - z^2)$.	K

Viewpoint	Representation	Status
Moment-generating product	$M(s) = \prod_{n \geq 0} \sinh(s/2^{n+1})/(s/2^{n+1})$.	K
Moment series	$M(s) = \sum_{m \geq 0} \mu_m s^m / m!$, with $\mu_m = \mathcal{B}_m(\kappa_1, \dots, \kappa_m)$.	K
Cumulants	$\kappa_{2m} = B_{2m}/[2m(1 - 4^{-m})]$, $\kappa_{2m+1} = 0$.	K
Critical cosine series	$u(x) = \frac{1}{2} + \sum_{k \geq 0} \Phi(k + \frac{1}{2}) \cos((2k + 1)\pi x)$.	K
Partition of unity	$\sum_{j \in \mathbb{Z}} u(x - j) = 1$.	K
Twisted Zak transform	$\sum_n u(x + n) e^{-2\pi i n \alpha} = \sum_k \Phi(k + \alpha) e^{2\pi i (k + \alpha)x}$.	D
Legendre expansion	$u(x) = \sum_{n \geq 0} a_{2n} P_{2n}(x)$, $a_{2n} = \frac{4n+1}{2} \int_{-1}^1 u(x) P_{2n}(x) dx$; all coefficients reduce to moments.	K
Haar, Walsh, and Schauder	Exact rational multiresolution expansions and finite Walsh products are developed in [14].	K
Cauchy transform	$R(z) = \int u(x)/(z - x) dx = \sum_{m \geq 0} \mu_m z^{-m-1}$.	K
Jacobi fraction	$R(z) = 1/(z - a_0 - b_1/(z - a_1 - b_2/(\dots)))$, with coefficients determined by the moment Hankel determinants.	K
Fredholm determinant	$\Phi(z) = \det(I - z^2 T)$ for eigenvalues n^{-2} of multiplicity $1 + \nu_2(n)$.	K
Spectral zeta	$\sum_{n \geq 1} (1 + \nu_2(n)) n^{-s} = \zeta(s)/(1 - 2^{-s})$.	K
Heat trace	$K(t) = \sum_{n \geq 1} (1 + \nu_2(n)) e^{-tn^2}$ and $\int_0^\infty t^{s/2-1} K(t) dt = \Gamma(s/2) \zeta(s)/(1 - 2^{-s})$.	D
Polyphase product	$\Phi(z) = \prod_{a=0}^{r-1} \Psi_r(z/2^a)$.	N
Polyphase convolution	$u = *_{a=0}^{r-1} (2^a g_r(2^a \cdot))$.	N
Radon section	$u(t) = \ w_r\ ^{-1} \int_{w_r \cdot y=t} \prod_a g_r(y_a) d\sigma$.	N
Averaging product	$u = (\prod_{j \geq 0} \mathcal{U}_{2^{-j-1}}) \delta_0$.	N
Differential exponential	$u = \exp(\sum_{m \geq 1} B_{2m} D^{2m} / [2m(2m)!(1 - 4^{-m})]) \delta_0$, interpreted as in Appendix 9 .	N

12.2 The bounded and inverse Fabius functions

Viewpoint	Representation	Status
CDF	$F(x) = \mathbb{P}(X \leq 2x - 1)$.	K
Fold	$F(x) = u(x - 1)$ and $F(1 - x) = u(x)$.	K
Symmetry	$F(1 - x) = 1 - F(x)$.	K
Differential equation	$F'(x) = 2F(2x)$ on $0 \leq x \leq 1/2$.	K
Density Fourier integral	$F(x) = \int_{\mathbb{R}} \Phi(z) e^{2\pi i z(x-1)} dz$.	K

Viewpoint	Representation	Status
CDF sine integral	$F(x) = \frac{1}{2} + \pi^{-1} \int_0^\infty \Phi(z) \sin(2\pi z(2x - 1)) \, dz/z.$	K
Product integral	Replace Φ in either preceding integral by its sinc, cosine, Gamma, canonical, or polyphase product.	D
Laplace–Stieltjes	$\int_0^1 e^{sx} \, dF(x) = e^{s/2} M(s/2).$	K
Ordinary Laplace transform	$\int_0^1 e^{sx} F(x) \, dx = (e^s - e^{s/2} M(s/2))/s.$	K
Bromwich inversion	$F(x) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} e^{s(x-1/2)} M(s/2) \, ds/s.$	D
Fourier image on $[0, 1]$	$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = [e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}]/(2\pi i \xi).$	D
Cosine transform	Equation (807).	D
Sine transform	Equation (808).	D
Half-integer cosine series	$F(x) = \frac{1}{2} - \sum_{k \geq 0} \Phi(k + \frac{1}{2}) \cos((2k + 1)\pi x).$	K
Twisted Poisson family	Equation (801), valid for every $\alpha \notin \mathbb{Z}$.	N
Zero-slope sine series	$F(x) = x + \pi^{-1} \sum_{k \geq 1} \Phi'(k) \sin(2\pi k x).$	N
Half-integer slope series	$F(x) = x - \pi^{-1} \sum_{m \geq 0} \Phi(m + \frac{1}{2}) \sin(2\pi(2m + 1)x)/(2m + 1).$	N
Moment integral	$\int_0^1 x^p F(x) \, dx = (1 - \mathbb{E}Y^{p+1})/(p + 1), \Re p > -1.$	K
Polyphase volume	$F(x) = \int_{w_r \cdot y \leq 2x-1} \prod_a g_r(y_a) \, dy.$	N
Polyphase section	$F(x) = \ w_r\ ^{-1} \int_{w_r \cdot y = x-1} \prod_a g_r(y_a) \, d\sigma.$	N
Quantile level set	$G(y) = \int_0^1 \mathbf{1}_{\{F(x) \leq y\}} \, dx$ for $0 < y < 1$.	D
Quantile transport	$\int_0^1 h(G(y)) \, dy = \mathbb{E}h(Y).$	K
Quantile moments	$\int_0^1 G(y)^m \, dy = \mathbb{E}Y^m.$	K
Quantile MGF	$\int_0^1 e^{sG(y)} \, dy = e^{s/2} M(s/2).$	K
Inverse derivative	$G'(y) = 1/[2u(2G(y) - 1)].$	K
Radon quantile geometry	$\mathcal{C}_{w_r p_r}(2G(y) - 1) = \mathcal{R}_{w_r p_r}(G(y) - 1) = y.$	N
Area duality	$\int_0^1 F(x) \, dx + \int_0^1 G(y) \, dy = 1.$	K
Dyadic arithmetic	Finite Gaussian- q -binomial–Thue–Morse formulas for $F(m/2^n)$ are developed in [10, 11].	K
Endpoint Lambert- W	The lower-branch Lambert coordinate and its periodic all-orders expansion are established in the audited asymptotic reports.	K

12.3 Signed and unsigned Thue–Morse representations

Viewpoint	Representation	Status
Binary definition	$\varepsilon_n = (-1)^{s_2(n)}, \tau(n) = (1 - \varepsilon_n)/2.$	K
Recursion	$\varepsilon_{2n} = \varepsilon_n, \varepsilon_{2n+1} = -\varepsilon_n$; equivalently $\tau(2n) = \tau(n), \tau(2n+1) = 1 - \tau(n).$	K
Floor-sum parity	$\varepsilon_n = (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor}.$	K
Factorial valuation	$\varepsilon_n = (-1)^{n - \nu_2(n!)}.$	K
Finite coefficient product	$\varepsilon_n = [z^n] \prod_{j=0}^{N-1} (1 - z^{2^j})$ for $n < 2^N.$	K
Block Fourier product	$\sum_{n < 2^N} \varepsilon_n e^{2\pi i n \theta} = \prod_{j < N} (1 - e^{2\pi i 2^j \theta}).$	K
Prouhet moments	$\sum_{n < 2^N} \varepsilon_n n^d = 0$ for $d < N.$	K
Rvachev lobe sign	$\varepsilon_n = \operatorname{sgn} \Phi(n + \frac{1}{2}).$	K
Odd zero slope	$\varepsilon_n = -\operatorname{sgn} \Phi'(2n+1).$	N
Sparse lobe product	$\varepsilon_n = \prod_{a=0}^{r-1} \operatorname{sgn} \Psi_r((n+1/2)/2^a).$	N
Base-2 ^r digit factorization	$\varepsilon_n = \prod_{a=0}^{r-1} (-1)^{s_{2^r}(\lfloor n/2^a \rfloor)}.$	N
Unsigned base factorization	$\tau(n) \equiv \sum_{a < r} s_{2^r}(\lfloor n/2^a \rfloor) \pmod{2}.$	N
Spline jump	$\varepsilon_k = q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}.$	N
Unsigned spline jump	$\tau(k) = \frac{1}{2} (1 - q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}).$	N
Distributional comb	$\sum_{k < 2^N} \varepsilon_k \delta_{x_k} = q^{\binom{N}{2}} D^N g_{r,N}.$	N
Fourier deconvolution	Equation (746).	N
Cardinal integral	$\varepsilon_k = (-1)^N q^{\binom{N}{2}} \int g_{r,N}(x) \varphi_k^{(N)}(x) dx.$	N
Spatial scan	The normalized jumps of $D^{N-1} g_{r,N}$, ordered from left to right, are $\varepsilon_0, \dots, \varepsilon_{2^N-1}.$	N
Signed Laplace product	$\sum_{k < 2^N} \varepsilon_k e^{tx_k} = (-2)^N \prod_{j < N} \sinh(tq^j/2).$	N
Unsigned total product	$\sum_{k < 2^N} e^{tx_k} = 2^N \prod_{j < N} \cosh(tq^j/2).$	N
Evil/odious Fourier products	Equations (753) and (754).	N
All surviving powers	$\sum \varepsilon_k x_k^{N+m} = (-1)^N q^{\binom{N}{2}} (N+m)! \mu_m^{(r,N)} / m!.$	N
Bell–Bernoulli powers	Substitute (763) into the complete Bell polynomial in (765).	N
Moment limit	Equation (773); $r = 1$ gives all up-moments.	N
Pascal, valuation, and automatic atlas	Many additional binomial, Lucas–Kummer, Catalan, determinant, Mahler, Dirichlet, Walsh, and rarefied-sum formulas are catalogued in [11].	K

How to read the concordance

The atlas is deliberately redundant across mathematical languages. For example, the sinc product, random series, averaging operator, cumulant series, canonical determinant, and Radon section are not six unrelated objects; they are six realizations of one law. The value of the concordance is precisely that a calculation difficult in one realization may become elementary in another.

13 Rigidity and extraction of the polyphase factor

The sparse factor is not an arbitrary choice among many possible entire factors. Once all r phases are required to be dyadic dilates of one normalized entire function, the factor is forced.

Theorem 13.1 (Analytic rigidity of the common polyphase factor). *Let H be entire with $H(0) = 1$ and suppose*

$$\Phi(z) = \prod_{a=0}^{r-1} H(z/2^a) \quad (817)$$

for all $z \in \mathbb{C}$. Then

$$H(z) = \Psi_r(z). \quad (818)$$

In particular, Ψ_r is the unique normalized entire characteristic function whose first r dyadic dilates convolve to the up-law.

Proof. Near zero, choose the analytic logarithms with value zero. If $\log H(z) = \sum_{n \geq 1} h_n z^n$, then

$$\log \Phi(z) = \sum_{n \geq 1} h_n \left(\sum_{a=0}^{r-1} 2^{-an} \right) z^n.$$

The finite geometric multiplier never vanishes. Hence every h_n is uniquely determined by the Taylor series of $\log \Phi$. Odd coefficients vanish, and for $n = 2m$ one obtains

$$h_{2m} = -\frac{\zeta(2m)}{m(1 - 2^{-2rm})},$$

the coefficients of (681). Thus $H = \Psi_r$ near zero; the identity theorem extends equality to the plane. $\square$

There is also an explicit extraction product using only the full Fourier image. Since $\Phi(z)/\Phi(z/2) = \text{sinc}_\pi(z)$,

$$\Psi_r(z) = \prod_{m=0}^{\infty} \frac{\Phi(z/2^{rm})}{\Phi(z/2^{r(m+1)})}. \quad (819)$$

Apparent singularities in individual ratios at common zeros are removable; the ratio is globally the entire sinc factor. Formula (819) is a multiplicative scale decimator: it deletes all Fourier scales except those congruent to zero modulo r .

Conversely, the full product is reconstructed by the finite interpolation (667). Therefore

$$\Phi \xrightarrow{\text{scale decimation}} \Psi_r \xrightarrow{r \text{ dilates and multiplication}} \Phi \quad (820)$$

is an exact analysis–synthesis cycle, analogous to a polyphase filter bank but in the multiplicative algebra of entire characteristic functions.

Contribution IX: uniqueness

The modular-scale component is uniquely determined by the requirement that one entire factor and its first $r - 1$ dyadic dilates multiply to the Rvachev Fourier image. It can be extracted from the full image by an infinite product of adjacent-scale ratios and synthesized back by a finite product.

14 Numerical and exact verification

All identities in the main theorem chain are proved symbolically. The numerical work serves three narrower purposes: detecting normalization errors, checking complex-valued product regrouping at high precision, and making the jump-measure geometry visible. The file `numerical_experiments.py` is included in the archive and uses only Python, mpmath, SymPy, NumPy, and Matplotlib.

14.1 Verification protocol

The script performs four independent checks.

1. It evaluates $\Phi(z)$ and $\prod_{a=0}^{r-1} \Psi_r(z/2^a)$ at real and genuinely complex points for $r = 2, 3, 4, 5$, using 90 decimal digits.
2. It compares the direct finite atomic transform $\sum_k \varepsilon_k e^{-2\pi i z x_k}$ with $(2\pi i z)^N q^{\binom{N}{2}} \Psi_r(z) / \Psi_r(q^N z)$.
3. It constructs finite moments from the hyperbolic MGF in exact rational arithmetic and independently reconstructs them from the signed high power sum (759).
4. It verifies the cumulant recombination (691) symbolically for orders 2 through 12 and $r = 2, 3, 4, 5$.

Check family	Number of rows	Maximum reported residual
Polyphase product, 90-digit arithmetic	16	7.37×10^{-91}
Fourier deconvolution quotient	9	3.64×10^{-90}
Moment/power-sum reconstruction	18	exactly 0
Cumulant recombination	24	exactly 0

Table 19: Verification summary. The two numerical residuals are at the level of the selected mpmath precision; the other tables use exact SymPy rationals.

Representative exact rows are

(r, N, m)	$\mu_m^{(r,N)}$ from $M_{r,N}$	from signed powers	difference
(1, 6, 2)	455/4096	455/4096	0
(1, 6, 4)	2359721/83886080	same	0
(2, 5, 4)	3161984903/206158430208	same	0
(3, 4, 6)	60145469786596675/24211351596743786496	same	0

The full values are in `power_sum_moment_checks.csv`.

14.2 Visual factorization

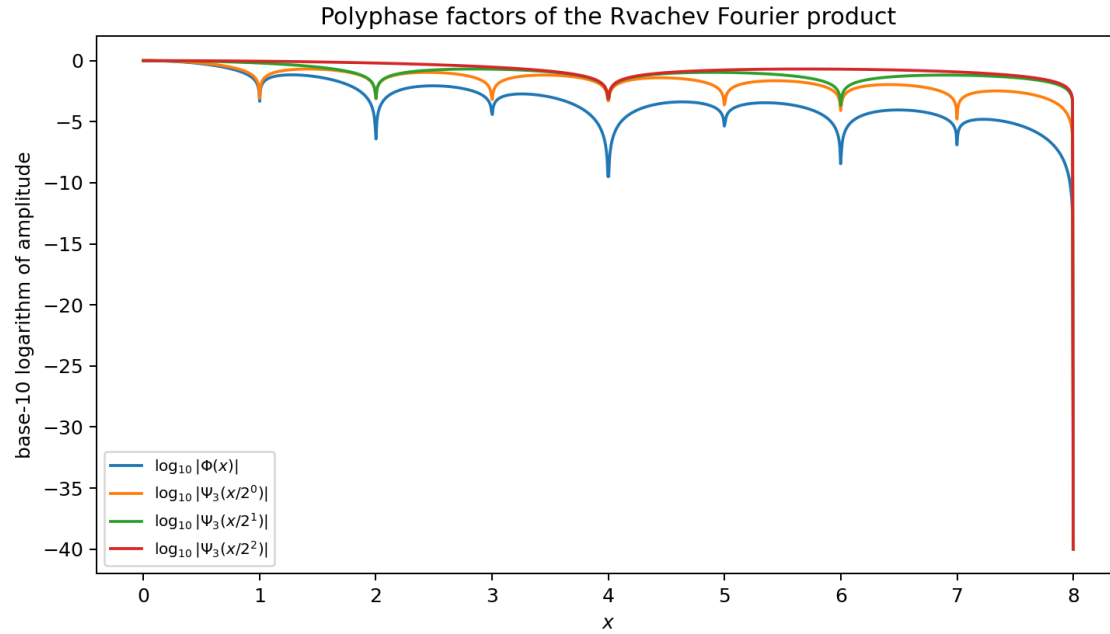


Figure 13: For $r = 3$, the logarithmic magnitude of the full Fourier image and its three sparse factors. At each integer zero the component multiplicities add to the multiplicity of Φ . The graph clips the logarithm near zeros; the product identity itself was checked at high precision away from and near the zero lattice.

The jump-comb visualization in Fig. 12 is produced from the exact subset sums, not by differentiating noisy sampled data. This matters: the highest classical derivative is piecewise constant and its jumps are mathematically exact Dirac masses in the next distributional derivative.

14.3 Reproduction

From the extracted archive directory, run

```
python numerical_experiments.py
```

The script writes

- `polyphase_product_checks.csv`;
- `fourier_deconvolution_checks.csv`;
- `power_sum_moment_checks.csv`;
- `polyphase_cumulant_checks.csv`;
- `operator_coefficients.csv`;
- the two PDF/PNG figures used in this report; and
- `verification_log.txt`.

No network access, random seed, or machine-dependent numerical integration is required.

Warning 14.1. A tiny residual is not used as a proof of an infinite-product identity. The proofs precede the experiments. The checks are valuable because the formulas contain several easily misplaced

factors: 2π , the centered support radius, $q^{\binom{N}{2}}$, the derivative sign $(-1)^N$, and the distinction between $\Psi_r(q^N z)$ and $\Psi_r(z/2^N)$.

15 Conjectures and frontier directions

The statements in this section are separated sharply from the proved results. Some are precise conjectures; others are research programs where fixing the correct normalization is itself part of the work. All concern structures created or clarified by the polyphase decomposition.

15.1 Uniform all-orders deconvolution

Define coefficients $a_{r,\ell}$ by the formal identity

$$\exp\left(-\sum_{m \geq 1} c_{r,m} t^m D^{2m}\right) = \sum_{\ell \geq 0} a_{r,\ell} t^\ell D^{2\ell}. \quad (821)$$

Thus

$$a_{r,0} = 1, \quad a_{r,1} = -c_{r,1}, \quad a_{r,2} = -c_{r,2} + \frac{c_{r,1}^2}{2}. \quad (822)$$

Conjecture 15.1 (Strong spline-tail expansion). For fixed $r, m, L \geq 0$,

$$\left\| D^m g_{r,N} - \sum_{\ell=0}^{L-1} a_{r,\ell} q^{2\ell N} D^{m+2\ell} g_r \right\|_\infty = \mathcal{O}_{r,m,L}(q^{2LN}) \quad (823)$$

for all sufficiently large N . In particular,

$$q^{-2N}(g_{r,N}^{(m)} - g_r^{(m)}) \longrightarrow -c_{r,1} g_r^{(m+2)} \quad (824)$$

uniformly.

The local Fourier expansion proves the coefficient prediction. A proof of the uniform remainder must use the exact quotient $\Psi_r(z)/\Psi_r(q^N z)$ to control the high-frequency region where the formal reciprocal has poles. Compact support and superalgebraic Fourier decay make the conjecture plausible, but the cancellation near the moving zero lattice should be handled explicitly rather than hidden in a Taylor argument.

15.2 Nondegeneracy of sparse Mellin harmonics

After subtracting the elementary local and mean terms, let P_r denote the periodic fluctuation in the regularized asymptotics of $\log |\Psi_r(t)|$ or $\log M_r(t)$. Normalize its period to r :

$$P_r(\theta) = \sum_{k \in \mathbb{Z}} c_{r,k} e^{2\pi i k \theta / r}. \quad (825)$$

Conjecture 15.2 (Mellin-harmonic nondegeneracy). Apart from coefficients forced to vanish by parity or the chosen regularization, $c_{r,k} \neq 0$ for every $k \neq 0$. Consequently the polyphase projection

$$P_1(\theta) = r \sum_{\ell \in \mathbb{Z}} c_{r,r\ell} e^{2\pi i \ell \theta} \quad (826)$$

loses harmonics only through the exact roots-of-unity cancellation of [Theorem 5.1](#), not through accidental zeros of Mellin residues.

A practical attack is to derive a rigorously convergent expression for the regularized Mellin transform $H(s)$ and certify $H(2\pi ik/(r \log 2)) \neq 0$ with interval arithmetic. The decomposition allows one to work at the longer period $r \log 2$, where individual modes may be easier to separate numerically.

15.3 Sparse endpoint asymptotics

The component support radius is $A_r = 1/[2(1 - 2^{-r})]$. General small-deviation heuristics for geometrically weighted sums predict

$$\log g_r(A_r - t) \sim -\frac{(\log(1/t))^2}{2r \log 2}, \quad t \downarrow 0. \quad (827)$$

The full Rvachev endpoint theory in the corpus contains much finer lower Lambert– W and periodic structure.

Conjecture 15.3 (Sparse Lambert– W transseries). For each $r \geq 1$, the endpoint $g_r(A_r - t)$ has an all-orders transseries in a lower-branch Lambert coordinate whose algebraic coefficients are accompanied by a nonconstant periodic function of logarithmic phase with fundamental period $r \log 2$. Under convolution of the r polyphase dilates, the periodic coefficients combine by the Mellin projector of [Corollary 5.2](#) to give the ordinary period-log 2 Rvachev expansion.

This conjecture is stronger than merely replacing $\log 2$ by $r \log 2$ in a leading term. The saddle location, the determinant factor, and the phase origin must all be recomputed for M_r . The reward is an “analysis then synthesis” derivation of the full endpoint fluctuation from sparser components.

15.4 Radon tomography and section algorithms

The equality

$$\mathcal{C}_{w_r p_r}(2x - 1) = \mathcal{R}_{w_r p_r}(x - 1) \quad (828)$$

invites several geometric questions.

Open problem 15.4 (Direct section computation). Develop a certified quadrature for F and G based on the section integral (712), compare its conditioning with direct Fourier inversion, and determine the optimal choice of r as a function of the target point and precision.

For $r = 2$, the integrand is a product of two copies of a broader, flatter component g_2 . Near the endpoints of F , a larger r may distribute the small-deviation event across a higher-dimensional section and improve numerical stability; in the interior, the dimensional cost may dominate. This tradeoff is concrete and testable.

Open problem 15.5 (Quantile tomography). Use

$$\mathcal{C}_{w_r p_r}(2G(y) - 1) = \mathcal{R}_{w_r p_r}(G(y) - 1) = y \quad (829)$$

to construct monotone Newton or safeguarded secant schemes in which both the residual and derivative are computed as sections of the same product density. Investigate whether the two-level geometry yields a posteriori error bounds stronger than generic inverse-function estimates.

15.5 Orthogonal polynomials and sparse spectral data

Let $(P_n^{(r)})$ be the monic orthogonal polynomials for $g_r(x) dx$. Symmetry gives

$$P_{n+1}^{(r)}(x) = xP_n^{(r)}(x) - \beta_n^{(r)}P_{n-1}^{(r)}(x). \quad (830)$$

The moments are explicit Bell–Bernoulli expressions, so every finite $\beta_n^{(r)}$ is an exact rational number.

Open problem 15.6 (Endpoint-flat Jacobi asymptotics). Determine the asymptotic expansion of $\beta_n^{(r)}$ beyond the universal Nevai limit $A_r^2/4$. In particular, test whether the infinitely flat, log-periodic endpoint behavior of g_r creates logarithmic or periodic corrections to the usual algebraic recurrence-coefficient asymptotics.

The polyphase convolution gives a second route: the Jacobi data of u should be constrained by the r component moment sequences and their scaled free cumulant-like convolution algebra. No simple formula is expected, but exact Hankel determinants at moderate order may reveal a stable pattern.

15.6 Twisted scale characters

The ordinary polyphase split uses the trivial character of the cyclic group $\mathbb{Z}/r\mathbb{Z}$. Near the origin, where logarithms are single-valued, introduce

$$L_{r,\ell}(z) = \sum_{a=0}^{r-1} \omega^{-a\ell} \log \Psi_r(z/2^a), \quad \omega = e^{2\pi i/r}. \quad (831)$$

Then $L_{r,0} = \log \Phi$, while the nontrivial $L_{r,\ell}$ isolate scale characters which cancel in the ordinary product.

Open problem 15.7 (Scale-character continuation). Find natural single-valued analytic or meromorphic objects whose local logarithms are $L_{r,\ell}$, describe their zero divisors, and determine whether they encode the discarded Mellin harmonics directly. A successful construction would diagonalize modular scale refinement in the same way that the finite Fourier transform diagonalizes cyclic translation.

Complex powers of factors cannot simply be written globally because zeros create monodromy. A divisor-valued or line-bundle formulation may be the right language.

15.7 General bases and residue subsets

For $b > 1$, the formal analogue

$$\Psi_{b,r}(z) = \prod_{m \geq 0} \text{sinc}_\pi(z/b^{rm}) \quad (832)$$

has a base- b polyphase decomposition. Much of the generic base- b theory is already present in the corpus, so the frontier is not the definition itself. The new questions are structural:

- classify residue subsets $A \subset \{0, \dots, r-1\}$ for which the partial product is a characteristic function with especially simple compact support or refinement equation;
- determine the lobe-sign automatic sequence when b is not even, where the parity simplification in (811) changes;
- replace binary signs by roots of unity and recover generalized Prouhet–Thue–Morse words from distributional derivatives of complex or vector-valued box splines;
- identify the exact Mellin character projector for arbitrary scale residue weights.

15.8 Arithmetic of sparse knot limits

All finite knot locations, cumulants, moments, and power sums are rational for $q = 2^{-r}$. The limiting density values are subtler.

Open problem 15.8 (Sparse-adic values). Determine the arithmetic nature of $g_r(x)$ and its derivatives at rational points whose denominators are powers of 2^r . Establish either a rationality theorem with denominator bounds, analogous to the dyadic Fabius theory, or a sharp obstruction showing why the self-CDF mechanism at $r = 1$ is exceptional.

The plateau present for $r \geq 2$ supplies many trivial rational values, but the transition regions carry the real arithmetic content. The nested factorization (723) may transfer information between different r .

15.9 Formalization roadmap

The new theorem chain is unusually well suited to formal verification because most proofs reduce to finite reindexing, characteristic functions, and polynomial distribution identities.

A practical Lean sequence is:

1. define Ψ_r as a locally uniformly convergent entire product and prove the partition $n = a + rm$;
2. construct Z_r from the existing weighted-uniform probability infrastructure and identify its characteristic function;
3. prove the finite and infinite convolution factorizations;
4. formalize zero multiplicities using ν_2 and Euclidean division;
5. define $g_{r,N}$ as a finite convolution, then prove the truncated-power formula and $D^N g_{r,N}$ as a finite signed measure;
6. prove the bit-reversal ordering and parity invariance;
7. derive the power-sum identity by distributional integration by parts, or first in the polynomial finite-difference algebra and then transfer;
8. connect finite cumulants to Bell polynomials and Bernoulli numbers;
9. formalize the base- 2^r lobe-sign theorem using the valuation-counting identity;
10. treat Radon/coarea statements last, since they require the heaviest measure-geometric library support.

The finite jump and power-sum layer can be formalized without first developing all of infinite-product complex analysis, making it a good initial target.

Highest-value next experiments

The most decisive near-term computations are: interval certification of the first sparse Mellin harmonics; high-order exact Jacobi coefficients for g_r ; uniform-error measurements for (823); and a direct comparison of Fourier versus Radon-section evaluation of F and G . Each experiment tests a precise structural prediction and can feed directly into a proof attempt.

16 Conclusion

The Fabius–Rvachev system is unusually representation-rich because a single object lies at the intersection of compactly supported smooth analysis, probability, automatic sequences, infinite products, spectral determinants, and digital asymptotics. The audited repository already develops most of the classical and several frontier realizations. The main purpose of this report has been to add a layer that cuts across those existing themes rather than repeat them.

The central operation is Euclidean division of the Fourier scale index. From that elementary reindexing follow:

1. a sparse characteristic function Ψ_r , its compact density g_r , and a unique polyphase convolution factorization of u ;
2. staircase cosine exponents, quotient-valued zero multiplicities, Gamma and canonical products, a sparse spectral zeta, exact determinant splitting, and cumulant recombination;
3. a roots-of-unity projector on Mellin poles and logarithmic-scale harmonics;
4. a product-density Radon representation in which a Fabius value is both a half-space volume and a parallel hyperplane section;
5. finite sparse splines whose highest derivative jumps, distributional derivative, Fourier quotient, and cardinal integrals all recover Thue–Morse signs;
6. an exact left-to-right reading of the ordinary Thue–Morse word from the spatially ordered jumps;
7. Bell–Bernoulli formulas for every surviving signed power sum and for the separate evil and odious power sums;
8. an infinite product of uniform averaging operators with an exact Bernoulli/zeta differential logarithm;
9. a twisted Poisson family and a zero-slope sine series for F ;
10. a factorization of binary Thue–Morse parity into overlapping base- 2^r digit-parity characters, together with recovery from odd zero slopes of Φ .

Several of these statements are short once the correct object is introduced. That is a strength rather than a weakness: the polyphase factor exposes a common mechanism behind formulas which otherwise look unrelated. The next stage should combine the sparse Mellin analysis, endpoint Lambert– W coordinates, and the exact finite jump calculus. Those three layers offer a credible route from computer-assisted experimentation to theorem-level control of periodic asymptotics and approximation errors.

A Corpus audit boundary

The source boundary is the repository commit

19ad531146cda7bc11a206827566db7f5753d5b3.

The branch advanced while the audit was in progress; pinning the hash prevents later reorganizations from silently changing the novelty comparison.

The recursive inventory covered current expositions, the living frontier synthesis, active thematic drafts, frozen archive lineages, and corrected paper sources. The principal exclusion documents were:

Corpus component	Main themes excluded from novelty claims
Primary exposition	Characterization, probability, Fourier product, dyadic values, derivatives, partition of unity, arithmetic.
Thue–Morse atlas	Signed/unsigned formulas, automatic and Mahler forms, binomial and valuation formulas, Prouhet identities, rarefied sums.
Finite-block frontier	Exact tails, confluent cancellation, finite products, extrapolation, spline approximants.
Representation frontier	Resolvents, logarithmic fixed points, Jacobi fractions, determinants, transform families.
Multiresolution report	Haar, Walsh, Faber–Schauder, beta and sinc mixtures, rational finite systems.
Inverse/sampling report	Quantile transport, inverse self-sampling, level-set and transform formulas.
Exponent/ q report	Newton exponent sequences, Gaussian q -binomial structure, Bell and Bernoulli coefficient algebra.
Spectral/arithmetic report	Zero jets, spectral zeta and determinants, half-integer spectra, arithmetic rays.
Integration reports	Integer/fractional transforms, repeated antiderivatives, monomial products, inverse integrals.
Fourier-decay campaign	Sharp decay scales, orbit dependence, log-periodic fluctuations, audit of competing asymptotics.

The label Ntherefore means “not located in this pinned TeX corpus after targeted full-tree searches.” It is not a claim that no equivalent formula has ever appeared in the worldwide literature. Targeted public-web searches for the exact combinations “polyphase Rvachev”, “scale-residue sinc factorization”, “Thue–Morse box-spline jumps”, and “Fabius hyperplane section” did not reveal a direct antecedent, but a formal priority claim would require a dedicated bibliographic study across several adjacent literatures.

B Archive contents and reproducibility

The ZIP archive accompanying this report contains:

- `fabius_rvachev_polyphase_representations.tex`, the source;
- `fabius_rvachev_polyphase_representations.pdf`, the rendered report;
- `numerical_experiments.py`, with detailed comments and no network dependency;
- five CSV verification tables;
- two figures in both PNG and PDF form;
- `verification_log.txt`;
- `README.md` with build commands; and
- `MANIFEST.sha256` for integrity checking.

A standard build is

```
python numerical_experiments.py
latexmk -pdf -interaction=nonstopmode \
    fabius_rvachev_polyphase_representations.tex
```

The report itself does not require rerunning the experiments if the included figures are retained.

C Selected identities in machine-friendly form

For use in symbolic systems, the main new identities can be summarized without prose as follows. Put $q = 2^{-r}$.

$$\Psi_r(z) = \prod_{m \geq 0} \operatorname{sinc}_\pi(q^m z), \quad (833)$$

$$\Phi(z) = \prod_{a=0}^{r-1} \Psi_r(2^{-a} z), \quad (834)$$

$$\log \Psi_r(z) = - \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^{2m})} z^{2m}, \quad (835)$$

$$\operatorname{ord}_{z=n} \Psi_r = 1 + \left\lfloor \frac{\nu_2(n)}{r} \right\rfloor, \quad (836)$$

$$\mathcal{Z}_r(s) = \frac{\zeta(s)}{1 - 2^{-rs}}, \quad (837)$$

$$g_{r,N}(x) = \frac{q^{-\binom{N}{2}}}{(N-1)!} \sum_{k < 2^N} \varepsilon_k (x - x_{r,N,k})_+^{N-1}, \quad (838)$$

$$D^N g_{r,N} = q^{-\binom{N}{2}} \sum_{k < 2^N} \varepsilon_k \delta_{x_{r,N,k}}, \quad (839)$$

$$\varepsilon_k = q^{\binom{N}{2}} [D^{N-1} g_{r,N}]_{x_{r,N,k}}, \quad (840)$$

$$\widehat{g_{r,N}}(z) = \Psi_r(z) / \Psi_r(q^N z), \quad (841)$$

$$\sum_{k < 2^N} \varepsilon_k x_{r,N,k}^{N+m} = (-1)^N q^{\binom{N}{2}} \frac{(N+m)!}{m!} \mu_m^{(r,N)}, \quad (842)$$

$$\varepsilon_n = \prod_{a=0}^{r-1} (-1)^{s_{2^r}(\lfloor n/2^a \rfloor)}, \quad (843)$$

$$\varepsilon_n = -\operatorname{sgn} \Phi'(2n+1). \quad (844)$$

These formulas use the Fourier convention (629); a computer algebra implementation using another convention must modify every 2π factor consistently.

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Part VI

Sampling, Padé, Mellin, Resolvent, and Product–Integral Representations

Abstract

This report gives a comprehensive, snapshot-relative audit of the recursive LaTeX corpus under `Analysis/FabiusFunction/docs`. The audit covers the 67-path pinned ledger at commit `32d6d36c51d803289e6d6a0dc0c37753766eba47`, the living primary and frontier manuscripts, the Thue–Morse formula atlases, the Fourier-decay campaign, the inverse-function and self-sampling drafts, the archive lineages, and the paper-source transcriptions. Existing results are organized into a representation atlas spanning probability laws, infinite convolutions, finite Thue–Morse splines, sinc and cosine products, Fourier inversion, gamma and zero products, moment/cumulant/Bell algebra, q-binomial formulas, exact dyadic arithmetic, Lambert– W end-point asymptotics, Cauchy transforms, one-sided transforms, Fredholm determinants, and Jacobi continued fractions. Repository results, immediate recombinations, and new corpus-relative results are explicitly distinguished.

The first-wave frontier constructions are retained: the Cauchy–Stieltjes transform of Rvachev’s up-function, its logarithmic dyadic fixed point, the functional-differential law

$$R'(z) = 2\{R(2z + 1) - R(2z - 1)\},$$

the all-orders Thue–Morse derivative tree, finite logarithmic resolvents, exact transforms of F , one-sided up, and F^{-1} , the cotangent/digamma/Hurwitz-zeta hierarchy for Φ'/Φ , a trace-class Fredholm determinant for Φ , and the positive Jacobi fraction of the up-law.

The second wave develops several different representation mechanisms. Compact support and the vanishing integer samples of Φ give the sparse half-lattice reconstruction

$$\Phi(z) = \operatorname{sinc}_\pi(2z) + \sum_{k \geq 0} \Phi(k + \tfrac{1}{2}) [\operatorname{sinc}_\pi(2z - 2k - 1) + \operatorname{sinc}_\pi(2z + 2k + 1)],$$

and hence a convergent Mittag–Leffler expansion whose residues recover both signed and unsigned Thue–Morse. Fourier duality gives odd-harmonic cosine series for up and F and the infinite weighted Prouhet identities

$$\sum_{k \geq 0} (-1)^{s_2(k)} |\Phi(k + \tfrac{1}{2})| (2k + 1)^{2r} = \begin{cases} \frac{1}{2}, & r = 0, \\ 0, & r \geq 1. \end{cases}$$

A Bose–Malmsten kernel converts the sinc product into an exponential integral and yields all logarithmic derivatives. Mellin transforms of up and F , the quantile moments of F^{-1} , and the generalized binomial series of the up-moments are unified by one entire function.

For the signed Thue–Morse generating function $E(z)$, a new formal orthogonality analysis proves

$$p_{2n}(x) = p_n(x^2), \quad a_n = (-1)^{n+1},$$

and gives an exact nonlinear dyadic recursion for the remaining Jacobi coefficients. At $N = 2^m$, $p_N(x) = x^N + 1$, producing the closed Padé approximant

$$E(z) = \frac{\prod_{j < m} (1 - z^{2^j})}{1 + z^N} + O(z^{2N}).$$

Its roots-of-minus-one partial fractions furnish an N -node complex quadrature that reproduces $2N$ consecutive Thue–Morse signs. Krawtchouk, contour, residue, integral, and antiperiodic Fourier representations of the signed and unsigned words are collected as well. A fully commented Python program, exact rational data, high-precision residuals, and four reproducible figures accompany the report. Every theorem labelled as new is proved here; conjectures are clearly separated from established statements.

1 Executive summary and novelty boundary

1.1 What was audited

The repository contains several kinds of LaTeX source that must not be conflated:

- 1. a living primary synthesis of the formalized and analytic theory;
- 2. a large consolidated non-formalized frontier volume;
- 3. current draft reports on arithmetic rays, half-integer spectra, self-sampling, inverse asymptotics, Newton/exponent-sequence hierarchies, Fourier decay, and related topics;
- 4. archived earlier lineages, retained because some contain historically distinctive formulas even when superseded;
- 5. seven paper-source transcriptions or corrected transcriptions.

The repository’s own pinned audit ledger records 67 LaTeX paths at the following commit:

32d6d36c51d803289e6d6a0dc0c37753766eba47.

The live tree inspected on 27 August 2026 contains additional current drafts, notably the self-sampling, inverse, and Newton/Rvachev reports. Appendix B records the pinned ledger and the live additions used in this study.

1.2 Non-duplication protocol

Every formula in the atlas is tagged conceptually as follows.

Tag	meaning
K	already explicit in the audited repository or classical source; included for orientation
D	an immediate transform or recombination of known identities; useful but not claimed as new
N	apparently absent from the audited corpus; proved or conjectured here

The word “new” throughout this report means *new relative to the audited repository*. No claim of global bibliographic priority is made without a separate literature search. In particular, the current frontier report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as an open direction. The present report constructs that fraction and develops its consequences.

1.3 Main new results

The most substantial additions are:

- 1. the Cauchy transform R and its moment, Laplace-product, Fourier–Laplace, boundary-value, and logarithmic fixed-point representations;
- 2. the exact resolvent equation and its all-orders Thue–Morse derivative tree;

3. a finite-level 2^N -term logarithmic formula converging to R , paired by boundary jumps with the finite Thue–Morse spline density;
4. the Jacobi J -fraction of R , the contracted Stieltjes S -fraction of the even moments, exact rational coefficients through order 40, Gaussian quadrature convergents, and the theorem $\beta_n \rightarrow 1/4$;
5. a Fredholm determinant realization of the Fourier image and a trace-resolvent interpretation of its logarithmic derivative;
6. cotangent, tangent, valuation-weighted partial-fraction, digamma, and Hurwitz-zeta representations of the logarithmic derivatives of the Fourier image;
7. exact one-sided Laplace and Fourier transforms of up , exact transforms of F on $[0, 1]$, and a master pushforward identity for F^{-1} .

2 Normalization and structural dictionary

2.1 The functions

Let $F : [0, 1] \rightarrow [0, 1]$ denote the bounded Fabius function [4, 2], extended by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$. On the unit interval it satisfies

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x), \quad (845)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (846)$$

The Rvachev function is written

$$\rho(x) := \text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (847)$$

Then ρ is even, strictly positive on $(-1, 1)$, normalized by $\int \rho = 1$, flat at ± 1 , and satisfies

$$\rho'(x) = 2(\rho(2x+1) - \rho(2x-1)). \quad (848)$$

2.2 Fourier convention and basic transforms

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i \xi x} d\xi. \quad (849)$$

Set

$$\Phi(\xi) := \widehat{\rho}(\xi), \quad \text{sinc}_{\pi}(z) := \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_{\pi}(0) = 1. \quad (850)$$

The classical products are

$$\Phi(z) = \prod_{n=0}^{\infty} \text{sinc}_{\pi}\left(\frac{z}{2^n}\right) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi z}{2^m}\right). \quad (851)$$

The scaling law is

$$\Phi(z) = \text{sinc}_{\pi}(z) \Phi(z/2). \quad (852)$$

2.3 The probability model

Let $U_1, U_2, \dots$ be independent random variables, uniform on $[-1, 1]$, and set

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (853)$$

Then X has density ρ . Equivalently,

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (854)$$

where $U \sim \text{Unif}[-1, 1]$ and X' is an independent copy of X . The moment generating function is

$$M(t) := \mathbb{E}e^{tX} = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} = \Phi\left(\frac{it}{2\pi}\right). \quad (855)$$

The even moments are

$$\mu_{2n} := \mathbb{E}X^{2n} = \int_{-1}^1 x^{2n} \rho(x) \, dx, \quad \mu_{2n+1} = 0. \quad (856)$$

Finally, if

$$Y = \frac{X + 1}{2}, \quad (857)$$

then F is the distribution function of Y , and

$$F'(x) = 2\rho(2x - 1), \quad 0 < x < 1. \quad (858)$$

3 Audited representation atlas

This section compresses the principal representation families already present in the repository and places the new formulas beside them. Proofs of the new entries begin in Section 4.

3.1 Real-space and probability representations of ρ

Infinite convolution (K). Since $2^{-k}U_k$ is uniform on $[-2^{-k}, 2^{-k}]$,

$$\rho = \bigstar_{k=1}^{\infty} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]} \right). \quad (859)$$

This is the real-space counterpart of the sinc product.

Infinite cube integral (D). For every bounded continuous test function g ,

$$\int_{-1}^1 g(x) \rho(x) \, dx = \int_{[-1, 1]^{\mathbb{N}}} g\left(\sum_{k \geq 1} 2^{-k} u_k\right) \prod_{k \geq 1} \frac{du_k}{2}. \quad (860)$$

Formally, this may be written as a Dirac-delta integral for $\rho(x)$.

Finite Thue–Morse splines (K). Put

$$X_N = \sum_{k=1}^N 2^{-k} U_k, \quad a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \quad \varepsilon_j = (-1)^{\text{wt}_2(j)}. \quad (861)$$

The density ρ_N of X_N is

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (x + a_{N,j})_+^{N-1}. \quad (862)$$

For each fixed derivative order, $\rho_N \rightarrow \rho$ uniformly after N is sufficiently large relative to that order. The repository develops several centered and one-sided variants with explicit error control.

Fourier inversion and cosine form (K). Because Φ is real and even,

$$\rho(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (863)$$

$$\rho(x) = 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (864)$$

The audited corpus also contains Poisson-summation formulas, Legendre expansions, periodized cosine series, dyadic spline limits, and endpoint transseries. They are not rederived here.

3.2 Product, zero, gamma, and exponential representations of Φ

Canonical zero product (K). Let $\nu_2(n)$ be the exponent of 2 in n . The nonzero integer n is a zero of multiplicity $1 + \nu_2(|n|)$, and

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}. \quad (865)$$

Gamma product (D). Euler’s reflection identity gives

$$\Phi(z) = \prod_{k=0}^{\infty} \frac{1}{\Gamma(1 + z/2^k) \Gamma(1 - z/2^k)}. \quad (866)$$

This is the natural entry point for the Barnes and multiple-gamma compressions studied in current repository drafts.

Spectral-zeta exponential (K). For $|z| < 1$,

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 2^{-2m})} z^{2m}. \quad (867)$$

The coefficient sequence is the spectral zeta function

$$Z_{\Phi}(s) = \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (868)$$

Fredholm determinant (N). Section 9 upgrades (865) to

$$\Phi(z) = \det(I - z^2 \mathcal{K}), \quad (869)$$

for an explicit positive trace-class diagonal operator $\mathcal{K}$. This makes (868) literally the trace zeta function of $\mathcal{K}$.

3.3 Moment, cumulant, and Bell-polynomial representations

The moment generating function has the convergent Taylor series

$$M(t) = \sum_{n=0}^{\infty} \mu_{2n} \frac{t^{2n}}{(2n)!}. \quad (870)$$

Its logarithm is

$$\log M(t) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(4^m - 1)} t^{2m}, \quad (871)$$

so all moments are complete Bell polynomials in explicit Bernoulli/Mersenne cumulants. These formulas, together with reciprocal Bell expansions for $1/\Phi$, are developed extensively in the repository.

3.4 Transform representations of F and F^{-1}

The CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(\xi) \frac{\sin(2\pi(2x-1)\xi)}{\xi} d\xi, \quad 0 < x < 1. \quad (872)$$

Section 7 adds closed Laplace and Fourier transforms of F as a function, as well as transforms of its derivative measure and its inverse.

3.5 A compact atlas table

Table 20: Representation atlas. Tags refer to the audited-corpus novelty boundary.

Object	Representation family	Tag	Location
ρ	fold of F , compact-support DDE, infinite convolution, random dyadic sum	K	§2–3
ρ	finite Thue–Morse truncated-power splines	K	(862)
ρ	Fourier inversion, cosine integral, Poisson and Legendre series	K	(863)
ρ	boundary jump of a finite logarithmic resolvent sum	N	Thm. 6.1
Φ	sinc product and weighted cosine product	K	(851)
Φ	canonical integer-zero product	K	(865)
Φ	gamma/Barnes products	K/D	(866)
Φ	zeta-exponential series	K	(867)
Φ	trace-class Fredholm determinant	N	Thm. 9.2
Φ'/Φ	cotangent dyadic sum	N	(904)
Φ'/Φ	weighted tangent dyadic sum	N	(905)
Φ'/Φ	valuation-weighted partial fractions	N	(906)

Object	Representation family	Tag	Location
$(\log \Phi)^{(r)}$	digamma and Hurwitz-zeta hierarchy	N	Thm. 8.2
M	hyperbolic-sinc product, moment series, Bell cumulants	K	(855)–(871)
M	positive Fredholm determinant	N	(915)
R	moment Laurent series	N	(875)
R	Laplace transform of the hyperbolic product	N	(876)
R	one-sided Fourier–Laplace integral	N	(877)
R	logarithmic fixed-point expectation	N	(880)
R	dyadic functional-differential equation	N	(881)
$R^{(n)}$	finite Thue–Morse affine orbit of R	N	(882)
R	finite 2^N -logarithm limit	N	(885)
R	Jacobi continued fraction	N	(926)
R	Gaussian-quadrature partial fractions	N	(935)
$\sum \mu_{2n} t^n$	Stieltjes continued fraction and product integral	N	(928)
F	CDF sine-product integral	D	(872)
F	exact Laplace transform on $[0, 1]$	N	(889)
F	exact Fourier transform on $[0, 1]$	N	(890)
$\rho _{[0,1]}$	exact one-sided Laplace/Fourier/sine transforms	N	Thm. 7.2
F^{-1}	pushforward expectation, Laplace, Fourier, and Cauchy transforms	N	Thm. 7.3

4 The Cauchy–Stieltjes transform

Definition 4.1. For $z \in \mathbb{C} \setminus [-1, 1]$, define

$$R(z) = \int_{-1}^1 \frac{\rho(x)}{z - x} dx = \mathbb{E} \frac{1}{z - X}. \quad (873)$$

The transform R is odd because ρ is even. It is an anti-Herglotz function: when $\operatorname{Im} z > 0$,

$$\operatorname{Im} R(z) = -\operatorname{Im} z \int_{-1}^1 \frac{\rho(x)}{|z - x|^2} dx < 0. \quad (874)$$

In particular, R has no zeros off the cut. For real $z > 1$, $R(z) > 0$; for $z < -1$, $R(z) < 0$.

Theorem 4.2 (Four complementary representations of R). *The following formulas hold.*

(i) For $|z| > 1$,

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_{2n}}{z^{2n+1}}. \quad (875)$$

(ii) For $\operatorname{Re} z > 1$,

$$R(z) = \int_0^{\infty} e^{-zt} \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} dt. \quad (876)$$

(iii) For $\operatorname{Im} z > 0$,

$$R(z) = -2\pi i \int_0^{\infty} e^{2\pi i z \xi} \Phi(\xi) d\xi. \quad (877)$$

For $\operatorname{Im} z < 0$, take the conjugate counterpart.

(iv) For $-1 < x < 1$, the non-tangential boundary values satisfy

$$\rho(x) = -\frac{1}{\pi} \lim_{y \downarrow 0} \operatorname{Im} R(x + iy), \quad (878)$$

$$\operatorname{PV} \int_{-1}^1 \frac{\rho(t)}{x-t} dt = \lim_{y \downarrow 0} \operatorname{Re} R(x + iy) = 2\pi \int_0^\infty \Phi(\xi) \sin(2\pi x \xi) d\xi. \quad (879)$$

Proof. For $|z| > 1$, expand $(z - X)^{-1} = z^{-1} \sum_{m \geq 0} (X/z)^m$, use $|X| \leq 1$, and then symmetry. This gives (875).

For $\operatorname{Re} z > 1$,

$$\frac{1}{z - X} = \int_0^\infty e^{-(z-X)t} dt.$$

Fubini's theorem is justified because $X \leq 1$. Taking expectation and using (855) yields (876).

For $\operatorname{Im} z > 0$,

$$\frac{1}{z - x} = -2\pi i \int_0^\infty e^{2\pi i(z-x)\xi} d\xi.$$

The exponential damping from $\operatorname{Im} z > 0$, together with the rapid decay of Φ , permits Fubini and gives (877). Finally, Plemelj's boundary formula gives (878); taking real parts of (877) gives (879). $\square$

Remark 4.3. The four formulas emphasize different structures. The Laurent series exposes exact rational moments; the Laplace integral exposes the hyperbolic product; the Fourier–Laplace integral exposes spectral decay; and the boundary values recover both ρ and its Hilbert transform. Thus R is a single analytic object joining the moment, product, and inversion theories.

5 A dyadic functional equation for the resolvent

5.1 Logarithmic fixed-point identity

Theorem 5.1 (Logarithmic fixed point). *For real $z > 1$,*

$$R(z) = \int_{-1}^1 \rho(y) \log \left(\frac{2z + 1 - y}{2z - 1 - y} \right) dy. \quad (880)$$

The identity extends by analytic continuation to any simply connected subdomain of $\mathbb{C} \setminus [-1, 1]$, with the logarithm chosen by continuation from large positive z .

Proof. Use the fixed point (854) and condition on X' :

$$\begin{aligned} R(z) &= \mathbb{E} \left[\int_{-1/2}^{1/2} \frac{du}{z - u - X'/2} \right] \\ &= \mathbb{E} \log \left(\frac{z + 1/2 - X'/2}{z - 1/2 - X'/2} \right), \end{aligned}$$

which is (880) after multiplying numerator and denominator by 2. $\square$

5.2 Functional-differential equation

Theorem 5.2 (Resolvent equation). *For every $z \in \mathbb{C} \setminus [-1, 1]$,*

$$R'(z) = 2(R(2z + 1) - R(2z - 1)). \quad (881)$$

Proof. Differentiate (880) first for $z > 1$:

$$\begin{aligned} R'(z) &= \mathbb{E} \left(\frac{2}{2z+1-X'} - \frac{2}{2z-1-X'} \right) \\ &= 2(R(2z+1) - R(2z-1)). \end{aligned}$$

Both sides are holomorphic on $\mathbb{C} \setminus [-1, 1]$, so the identity theorem gives the stated continuation. $\square$

This is the transform-side analogue of the Rvachev equation (848). It is not merely a Fourier rephrasing: it acts on the spectral parameter of a Cauchy transform and therefore interacts directly with moments, Padé approximation, and orthogonal polynomials.

5.3 All-order Thue–Morse derivative tree

Theorem 5.3 (Thue–Morse derivative orbit). *For every integer $n \geq 0$,*

$$R^{(n)}(z) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (882)$$

Consequently,

$$\int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx = \frac{(-1)^n 2^{n(n+1)/2}}{n!} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (883)$$

Proof. The case $n = 0$ is trivial and $n = 1$ is (881). Assume the formula at order n . Differentiating contributes a factor 2^n , and then applying (881) contributes another factor 2. The exponent therefore changes by $n + 1$, from $n(n + 1)/2$ to $(n + 1)(n + 2)/2$.

For the shifts, write $a_{n,j} = 2^n - 1 - 2j$. The two children are

$$2a_{n,j} + 1 = 2^{n+1} - 1 - 2(2j), \quad 2a_{n,j} - 1 = 2^{n+1} - 1 - 2(2j + 1).$$

The signs obey $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$, exactly matching the subtraction in (881). This proves (882). Formula (883) follows from

$$R^{(n)}(z) = (-1)^n n! \int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx.$$

$\square$

Taking boundary imaginary parts in (882) recovers the familiar iterated derivative formula for ρ . The new point is that the same finite Thue–Morse orbit controls every higher Cauchy kernel, not only the density derivatives.

5.4 Uniqueness in the Stieltjes class

Proposition 5.4. *Among Cauchy transforms of probability measures supported on $[-1, 1]$, R is the unique solution of (881) satisfying*

$$R(z) = \frac{1}{z} + \mathcal{O}(z^{-3}) \quad (z \rightarrow \infty). \quad (884)$$

Proof. Let $G(z) = \sum_{r \geq 0} m_r z^{-r-1}$ be the Laurent expansion at infinity of any such Cauchy transform. The normalization gives $m_0 = 1$ and $m_1 = 0$. Substitute this expansion into (881). In the coefficient of z^{-r-2} , the contribution containing m_r is

$$-(r+1)m_r + \frac{r+1}{2^r}m_r = -(r+1)(1-2^{-r})m_r,$$

while every other contribution contains only $m_0, \dots, m_{r-2}$. Since the displayed factor is nonzero for $r \geq 1$, the differential equation determines all moments recursively. The transform R is one solution, so G and R have identical moments. Probability measures on the compact interval $[-1, 1]$ are determinate by their moments (equivalently, polynomials are dense in $C[-1, 1]$); hence the two measures and their Cauchy transforms coincide. $\square$

6 Finite logarithmic resolvents and the spline boundary jump

The finite random sum X_N has two exact representations: its density is a Thue–Morse spline, while its Cauchy transform is a Thue–Morse sum of logarithms. The second formula appears to be absent from the audited corpus.

Theorem 6.1 (Finite logarithmic formula). *Let*

$$R_N(z) = \mathbb{E} \frac{1}{z - X_N}, \quad X_N = \sum_{k=1}^N 2^{-k} U_k.$$

For $z \notin [-1 + 2^{-N}, 1 - 2^{-N}]$, with logarithms continued consistently from $z = +\infty$,

$$R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j(z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}). \quad (885)$$

Moreover, $R_N \rightarrow R$ locally uniformly on $\mathbb{C} \setminus [-1, 1]$, so

$$R(z) = \lim_{N \rightarrow \infty} \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j(z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}). \quad (886)$$

Finally, the upper boundary jump of (885) is exactly the spline (862):

$$\rho_N(x) = -\frac{1}{\pi} \text{Im } R_N(x + i0). \quad (887)$$

Proof. For $a > 0$, convolution with the uniform density on $[-a, a]$ sends a Cauchy transform $G'(z)$ to

$$\frac{1}{2a} \int_{-a}^a G'(z - u) \, du = \frac{G(z + a) - G(z - a)}{2a}. \quad (888)$$

Start from the point mass at zero, whose Cauchy transform is $1/z$. An N -fold antiderivative is

$$H_N(z) = \frac{z^{N-1} \text{Log } z}{(N-1)!}; \quad H_N^{(N)}(z) = \frac{1}{z}.$$

Applying (888) successively with $a_k = 2^{-k}$ gives

$$R_N(z) = \frac{1}{\prod_{k=1}^N 2a_k} \sum_{\sigma \in \{\pm 1\}^N} \left(\prod_{k=1}^N \sigma_k \right) H_N \left(z + \sum_{k=1}^N \sigma_k a_k \right).$$

Since

$$\frac{1}{\prod_{k=1}^N 2a_k} = 2^{N(N-1)/2},$$

and binary subsets identify the shifts and signs with $a_{N,j}$ and ε_j , this is (885).

The residual tail $X - X_N$ has support in $[-2^{-N}, 2^{-N}]$, so $X_N \rightarrow X$ almost surely and uniformly in support. The kernels $(z - x)^{-1}$ are uniformly continuous on compact subsets away from $[-1, 1]$; hence $R_N \rightarrow R$ locally uniformly.

For the boundary value, $\text{Im Log}(x + a + i0) = \pi$ when $x + a < 0$ and zero when $x + a > 0$. The alternating finite difference annihilates every polynomial of degree less than N , so the sum of the full powers $(x + a_{N,j})^{N-1}$ is zero. Replacing the negative-part sum by minus the positive-part sum yields exactly (862). This proves (887). $\square$

Remark 6.2 (Two complementary approximation geometries). The finite-convolution approximant R_N is a sum of logarithms and preserves the dyadic construction. The Jacobi approximants in Section 10 are rational functions and preserve moments. Both converge locally uniformly off the support, but their poles, branch cuts, and error mechanisms are completely different. This duality suggests hybrid algorithms: use finite convolution to resolve endpoint geometry and a short Jacobi fraction to compress the remaining moment error.

7 Exact transforms of F , one-sided up, and F^{-1}

7.1 The Laplace and Fourier transforms of F

Theorem 7.1 (Transform of the Fabius function). *For complex s , with the value at $s = 0$ understood by continuity,*

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} M(s/2) - e^{-s}}{s}. \quad (889)$$

Equivalently, for real or complex ξ ,

$$\int_0^1 F(x) e^{-2\pi i \xi x} \, dx = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (890)$$

At $\xi = 0$, the value is $1/2$.

Proof. The distributional Laplace transform of Y is

$$\int_0^1 e^{-sx} \, dF(x) = \mathbb{E} e^{-sY} = e^{-s/2} M(s/2). \quad (891)$$

Integration by parts gives

$$\int_0^1 e^{-sx} \, dF(x) = e^{-s} + s \int_0^1 e^{-sx} F(x) \, dx,$$

which is (889). Substitute $s = 2\pi i \xi$ and use $M(\pi i \xi) = \Phi(\xi/2)$. $\square$

Separating real and imaginary parts gives the explicit sine and cosine transforms

$$\int_0^1 F(x) \cos(2\pi \xi x) \, dx = \frac{\sin(2\pi \xi) - \Phi(\xi/2) \sin(\pi \xi)}{2\pi \xi}, \quad (892)$$

$$\int_0^1 F(x) \sin(2\pi \xi x) \, dx = \frac{\Phi(\xi/2) \cos(\pi \xi) - \cos(2\pi \xi)}{2\pi \xi}. \quad (893)$$

7.2 One-sided transforms of the up-function

On $[0, 1]$, $\rho(x) = F(1 - x) = 1 - F(x)$. Therefore the previous theorem immediately yields the following closed forms.

Theorem 7.2 (One-sided transform package). *For complex s , with the removable value at zero,*

$$\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} M(s/2)}{s}. \quad (894)$$

Consequently,

$$\int_0^1 e^{-2\pi i \xi x} \rho(x) \, dx = \frac{1 - e^{-\pi i \xi} \Phi(\xi/2)}{2\pi i \xi}. \quad (895)$$

The half-line cosine and sine transforms are

$$\int_0^1 \rho(x) \cos(2\pi \xi x) \, dx = \frac{1}{2} \Phi(\xi), \quad (896)$$

$$\int_0^1 \rho(x) \sin(2\pi \xi x) \, dx = \frac{1 - \Phi(\xi/2) \cos(\pi \xi)}{2\pi \xi}. \quad (897)$$

The sum of (894) at s and $-s$ recovers $M(s)$ precisely because

$$M(s) = \frac{2 \sinh(s/2)}{s} M(s/2), \quad (898)$$

which is the hyperbolic continuation of (852).

7.3 The inverse Fabius function as a transport map

Let $I = F^{-1}$ on $[0, 1]$. Since I pushes Lebesgue measure forward to the law of Y , every integral of a composition with I reduces to the up-law.

Theorem 7.3 (Inverse pushforward master identity). *For every Borel function g for which one side is integrable,*

$$\int_0^1 g(I(u)) \, du = \int_0^1 g(x) \, dF(x) = \int_{-1}^1 g\left(\frac{x+1}{2}\right) \rho(x) \, dx. \quad (899)$$

In particular,

$$\int_0^1 e^{-sI(u)} \, du = e^{-s/2} M(s/2), \quad (900)$$

$$\int_0^1 e^{-2\pi i \xi I(u)} \, du = e^{-\pi i \xi} \Phi(\xi/2), \quad (901)$$

$$\int_0^1 \frac{du}{z - I(u)} = 2R(2z - 1), \quad z \in \mathbb{C} \setminus [0, 1], \quad (902)$$

$$\int_0^1 I(u)^n \, du = 2^{-n} \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mu_{2r}. \quad (903)$$

Proof. If U is uniform on $[0, 1]$, then $I(U)$ has CDF F , hence the same law as Y . This proves (899). The remaining formulas follow by taking $g(x) = e^{-sx}$, $e^{-2\pi i \xi x}$, $(z - x)^{-1}$, and x^n , respectively, and then using (857) and symmetry of X . $\square$

Remark 7.4. Formula (902) transports the entire Jacobi-fraction theory of R to the inverse Fabius function. Thus a function whose pointwise inverse asymptotics involve Lambert W nevertheless has a globally simple Cauchy transform when sampled with uniform measure in its range variable.

8 Logarithmic derivatives of the Fourier image

8.1 Three meromorphic expansions

Theorem 8.1 (Cotangent, tangent, and zero-set formulas). *Away from the integer zero set of Φ ,*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} \left[\frac{\pi}{2^k} \cot\left(\frac{\pi z}{2^k}\right) - \frac{1}{z} \right], \quad (904)$$

$$= -\pi \sum_{m=1}^{\infty} \frac{m}{2^m} \tan\left(\frac{\pi z}{2^m}\right), \quad (905)$$

$$= -2z \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^2 - z^2}. \quad (906)$$

All three series converge locally normally on compact subsets avoiding the poles.

Proof. Differentiate the logarithm of each of the three products (851) and (865). For large k , the bracket in (904) and the term in (905) are $\mathcal{O}(4^{-k})$ uniformly on compact sets. The terms in (906) are bounded by a constant times $(1 + \nu_2(n))/n^2$, which is summable by (868) at $s = 2$. $\square$

The three forms have different numerical regimes. The tangent sum is usually the fastest away from its poles; the cotangent sum tracks the original sinc factors; and the zero-set sum exposes multiplicities and supports rigorous rational tail bounds.

8.2 Digamma and Hurwitz-zeta hierarchy

The valuation weight has the divisor decomposition

$$1 + \nu_2(n) = \sum_{k=0}^{\infty} \mathbf{1}_{2^k | n}. \quad (907)$$

This turns the zero sum into a dyadic superposition of classical special functions.

Theorem 8.2 (Digamma and Hurwitz-zeta representations). *Let $\psi = \Gamma'/\Gamma$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} 2^{-k} \left[\psi\left(1 - \frac{z}{2^k}\right) - \psi\left(1 + \frac{z}{2^k}\right) \right]. \quad (908)$$

For every integer $r \geq 2$,

$$\frac{d^r}{dz^r} \log \Phi(z) = (r-1)! \sum_{k=0}^{\infty} 2^{-kr} \left[-\zeta\left(r, 1 - \frac{z}{2^k}\right) + (-1)^{r-1} \zeta\left(r, 1 + \frac{z}{2^k}\right) \right]. \quad (909)$$

Proof. From (865), for $r \geq 1$, paired symmetrically when $r = 1$,

$$(\log \Phi)^{(r)}(z) = (r-1)! \sum_{n \geq 1} (1 + \nu_2(n)) \left[-\frac{1}{(n-z)^r} + \frac{(-1)^{r-1}}{(n+z)^r} \right]. \quad (910)$$

Insert (907), write $n = 2^k m$, and sum over $m \geq 1$. For $r \geq 2$, the inner sums are Hurwitz zeta functions. For $r = 1$, the convergent paired sum is the standard digamma difference. $\square$

Corollary 8.3 (Local jet data at nonzero zeros). *Let $N \neq 0$ be an integer and $d_N = 1 + \nu_2(|N|)$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \frac{d_N}{z - N} + H_N(z), \quad (911)$$

where H_N is holomorphic near N , and every derivative of H_N can be written as an absolutely convergent valuation-weighted Hurwitz-zeta sum. Thus the complete local Taylor jet of $\Phi(z)/(z - N)^{d_N}$ is algorithmically explicit.

9 A Fredholm determinant for the Fourier product

The canonical product can be realized as an ordinary Fredholm determinant, not merely as a formal spectral product.

Definition 9.1. Let

$$\mathcal{I} = \{(n, r) : n \geq 1, 1 \leq r \leq 1 + \nu_2(n)\}, \quad (912)$$

and define the positive diagonal operator $\mathcal{K}$ on $\ell^2(\mathcal{I})$ by

$$\mathcal{K}e_{n,r} = \frac{1}{n^2}e_{n,r}. \quad (913)$$

Theorem 9.2 (Fredholm determinant representation). *The operator $\mathcal{K}$ is trace class and*

$$\boxed{\Phi(z) = \det(I - z^2\mathcal{K})}, \quad (914)$$

$$\boxed{M(t) = \det\left(I + \frac{t^2}{4\pi^2}\mathcal{K}\right)}. \quad (915)$$

Furthermore,

$$\mathrm{tr}(\mathcal{K}^m) = \frac{\zeta(2m)}{1 - 2^{-2m}}, \quad (916)$$

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \, \mathrm{tr}(\mathcal{K}(I - z^2\mathcal{K})^{-1}). \quad (917)$$

Proof. By (868),

$$\mathrm{tr} \mathcal{K} = \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^2} = \frac{\zeta(2)}{1 - 2^{-2}} < \infty,$$

so $\mathcal{K}$ is trace class. Its Fredholm determinant is the product over its eigenvalues, with multiplicity:

$$\det(I - z^2\mathcal{K}) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1 + \nu_2(n)} = \Phi(z).$$

Substituting $z = it/(2\pi)$ gives (915). Formula (916) is again (868), and differentiating the Fredholm logarithm gives (917). $\square$

Remark 9.3. This representation converts several scalar identities into operator identities. For example, the zeta expansion (867) is the standard trace expansion

$$\log \det(I - z^2\mathcal{K}) = - \sum_{m \geq 1} \frac{z^{2m}}{m} \mathrm{tr}(\mathcal{K}^m).$$

It also suggests perturbation theory for generalized exponent sequences: changing the multiplicity sequence changes only the diagonal spectral multiplicities.

10 Jacobi continued fractions and exact rational coefficients

10.1 Exact moment recurrence

Proposition 10.1 (Moment recurrence). *The even moments satisfy $\mu_0 = 1$ and, for $n \geq 1$,*

$$(4^n - 1)\mu_{2n} = \sum_{j=1}^n \binom{2n}{2j} \frac{\mu_{2n-2j}}{2j+1}. \quad (918)$$

In particular, every μ_{2n} is rational.

Proof. Raise (854) to the $2n$ -th power and take expectations:

$$4^n \mu_{2n} = \sum_{j=0}^n \binom{2n}{2j} \mathbb{E} U^{2j} \mu_{2n-2j}.$$

Since $\mathbb{E} U^{2j} = 1/(2j+1)$, the $j=0$ term is μ_{2n} . Move it to the left. $\square$

10.2 Monic orthogonal polynomials

Let P_n be the monic orthogonal polynomials for the probability measure $\rho(x) dx$, and let

$$h_n = \int_{-1}^1 P_n(x)^2 \rho(x) dx. \quad (919)$$

Evenness forces the diagonal Jacobi parameters to vanish, so

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \quad \beta_n = \frac{h_n}{h_{n-1}} > 0. \quad (920)$$

The first polynomials are

$$P_0(x) = 1, \quad (921)$$

$$P_1(x) = x, \quad (922)$$

$$P_2(x) = x^2 - \frac{1}{9}, \quad (923)$$

$$P_3(x) = x^3 - \frac{19}{75}x, \quad (924)$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \quad (925)$$

10.3 The Jacobi and Stieltjes fractions

Theorem 10.2 (Jacobi continued fraction). *The Cauchy transform has the locally uniform continued fraction*

$$R(z) = \frac{1}{z - \frac{\beta_1}{z - \frac{\beta_2}{z - \frac{\beta_3}{z - \ddots}}}}. \quad (926)$$

The coefficients are positive rational numbers.

Proof. Positivity follows from $\beta_n = h_n/h_{n-1}$, and rationality follows because Gram–Schmidt applied to the rational moment functional (918) uses only rational operations. Markov’s theorem for compactly supported positive measures identifies the continued fraction generated by (920) with the Cauchy transform and gives local uniform convergence off the support [7, 6]. $\square$

Define the ordinary even-moment generating function

$$\mathcal{O}(t) = \sum_{n=0}^{\infty} \mu_{2n} t^n = \int_{-1}^1 \frac{\rho(x)}{1 - tx^2} dx, \quad |t| < 1. \quad (927)$$

Substituting $t = z^{-2}$ into (926) gives the contracted fraction.

Corollary 10.3 (Stieltjes fraction and product integral). *For $|t| < 1$,*

$$\mathcal{O}(t) = \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \frac{\beta_3 t}{1 - \ddots}}}} \quad (928)$$

$$\mathcal{O}(t) = \int_0^{\infty} e^{-u} \prod_{k=1}^{\infty} \frac{\sinh(\sqrt{t} u/2^k)}{\sqrt{t} u/2^k} du. \quad (929)$$

Proof. The continued fraction follows from $\mathcal{O}(z^{-2}) = zR(z)$. For the integral, use

$$\frac{1}{1 - tx^2} = \int_0^{\infty} e^{-u} \cosh(\sqrt{t} xu) du$$

and then average over X , obtaining $M(\sqrt{t} u)$. $\square$

For $s \geq 0$, $\mathcal{O}(-s)$ is completely monotone:

$$(-1)^m \frac{d^m}{ds^m} \mathcal{O}(-s) = m! \int_{-1}^1 \frac{x^{2m} \rho(x)}{(1 + sx^2)^{m+1}} dx > 0. \quad (930)$$

Thus the even moments form a Stieltjes moment sequence, and the fraction (928) belongs to the classical positive Stieltjes class.

10.4 Hankel determinants

Let

$$\Delta_n = \det[\mu_{i+j}]_{i,j=0}^{n-1}, \quad \Delta_0 = 1. \quad (931)$$

Then

$$h_n = \frac{\Delta_{n+1}}{\Delta_n}, \quad \boxed{\beta_n = \frac{\Delta_{n+1} \Delta_{n-1}}{\Delta_n^2}}. \quad (932)$$

Because odd moments vanish, each Δ_n factors after a parity permutation into an even-even and odd-odd moment determinant. This parity factorization is useful for exact arithmetic and should be exploited in future Lean formalization.

10.5 Exact coefficients and numerical evidence

The accompanying Python program computes the moments by (918) and then uses the exact symmetric Stieltjes procedure (920). Selected results are:

n	μ_{2n}	decimal
0	1	1
1	$\frac{1}{9}$	0.111111111111
2	$\frac{19}{675}$	0.0281481481481
3	$\frac{583}{59535}$	0.00979255899891
4	$\frac{132809}{32531625}$	0.00408245822334
5	$\frac{46840699}{24405225075}$	0.00191928977733
6	$\frac{4068990560161}{4133856862760625}$	0.000984308527181
7	$\frac{1204567303451311}{2232691548877164375}$	0.000539513532022
8	$\frac{4146897304424408411}{13301767332333178846875}$	0.00031175536309
9	$\frac{18814360006695807527868793}{100028040755473167511640090625}$	0.000188090857969
10	$\frac{21431473463327429953796293981397}{182171989134769427819794434994453125}$	0.000117644175513

n	β_n (exact when compact)	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.168435374150
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288
6	see data file	0.199759514028
7	see data file	0.205964295192
8	see data file	0.211147954951
9	see data file	0.215261988744
10	see data file	0.218406233337
11	see data file	0.221259971300
12	see data file	0.223692820062

The full numerators and denominators through $n = 40$ are in `up_jacobi_coefficients.csv`. Their size grows rapidly: by moderate order the reduced integers contain thousands of decimal digits. This arithmetic complexity is not a floating-point artifact.

10.6 The limiting coefficient

Theorem 10.4 (Nevai-class limit). *The Jacobi coefficients satisfy*

$$\lim_{n \rightarrow \infty} \beta_n = \frac{1}{4}. \quad (933)$$

Proof. The measure $\rho(x) dx$ has essential support $[-1, 1]$, and its density is strictly positive for every $x \in (-1, 1)$. The Denisov–Rakhmanov theorem therefore places its orthonormal Jacobi matrix

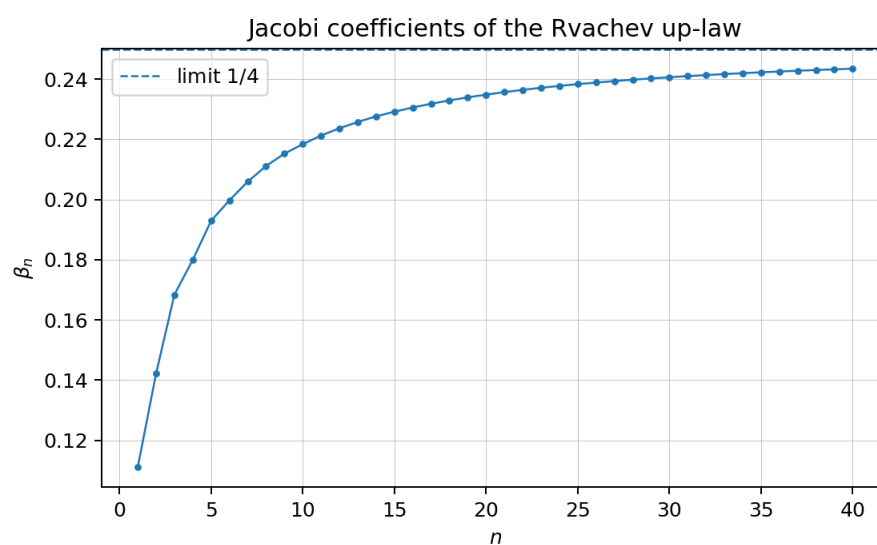


Figure 14: Exact Jacobi coefficients converted to decimals. The dashed line is the proved limit $1/4$.

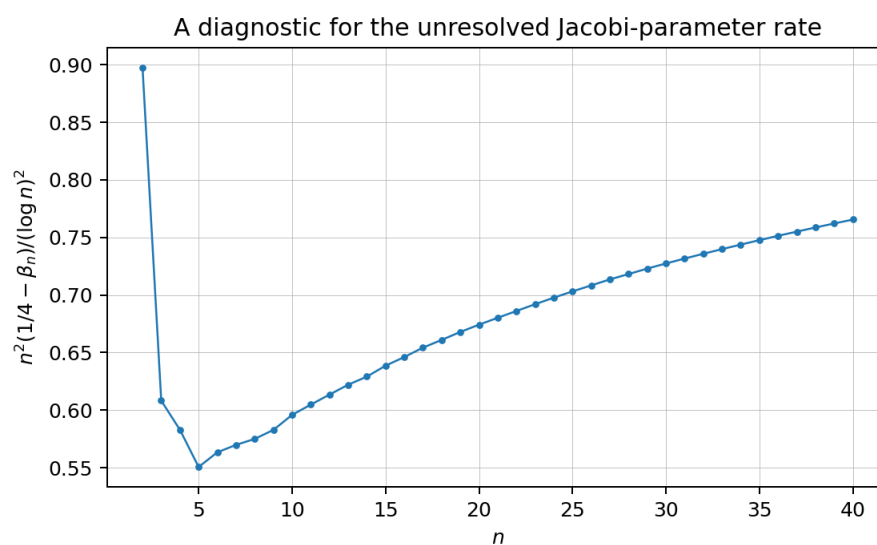


Figure 15: The diagnostic quantity $n^2(1/4 - \beta_n)/(\log n)^2$. The range $n \leq 40$ is evidence, not an asymptotic proof.

in the Nevai class of $[-1, 1]$ [3, 6]: the off-diagonal orthonormal coefficients tend to $1/2$, and the diagonal coefficients tend to zero. In the monic recurrence (920), β_n is the square of the off-diagonal orthonormal coefficient, hence tends to $1/4$. $\square$

10.7 Padé and Gaussian-quadrature representations

Let Q_{N-1} be the polynomial of the second kind

$$Q_{N-1}(z) = \int_{-1}^1 \frac{P_N(z) - P_N(x)}{z - x} \rho(x) \, dx. \quad (934)$$

The N -th Jacobi convergent is Q_{N-1}/P_N . If $x_{1,N}, \dots, x_{N,N}$ are the zeros of P_N , then

$$\frac{Q_{N-1}(z)}{P_N(z)} = \sum_{j=1}^N \frac{w_{j,N}}{z - x_{j,N}}, \quad w_{j,N} > 0, \quad \sum_j w_{j,N} = 1. \quad (935)$$

This is the Cauchy transform of the N -node Gaussian quadrature measure for ρ , and it matches moments through degree $2N - 1$. The exact error identity is

$$R(z) - \frac{Q_{N-1}(z)}{P_N(z)} = \frac{1}{P_N(z)^2} \int_{-1}^1 \frac{P_N(x)^2}{z - x} \rho(x) \, dx. \quad (936)$$

Hence every Gaussian resolvent approximant is a strict lower bound for $R(z)$ when $z > 1$, and a strict upper bound when $z < -1$.

The first nontrivial approximants are completely explicit:

$$R_1^G(z) = \frac{1}{z}, \quad (937)$$

$$R_2^G(z) = \frac{z}{z^2 - 1/9} = \frac{1}{2} \left(\frac{1}{z - 1/3} + \frac{1}{z + 1/3} \right), \quad (938)$$

$$R_3^G(z) = \frac{32/57}{z} + \frac{25/114}{z - \sqrt{19/75}} + \frac{25/114}{z + \sqrt{19/75}}. \quad (939)$$

Thus the continued fraction gives a hierarchy of positive finite atomic models for the up-law, complementary to the continuous finite-convolution models of Section 5.

11 Further combined representations

11.1 The Cauchy transform of the Fabius measure

Let

$$S(z) = \int_0^1 \frac{dF(x)}{z - x} = \mathbb{E} \frac{1}{z - Y}. \quad (940)$$

The affine relation (857) gives

$$S(z) = 2R(2z - 1). \quad (941)$$

Consequently, every representation of R immediately yields one for S . For example, for $|2z - 1| > 1$,

$$S(z) = 2 \sum_{n=0}^{\infty} \frac{\mu_{2n}}{(2z - 1)^{2n+1}}, \quad (942)$$

and S has a Jacobi fraction obtained by the affine change of variable in (926).

11.2 A determinant form of the Fabius Laplace transform

Combining (889) with (915) gives

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right) - e^{-s}}{s}. \quad (943)$$

Similarly,

$$\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right)}{s}. \quad (944)$$

These are literal combinations of a compact-support transform with a trace-class spectral determinant.

11.3 A resolvent–product–continued-fraction triangle

For $|t| < 1$, the same analytic function $\mathcal{O}(t)$ has three structurally different representations:

$$\begin{aligned} \mathcal{O}(t) &= \sum_{n \geq 0} \mu_{2n} t^n \\ &= \int_0^\infty e^{-u} \prod_{k \geq 1} \frac{\sinh(\sqrt{t} u / 2^k)}{\sqrt{t} u / 2^k} \, du \\ &= \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \ddots}}}. \end{aligned} \quad (945)$$

The first line is arithmetic, the second is multiplicative and analytic, and the third is orthogonal-polynomial/geometric. Equality of the three creates a direct bridge among the Bernoulli–Bell moment formulas, the infinite hyperbolic product, and the exact rational Jacobi parameters.

12 Half-lattice Shannon reconstruction and a convergent Mittag–Leffler formula

The corpus already proves that the integer zeros of Φ and the signs of its half-integer samples are controlled by binary arithmetic. A different question is whether those samples reconstruct the whole Fourier image. Compact support supplies an exact answer. The result is especially sparse because all nonzero integer samples vanish.

For this section write

$$c_k := \Phi \left(k + \frac{1}{2} \right), \quad a_k := 2k + 1, \quad A_k := |c_k| \quad (k \geq 0). \quad (946)$$

The half-integer sign theorem in the audited corpus gives

$$c_k = \varepsilon_k A_k, \quad \varepsilon_k = (-1)^{\text{wt}_2(k)}, \quad A_k > 0. \quad (947)$$

Because $\rho \in C_c^\infty([-1, 1])$, integration by parts shows that $c_k = \mathcal{O}_N(k^{-N})$ for every N . Every series below therefore converges absolutely, together with all termwise derivatives, on compact sets away from displayed poles.

Theorem 12.1 (Sparse half-lattice sampling). *For every $z \in \mathbb{C}$,*

$$\Phi(z) = \operatorname{sinc}_\pi(2z) + \sum_{k=0}^{\infty} c_k \{ \operatorname{sinc}_\pi(2z - a_k) + \operatorname{sinc}_\pi(2z + a_k) \}. \quad (948)$$

The series converges locally uniformly on $\mathbb{C}$, hence represents Φ as an entire function and may be differentiated term by term to arbitrary order.

Proof. With the Fourier convention (849), the inverse Fourier transform of the entire function Φ is supported in $[-1, 1]$. Thus Φ lies in the Paley–Wiener space of bandwidth one. The Whittaker–Shannon theorem [11] at the critical spacing $1/2$ gives

$$\Phi(z) = \sum_{n \in \mathbb{Z}} \Phi(n/2) \operatorname{sinc}_\pi(2z - n). \quad (949)$$

Here $\Phi(0) = 1$, while $\Phi(n) = 0$ for every nonzero integer n because the first factor $\operatorname{sinc}_\pi(z)$ in (851) vanishes. The surviving indices are therefore $n = 0$ and the odd integers. Since Φ is even, $\Phi(-(2k+1)/2) = c_k$, and pairing positive and negative odd indices yields (948). Rapid decrease of the samples upgrades the usual Paley–Wiener convergence to absolute local uniform convergence. $\square$

The paired cardinal kernel has a rational form:

$$\operatorname{sinc}_\pi(2z - a) + \operatorname{sinc}_\pi(2z + a) = -\frac{4z \sin(2\pi z)}{\pi(4z^2 - a^2)} \quad (a \in 2\mathbb{Z} + 1). \quad (950)$$

Substitution gives a genuine, rapidly convergent partial-fraction expansion.

Corollary 12.2 (Half-lattice Mittag–Leffler expansion). *As an identity of meromorphic functions,*

$$\frac{\pi\Phi(z)}{\sin(2\pi z)} = \frac{1}{2z} - 4z \sum_{k=0}^{\infty} \frac{c_k}{4z^2 - a_k^2}. \quad (951)$$

The only poles of the right-hand side are simple poles at the nonzero half-integers; at $z = k + \frac{1}{2}$,

$$\operatorname{Res}_{z=k+1/2} \frac{\pi\Phi(z)}{\sin(2\pi z)} = -\frac{c_k}{2}. \quad (952)$$

Consequently the signed and unsigned Thue–Morse sequences admit the residue representations

$$\varepsilon_k = -\operatorname{sgn} \left(\operatorname{Res}_{z=k+1/2} \frac{\pi\Phi(z)}{\sin(2\pi z)} \right), \quad (953)$$

$$\tau_k = \frac{1}{2} \left[1 + \operatorname{sgn} \left(\operatorname{Res}_{z=k+1/2} \frac{\pi\Phi(z)}{\sin(2\pi z)} \right) \right]. \quad (954)$$

Remark 12.3 (Why this differs from the zero-set expansion). The audited corpus and the first-wave report contain valuation-weighted partial fractions for Φ'/Φ , whose poles are the integer zeros of Φ . Formula (951) is structurally different: it reconstructs Φ itself from its nonzero half-lattice samples, and its poles are introduced by division by $\sin(2\pi z)$. The residues carry the Thue–Morse signs rather than the integer-zero multiplicities.

Corollary 12.4 (Derivative cardinal series). *For every integer $r \geq 0$,*

$$\Phi^{(r)}(z) = 2^r \operatorname{sinc}_\pi^{(r)}(2z) + 2^r \sum_{k \geq 0} c_k \left\{ \operatorname{sinc}_\pi^{(r)}(2z - a_k) + \operatorname{sinc}_\pi^{(r)}(2z + a_k) \right\}. \quad (955)$$

In particular, every Taylor coefficient and every moment of the up-law can be recovered from the single signed sample sequence $(c_k)_{k \geq 0}$.

13 Odd-harmonic cosine series and spectral Prouhet cancellation

Sampling in frequency has a dual real-space formulation. Periodize ρ with period 2. Its Fourier coefficient at the integer frequency n is $\frac{1}{2}\Phi(n/2)$. The even frequencies disappear, leaving only odd harmonics.

Theorem 13.1 (Sparse Fourier series for ρ and F). *For $-1 \leq x \leq 1$,*

$$\rho(x) = \frac{1}{2} + \sum_{k=0}^{\infty} c_k \cos(a_k \pi x). \quad (956)$$

For $0 \leq x \leq 1$,

$$F(x) = \frac{1}{2} - \sum_{k=0}^{\infty} c_k \cos(a_k \pi x). \quad (957)$$

Both series converge absolutely in every C^r -norm. More explicitly, for $r \geq 0$,

$$\rho^{(2r)}(x) = (-1)^r \pi^{2r} \sum_{k=0}^{\infty} c_k a_k^{2r} \cos(a_k \pi x), \quad (958)$$

$$\rho^{(2r+1)}(x) = (-1)^{r+1} \pi^{2r+1} \sum_{k=0}^{\infty} c_k a_k^{2r+1} \sin(a_k \pi x). \quad (959)$$

The corresponding formulas for F differ only by the sign visible in (957).

Proof. The period-two Fourier series of ρ is

$$\rho(x) = \frac{1}{2} \sum_{n \in \mathbb{Z}} \Phi(n/2) e^{\pi i n x}.$$

As in the proof of Theorem 12.1, the only nonzero even-frequency coefficient is the constant coefficient. Pairing the odd positive and negative frequencies gives (956). On $[0, 1]$, the folding relation $\rho(x) = F(1 - x) = 1 - F(x)$ gives (957). Rapid decay of c_k justifies all derivatives. $\square$

Formula (956) converts flatness at the center into infinitely many exact weighted Thue–Morse cancellations.

Corollary 13.2 (Spectral Prouhet identities). *With $A_k = |c_k|$,*

$$\sum_{k=0}^{\infty} \varepsilon_k A_k = \frac{1}{2}, \quad \sum_{k=0}^{\infty} \varepsilon_k A_k a_k^{2r} = 0 \quad (r \geq 1). \quad (960)$$

All sums are absolutely convergent. Equivalently, for $r \geq 1$, the unsigned sequence satisfies

$$\sum_{k \geq 0} \tau_k A_k a_k^{2r} = \frac{1}{2} \sum_{k \geq 0} A_k a_k^{2r}. \quad (961)$$

Proof. At $x = 0$, $\rho(0) = 1$, while $\rho^{(m)}(0) = 0$ for every $m \geq 1$: the up-function has a flat maximum at the origin. Set $x = 0$ in (956) and (958). Then use $c_k = \varepsilon_k A_k$. Finally substitute $\varepsilon_k = 1 - 2\tau_k$ to obtain (961). $\square$

These identities resemble the finite Prouhet cancellation $\sum_{n < 2^m} \varepsilon_n n^r = 0$ for $r < m$, but the mechanism and weights are different. Here one fixed infinite Thue–Morse word cancels *every* positive even moment after being weighted by the half-lattice Fourier amplitudes A_k . The weights themselves encode the smooth compactly supported fixed point.

Corollary 13.3 (Exact dyadic cosine-product sums). *For integers $m \geq 0$ and $0 \leq a \leq 2^m$,*

$$F\left(\frac{a}{2^m}\right) = \frac{1}{2} - \sum_{k=0}^{\infty} \varepsilon_k A_k \cos\left(\frac{a(2k+1)\pi}{2^m}\right). \quad (962)$$

The left side is rational by the exact dyadic arithmetic developed in the repository. Thus (962) is an infinite trigonometric-product representation of a rational number, with

$$A_k = \left| \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{k+1/2}{2^j}\right) \right|. \quad (963)$$

A particularly direct new representation of both Thue–Morse forms follows from the Fourier coefficient itself:

$$\varepsilon_k = \operatorname{sgn} \int_{-1}^1 \rho(x) \cos(a_k \pi x) \, dx, \quad (964)$$

$$\tau_k = \frac{1}{2} \left(1 - \operatorname{sgn} \int_{-1}^1 \rho(x) \cos(a_k \pi x) \, dx \right). \quad (965)$$

The significance is not merely that a sign can be hidden inside an integral: the integrand uses the unique smooth dyadic probability fixed point, and the entire sign word is recovered from its odd Fourier spectrum.

14 Bose–Malmsten product-integrals for the Fourier image

The infinite sinc product can be converted into an integral over the classical Bose kernel $(e^t - 1)^{-1}$. Define the dyadic hyperbolic kernel

$$H(u) := \sum_{j=0}^{\infty} (\cosh(u/2^j) - 1). \quad (966)$$

It is entire and obeys

$$H(u) = \cosh u - 1 + H(u/2), \quad (967)$$

$$H(u) = \sum_{r=1}^{\infty} \frac{u^{2r}}{(2r)!(1 - 4^{-r})}. \quad (968)$$

Theorem 14.1 (Bose–Malmsten representation). *In the zero-free strip $|\Re z| < 1$, with the branch normalized by $\log \Phi(0) = 0$,*

$$\log \Phi(z) = -2 \int_0^{\infty} \frac{H(z t)}{t(e^t - 1)} \, dt. \quad (969)$$

Equivalently,

$$\Phi(z) = \exp \left\{ -2 \int_0^{\infty} \frac{H(z t)}{t(e^t - 1)} \, dt \right\}. \quad (970)$$

Proof. For $|\Re w| < 1$, Euler’s product for the sine and Frullani’s integral give

$$\begin{aligned}\log \operatorname{sinc}_{\pi}(w) &= \sum_{n=1}^{\infty} \log \left(1 - \frac{w^2}{n^2}\right) \\ &= -2 \int_0^{\infty} \frac{\cosh(wt) - 1}{t(e^t - 1)} dt.\end{aligned}$$

Indeed, $\int_0^{\infty} e^{-nt}(\cosh(wt) - 1) dt/t = -\frac{1}{2} \log(1 - w^2/n^2)$. Sum this identity over $w = z/2^j$. Absolute convergence in compact substrips permits Fubini’s theorem, and the sum inside the integral is precisely $H(zt)$. $\square$

Corollary 14.2 (All logarithmic derivatives). *For $q \geq 1$ and $|\Re z| < 1$,*

$$\frac{d^q}{dz^q} \log \Phi(z) = -2 \int_0^{\infty} \frac{t^{q-1}}{e^t - 1} \sum_{j=0}^{\infty} 2^{-jq} \chi_q(zt/2^j) dt, \quad (971)$$

where $\chi_q = \sinh$ for odd q and $\chi_q = \cosh$ for even q . In particular,

$$\left. \frac{d^{2r}}{dz^{2r}} \log \Phi(z) \right|_{z=0} = -\frac{2(2r-1)!\zeta(2r)}{1-4^{-r}}. \quad (972)$$

Expanding H and using $\int_0^{\infty} t^{s-1}/(e^t - 1) dt = \Gamma(s)\zeta(s)$ yields the local zeta series

$$\log \Phi(z) = -\sum_{r=1}^{\infty} \frac{\zeta(2r)}{r(1-4^{-r})} z^{2r}, \quad |z| < 1. \quad (973)$$

This recovers the cumulant/Bernoulli expansion from a single positive-kernel integral and makes complete monotonicity on the imaginary axis transparent.

The strip restriction can be removed by retaining finitely many sinc factors.

Corollary 14.3 (Global finite-product–integral hybrid). *For any $z \in \mathbb{C}$, choose an integer $M \geq 0$ such that $|\Re(z/2^M)| < 1$. Then*

$$\Phi(z) = \left[\prod_{j=0}^{M-1} \operatorname{sinc}_{\pi}(z/2^j) \right] \exp \left\{ -2 \int_0^{\infty} \frac{H(z t/2^M)}{t(e^t - 1)} dt \right\}. \quad (974)$$

If a finite factor vanishes, both sides are interpreted by continuity and equal zero. For $z = k + \frac{1}{2}$, this gives a finite-product–integral representation of every spectral weight c_k , and hence of every sign in (953).

Combining (974) with Fourier inversion gives a nested integral representation of the up-function:

$$\rho(x) = \int_{\mathbb{R}} e^{2\pi i x t} \left[\prod_{j=0}^{M(t)-1} \operatorname{sinc}_{\pi}(t/2^j) \right] \exp \left\{ -2 \int_0^{\infty} \frac{H(tu/2^{M(t)})}{u(e^u - 1)} du \right\} dt, \quad (975)$$

where any measurable integer choice $M(t) > \log_2^+ |t|$ is admissible. Although this form is not intended for routine computation, it is an explicit integral–finite-product–exponential combination of the type requested in the present investigation.

15 Mellin and quantile representations

Let X have density ρ , let $Y = (1 + X)/2$, and let $I = F^{-1}$. Write

$$\mathcal{M}_Y(s) := \mathbb{E}Y^s = \int_0^1 y^s \, dF(y). \quad (976)$$

The endpoint flatness of F' implies that $y^{-A}F'(y)$ is integrable for every real A . Hence $\mathcal{M}_Y$ is an entire function of s , not merely a right-half-plane Mellin transform.

Theorem 15.1 (Mellin–quantile master package). *For all $s \in \mathbb{C}$,*

$$\boxed{\mathcal{M}_Y(s) = \int_0^1 I(u)^s \, du.} \quad (977)$$

For $\Re s > 0$,

$$\boxed{\int_0^1 x^{s-1} \rho(x) \, dx = \frac{\mathcal{M}_Y(s)}{s},} \quad (978)$$

$$\boxed{\int_0^1 x^{s-1} F(x) \, dx = \frac{1 - \mathcal{M}_Y(s)}{s}.} \quad (979)$$

The second right-hand side extends to an entire function at $s = 0$; the first has the expected simple pole of residue one there.

Proof. Equation (977) is the inverse-pushforward identity (899) with $g(y) = y^s$. Its convergence for arbitrary complex s follows from the flat density at zero. Since $\rho(x) = 1 - F(x)$ on $[0, 1]$, integration by parts gives

$$\int_0^1 x^{s-1} F(x) \, dx = \frac{1}{s} - \frac{1}{s} \int_0^1 x^s \, dF(x) = \frac{1 - \mathcal{M}_Y(s)}{s}.$$

Subtracting from $\int_0^1 x^{s-1} \, dx = 1/s$ gives (978). $\square$

Let $\mu_{2j} = \mathbb{E}X^{2j}$ be the even up-moments from (856).

Theorem 15.2 (Generalized binomial moment series). *For $\Re s > 0$,*

$$\boxed{\mathcal{M}_Y(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \mu_{2j}.} \quad (980)$$

The series is absolutely convergent. For a nonnegative integer n , it terminates and reduces to

$$\mathcal{M}_Y(n) = 2^{-n} \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \mu_{2j}. \quad (981)$$

Moreover,

$$\mathcal{M}_Y(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \frac{(-1)^j \Phi^{(2j)}(0)}{(2\pi)^{2j}}. \quad (982)$$

Proof. For $\Re s > 0$, the generalized binomial coefficients are absolutely summable, and $(1 + X)^s = \sum_{n \geq 0} \binom{s}{n} X^n$ may be averaged termwise because $|X| \leq 1$. Symmetry kills the odd moments. The Fourier-jet form follows from $\Phi^{(2j)}(0) = (-1)^j (2\pi)^{2j} \mu_{2j}$. $\square$

Differentiation at $s = 0$ supplies logarithmic inverse moments that are difficult to see pointwise:

$$\mathcal{M}_Y^{(r)}(0) = \int_0^1 \log^r I(u) \, du = \int_0^1 \log^r y \, dF(y) \quad (r \geq 0). \quad (983)$$

Thus every logarithmic moment of the inverse Fabius function is a derivative of one entire Mellin function. Together with the Bell-polynomial formulas for μ_{2j} , (980) forms a Mellin–Bell–Fourier representation chain.

16 Formal Thue–Morse orthogonality and nonlinear Jacobi renormalization

The signed Thue–Morse generating function

$$E(z) := \sum_{n=0}^{\infty} \varepsilon_n z^n = \prod_{j=0}^{\infty} (1 - z^{2^j}), \quad |z| < 1, \quad (984)$$

can itself be regarded as a moment generating function, but for an indefinite formal functional rather than a positive measure. Define

$$\mathcal{L}[x^n] := \varepsilon_n. \quad (985)$$

The nonvanishing Hankel theorem of Allouche–Peyrière–Wen–Wen [9, 10] guarantees that $\mathcal{L}$ is quasi-definite: there is a unique monic formal orthogonal polynomial p_n of every degree. Put

$$h_n := \mathcal{L}[p_n^2], \quad (986)$$

$$p_{n+1}(x) = (x - a_n)p_n(x) - b_n p_{n-1}(x), \quad (987)$$

$$p_{-1} = 0, \quad p_0 = 1, \quad b_n = h_n/h_{n-1} \quad (n \geq 1). \quad (988)$$

Unlike the positive Jacobi matrix of Section 10, the numbers b_n here may have either sign.

The binary recursion gives two identities valid for every polynomial f :

$$\mathcal{L}[f(x^2)] = \mathcal{L}[f(x)], \quad \mathcal{L}[xf(x^2)] = -\mathcal{L}[f(x)]. \quad (989)$$

They force a complete renormalization law.

Theorem 16.1 (Quadratic decimation of the formal orthogonal polynomials). *For every $n \geq 0$,*

$$p_{2n}(x) = p_n(x^2), \quad (990)$$

$$h_{2n} = h_n, \quad (991)$$

$$a_n = (-1)^{n+1}. \quad (992)$$

Every odd polynomial has a factor $x + 1$: there is a monic polynomial q_n such that

$$p_{2n+1}(x) = (x + 1)q_n(x^2). \quad (993)$$

The off-diagonal coefficients are determined from $b_1 = -2$ by the nonlinear dyadic recursion

$$b_{2n-1}b_{2n} = b_n, \quad (994)$$

$$b_{2n} + b_{2n+1} = (-1)^{n+1} - 1 \quad (n \geq 1). \quad (995)$$

Equivalently,

$$b_{2n} = \frac{b_n}{b_{2n-1}}, \quad b_{2n+1} = (-1)^{n+1} - 1 - b_{2n}. \quad (996)$$

Proof. The polynomial $p_n(x^2)$ is monic of degree $2n$. For $j < n$,

$$\begin{aligned}\mathcal{L}[p_n(x^2)x^{2j}] &= \mathcal{L}[p_n(x)x^{2j}] = 0, \\ \mathcal{L}[p_n(x^2)x^{2j+1}] &= -\mathcal{L}[p_n(x)x^{2j}] = 0,\end{aligned}$$

by (989). These are all monomials of degree below $2n$, so uniqueness of the monic orthogonal polynomial proves (990). The norm identity follows immediately.

For even indices,

$$a_{2n} = \frac{\mathcal{L}[xp_{2n}^2]}{h_{2n}} = -\frac{\mathcal{L}[p_n^2]}{h_n} = -1.$$

Since $a_0 = -1$, $p_1 = x + 1$. The recurrence at an even index now shows inductively that p_{2n+1} is divisible by $x + 1$, proving (993). Write $p_{2n+1} = (x + 1)q_n(x^2)$ and set $f = q_n^2$. The moment rules imply

$$\begin{aligned}\mathcal{L}[(x + 1)^2 f(x^2)] &= \mathcal{L}[x f(x)] - \mathcal{L}[f(x)], \\ \mathcal{L}[x(x + 1)^2 f(x^2)] &= \mathcal{L}[x f(x)] - \mathcal{L}[f(x)].\end{aligned}$$

Their quotient is one, so $a_{2n+1} = 1$. This proves (992).

Finally use the recurrences at indices $2n - 1, 2n, 2n + 1$. From

$$p_{2n} = (x - 1)p_{2n-1} - b_{2n-1}p_{2n-2}$$

we obtain

$$(x - 1)p_{2n-1} = p_n(x^2) + b_{2n-1}p_{n-1}(x^2).$$

Substitute this and $p_{2n+1} = (x + 1)p_n(x^2) - b_{2n}p_{2n-1}$ into the recurrence for p_{2n+2} . The result is

$$p_{2n+2} = [x^2 - 1 - b_{2n} - b_{2n+1}]p_n(x^2) - b_{2n-1}b_{2n}p_{n-1}(x^2).$$

Compare with $p_{n+1}(x^2) = [x^2 - a_n]p_n(x^2) - b_n p_{n-1}(x^2)$ to obtain (994)–(995). Directly, $b_1 = h_1/h_0 = \mathcal{L}[(x + 1)^2] = -2$. $\square$

The first coefficients are

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
b_n	-2	1	-1	-1	-1	1	-1	1	-3	$\frac{1}{3}$	$-\frac{1}{3}$	-3	1	-1	1	1.

(997)

The nonlinear recursion computes the whole sequence using rational arithmetic without any Hankel determinant. It also proves that the formal Jacobi fraction of E has exactly alternating diagonal entries:

$$E(z) = \frac{1}{1 + z - \frac{b_1 z^2}{1 - z - \frac{b_2 z^2}{1 + z - \frac{b_3 z^2}{1 - z - \ddots}}}}. \quad (998)$$

The equality is an identity of formal power series and hence analytic for sufficiently small $|z|$.

Corollary 16.2 (Dyadic formal orthogonal polynomials). *For $N = 2^m$,*

$$\boxed{p_N(x) = x^N + 1, \quad h_N = -2.} \quad (999)$$

Proof. Iterate (990) from $p_1 = x + 1$; iterate (991) from $h_1 = -2$. $\square$

17 Explicit dyadic Padé approximants and antiperiodic quadrature

Let

$$P_m(z) := \sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad N := 2^m. \quad (1000)$$

The Mahler/block factorization is $E(z) = P_m(z)E(z^N)$. The elementary approximation $E(w) = 1 - w - w^2 + \mathcal{O}(w^3)$ has a surprisingly strong dyadic lift.

Theorem 17.1 (Closed dyadic Padé approximant). *For every $m \geq 0$,*

$$\boxed{E(z) - \frac{P_m(z)}{1 + z^N} = -2z^{2N} + \mathcal{O}(z^{2N+1})}. \quad (1001)$$

Thus $P_m(z)/(1 + z^N)$ is the Padé approximant of type $[N - 1/N]$ to $E(z)$. The exact error identity is

$$E(z) - \frac{P_m(z)}{1 + z^N} = P_m(z) \left(E(z^N) - \frac{1}{1 + z^N} \right). \quad (1002)$$

In the standard Padé–orthogonality correspondence [12, 13], the denominator $1 + z^N$ is the reciprocal polynomial of the formal orthogonal polynomial $p_N(x) = x^N + 1$ in (999).

Proof. Use $E(z) = P_m(z)E(z^N)$. At the base scale,

$$E(w) - \frac{1}{1 + w} = -2w^2 + \mathcal{O}(w^3).$$

Set $w = z^N$ and multiply by $P_m(z) = 1 + \mathcal{O}(z)$. The numerator and denominator degrees are $N - 1$ and N , respectively, so matching through degree $2N - 1$ is exactly the defining Padé condition. $\square$

The poles lie at the roots of -1 , and partial fractions turn the approximant into a finite complex exponential formula for the sequence itself.

Theorem 17.2 (Antiperiodic Gaussian quadrature). *Let*

$$\Lambda_N := \{\lambda \in \mathbb{C} : \lambda^N = -1\}, \quad w_\lambda := \frac{1}{N} P_m(\lambda^{-1}). \quad (1003)$$

Then

$$\frac{P_m(z)}{1 + z^N} = \sum_{\lambda \in \Lambda_N} \frac{w_\lambda}{1 - \lambda z}. \quad (1004)$$

Consequently, for every integer $0 \leq r < 2N$,

$$\boxed{\varepsilon_r = \frac{1}{N} \sum_{\lambda^N = -1} P_m(\lambda^{-1}) \lambda^r}. \quad (1005)$$

The unsigned form is

$$\tau_r = \frac{1}{2} \left[1 - \frac{1}{N} \sum_{\lambda^N = -1} P_m(\lambda^{-1}) \lambda^r \right] \quad (0 \leq r < 2N). \quad (1006)$$

Proof. The poles of $P_m(z)/(1 + z^N)$ are $z = \lambda^{-1}$. Multiplying by $1 - \lambda z$ and taking the limit gives the coefficient $P_m(\lambda^{-1})/N$, proving (1004). Expand every term as a geometric series. The Padé matching in (1001) says that the first $2N$ coefficients equal ε_r , which gives (1005). $\square$

Formula (1005) is an N -node complex Gaussian quadrature for the formal Thue–Morse moment functional: it reproduces $2N$ consecutive moments. It can also be verified directly from the anti-periodicity $\varepsilon_{N+r} = -\varepsilon_r$ for $0 \leq r < N$ and $\lambda^N = -1$, but the Padé derivation explains why the same nodes and weights are canonical.

The unsigned generating function

$$T(z) = \sum_{n \geq 0} \tau_n z^n = \frac{1}{2} \left(\frac{1}{1-z} - E(z) \right) \quad (1007)$$

has the simultaneous rational approximation

$$T(z) = \frac{1 + z^N - (1-z)P_m(z)}{2(1-z)(1+z^N)} + \mathcal{O}(z^{2N}). \quad (1008)$$

17.1 A Krawtchouk and contour avatar

There is a second orthogonal-polynomial representation of the signs, independent of the formal Jacobi system above. Fix any m with $n < 2^m$, and put $r = \text{wt}_2(n)$. The top-degree Krawtchouk coefficient [14] gives

$$\varepsilon_n = [u^m](1-u)^r(1+u)^{m-r} = {}_2F_1(-m, -r; -m; 2) = (-1)^r. \quad (1009)$$

Equivalently,

$$\varepsilon_n = \frac{1}{2\pi i} \oint_{|u|=\delta} \frac{(1-u)^r(1+u)^{m-r}}{u^{m+1}} du, \quad (1010)$$

with $0 < \delta < 1$, and

$$\tau_n = \frac{1}{2} (1 - [u^m](1-u)^r(1+u)^{m-r}). \quad (1011)$$

The coefficient of u^m has only one contributing binomial term, namely the choice of all r minus signs and all $m-r$ plus signs. Although elementary after the fact, this places the signed and unsigned words at an extremal Krawtchouk degree and links their binary digit statistic to a classical hypergeometric family.

18 Numerical experiments and reproducibility

18.1 Exact arithmetic

The script `rvachev_frontier_experiments.py` performs the following operations.

1. It computes $\mu_0, \mu_2, \dots$ exactly using (918).
2. It constructs the monic orthogonal polynomials using the symmetric three-term recurrence and exact rational inner products.
3. It writes all exact β_n through a user-selected order to CSV.
4. It checks (881) independently using a long moment expansion.
5. It compares the cotangent, tangent, and zero-set formulas for Φ'/Φ .
6. It generates Figures 14 and 15.

The default order is 40. On the reference environment the full run takes tens of seconds; the exact integers, rather than floating-point work, dominate at higher order.

18.2 Numerical verification of the resolvent equation

At 45-decimal mpmath precision, using moment expansions through orders 160 and 220, the computed values were:

z	$R'(z)$	$2(R(2z+1) - R(2z-1))$	relative discrepancy
1.25	-0.8312850091714830623446214	same	3.6×10^{-46}
1.75	-0.3679189599146538753803797	same	below printed precision
2.50	-0.1691589760955563001740583	same	4.4×10^{-47}

At $z = 1.25$, the change between the two truncation lengths was about 3.8×10^{-41} , so the identity error is below the visible truncation scale.

18.3 Numerical comparison of logarithmic derivatives

The cotangent and tangent formulas agreed to roughly 44 decimal places at the tested nonzero points. The zero-set formula, truncated at 10^5 , differed by 5.2×10^{-6} , 1.5×10^{-5} , and 2.9×10^{-5} at $z = 0.13, 0.37, 0.73$, respectively, consistent with its algebraic tail. The data file `numerical_checks.txt` records all values.

18.4 Second-wave sampling, Mellin, and Padé checks

The companion program `frontier_sampling_pade_experiments.py` independently checks the new formulas at 80-decimal precision and generates all data in `generated/data`. The principal residuals were:

- sparse Shannon and Mittag–Leffler formulas: between 10^{-20} and 10^{-18} with 128 half-lattice coefficients;
- the Bose–Malmsten integral: below 5.1×10^{-15} at real and complex test points;
- the nonlinear formal-Jacobi recursion and $p_{2^m} = x^{2^m} + 1$: exact through degree 96 using rational arithmetic;
- roots-of-minus-one quadrature: below 5×10^{-79} for all moments through level $N = 64$;
- Mellin moment series versus direct cosine-density quadrature: between 10^{-13} and 2.1×10^{-14} for the displayed tests.

No numerical observation is used inside a proof. The exact checks are regression guards against normalization, scaling, and sign errors.

19 Conjectures and frontier problems

The following conjectures are deliberately separated from the proved results.

Conjecture 19.1 (Strict Jacobi monotonicity). The exact Jacobi coefficients are strictly increasing:

$$0 < \beta_1 < \beta_2 < \cdots < \frac{1}{4}. \quad (1012)$$

Exact arithmetic verifies the inequalities through $n = 40$. The upper limit is proved by Theorem 10.4, but monotonicity does not follow from the Denisov–Rakhmanov theorem and appears to require a Fabius-specific argument.

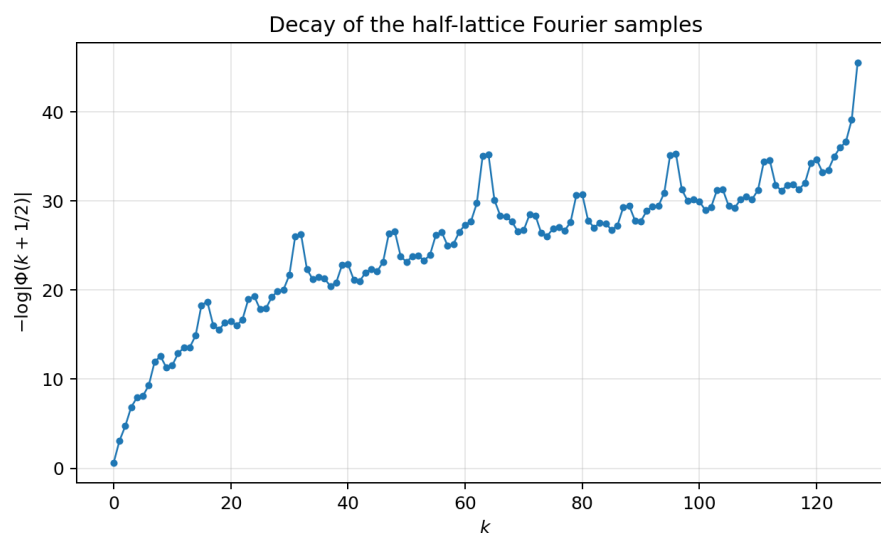


Figure 16: The half-lattice samples decay faster than every power, while their signs are the signed Thue–Morse word. The local irregularity reflects the binary orbit structure.

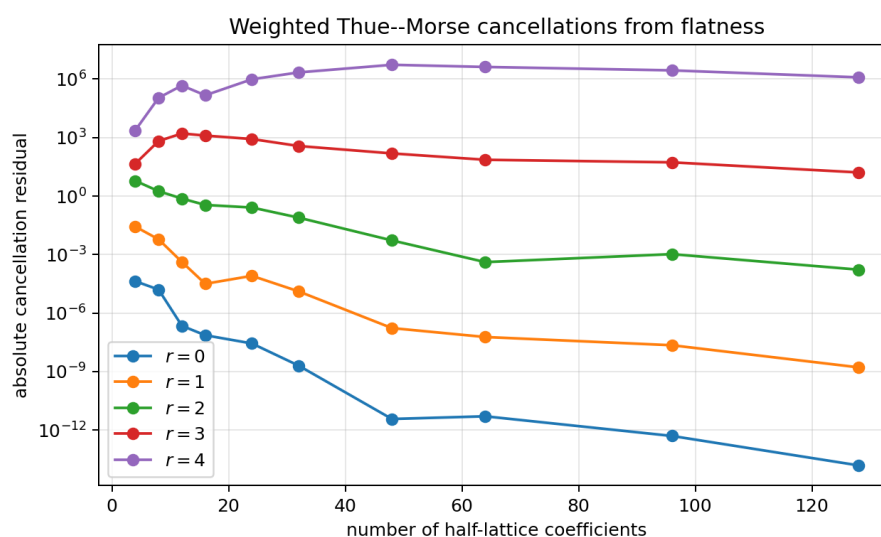


Figure 17: Truncation residuals in the spectral Prouhet identities (960). Higher powers initially amplify the tail before the super-polynomial Fourier decay dominates.

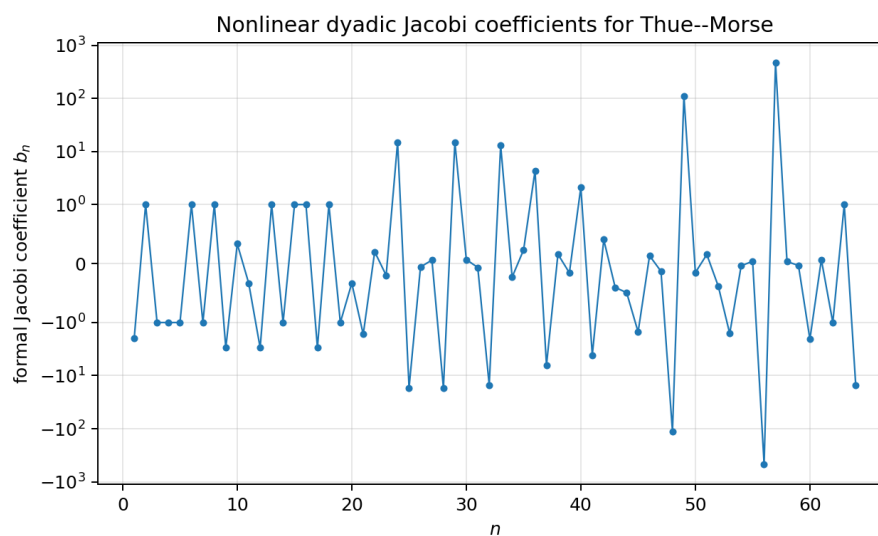


Figure 18: The indefinite formal Jacobi coefficients b_n , shown on a symmetric logarithmic scale. Their nonlinear dyadic recursion creates simultaneous very large and very small rational values.

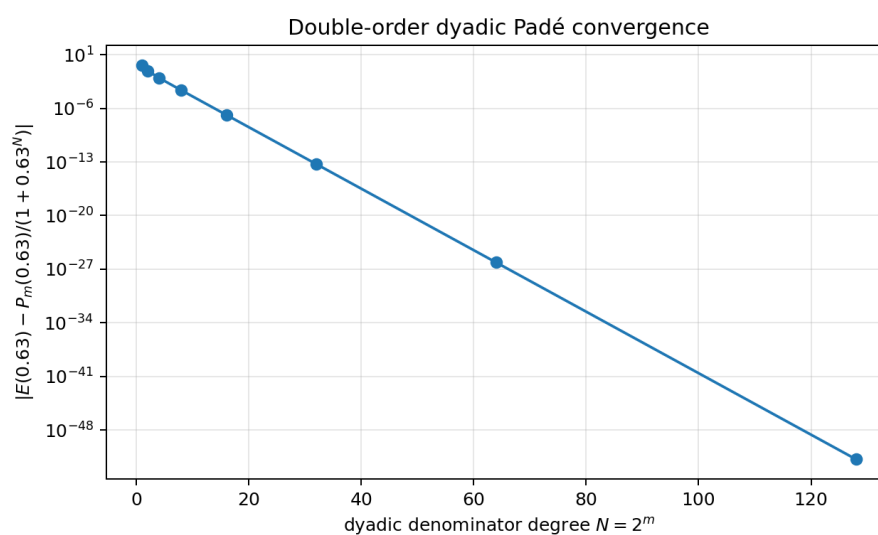


Figure 19: Error of the explicit dyadic Padé approximant at $z = 0.63$. The order doubles with each dyadic level, as proved in Theorem 17.1.

Conjecture 19.2 (Logarithmically corrected recurrence rate). There exists a constant $C_{\text{up}} \in (0, \infty)$ such that

$$\frac{1}{4} - \beta_n \sim C_{\text{up}} \frac{(\log n)^2}{n^2}. \quad (1013)$$

The scaled quantity in Figure 15 rises from approximately 0.674 at $n = 20$ to 0.766 at $n = 40$. This is too short a range to determine a constant or even to exclude a different slowly varying factor. The conjecture is motivated jointly by the data and by the squared-logarithmic endpoint scale of ρ . A Riemann–Hilbert or strong Szegő analysis adapted to the endpoint flatness is the natural next step.

Conjecture 19.3 (Dyadically regularized Mittag–Leffler expansion). Let $d_n = 1 + \nu_2(|n|)$, and let $\text{PP}_n(z)$ be the principal part of $1/\Phi(z)$ at the nonzero integer n . There exists a canonical dyadic grouping and a finite genus of subtraction polynomials such that

$$\frac{1}{\Phi(z)} = E(z) + \sum_{n \in \mathbb{Z} \setminus \{0\}}^{\text{dyadic}} (\text{PP}_n(z) - \text{canonical subtraction at } n), \quad (1014)$$

where E is an explicitly controlled entire function, ideally constant or of very low order.

The local principal parts are computable from Theorem 8.2; the unresolved issue is a global summation scheme compatible with the unbounded dyadic multiplicities.

Open problem 19.4 (Arithmetic of β_n). Determine sharp denominator-clearing factors for β_n , h_n , and the parity Hankel determinants. The first coefficients display repeated migration of prime factors between consecutive norms, suggesting that factored norm variables may be arithmetically simpler than reduced β_n themselves.

Open problem 19.5 (Endpoint-sensitive Gaussian nodes). Determine the asymptotic distance of the extreme zero of P_n from 1, and compare it with the Chebyshev scale n^{-2} . The density is positive almost everywhere but flatter than every power at the endpoint, so the leading equilibrium distribution is classical while the first correction should retain the Lambert– W /log-square endpoint signature.

Open problem 19.6 (A coupled resolvent hierarchy for exponent sequences). The repository’s Newton/Rvachev theory replaces the unit exponent sequence in the sinc product by general integer sequences. Construct the corresponding Cauchy transforms and derive coupled affine functional equations when the exponent sequence is closed under a Pascal or shift operator. The Fredholm model of Section 9 suggests that the Fourier side is easy; the real-space fixed-point law is the main obstacle.

Open problem 19.7 (Fractional resolvent dynamics). Equation (881) gives every integer derivative as a finite Thue–Morse orbit. Seek a fractional-order interpolation

$$D_z^\alpha R(z) = \int_{\Gamma_\alpha} K_\alpha(j) R(2^\alpha z + \text{affine shift}(j)) \, dj$$

that reduces to (882) for $\alpha \in \mathbb{N}$. A semigroup built from the two affine branches $z \mapsto 2z \pm 1$ may provide the correct operator calculus.

Open problem 19.8 (Formal verification roadmap). A Lean development can be staged as follows:

1. define the Cauchy transform of the canonical up probability measure;
2. prove the logarithmic fixed-point identity first on $(1, \infty)$;
3. differentiate under the integral and formalize (881);
4. prove the Thue–Morse derivative tree by induction on binary children;
5. define rational moments and the monic Stieltjes procedure;

6. construct finite Jacobi convergents and prove the Gaussian error identity;
7. postpone the Denisov–Rakhmanov limit, which depends on substantial external orthogonal-polynomial theory.

Conjecture 19.9 (Unbounded formal-Jacobi renormalization). For the rational sequence defined by (996),

$$\limsup_{n \rightarrow \infty} |b_n| = \infty, \quad \liminf_{n \rightarrow \infty} |b_n| = 0.$$

Exact computation through several thousand indices shows both effects on every sufficiently large dyadic block. The identity $\prod_{n=2}^{2^m} b_n = 1$, which follows from $h_{2^m} = h_1$, forces a geometric balance but does not by itself prove either extreme.

Conjecture 19.10 (Dyadic weak limit for logarithmic Jacobi coefficients). The empirical measures

$$\nu_m = 2^{-m} \sum_{n=1}^{2^m} \delta_{m^{-2} \log |b_n|}$$

converge weakly to a compactly supported, non-degenerate probability measure. A more refined version should replace the factor m^{-2} by the true height scale and should admit a transfer-operator description induced by (996).

Conjecture 19.11 (No anomalous spectral cancellation). For each fixed $r \geq 0$, the tail in (960) has the same logarithmic order as its largest term along a set of dyadic blocks of density one. Equivalently, apart from exceptional binary rays, the exact global cancellation does not create a second systematic exponential scale beyond the known decay scale of $|\Phi(k + \frac{1}{2})|$.

Open problem 19.12 (Formalization package). Formalize Theorems 12.1, 14.1, 16.1, and 17.1 in Lean. The last two require only polynomial algebra, quasi-definiteness imported from finite Hankel determinants, and the binary recursions $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$; they are the most immediate targets. The sampling theorem requires a reusable Paley–Wiener library.

20 Promising research program

20.1 Analytic program

The Cauchy transform should be treated as a primary Fabius–Rvachev object, on the same level as F , ρ , and Φ . Immediate goals are:

1. derive quantitative bounds for R near the cut from the sharp Fourier decay;
2. study the Hilbert transform (879), including its zeros and endpoint asymptotics;
3. use (881) to continue asymptotic expansions along dyadic affine rays;
4. compare finite-logarithmic and Jacobi rational approximants uniformly near the support.

20.2 Orthogonal-polynomial program

The exact J -fraction opens a separate research direction:

1. prove or disprove Conjecture 19.1;
2. determine strong asymptotics of P_n , h_n , and β_n ;

3. construct Fabius-specific Gauss–Kronrod or nested positive rules;
4. investigate whether the dyadic moment recurrence induces a nonlinear recurrence or integrable system for β_n ;
5. study the arithmetic Galois structure of Gaussian nodes and exact weights.

20.3 Operator and special-function program

The Fredholm determinant and Hurwitz-zeta hierarchy suggest:

1. regularized determinants and spectral shifts for deformed multiplicity sequences;
2. a Barnes-type compression of the determinant rather than only of the scalar product;
3. exact trace identities for logarithmic derivatives at rational and half-integer arguments;
4. meromorphic continuation of dyadically weighted spectral sums with local factors dictated by $\nu_2(n)$.

20.4 Inverse-function program

The pushforward formulas (899)–(902) show that many global integral questions about F^{-1} are easier than pointwise questions. Promising directions include:

1. exact orthogonal expansions of $I(u)$ using its known pushforward moments;
2. entropy and transport functionals of I , reduced to integrals against ρ ;
3. combining the global Cauchy transform with local Lambert– W endpoint inversions;
4. rational and Gaussian quadrature for inverse-function integrals without directly evaluating I .

21 Conclusion

The audited repository already contains most of the classical and modern representation theory one would first seek: convolution laws, sinc and cosine products, integer-zero products, moment and cumulant series, Thue–Morse splines, q-binomial dyadic values, Poisson and Legendre expansions, inverse asymptotics, and Fourier-decay envelopes. The most productive way to extend that corpus is therefore to introduce a new organizing transform rather than another isolated formula.

The Cauchy transform R accomplishes this. It joins the moment sequence to an analytic function, converts the dyadic law into a functional-differential equation, turns finite convolutions into logarithmic Thue–Morse sums, and generates a Jacobi continued fraction with exact rational coefficients. The Fourier image simultaneously acquires an operator determinant and a special-function logarithmic-derivative calculus. Closed one-sided transforms and inverse pushforward identities complete a network in which real-space, frequency-space, moment, orthogonal-polynomial, and operator representations are mutually convertible.

The proved limit $\beta_n \rightarrow 1/4$ anchors the new continued-fraction theory. The exact order-40 data, the monotonicity conjecture, the endpoint-sensitive rate problem, and the regularized Mittag–Leffler problem provide concrete frontier targets suitable for both analysis and formalization.

A Reproducibility instructions

From the directory containing this report, run the two independent experiment suites:

```
python3 rvachev_frontier_experiments.py \
  --order 40 --digits 45 --output-dir first_wave_generated

python3 frontier_sampling_pade_experiments.py \
  --output-dir generated
```

The first command regenerates the exact up-law moments, the positive Jacobi coefficients, `generated_tables.tex`, the resolvent/logarithmic-derivative check log, and the two first-wave figures. The second command regenerates the half-lattice samples, weighted Prouhet checks, exact formal-Jacobi coefficients, dyadic Padé/quadrature checks, Mellin and Malmsten checks, and all four second-wave figures under `generated/data` and `generated/figures`.

The exact arithmetic uses Python's `fractions.Fraction`. The numerical checks use `mpmath` and `scipy`; plotting uses `matplotlib`. Both scripts contain detailed comments, normalization statements, argument validation, and exact regression checks. Packaged root-level copies of the LaTeX fragment and figure PDFs make the report compile without rerunning either program.

B Corpus audit ledger

The pinned 67-path inventory and the additional live draft families used in the audit are listed in the accompanying source fragment `corpus_inventory.tex`.

Pinned 67-source ledger

No.	Path relative to Analysis/FabiusFunction/docs
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
6	archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
7	archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
8	archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
9	archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
10	archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
11	archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
12	archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex

No.	Path relative to Analysis/FabiusFunction/docs
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15	archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_ Up.tex
16	archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up. tex
17	archive/primary-exposition/Fabius_function_and_up_function/Fabius_fun ction_and_up_function.tex
18	archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Fu nction_Missing_Parts.tex
19	archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Func tion_Missing_Parts.tex
20	archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functi ons.tex
21	archive/q-calculus/fabius-q-special-functions/fabius-q-special-functio ns.tex
22	archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyad ic_Asymptotic_Bridge.tex
23	archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Repr esentations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
24	archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabiu s_Dyadic_Formulae_to_Asymptotics.tex
25	archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q _Connections.tex
26	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_T hue_Morse_Convergence_Rate.tex
27	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_ Thue_Morse_Convergence_Rates.tex
28	archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exp osition_Gap_Register.tex
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30	archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvache v_Up_from_Repeated_Integration.tex
31	archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Form ula_Atlas.tex
32	archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Fo rmula_Atlas_for_the_Thue_Morse_Sequence.tex
33	archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Form ula_Atlas.tex
34	archive/research-frontiers/non-formalized-research-frontiers/semi-for malized-research-frontiers.tex
35	archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non _Elementarity_of_the_Fabius_Function.tex
36	archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_A symptotics.tex
37	archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-i nverse-dyadic-closed-form.tex

No.	Path relative to Analysis/FabiusFunction/docs
38	semi-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex
39	semi-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex
40	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex
41	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex
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43	semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex
44	semi-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex
45	semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
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49	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
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54	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
55	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
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58	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
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No. Path relative to Analysis/FabiusFunction/docs

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61 papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
62 papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
63 papers/arXiv-1502.06738v2/Lacunary_products.tex
64 papers/arXiv-1702.05442v1/09-Function.tex
65 papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66 papers/arXiv-2211.13570v1/document.tex
67 papers/ubiq15/ubiq15-corrected.tex

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Live additions checked after the pinned ledger

- semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
- semi-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex
- semi-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex
- semi-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/ (report source and auxiliaries)

Additional auxiliary TeX fragments inside these live report packages were checked through their parent report structure. The pinned ledger remains the stable reproducible count; the live tree changes during active research and therefore should not be assigned a permanent count without a fresh commit-pinned traversal.

C Selected exact data

The first eleven even moments and first twelve Jacobi coefficients are typeset in the main text. The CSV files contain the exact data through order 40. Selected late-order decimal values are:

n	β_n	$1/4 - \beta_n$	$n^2(1/4 - \beta_n)/(\log n)^2$
12	0.223692820061698	0.026307179938302	0.6135029154
15	0.229185734718218	0.020814265281782	0.6386018638
20	0.234873012373992	0.015126987626008	0.6742274757
25	0.238343193605764	0.011656806394236	0.7031551379
30	0.240650640426945	0.009349359573055	0.7273789024
35	0.242285352613889	0.007714647386111	0.7476320691
40	0.243489365492140	0.006510634507860	0.7655161747

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Part VII

Unit-Circle, Bessel, and Spectral–Monodromy Representations

Abstract

This report audits the mathematical $\text{\LaTeX}$ corpus under <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>, including its subdirectories, source inventories, historical snapshots, imported papers, and living frontier reports. It first consolidates the repository’s established probabilistic, convolutional, spline, Fourier, product, moment, resolvent, continued-fraction, multiresolution, integral-transform, and inverse-function representations. It then develops a repo-relative new coordinate system based on the probability–Chebyshev moments

$$c_n = \int_{-1}^1 T_n(x) \text{up}(x) \, dx.$$

These moments lift the Rvachev probability measure to a symmetric probability measure on the unit circle. The lift yields an exact Carathéodory function, a Joukowski representation of the Cauchy transform, weighted Chebyshev boundary series for up, phase series for the Fabius function and its inverse, rational Schur–Verblunsky parameters, a CMV resolvent, Toeplitz determinants, and a Szegő product.

Jacobi–Anger expansions produce entire Bessel-series identities equal to the infinite sinc product and its canonical dyadic zero product, while modified Bessel series equal the hyperbolic moment-generating product. A Laplace–Bessel–Joukowski chain unifies these identities with the Cauchy transform. For the signed and unsigned Thue–Morse sequences, we derive argument-principle formulas from the multiplicities of the integer zeros of the Fourier image, local residue recurrences, a logarithmic-monodromy recovery formula for finite blocks, and a smooth Abel–Poisson regularization of the Rademacher sign product.

Every theorem marked “repo-relative new” is proved here and was not located by exact-term or structural search in the audited repository. This phrase is deliberately narrower than a claim of worldwide priority. Conjectures and a research program are stated separately. The accompanying Python program uses exact rational arithmetic and high-precision numerics to verify the coefficient pipelines and analytic identities.

Status and reproducibility

The archive includes the full $\LaTeX$ source, exact coefficient CSVs, generated tables, three convergence/parameter figures, a commented Python program, a numerical verification log, and a source-overlap ledger. The direct Schur algorithm and the independent Geronimus reconstruction agree exactly through the checked orders; the spectral Thue–Morse formulas pass for $0 \leq n \leq 256$; and three independent Cauchy-transform evaluations agree to the precision recorded in [Appendix 9](#).

1 Scope, conventions, and nonduplication protocol

1.1 Normalized objects

We use the Rvachev probability density

$$\rho(x) := \text{up}(x), \quad \rho \geq 0, \quad \text{supp } \rho = [-1, 1], \quad \int_{-1}^1 \rho(x) \, dx = 1,$$

which is even and C^∞ , with every derivative flat at the endpoints. Its associated random variable admits the convergent series

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k, \quad U_k \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1]. \quad (1015)$$

Set

$$Y = \frac{X+1}{2}, \quad F(u) = \mathbb{P}(Y \leq u) = \int_{-1}^{2u-1} \rho(x) \, dx, \quad I(v) = F^{-1}(v).$$

Thus, on $0 < u < 1$,

$$F'(u) = 2\rho(2u-1). \quad (1016)$$

The Fourier convention is

$$\Phi(\xi) = \int_{-1}^1 \rho(x) e^{-2\pi i \xi x} \, dx, \quad \text{sinc}_\pi z = \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_\pi 0 = 1. \quad (1017)$$

The moment-generating and Cauchy transforms are

$$M(t) = \mathbb{E} e^{tX}, \quad R(z) = \int_{-1}^1 \frac{\rho(x)}{z-x} \, dx, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (1018)$$

For the Thue–Morse sequence we use

$$\epsilon_n = (-1)^{s_2(n)}, \quad \tau_n = \frac{1 - \epsilon_n}{2} \in \{0, 1\}, \quad (1019)$$

where $s_2(n)$ is the sum of the binary digits of n .

1.2 Corpus audit

The current repository manifest organizes the documentation into Fourier-decay, Thue–Morse, exponent and q -series, spectral/arithmetic, integration and transform, inverse/sampling, representation, and frontier campaigns. The repository’s own pinned corpus inventory enumerates sixty-seven source documents plus later living additions, while the multiresolution report records a source-level audit of eighty-nine $\text{\LaTeX}$ paths at commit 924652f4a9a656fd957d2936b2f053747c6f6c47. The consolidated frontier volume, the current manifest, and the overlap-sensitive living reports were used jointly so that historical duplicates did not masquerade as independent novelty.

The following families are already present and are therefore treated as baseline rather than novelty: the infinite sinc product; the canonical zero product; finite box-convolution splines; finite Thue–Morse truncated-power formulas; Cauchy and logarithmic resolvents; moment/cumulant and Bell–Bernoulli formulas; transforms of F and F^{-1} ; digamma, Hurwitz-zeta, and cotangent logarithmic derivatives; Fredholm determinants; Jacobi and Stieltjes continued fractions; Haar, Schauder, Walsh, and beta-mixture expansions; fractional Volterra integrals; Mellin/Newton moments; inverse endpoint

asymptotics; the half-integer sign formula; dyadic cardinal reproduction; and the non-holonomicity theorem from unbounded dyadic zero multiplicities.

The exact terms and structures *Carathéodory function*, *OPUC*, *Verblunsky coefficient*, *CMV matrix*, *Joukowski transform*, *probability–Chebyshev moment*, *Bessel product-series*, *argument-principle Thue–Morse formula*, and *logarithmic-monodromy encoding* were not found in the current consolidated volume or in the relevant living reports. In particular, the multiresolution report’s density-Chebyshev coefficients are different from the probability-Chebyshev moments introduced in [Appendix 3](#).

Repository baseline

The audit is documented in `corpus_audit.txt`. Throughout the report, “new” means “not located in the audited PROVEIt subtree.” Standard external machinery—Jacobi–Anger expansions, the Schur algorithm, Geronimus relations, Szegő’s theorem, CMV matrices, Shannon sampling, and the argument principle—is cited rather than claimed as new.

1.3 Proof-status convention

A theorem or proposition without qualification is proved in this report. A boxed “repository baseline” records known material used as input. Conjectures are numbered and visually separated. Numerical evidence is never promoted to a proof; exact rational calculations are identified as such.

2 Established representation atlas

This section compresses the repository’s existing representation families into one normalization. It is included both for completeness and to make the genuinely new bridges in later sections transparent.

2.1 Random series, convolutions, and refinement

From (1015), X satisfies the distributional fixed point

$$X \stackrel{d}{=} \frac{X + U}{2}, \quad U \sim \text{Unif}[-1, 1], \quad U \perp X. \quad (1020)$$

Consequently ρ is an infinite convolution of normalized boxes,

$$\rho = \bigstar_{k \geq 1} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]} \right), \quad (1021)$$

and the finite truncation $X_N = \sum_{k=1}^N 2^{-k} U_k$ has a compactly supported polynomial-spline density ρ_N . The refinement and differential identities can be written

$$\rho(x) = \int_{-1}^1 \rho(2x - u) \, du, \quad (1022)$$

$$\rho'(x) = 2(\rho(2x + 1) - \rho(2x - 1)). \quad (1023)$$

They imply the familiar Fabius functional-differential relations after the affine change $x = 2u - 1$.

2.2 Finite Thue–Morse splines

Let

$$a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \quad 0 \leq j < 2^N.$$

The finite convolution density has the exact truncated-power form

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \epsilon_j (x + a_{N,j})_+^{N-1}. \quad (1024)$$

The cancellation of all polynomial pieces below degree N is a Prouhet–Thue–Morse phenomenon. Equivalently,

$$P_N(z) := \prod_{r=0}^{N-1} (1 - z^{2^r}) = \sum_{j=0}^{2^N-1} \epsilon_j z^j. \quad (1025)$$

The corresponding unsigned block has

$$\sum_{j=0}^{2^N-1} \tau_j z^j = \frac{1}{2} \left(\frac{1 - z^{2^N}}{1 - z} - P_N(z) \right). \quad (1026)$$

2.3 Fourier, hyperbolic, and canonical products

Independence in (1015) gives

$$\Phi(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi} \left(\frac{\xi}{2^j} \right), \quad (1027)$$

$$M(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}. \quad (1028)$$

Rearranging Euler products yields

$$\Phi(\xi) = \prod_{m=1}^{\infty} \left(1 - \frac{\xi^2}{m^2} \right)^{1+\nu_2(m)}, \quad (1029)$$

$$M(t) = \prod_{m=1}^{\infty} \left(1 + \frac{t^2}{4\pi^2 m^2} \right)^{1+\nu_2(m)}. \quad (1030)$$

Thus Φ has a zero at every nonzero integer m , of multiplicity

$$d_m = 1 + \nu_2(|m|). \quad (1031)$$

Its logarithmic derivative is the normally convergent meromorphic series

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^2 - z^2}. \quad (1032)$$

The audited repository develops several equivalent cotangent, tangent, digamma, and Hurwitz-zeta rearrangements of this expression.

2.4 Moments, cumulants, Bell polynomials, and Bernoulli numbers

Write $\mu_n = \mathbb{E}X^n$. Symmetry gives $\mu_{2m+1} = 0$. Applying (1020) gives the exact triangular recurrence

$$\mu_n = \frac{1}{2^n - 1} \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} \binom{n}{k} \frac{\mu_{n-k}}{k+1} \quad (n \geq 2 \text{ even}). \quad (1033)$$

The cumulants are

$$\kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0, \quad (1034)$$

and therefore

$$\mu_n = Y_n(\kappa_1, \dots, \kappa_n), \quad (1035)$$

where Y_n is the complete exponential Bell polynomial. These formulas make all moments rational and expose the Bernoulli and dyadic denominator structure.

2.5 Cauchy, logarithmic, determinant, and continued-fraction forms

For $z \notin [-1, 1]$, the Cauchy transform has the moment expansion

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1, \quad (1036)$$

and, for $\Re z > 1$, the Laplace form

$$R(z) = \int_0^{\infty} e^{-zt} M(t) dt. \quad (1037)$$

Differentiating under the integral or using (1023) gives

$$R'(z) = 2(R(2z+1) - R(2z-1)). \quad (1038)$$

The finite spline transform possesses the logarithmic formula

$$R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \epsilon_j (z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}), \quad (1039)$$

with polynomial ambiguities canceled by Prouhet identities. The living representation report also gives Fredholm determinant and Jacobi/Stieltjes continued-fraction forms. If P_n are the monic symmetric orthogonal polynomials,

$$P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \quad \beta_n > 0, \quad (1040)$$

then

$$R(z) = \frac{1}{z - \frac{\beta_1}{z - \frac{\beta_2}{z - \ddots}}}. \quad (1041)$$

2.6 Fabius and inverse-function transform forms

The pushforward identity $Y = (X + 1)/2$ yields

$$\int_0^1 e^{-2\pi i \xi u} dF(u) = e^{-\pi i \xi} \Phi(\xi/2). \quad (1042)$$

For $\xi \neq 0$, integration by parts gives

$$\int_0^1 F(u) e^{-2\pi i \xi u} du = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (1043)$$

Since $I(V)$ has distribution F when $V \sim \text{Unif}[0, 1]$,

$$\int_0^1 e^{-2\pi i \xi I(v)} dv = e^{-\pi i \xi} \Phi(\xi/2). \quad (1044)$$

The repository further contains finite dyadic antiderivatives, complex powers, fractional Volterra operators, Mellin/Newton moments, inverse primitives and higher derivatives, Stieltjes/log-potential transforms, autocorrelation identities, and endpoint asymptotics.

Repository baseline

Equations (1021)–(1044) form the input atlas. Later sections do not claim these identities as new; they reorganize them through a unit-circle probability lift and derive consequences absent from the audited subtree.

3 Probability–Chebyshev coordinates

Definition 3.1. For $n \geq 0$, define the *probability–Chebyshev moment*

$$c_n := \mathbb{E}[T_n(X)] = \int_{-1}^1 T_n(x) \rho(x) dx, \quad (1045)$$

where $T_n(\cos \theta) = \cos(n\theta)$.

Because ρ is even, $c_{2m+1} = 0$. These numbers are not the unweighted Chebyshev coefficients of the density: the integration measure in (1045) is $\rho(x) dx$, not $dx/\sqrt{1-x^2}$.

Theorem 3.2 (Exact Bell–Bernoulli formula). For $m \geq 1$,

$$c_{2m} = m \sum_{k=0}^m (-1)^{m-k} \frac{(m+k-1)!}{(m-k)!(2k)!} 2^{2k} \mu_{2k}. \quad (1046)$$

Equivalently, with κ_{2j} as in (1034),

$$c_{2m} = m \sum_{k=0}^m (-1)^{m-k} \frac{(m+k-1)!}{(m-k)!(2k)!} 2^{2k} Y_{2k}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2k}). \quad (1047)$$

In particular every c_n is rational.

Proof. For $m \geq 1$,

$$T_{2m}(x) = m \sum_{k=0}^m (-1)^{m-k} \frac{(m+k-1)!}{(m-k)!(2k)!} (2x)^{2k}.$$

Integrating termwise against $\rho(x) dx$ proves (1046); substituting (1035) and (1034) proves (1047). $\square$

The first terms are

$$\begin{aligned} c_0 &= 1, & c_2 &= -\frac{7}{9}, & c_4 &= \frac{227}{675}, & c_6 &= -\frac{11237}{297675}, \\ c_8 &= -\frac{8241761}{227721375}, & c_{10} &= \frac{171586753}{8135075025}, & c_{12} &= -\frac{472381023339557}{45472425490366875}. \end{aligned}$$

The repeated negative sign at c_6, c_8 warns against a naive alternating-sign conjecture.

Proposition 3.3 (Generating function). *For $|w| < 1$,*

$$G(w) := \sum_{n=0}^{\infty} c_n w^n = \int_{-1}^1 \frac{1-wx}{1-2wx+w^2} \rho(x) dx, \quad (1048)$$

$$= 1 + \sum_{m=1}^{\infty} c_{2m} w^{2m}. \quad (1049)$$

Proof. Integrate the standard generating function $\sum_{n \geq 0} T_n(x) w^n = (1-wx)/(1-2wx+w^2)$ against the probability measure. $\square$

For any $0 < r < 1$, Cauchy coefficient extraction gives

$$c_n = \frac{1}{2\pi r^n} \int_0^{2\pi} G(re^{i\theta}) e^{-in\theta} d\theta. \quad (1050)$$

This will become an ordinary circle Fourier coefficient in the next section.

Repo-relative new result

The sequence c_n is one rational coordinate system controlling a positive-real Carathéodory function, the Joukowski pullback of R , a weighted Chebyshev expansion of ρ , phase series for F and F^{-1} , Bessel expansions of Φ and M , and rational OPUC/CMV data. This unification is the central structural result of the report.

4 Unit-circle lift and Carathéodory–Joukowski bridge

4.1 A canonical probability measure on the circle

Define a probability measure ν on $\mathbb{T}$ by

$$d\nu(\theta) = \frac{1}{2} \rho(\cos \theta) |\sin \theta| d\theta, \quad 0 \leq \theta < 2\pi. \quad (1051)$$

The map $\theta \mapsto x = \cos \theta$ pushes ν forward to $\rho(x) dx$. Moreover, ν is invariant under conjugation and the half-turn $\zeta \mapsto -\zeta$.

Proposition 4.1 (Fourier moments of the lift). *For every $n \in \mathbb{Z}$,*

$$\int_{\mathbb{T}} \zeta^{-n} d\nu(\zeta) = c_{|n|}. \quad (1052)$$

Proof. Imaginary parts cancel under conjugation, while $\cos(n \arccos x) = T_n(x)$. The two preimages of each $x \in (-1, 1)$ contribute equally. $\square$

Definition 4.2. The Carathéodory function of ν is

$$C(w) = \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} d\nu(\zeta), \quad w \in \mathbb{D}. \quad (1053)$$

It is analytic in $\mathbb{D}$, satisfies $C(0) = 1$, and has positive real part.

Theorem 4.3 (Carathéodory representation). For $|w| < 1$,

$$C(w) = \int_{-1}^1 \frac{1-w^2}{1-2wx+w^2} \rho(x) dx, \quad (1054)$$

$$= 1 + 2 \sum_{n=1}^{\infty} c_n w^n = 1 + 2 \sum_{m=1}^{\infty} c_{2m} w^{2m}, \quad (1055)$$

$$= 2G(w) - 1. \quad (1056)$$

Proof. Pair $e^{i\theta}$ and $e^{-i\theta}$ in (1053); their average kernel is $(1-w^2)/(1-2w \cos \theta + w^2)$. The geometric expansion $(\zeta + w)/(\zeta - w) = 1 + 2 \sum_{n \geq 1} w^n \zeta^{-n}$ and Proposition 4.1 give the series. $\square$

The positive harmonic regularization

$$W_r(\theta) := \Re C(re^{i\theta}) = 1 + 2 \sum_{m=1}^{\infty} c_{2m} r^{2m} \cos(2m\theta) \quad (1057)$$

preserves positivity for every $0 \leq r < 1$, unlike an arbitrary boundary truncation.

4.2 Joukowski uniformization

Let

$$\mathcal{J}(w) = \frac{1}{2}(w + w^{-1}), \quad 0 < |w| < 1, \quad (1058)$$

with inverse branch

$$w(z) = z - \sqrt{z^2 - 1}, \quad \sqrt{z^2 - 1} \sim z \quad (z \rightarrow \infty). \quad (1059)$$

Theorem 4.4 (Exact Joukowski bridge). For $0 < |w| < 1$,

$$C(w) = \frac{1-w^2}{2w} R(\mathcal{J}(w)). \quad (1060)$$

Equivalently, for $z \in \mathbb{C} \setminus [-1, 1]$,

$$R(z) = \frac{2w(z)}{1-w(z)^2} C(w(z)), \quad (1061)$$

$$= \frac{1 + 2 \sum_{m=1}^{\infty} c_{2m} w(z)^{2m}}{\sqrt{z^2 - 1}}. \quad (1062)$$

Proof. Since $\mathcal{J}(w) - x = (1-2wx+w^2)/(2w)$, compare (1054) with the definition of R . On the chosen branch, $\sqrt{z^2 - 1} = (1-w^2)/(2w)$. $\square$

The ordinary moment series is centered at infinity in powers of z^{-1} . By contrast, (1062) uses the conformal coordinate of the slit plane, so it remains geometrically adapted even near the support.

4.3 Boundary recovery of ρ

Set

$$W(\theta) = \pi \rho(\cos \theta) |\sin \theta|. \quad (1063)$$

This is the density of ν relative to $d\theta/(2\pi)$. Endpoint flatness makes it a C^∞ , π -periodic function.

Theorem 4.5 (Weighted Chebyshev boundary series). *For every θ ,*

$$W(\theta) = \Re C(e^{i\theta}) = 1 + 2 \sum_{m=1}^{\infty} c_{2m} \cos(2m\theta), \quad (1064)$$

with convergence in C^k for every fixed k . Consequently, for $|x| < 1$,

$$\boxed{\rho(x) = \frac{1 + 2 \sum_{m=1}^{\infty} c_{2m} T_{2m}(x)}{\pi \sqrt{1 - x^2}}.} \quad (1065)$$

Proof. The boundary theorem for Poisson integrals gives $\Re C(e^{i\theta}) = W(\theta)$. Smoothness implies uniform convergence of all fixed derivatives. Substitute $x = \cos \theta$. $\square$

The harmonic conjugate is

$$\Im C(e^{i\theta}) = 2 \sum_{m=1}^{\infty} c_{2m} \sin(2m\theta), \quad (1066)$$

$$= \text{PV} \int_0^{2\pi} \cot\left(\frac{\theta - \varphi}{2}\right) d\nu(\varphi). \quad (1067)$$

Thus one disk-analytic function packages the density and its circular Hilbert transform.

4.4 Phase series for F and F^{-1}

For $0 < u < 1$, put

$$\theta(u) = \arccos(2u - 1) \in (0, \pi). \quad (1068)$$

Theorem 4.6 (Fabius phase series). *For $0 \leq u \leq 1$,*

$$\boxed{F(u) = 1 - \frac{\theta(u)}{\pi} - \frac{1}{\pi} \sum_{m=1}^{\infty} \frac{c_{2m}}{m} \sin(2m\theta(u)).} \quad (1069)$$

For $0 < u < 1$,

$$F'(u) = \frac{\Re C(e^{i\theta(u)})}{\pi \sqrt{u(1-u)}} = \frac{1 + 2 \sum_{m \geq 1} c_{2m} T_{2m}(2u - 1)}{\pi \sqrt{u(1-u)}}. \quad (1070)$$

Proof. Substitute $x = \cos \varphi$ in $F(u) = \int_{-1}^{2u-1} \rho(x) dx$: $F(u) = \pi^{-1} \int_{\theta(u)}^{\pi} W(\varphi) d\varphi$. Integrate (1064) termwise and differentiate. $\square$

Corollary 4.7 (Inverse phase equation). *For each $v \in (0, 1)$, let $\Theta(v) \in (0, \pi)$ be the unique solution of*

$$v = 1 - \frac{\Theta}{\pi} - \frac{1}{\pi} \sum_{m=1}^{\infty} \frac{c_{2m}}{m} \sin(2m\Theta). \quad (1071)$$

Then

$$I(v) = \frac{1 + \cos \Theta(v)}{2}, \quad (1072)$$

$$I'(v) = \frac{\pi \sqrt{I(v)(1 - I(v))}}{\Re C(e^{i\Theta(v)})}. \quad (1073)$$

The explicit factors $1/\sqrt{u(1-u)}$ do not create endpoint singularities in F' : the numerator W is flat enough to cancel every algebraic divergence. In the inverse formula, $1/W$ isolates the true rapid endpoint growth and is compatible with the Lambert– W asymptotics in the repository.

4.5 Disk form of the resolvent differential equation

Set

$$A(w) = \frac{2wC(w)}{1-w^2} = R(\mathcal{J}(w)) \quad (1074)$$

and, as germs at $w = 0$,

$$q_{\pm}(w) = 2\mathcal{J}(w) \pm 1 - \sqrt{(2\mathcal{J}(w) \pm 1)^2 - 1}, \quad (1075)$$

choosing square roots so that $q_{\pm}(w) \rightarrow 0$.

Proposition 4.8 (Disk composition equation). *Near $w = 0$,*

$$\frac{d}{dw} \left(\frac{2wC(w)}{1-w^2} \right) = 4\mathcal{J}'(w) \left[\frac{q_+(w)C(q_+(w))}{1-q_+(w)^2} - \frac{q_-(w)C(q_-(w))}{1-q_-(w)^2} \right]. \quad (1076)$$

Proof. Differentiate $A(w) = R(\mathcal{J}(w))$, apply (1038), and use $R(\mathcal{J}(q)) = 2qC(q)/(1-q^2)$ for each inverse branch. $\square$

5 Bessel series equal to the dyadic products

The Jacobi–Anger expansions

$$e^{-iz \cos \theta} = J_0(z) + 2 \sum_{n=1}^{\infty} (-i)^n J_n(z) \cos(n\theta), \quad (1077)$$

$$e^{z \cos \theta} = I_0(z) + 2 \sum_{n=1}^{\infty} I_n(z) \cos(n\theta) \quad (1078)$$

turn the Fourier moments of ν into Bessel coefficients.

Theorem 5.1 (Bessel–sinc–canonical product identity). *For every $\xi \in \mathbb{C}$,*

$$\boxed{\Phi(\xi) = J_0(2\pi\xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(2\pi\xi).} \quad (1079)$$

The series is locally uniformly convergent, and

$$J_0(2\pi\xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(2\pi\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi} \left(\frac{\xi}{2^j} \right) \quad (1080)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{\xi^2}{n^2} \right)^{1+\nu_2(n)}. \quad (1081)$$

Proof. Under ν , $X = \cos \Theta$. Integrate (1077) with $z = 2\pi\xi$; odd moments vanish and $(-i)^{2m} = (-1)^m$. The product equalities are (1027) and (1029). $\square$

For every $r \geq 0$, termwise differentiation gives

$$\Phi^{(r)}(\xi) = (2\pi)^r \left[J_0^{(r)}(2\pi\xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}^{(r)}(2\pi\xi) \right]. \quad (1082)$$

At an integer zero $N \neq 0$, the first $1 + \nu_2(N)$ equations in this derivative ladder become exact Bessel cancellation identities. In particular,

$$J_0(2\pi N) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(2\pi N) = 0. \quad (1083)$$

Theorem 5.2 (Modified-Bessel–hyperbolic identity). *For every $t \in \mathbb{C}$,*

$$M(t) = I_0(t) + 2 \sum_{m=1}^{\infty} c_{2m} I_{2m}(t). \quad (1084)$$

Moreover,

$$I_0(t) + 2 \sum_{m=1}^{\infty} c_{2m} I_{2m}(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} \quad (1085)$$

$$= \prod_{n=1}^{\infty} \left(1 + \frac{t^2}{4\pi^2 n^2} \right)^{1+\nu_2(n)}. \quad (1086)$$

Proof. Integrate (1078) against ν and invoke the product formulas for M . $\square$

5.1 Laplace–Bessel–Joukowski chain

For $\Re z > 1$,

$$\int_0^{\infty} e^{-zt} I_n(t) dt = \frac{(z - \sqrt{z^2 - 1})^n}{\sqrt{z^2 - 1}}. \quad (1087)$$

Theorem 5.3 (Three-level transform identity). *For $\Re z > 1$, with $w = z - \sqrt{z^2 - 1}$,*

$$R(z) = \int_0^{\infty} e^{-zt} \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} dt \quad (1088)$$

$$= \int_0^{\infty} e^{-zt} \left[I_0(t) + 2 \sum_{m=1}^{\infty} c_{2m} I_{2m}(t) \right] dt \quad (1089)$$

$$= \frac{1 + 2 \sum_{m=1}^{\infty} c_{2m} w^{2m}}{\sqrt{z^2 - 1}}. \quad (1090)$$

Proof. Insert (1028) and (1084) into (1037); use dominated convergence and (1087). $\square$

5.2 Fabius and inverse-Fabius transforms

For $\xi \neq 0$,

$$\int_0^1 F(u) e^{-2\pi i \xi u} du = \frac{e^{-\pi i \xi} [J_0(\pi \xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(\pi \xi)] - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (1091)$$

The value at $\xi = 0$ is $1/2$. For the inverse pushforward,

$$\int_0^1 e^{-2\pi i \xi I(v)} dv = e^{-\pi i \xi} \left[J_0(\pi \xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(\pi \xi) \right]. \quad (1092)$$

Repo-relative new result

Equations (1080), (1081), (1085), and (1086) are direct series–product bridges: the same rational sequence c_{2m} expands both the Fourier image and the moment generating function in Bessel bases. No corresponding bridge was located in the audited repository.

6 Schur, Verblunsky, CMV, and Toeplitz representations

The Carathéodory function places the Rvachev measure in the theory of orthogonal polynomials on the unit circle (OPUC) [5, 6]. This creates a rational coefficient system distinct from, but explicitly coupled to, the repository’s Jacobi coefficients.

6.1 Schur algorithm and rationality

Define

$$f_0(w) = \frac{C(w) - 1}{w(C(w) + 1)}, \quad C(w) = \frac{1 + wf_0(w)}{1 - wf_0(w)}. \quad (1093)$$

Recursively set

$$\alpha_n = f_n(0), \quad f_{n+1}(w) = \frac{f_n(w) - \alpha_n}{w(1 - \alpha_n f_n(w))}. \quad (1094)$$

All functions and coefficients are real under the present symmetries.

Theorem 6.1 (Rational Verblunsky coefficients). *The Schur parameters α_n are the Verblunsky coefficients of ν . They satisfy*

$$\alpha_n \in \mathbb{Q} \cap (-1, 1), \quad \alpha_{2n} = 0 \quad (n \geq 0). \quad (1095)$$

The odd coefficients are obtained by exact rational operations from the moments c_{2m} .

Proof. The first assertion is the standard Carathéodory–Schur correspondence. Rationality follows because the Taylor coefficients of C are rational and every Schur step uses rational operations. The half-turn invariance $\nu(E) = \nu(-E)$ forces all odd circle moments and, equivalently, every even-index Verblunsky coefficient to vanish. Infinite support gives $|\alpha_n| < 1$. $\square$

The first nonzero values are

$$\begin{aligned} \alpha_1 &= -\frac{7}{9}, & \alpha_3 &= -\frac{17}{25}, & \alpha_5 &= -\frac{1849}{3087}, \\ \alpha_7 &= -\frac{3569379}{6492691}, & \alpha_9 &= -\frac{445076883510763}{886612917691413}, & \alpha_{11} &= -\frac{594580974953911037169143}{1270428317522023221443375}. \end{aligned}$$

The exact program computes much longer coefficients without floating-point contamination.

6.2 Wall–Schur continued fraction

Solving (1094) for f_n gives

$$f_n(w) = \frac{\alpha_n + wf_{n+1}(w)}{1 + \alpha_n wf_{n+1}(w)}. \quad (1096)$$

Iterating this Möbius recursion and inserting it into (1093) produces a Wall–Schur continued fraction for C . Truncating after N parameters by setting $f_N = 0$ gives a rational C_N with positive real part in $\mathbb{D}$. Therefore

$$R_N^{\text{Schur}}(z) = \frac{C_N(w(z))}{\sqrt{z^2 - 1}} \quad (1097)$$

is a positivity-preserving rational/conformal approximation to the Cauchy transform. It is geometrically different from truncating the real-line Jacobi fraction (1041).

6.3 Geronimus bridge to Jacobi coefficients

For the symmetric pushforward under $x = (\zeta + \zeta^{-1})/2$, the Geronimus relation simplifies because $\alpha_{2n} = 0$.

Theorem 6.2 (Exact Jacobi–Verblunsky conversion). *With $\alpha_{-1} = -1$, the monic Jacobi coefficients satisfy*

$$\boxed{\beta_{n+1} = \frac{1}{4}(1 - \alpha_{2n-1})(1 + \alpha_{2n+1})} \quad (n \geq 0). \quad (1098)$$

Equivalently,

$$\alpha_{2n+1} = \frac{4\beta_{n+1}}{1 - \alpha_{2n-1}} - 1. \quad (1099)$$

The first values

$$\beta_1 = \frac{1}{9}, \quad \beta_2 = \frac{32}{225}, \quad \beta_3 = \frac{619}{3675}, \quad \beta_4 = \frac{835232}{4640643} \quad (1100)$$

reproduce $\alpha_1, \alpha_3, \alpha_5, \alpha_7$. In the accompanying code, the β_n are generated by exact Gram–Schmidt from μ_n , while a direct formal Schur algorithm independently generates α_n ; the pipelines agree exactly through the checked indices.

6.4 CMV resolvent

Let $\mathcal{C}$ be the CMV matrix associated with $(\alpha_n)_{n \geq 0}$ on $\ell^2(\mathbb{N}_0)$, and let $e_0 = (1, 0, \dots)$. Its spectral measure at e_0 is ν .

Theorem 6.3 (CMV representation). *For $|w| < 1$,*

$$C(w) = \langle e_0, (\mathcal{C} + w)(\mathcal{C} - w)^{-1}e_0 \rangle. \quad (1101)$$

Consequently,

$$R(\mathcal{J}(w)) = \frac{2w}{1 - w^2} \langle e_0, (\mathcal{C} + w)(\mathcal{C} - w)^{-1}e_0 \rangle. \quad (1102)$$

Proof. Apply the spectral theorem to multiplication by ζ in $L^2(\nu)$, then express that unitary operator in the CMV basis. The second formula follows from (1060). $\square$

6.5 Toeplitz determinants and Szegő product

Let

$$D_N = \det[c_{j-k}]_{j,k=0}^{N-1}, \quad c_{-n} = c_n. \quad (1103)$$

These determinants are positive rationals.

Proposition 6.4 (Exact determinant product). *For $N \geq 1$,*

$$D_N = \prod_{j=0}^{N-2} (1 - \alpha_j^2)^{N-1-j}. \quad (1104)$$

Only odd j contribute.

The weight $W(\theta)$ is positive almost everywhere and $\log W \in L^1(\mathbb{T})$. The latter follows from the established endpoint asymptotics: although $\rho(\cos \theta)$ is superflat, its logarithm grows only polylogarithmically in $1/|\theta|$, which is integrable against $d\theta$. Szegő's theorem gives

$$\prod_{n=0}^{\infty} (1 - \alpha_n^2) = \exp \left(\frac{1}{2\pi} \int_0^{2\pi} \log W(\theta) d\theta \right). \quad (1105)$$

Rakhmanov's theorem also yields $\alpha_n \rightarrow 0$, because $W > 0$ almost everywhere.

Repo-relative new result

Equations (1101), (1102), (1104), and (1105) give operator, determinant, and entropy-product representations of the Rvachev law. The associated CMV coefficients are exact rationals and are linked to the known Jacobi coefficients by (1098).

7 Spectral, contour, and monodromy formulas for Thue–Morse

This section derives representations for both ϵ_n and τ_n by treating the integer zeros of Φ as spectral data. These formulas differ from the repository's coefficient, digit, half-integer-sign, trigonometric, and finite-product formulas.

7.1 Zero-counting encodes the full binary digit sum

Let

$$N_{\Phi}(n) := \sum_{m=1}^n (1 + \nu_2(m)), \quad N_{\Phi}(0) = 0. \quad (1106)$$

This is the number of positive zeros of Φ in $(0, n + 1/2)$, counted with multiplicity.

Theorem 7.1 (Spectral digit-sum identity). *For every $n \geq 0$,*

$$N_{\Phi}(n) = n + \nu_2(n!) = 2n - s_2(n). \quad (1107)$$

Thus the complete binary digit sum, not only its parity, is recovered by

$$s_2(n) = 2n - N_{\Phi}(n). \quad (1108)$$

Proof. The first equality follows from (1031). Legendre's formula for the 2-adic valuation gives $\nu_2(n!) = n - s_2(n)$. $\square$

Let Γ_n be any positively oriented contour enclosing precisely the positive integer zeros $1, \dots, n$, and no other zeros of Φ .

Corollary 7.2 (Argument-principle representations). *For $n \geq 0$,*

$$s_2(n) = 2n - \frac{1}{2\pi i} \oint_{\Gamma_n} \frac{\Phi'(z)}{\Phi(z)} dz, \quad (1109)$$

$$\epsilon_n = \exp \left(\frac{1}{2} \oint_{\Gamma_n} \frac{\Phi'(z)}{\Phi(z)} dz \right), \quad (1110)$$

$$\tau_n = \sin^2 \left(\frac{1}{4i} \oint_{\Gamma_n} \frac{\Phi'(z)}{\Phi(z)} dz \right). \quad (1111)$$

Proof. The argument principle gives the contour integral as $2\pi i N_\Phi(n)$. Since $N_\Phi(n) \equiv s_2(n) \pmod{2}$, exponentiation gives ϵ_n , while $\sin^2(\pi N_\Phi(n)/2)$ is its unsigned parity. $\square$

Substitution of (1032) turns the contour formulas into explicit meromorphic series integrals. Locally at a positive integer,

$$\operatorname{Res}_{z=n} \frac{\Phi'(z)}{\Phi(z)} = 1 + \nu_2(n), \quad (1112)$$

and hence

$$\frac{\epsilon_n}{\epsilon_{n-1}} = \exp \left(\pi i \operatorname{Res}_{z=n} \frac{\Phi'(z)}{\Phi(z)} \right) = (-1)^{1+\nu_2(n)}. \quad (1113)$$

Thus the local zero multiplicity records the carry chain when passing from $n-1$ to n .

7.2 Logarithmic monodromy of finite resolvents

The finite resolvent (1039) has branch points

$$z_{N,j} = -a_{N,j}, \quad 0 \leq j < 2^N. \quad (1114)$$

Let $\Delta_j R_N$ denote the change after one positive analytic continuation around $z_{N,j}$, holding all other logarithmic branches fixed.

Theorem 7.3 (Monodromy recovery of a finite Thue–Morse word). *For z near $z_{N,j}$,*

$$\Delta_j R_N(z) = 2\pi i \frac{2^{N(N-1)/2}}{(N-1)!} \epsilon_j (z - z_{N,j})^{N-1}. \quad (1115)$$

Consequently,

$$\epsilon_j = \frac{1}{2\pi i 2^{N(N-1)/2}} \left. \frac{d^{N-1}}{dz^{N-1}} \Delta_j R_N(z) \right|_{z=z_{N,j}}, \quad (1116)$$

and $\tau_j = (1 - \epsilon_j)/2$.

Proof. Only the j -th logarithm changes, by $2\pi i$, around $z_{N,j}$. This proves (1115). Differentiating $N-1$ times at the branch point cancels $(N-1)!$ and isolates ϵ_j . $\square$

The finite Thue–Morse word is therefore the normalized monodromy representation of a single multivalued analytic function. This interpretation is stronger than saying that ϵ_j merely occurs as an algebraic coefficient: the word is recovered from the analytic continuation operators themselves.

7.3 Abel–Poisson regularization of the Rademacher product

Define, for $0 < r < 1$,

$$H_r(\theta) := \frac{4}{\pi} \sum_{\ell=0}^{\infty} \frac{r^{2\ell+1}}{2\ell+1} \sin((2\ell+1)\theta) \quad (1117)$$

$$= \frac{2}{\pi} \operatorname{atan} 2(2r \sin \theta, 1 - r^2). \quad (1118)$$

This is the Abel/Poisson regularization of $\operatorname{sgn}(\sin \theta)$, and $H_r(\theta) \rightarrow \operatorname{sgn}(\sin \theta)$ away from multiples of π .

For $n \geq 1$, put $J_n = \lfloor \log_2(n + 1/2) \rfloor$; for $n = 0$, interpret the product below as empty.

Theorem 7.4 (Smooth finite-product representation). *For every $n \geq 0$,*

$$\epsilon_n = \lim_{r \uparrow 1} \prod_{j=0}^{J_n} H_r \left(\frac{\pi(n + 1/2)}{2^j} \right), \quad (1119)$$

$$\tau_n = \frac{1}{2} \left[1 - \lim_{r \uparrow 1} \prod_{j=0}^{J_n} H_r \left(\frac{\pi(n + 1/2)}{2^j} \right) \right]. \quad (1120)$$

Proof. The signed Thue–Morse sequence has the Rademacher sign product

$$\epsilon_n = \prod_{j \geq 0} \operatorname{sgn} \sin \left(\frac{\pi(n + 1/2)}{2^j} \right).$$

For $j > J_n$, the angle lies in $(0, \pi)$, so the factor is $+1$; only finitely many factors matter. Replace each by its Abel limit. $\square$

The accompanying computation shows uniform convergence over the first 64 symbols: the maximum errors for $r = 0.9, 0.99, 0.999, 0.9999$ are approximately 0.937, 0.237, 0.0257, 0.00259, respectively. The formula is especially useful when a smooth analytic surrogate is preferable to a discontinuous sign function.

7.4 A cardinal Bessel–Thue–Morse expansion

Since ρ is supported in $[-1, 1]$, the Paley–Wiener/Shannon formula at spacing $1/2$ is

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k/2) \operatorname{sinc}_\pi(2z - k). \quad (1121)$$

All nonzero integer samples vanish. If $A_n = |\Phi(n + 1/2)|$, the existing half-integer sign law $\operatorname{sgn} \Phi(n + 1/2) = \epsilon_n$ therefore yields

$$\Phi(z) = \operatorname{sinc}_\pi(2z) + \sum_{n=0}^{\infty} \epsilon_n A_n [\operatorname{sinc}_\pi(2z - (2n + 1)) + \operatorname{sinc}_\pi(2z + (2n + 1))]. \quad (1122)$$

This formula is included as a derived synthesis rather than a primary novelty claim, because the latest repository report already develops a broad cardinal-reproduction ladder.

Repo-relative new result

The contour formulas recover the full digit sum and both Thue–Morse normalizations from the zero divisor of Φ . The monodromy formula recovers each finite word from analytic continuation of R_N , while the Abel formula replaces discontinuous Rademacher signs by finite products of harmonic functions.

8 Compact representation atlas

The following table collects the principal forms used in the report. “Baseline” means already present in the audited repository; “new” means derived here and not located in the repository; “synthesis” means a new combination of known ingredients whose components are individually present.

Table 22: Representation atlas.

Object/type	Representation	Status and use
X , random series	$X = \sum_{k \geq 1} 2^{-k} U_k, U_k \sim \text{Unif}[-1, 1]$	Baseline; probability model
X , fixed point	$X \stackrel{d}{=} (X + U)/2$	Baseline; source of recurrences
ρ , convolution	$\rho = *_{k \geq 1} (2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]})$	Baseline; compact support and smoothness
ρ_N , spline	$\rho_N = 2^{N(N-1)/2} (N-1)!^{-1} \sum_{j < 2^N} \epsilon_j (x + a_{N,j})_+^{N-1}$	Baseline; finite exact form
ρ , refinement	$\rho(x) = \int_{-1}^1 \rho(2x - u) du$	Baseline; scaling
ρ , derivative	$\rho'(x) = 2[\rho(2x + 1) - \rho(2x - 1)]$	Baseline; differential-difference law
Φ , sinc product	$\prod_{j \geq 0} \text{sinc}_\pi(\xi/2^j)$	Baseline; Fourier image
Φ , zero product	$\prod_{n \geq 1} (1 - \xi^2/n^2)^{1+\nu_2(n)}$	Baseline; dyadic zero divisor
Φ , Bessel series	$J_0(2\pi\xi) + 2 \sum_{m \geq 1} (-1)^m c_{2m} J_{2m}(2\pi\xi)$	New; series–product bridge
$\Phi^{(r)}$	Termwise derivatives of the Bessel series	New; zero-multiplicity sum rules
M , hyperbolic product	$\prod_{k \geq 1} \sinh(t/2^k)/(t/2^k)$	Baseline; moment generating function
M , canonical product	$\prod_{n \geq 1} (1 + t^2/(4\pi^2 n^2))^{1+\nu_2(n)}$	Baseline rearrangement
M , Bessel series	$I_0(t) + 2 \sum_{m \geq 1} c_{2m} I_{2m}(t)$	New; modified-Bessel bridge
μ_n	Triangular rational recurrence (1033)	Baseline; exact moments
μ_n	Complete Bell polynomial in Bernoulli cumulants	Baseline; Bell–Bernoulli form
c_{2m}	Finite transform of μ_{2k} , (1046)	New; exact probability–Chebyshev coordinates
$G(w)$	$\int (1 - wx)(1 - 2wx + w^2)^{-1} \rho(x) dx$	New; moment generating function in degree
$C(w)$	$\int_{\mathbb{T}} (\zeta + w)(\zeta - w)^{-1} d\nu$	New; Carathéodory transform
$C(w)$, series	$1 + 2 \sum_{m \geq 1} c_{2m} w^{2m}$	New; positive-real disk series
$R(z)$, moments	$\sum_{n \geq 0} \mu_n z^{-n-1}$	Baseline; expansion at infinity
$R(z)$, Laplace	$\int_0^\infty e^{-zt} M(t) dt$	Baseline; integral transform
$R(z)$, Joukowski	$[1 + 2 \sum c_{2m} w(z)^{2m}]/\sqrt{z^2 - 1}$	New; conformal series
$R(z)$, Bessel–Laplace	$\int_0^\infty e^{-zt} [I_0(t) + 2 \sum c_{2m} I_{2m}(t)] dt$	New synthesis
$R(z)$, Jacobi CF	$1/(z - \beta_1/(z - \beta_2/(z - \dots)))$	Baseline; real-line continued fraction
$R(z)$, Schur/CMV	$C(w(z))/\sqrt{z^2 - 1}$, with C a CMV resolvent element	New; unitary operator representation
$\rho(x)$, weighted Cheb	$[1 + 2 \sum c_{2m} T_{2m}(x)]/[\pi\sqrt{1 - x^2}]$	New; boundary Fourier series
$F(u)$, phase series	$1 - \theta/\pi - \pi^{-1} \sum c_{2m} \sin(2m\theta)/m$	New; $\theta = \arccos(2u - 1)$

Object/type	Representation	Status and use
$F'(u)$	$\Re C(e^{i\theta})/[\pi\sqrt{u(1-u)}]$	New; positive boundary quotient
$I(v)$, phase	$I = (1 + \cos \Theta)/2$, with Θ solving (1071)	New; inverse trigonometric representation
$I'(v)$	$\pi\sqrt{I(1-I)}/\Re C(e^{i\Theta})$	New; inverse derivative
F , Fourier	$[e^{-\pi i\xi}\Phi(\xi/2) - e^{-2\pi i\xi}]/(2\pi i\xi)$	Baseline
F , Bessel Fourier	Insert the J_{2m} -series for $\Phi(\xi/2)$	New synthesis
I , pushforward	$\int_0^1 e^{-2\pi i\xi I(v)} dv = e^{-\pi i\xi}\Phi(\xi/2)$	Baseline
I , Bessel pushforward	Insert the J_{2m} -series in the preceding formula	New synthesis
OPUC data	Schur recursion with rational $\alpha_{2m} = 0$	New; exact unit-circle coefficients
Jacobi–OPUC	$\beta_{n+1} = \frac{1}{4}(1 - \alpha_{2n-1})(1 + \alpha_{2n+1})$	New application of Geronimus
Toeplitz	$D_N = \prod_{j=0}^{N-2} (1 - \alpha_j^2)^{N-1-j}$	New determinant representation
Szegő product	$\prod(1 - \alpha_n^2) = \exp[(2\pi)^{-1} \int \log W]$	New entropy/product invariant
$s_2(n)$, spectral	$2n - (2\pi i)^{-1} \oint \Phi'/\Phi$	New; full digit sum from zero divisor
ϵ_n , contour	$\exp[\frac{1}{2} \oint \Phi'/\Phi]$	New; signed Thue–Morse
τ_n , contour	$\sin^2[(4i)^{-1} \oint \Phi'/\Phi]$	New; unsigned Thue–Morse
Finite TM word	Normalized $(N-1)$ -st derivative of logarithmic monodromy	New; analytic-continuation encoding
ϵ_n , Abel	Boundary limit of a finite product of H_r -factors	New; smooth harmonic regularization
Φ , cardinal TM	Half-integer Shannon series with coefficients $\epsilon_n A_n$	Synthesis; overlaps repository cardinal program

9 Exact computations and numerical experiments

The file `numerical_experiments.py` is deliberately self-contained. It uses `fractions.Fraction` for exact moments, Chebyshev moments, Jacobi coefficients, and Verblunsky coefficients; `mpmath` for 70-digit analytic checks; and `matplotlib` for the figures. The code follows two independent coefficient routes:

1. ordinary moments $\rightarrow$ exact monic Gram–Schmidt $\rightarrow \beta_n \rightarrow \alpha_n$ through Geronimus;
2. ordinary moments $\rightarrow c_n \rightarrow C(w) \rightarrow \alpha_n$ through a formal exact Schur algorithm.

The two routes agree exactly through α_{15} in the automated assertion. The script also verifies $N_\Phi(n) = 2n - s_2(n)$ for $0 \leq n \leq 256$, the monodromy normalization for $1 \leq N \leq 9$, and all positivity/unit-disk conditions for the computed coefficients.

9.1 Bessel convergence

At 40 even terms, the pointwise errors against a 120-factor product were approximately

Table 23: First exact probability–Chebyshev moments.

m	$2m$	c_{2m}
0	0	1
1	2	$-\frac{7}{9}$
2	4	$\frac{227}{675}$
3	6	$-\frac{11237}{297675}$
4	8	$\frac{-8241761}{227721375}$
5	10	$\frac{171586753}{8135075025}$
6	12	$-\frac{472381023339557}{45472425490366875}$

Table 24: First exact nonzero Verblunsky coefficients.

n	α_n	decimal
1	$-\frac{7}{9}$	-0.777777777778
3	$-\frac{17}{25}$	-0.68
5	$-\frac{1849}{3087}$	-0.598963394882
7	$-\frac{3569379}{6492691}$	-0.549753407331
9	$-\frac{445076883510763}{886612917691413}$	-0.501996840594

Table 25: First exact monic Jacobi recurrence coefficients.

n	β_n	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.16843537415
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288

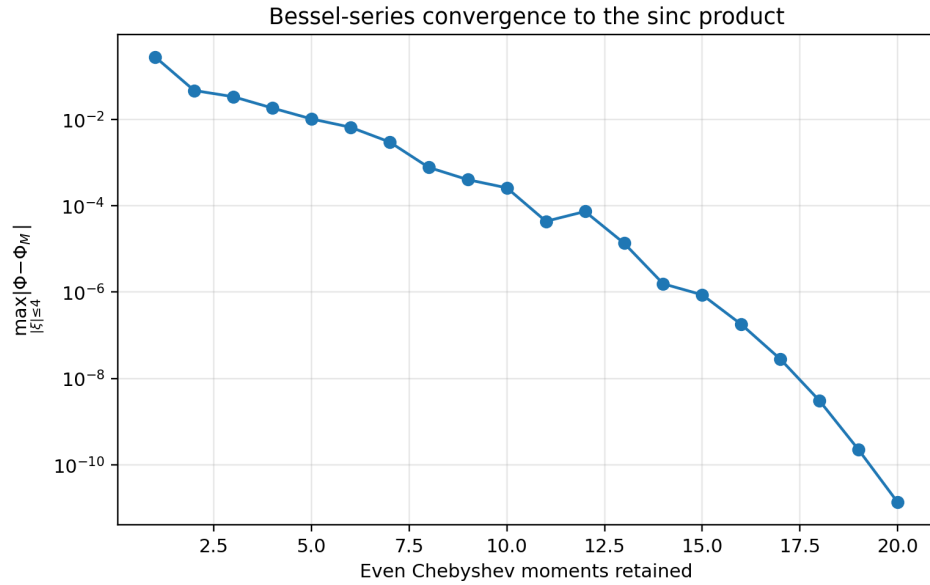


Figure 20: Maximum error on $|\xi| \leq 4$ of the first M even-order terms in (1079), measured against a 90-factor sinc product. At $M = 20$, the recorded maximum is approximately 1.35×10^{-11} .

ξ	absolute error
0.125	below the 70-digit working threshold
0.7	9.1×10^{-72}
1.75	5.5×10^{-69}
3.2	7.4×10^{-48}

The degradation at $\xi = 3.2$ is expected: a fixed order truncation must resolve more oscillatory Bessel content as $|\xi|$ grows.

9.2 Three independent Cauchy-transform evaluations

The Joukowski series, ordinary moment series, and direct Laplace-product integral agree as follows:

z	Joukowski/Chebyshev	moment-series error	Laplace-product error
1.5	0.7040034721763372090	1.71×10^{-22}	5.82×10^{-17}
2.0	0.5148540730138274802	5.75×10^{-33}	1.22×10^{-27}
3.0	0.3375690921956309530	1.21×10^{-47}	3.19×10^{-45}

At $z = 1.5$, the moment series is closer to its unit-circle boundary of convergence and the quadrature tail is harder; the Joukowski coordinate remains well adapted to the slit plane.

9.3 Verblunsky data

The figure is plotted from exact fractions, not from a fitted or floating-point recurrence. The largest fractions contain hundreds of digits; the CSV files preserve their exact numerators and denominators.

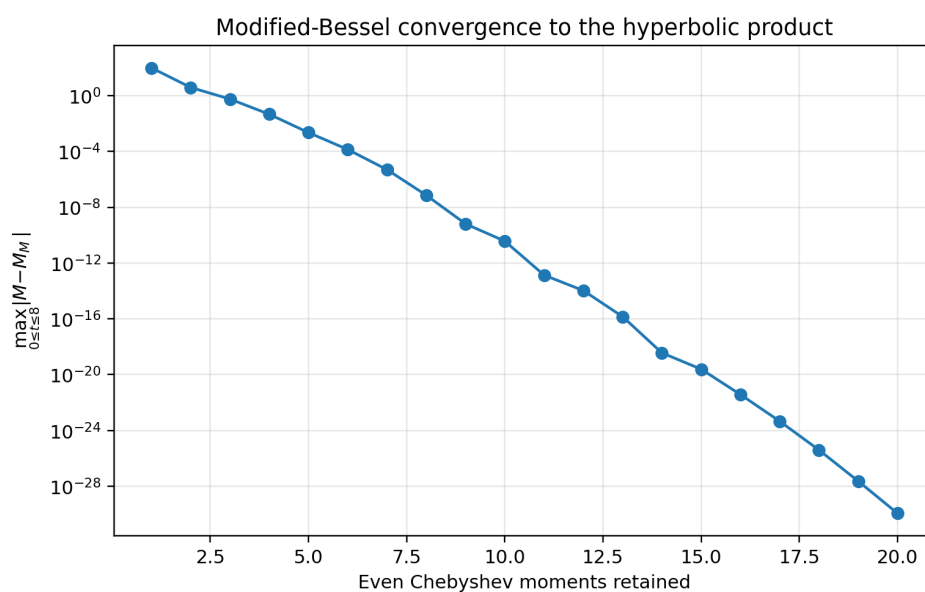


Figure 21: Maximum error on $0 \leq t \leq 8$ of the first M even-order terms in (1084), measured against a 90-factor hyperbolic product. At $M = 20$, the recorded maximum is approximately 1.14×10^{-30} .

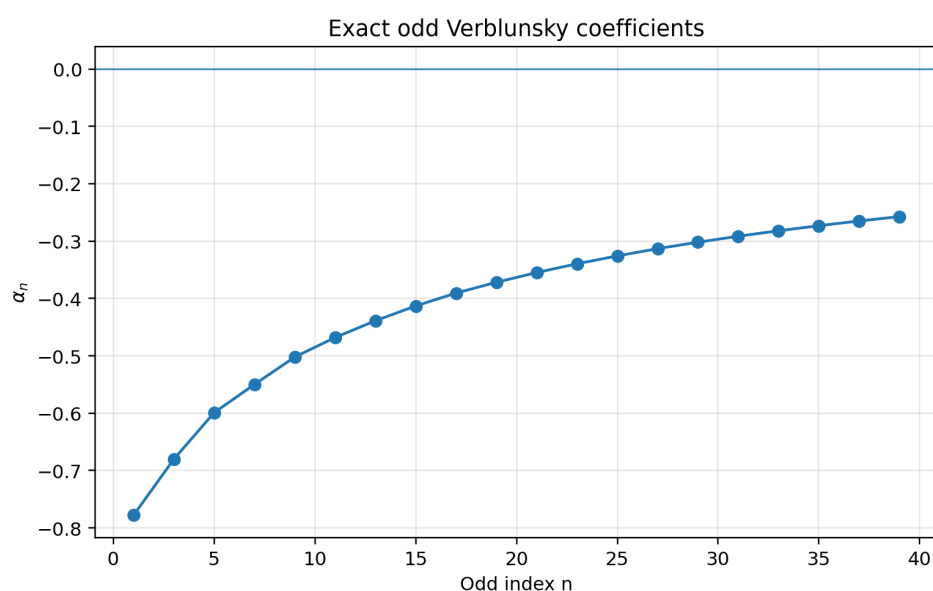


Figure 22: The first twenty exact odd Verblunsky coefficients. All even coefficients vanish. The displayed odd values are negative and increase toward zero, motivating [conjecture 10.1](#); Rakhmanov's theorem proves convergence to zero but not the observed sign and monotonicity.

9.4 Reproducibility boundaries

The experiments verify identities already proved symbolically; they do not establish the conjectures in [Appendix 10](#). Product truncation and quadrature errors are not given as formal interval bounds, so their decimal residuals should be read as consistency checks. A natural next step is to port the same recurrences to Arb or MPFI interval arithmetic.

10 Conjectures and frontier research program

Conjectural frontier

The statements in this section are unproved unless explicitly described as conditional. They are motivated by exact data, standard OPUC theory, and the known endpoint asymptotics, but they should not be cited as theorems.

Conjecture 10.1 (Sign and monotonicity of odd Verblunsky coefficients). For every $n \geq 0$,

$$-1 < \alpha_{2n+1} < \alpha_{2n+3} < 0. \quad (1123)$$

Thus the odd subsequence approaches zero strictly from below.

The first twenty odd coefficients support the conjecture. A proof may be accessible through total positivity of Toeplitz minors, a comparison principle for the Schur algorithm, or a special sign-regularity property of the circle weight W . Ordinary positivity of W is not enough; a genuinely stronger structural argument is needed.

Conjecture 10.2 (Logarithmic edge law). There is a constant $A > 0$ such that

$$\alpha_{2n-1} \sim -A \frac{\log n}{n}. \quad (1124)$$

If the Jacobi edge law has the form

$$\frac{1}{4} - \beta_{n+1} \sim C \frac{(\log n)^2}{n^2}, \quad (1125)$$

then the Geronimus relation predicts $A = 2\sqrt{C}$.

Indeed, writing $\gamma_n = -\alpha_{2n-1}$,

$$\frac{1}{4} - \beta_{n+1} = \frac{1}{4}(\gamma_{n+1} - \gamma_n + \gamma_n \gamma_{n+1}).$$

For $\gamma_n \asymp (\log n)/n$, the product term dominates the discrete derivative. This turns the real-line Jacobi edge problem into a unit-circle Schur asymptotic problem.

Conjecture 10.3 (Refined Szegő–Toeplitz asymptotics). The Toeplitz determinants D_N admit a strong asymptotic expansion whose first correction to the Szegő linear term is governed by the logarithmic endpoint-flatness law of ρ , and hence ultimately by the Lambert– W expansion appearing in the endpoint analysis.

The weight is positive and smooth but has infinite-order zeros. It lies outside the most naive strictly-positive smooth-weight regime, making it a promising test case for extending Fisher–Hartwig and strong Szegő methods to logarithmically superflat weights.

Conjecture 10.4 (Arithmetic structure of the Szegő constant). The entropy integral

$$\mathcal{S}_\rho = \frac{1}{2\pi} \int_0^{2\pi} \log(\pi \rho(\cos \theta) |\sin \theta|) d\theta \quad (1126)$$

has a representation in terms of the dyadic cumulant data, perhaps as a rapidly convergent series involving Bernoulli numbers, dyadic zeta factors, or a binary Barnes-type constant.

Conjecture 10.5 (Bessel zero hierarchy has a finite recursive basis). The derivative cancellation identities obtained from the zero of multiplicity $1 + \nu_2(N)$ at N can be reduced, using Bessel recurrences, to a triangular finite basis whose coefficient matrix has an explicit Thue–Morse or q -binomial factorization.

Conjecture 10.6 (Quantitative Abel convergence). For $n \geq 1$ and r sufficiently close to one,

$$\left| \epsilon_n - \prod_{j=0}^{J_n} H_r \left(\frac{\pi(n + 1/2)}{2^j} \right) \right| \leq C(1 - r) \sum_{j=0}^{J_n} \frac{1}{|\sin(\pi(n + 1/2)/2^j)|} \quad (1127)$$

with an absolute constant C , and the reciprocal-sine sum admits a sharp bound in terms of the binary carry structure of n .

10.1 Promising research directions

1. **Rigorous OPUC asymptotics.** Derive the odd Verblunsky asymptotics directly from the superflat weight using nonlinear steepest descent for the OPUC Riemann–Hilbert problem, or via fine spectral theory of the CMV matrix.
2. **Certified coefficient computation.** Combine exact rational recurrences with modular arithmetic and rational reconstruction. Hundreds of α_n and β_n should be feasible without storing enormous intermediate fractions.
3. **Fast inverse Fabius solvers.** Use the phase equation (1071) with FFT-evaluated trigonometric sums and safeguarded Newton steps. The positivity of $\Re C$ gives a natural monotonicity certificate.
4. **Entropy and determinant constants.** Numerically evaluate $\mathcal{S}_\rho$, compare it with products of exact $1 - \alpha_n^2$, and use integer-relation searches only after rigorous tail bounds are available.
5. **CMV/Jacobi scattering.** Treat the Rvachev Jacobi and CMV matrices as compact or slowly varying perturbations of the free models. The endpoint-flatness law should appear in reflection coefficients and spectral phase shifts.
6. **Base- b and q -Rvachev lifts.** For generalized products with scales b^{-k} or exponent sequences, construct the analogous circle lift, identify the Carathéodory function, and compare rationality/algebraicity of its Schur parameters.
7. **Monodromy categories.** Replace the binary finite spline by finite q -binomial or digit-weighted splines. Their logarithmic resolvents should carry a monodromy representation of the corresponding automatic word.
8. **De Branges and Hermite interpolation.** The integer zero divisor has unbounded multiplicity. Build a complete interpolation basis adapted to the values of $\Phi^{(r)}(n)$ for $0 \leq r \leq \nu_2(n)$, rather than ordinary simple-node Shannon sampling.
9. **Bessel sum-rule algebra.** Organize (1083) and its higher derivatives as identities in the Bessel differential algebra; search for finite closed combinations at powers of two.

10. **Mellin matching.** Match the unit-circle phase representation to the repository’s Mellin/Newton and Lambert– W endpoint expansions, obtaining uniform approximations from the center to the boundary.
11. **Formal verification.** The moment recurrence, exact Chebyshev transform, Carathéodory series, and contour zero count are suitable for Lean formalization. The analytic boundary and Szegő portions would require more functional-analysis infrastructure.
12. **Interval numerics.** Recompute the figures with ball arithmetic, certify the Schur parameters and Bessel identities, and establish explicit truncation bounds for the phase series and Abel product.

11 Conclusion

The audited repository already contains a remarkably broad representation theory: random series, splines, Thue–Morse coefficients, sinc and canonical products, exact moments, Bernoulli/Bell structure, resolvents, logarithmic finite transforms, determinant and continued-fraction forms, multiresolution expansions, integral transforms, inverse-function identities, and endpoint asymptotics. The main advance here is therefore not another isolated formula, but a new organizing bridge.

The probability–Chebyshev moments c_n lift the Rvachev law to the circle. From that lift, one analytic function C simultaneously yields the density, the Fabius phase series, the inverse phase equation, the Joukowski Cauchy transform, Schur and CMV data, and Toeplitz products. The same coefficients turn the infinite sinc and hyperbolic products into ordinary and modified Bessel series. On the arithmetic side, the integer zero divisor of Φ recovers the complete binary digit sum by the argument principle; finite logarithmic resolvents recover Thue–Morse words by monodromy; and Abel regularization gives a smooth finite-product representation.

These bridges place the Rvachev–Fabius system at an intersection of probability, Fourier analysis, special functions, OPUC, operator theory, complex monodromy, automatic sequences, and dyadic asymptotics. The conjectures above are concrete enough for exact computation and theoretical attack, yet broad enough to connect several currently separate campaigns in the repository.

A Proof details and convergence notes

A.1 Smoothness of the circle weight

Near $\theta = 0$, $1 - \cos \theta \asymp \theta^2$. Since ρ and all its derivatives vanish at $x = 1$, the composite $\rho(\cos \theta)|\sin \theta|$ and all its derivatives tend to zero at the origin. The same holds at π and 2π . This justifies the C^∞ periodic extension used in [Theorem 4.5](#), hence the superalgebraic decay of c_n .

A.2 Local uniform convergence of Bessel series

For $|z| \leq R$, the standard bound $|J_n(z)| \leq (R/2)^n/n! e^{R^2/4}$ is sufficient. Since $|c_n| \leq 1$, the series in [\(1079\)](#) converges uniformly on compact sets. An analogous factorial-in-order bound for I_n proves local uniform convergence of [\(1084\)](#). Termwise differentiation follows from Weierstrass convergence.

A.3 Log-integrability of the circle weight

The established endpoint analysis of $\rho(1 - h)$ has logarithm growing on the scale of a quadratic or mildly corrected quadratic polynomial in $\log(1/h)$. With $h \asymp \theta^2$, any finite power of $\log(1/\theta)$ is

integrable near zero:

$$\int_0^\varepsilon |\log^k \theta| \, d\theta < \infty.$$

The extra factor $|\sin \theta|$ contributes only $\log |\theta|$. Therefore $\log W \in L^1$, as required by Szegő’s theorem.

A.4 A direct expression for the circle measure

For any continuous h on $\mathbb{T}$,

$$\int_{\mathbb{T}} h(\zeta) \, d\nu(\zeta) = \int_{-1}^1 \frac{h(e^{i \arccos x}) + h(e^{-i \arccos x})}{2} \rho(x) \, dx. \tag{1128}$$

This formula is often the fastest route for transporting real-line identities to the circle.

B Reproducibility map

File	Purpose
<code>numerical_experiments.py</code>	Exact moments, c_n , Jacobi and Verblunsky coefficients; Bessel/product, Cauchy, spectral, Abel, and monodromy checks; figure and table generation.
<code>numerical_results.txt</code>	Human-readable verification transcript with decimal residuals.
<code>chebyshev_moments.csv</code>	Exact numerators, denominators, and decimal values of c_n .
<code>jacobi_coefficients.csv</code>	Exact β_n .
<code>schur_parameters.csv</code>	Exact α_n , including structural zeros.
<code>generated_tables.tex</code>	Tables inserted into this report.
<code>corpus_audit.txt</code>	Audit method, overlap exclusions, and novelty language.
<code>README.txt</code>	Reproduction and compilation instructions.

The exact coefficient computation starts from only (1033); all other rational quantities are consequences. This makes the computational chain auditable and suitable for independent reimplementa-tion.

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- [4] NIST Digital Library of Mathematical Functions, Chapters 10 and 11: Bessel functions and generating/Jacobi–Anger expansions, <https://dlmf.nist.gov/10>.
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Part VIII

Dyadic Multiresolution and Sampling Frontiers for the Fabius–Rvachev System

Abstract

The bounded Fabius distribution function F , Rvachev’s compactly supported function u_p , and the entire sinc product that is the Fourier transform of u_p possess unusually many mutually illuminating descriptions. This report first audits all 92 TeX paths recursively present under `Analysis/FabiusFunction/docs` at the pinned repository commit. Existing probability, convolution, functional–differential, Volterra, finite Thue–Morse spline, q -binomial, Fourier product, canonical-product, Fourier–Legendre, half-integer cosine, moment, Lambert– W , Bell-polynomial, and inverse-quantile representations are catalogued as prior corpus material, not reannounced as discoveries.

The main new theorem package is dyadic and multiresolution in nature. Exact rational values of F at dyadic points give a uniformly convergent Faber–Schauder expansion with rational coefficients, an exact-rational piecewise-affine limit for the inverse Fabius function, a Haar expansion of the density, and a locally uniformly convergent Haar–sinc² expansion of the infinite sinc product. Haar Parseval then gives a positive rational series for

$$A_2 = \int_0^1 F(x)^2 dx = \frac{1}{2\pi} \int_0^\infty \prod_{j \geq 1} \operatorname{sinc}^2\left(\frac{t}{2^j}\right) dt,$$

together with an all-orders dyadic tail expansion. This turns an arithmetically opaque sinc integral into monotone exact rational approximants and produces a high-order acceleration yielding about sixty stable decimal digits from level 16 dyadic data.

Several additional representation families are derived. Expanding every uniform coordinate into fair binary digits reorganizes the Fabius random series into independent binomial layers and proves the product identity

$$\prod_{j \geq 1} \operatorname{sinc}\left(\frac{t}{2^j}\right) = \prod_{n \geq 2} \cos\left(\frac{t}{2^n}\right)^{n-1}.$$

Paley–Wiener sampling reconstructs the same entire function solely from its odd π -lattice samples; the known even zero lattice then yields a hierarchy of new sample sum rules. Chebyshev and Gegenbauer coefficients become Bessel transforms of the sinc product. Dyadic Bernstein polynomials give exact-rational polynomial limits for F and up, while their Fourier images are finite sums of Kummer functions. Finally, finite approximants are identified with sublevel volumes and hyperplane sections of weighted cubes. Proofs, status labels, conjectures, a Lean formalization roadmap, and a fully commented Python verification program are included.

1 Audit boundary, status discipline, and contribution map

1.1 Pinned recursive corpus

The source boundary of this report is the Git commit

f7daf632f2db839fe091de0d080e22a9b50c286a

of Vladimir Reshetnikov’s ProveIt repository [16]. The audit root is `Analysis/FabiusFunction/docs`. A previously reconstructed pinned Git manifest contained 87 recursive TeX paths. During the audit the branch advanced by 25 commits; pinning the new head and comparing trees revealed five added TeX paths and no removals. Thus the current boundary contains exactly 92 TeX paths. The complete list, the five additions, and the modified-path ledger are included as `corpus_manifest_current.txt` in the accompanying archive.

The five additions are three large integral/transform manuscripts and two generated numerical fragments. They materially change the novelty boundary: finite and complex-power antiderivatives, fractional Volterra calculus, quantile beta transforms, Cauchy–Stieltjes hierarchies, inverse-Fabius resolvents, autocorrelations, order statistics, and a rational Jacobi continued fraction were already present by the pinned commit. Those results are therefore baseline here.

The audit did not treat archived revisions as independent corroboration. Historical copies were read for mathematically distinctive material, then deduplicated by lineage. Imported TeX transcriptions of the papers of Arias de Reyna and of work on lacunary products were included in the collision search. Targeted repository searches were also made for the exact phrases and formula shapes used below. No occurrence was found for the Faber–Schauder midpoint-defect expansion, the rational Haar energy series, the triangular independent-binomial decomposition, the odd-sample cardinal formula, the Chebyshev–Bessel sinc integral, the dyadic Bernstein–Kummer transform, or the weighted-cube section representation.

1.2 Meaning of the status labels

Three labels are used throughout.

- **Corpus baseline.** means that the result, perhaps in a different normalization, is explicit in the audited corpus or one of its imported sources.
- **New deduction relative to the pinned corpus.** means that a proof is supplied here and that no explicit counterpart was located in the pinned corpus after formula-level searches. It is not a claim of absolute publication priority.
- **Conjecture.** means that no proof is claimed. Exact computations and structural evidence are stated separately from the conjecture.

This distinction is especially important in the Fabius project: a new representation may be an elementary consequence of deep existing identities, while a visually similar formula may fail because it suppresses the nonconstant periodic phase in the endpoint asymptotics.

1.3 Existing material deliberately excluded from novelty claims

The following are treated as established inputs.

Layer	Audited prior material
Basic functions	Separation of the bounded CDF F , the compactly supported even bump up, and the signed global extension $\mathcal{F}$; uniqueness, smoothness, support, symmetry, strict shape, and flat endpoints.
Probability	Weighted sums of independent uniform coordinates, finite atomic and spline approximants, weak convergence, exact support bounds, and reflection.
Arithmetic	Rational moments, reciprocal-power values $F(2^{-n})$, arbitrary dyadic evaluation, Thue–Morse formulas, q -Pochhammer and Gaussian-binomial forms, denominator and 2-adic information.
Fourier analysis	Infinite sinc and cosine products, Fourier inversion, Poisson and half-integer sampling identities in physical space, integer zeros with multiplicity $1 + \nu_2(n)$, canonical products, sharp real-axis decay, spectral zeta and heat trace.
Approximation	Finite Thue–Morse splines, repeated integration, Fourier–Legendre expansion, exact Strang–Fix reproduction, positive quadrature, and Richardson acceleration of truncated products.
Integral transforms	Mellin, Laplace, Fourier, Cauchy, Stieltjes, logarithmic, correlation, spacing, order-statistic, inverse-quantile, and fractional-resolvent transforms.
Endpoint theory	Correct lower-Lambert coordinate, nonconstant periodic Gamma-zeta correction, all-order Bell-polynomial expansion, inverse-quantile reversion, and phase-aware Richardson extraction.

1.4 New theorem package

The report proves the following linked results.

1. An exact rational Faber–Schauder expansion of F , with sharp 4^{-J} interpolation scale, sign symmetry, and curvature recovery; inversion gives exact-rational piecewise-affine approximants to $G = F^{-1}$.
2. A rational Haar expansion of F' and up, and a corresponding Haar-sinc² series for the entire Fourier transform.
3. A positive rational series for $A_2 = \int_0^1 F^2$, a certified geometric tail bound, and a complete asymptotic expansion of its dyadic level energies and tails.
4. An independent-binomial layer representation obtained by regrouping the binary digits of the uniform coordinates, including a finite combinatorial CDF and probability polynomial.
5. An odd- π -lattice Shannon reconstruction of the sinc product and a hierarchy of sample identities forced by the higher even-zero multiplicities.
6. Chebyshev and Gegenbauer expansions whose coefficients are Bessel transforms of the sinc product; the Legendre specialization turns known rational coefficients into exact Bessel-product integrals.
7. Exact-rational dyadic Bernstein limits and finite Kummer-function formulas for their Fourier transforms.

8. Weighted-cube sublevel-volume and hyperplane-section limits unifying the probabilistic, spline, and Fourier-slice pictures.

2 Normalization and established representation atlas

2.1 The three functions

Let $F : \mathbb{R} \rightarrow [0, 1]$ denote the bounded Fabius distribution function [2, 4, 5]:

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (1129)$$

with

$$F'(x) = 2F(2x) \quad (0 < x < \tfrac{1}{2}), \quad F(x) + F(1-x) = 1 \quad (0 \leq x \leq 1). \quad (1130)$$

Rvachev's function is [3]

$$\text{up}(y) = \begin{cases} F(1 - |y|), & |y| \leq 1, \\ 0, & |y| > 1. \end{cases} \quad (1131)$$

It is even, nonnegative, C^∞ , supported on $[-1, 1]$, and normalized by $\int \text{up} = 1$. The conversion formulas are

$$F'(x) = 2 \text{up}(2x - 1), \quad F(x) = \text{up}(x - 1) \quad (0 \leq x \leq 1), \quad (1132)$$

and, for $0 \leq x \leq 1$,

$$\text{up}(x) = 1 - F(x) = F(1 - x). \quad (1133)$$

The signed global continuation $\mathcal{F}$ satisfies $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$ and agrees with F on $[0, 1]$. It is indispensable for global derivative masks, but the new multiresolution formulas below concern the bounded F and compactly supported up . We never silently substitute one global convention for another.

2.2 Probability and infinite convolution

Corpus baseline. Let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$ [1] and let $V_j = 2U_j - 1$, uniform on $[-1, 1]$. Then

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad Y = 2X - 1 = \sum_{j=1}^{\infty} \frac{V_j}{2^j}, \quad (1134)$$

$F(x) = \mathbb{P}(X \leq x)$, and Y has density up . Consequently

$$\text{up} = \bigstar_{j=1}^{\infty} (2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]}) \quad (1135)$$

in the sense of weak limits and, in fact, in the smooth-density topology available from the corpus.

The survival interpretation

$$\text{up}(t) = \mathbb{P}(X > t), \quad 0 \leq t \leq 1, \quad (1136)$$

immediately gives, for absolutely continuous H ,

$$\int_0^1 H'(t) \text{up}(t) dt = \mathbb{E}[H(X) - H(0)], \quad (1137)$$

$$\int_0^1 H'(t) F(t) dt = H(1) - \mathbb{E}H(X). \quad (1138)$$

This is the compact master identity behind many moment, Laplace, logarithmic, and resolvent integrals in the audited transform reports.

2.3 Functional–differential and Volterra forms

Corpus baseline. Integrating (1130) gives the primitive self-similarity

$$F(x/2) = \int_0^x F(t) dt, \quad 0 \leq x \leq 1. \quad (1139)$$

Repeated integration produces exact primitive ladders. The compact bump satisfies the global functional–differential equation

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1), \quad (1140)$$

and all higher derivatives are finite translates with Thue–Morse signs. The current corpus extends these identities to integer and fractional Volterra orders and to broad arbitrary-kernel transmutation formulas.

2.4 Fourier, Laplace, product, and canonical-product forms

We use the angular Fourier convention

$$\Phi(t) = \int_{-1}^1 \text{up}(y) e^{-ity} dy. \quad (1141)$$

The cycles-per-unit transform in the repository is $\widehat{\text{up}}(\xi) = \Phi(2\pi\xi)$. Throughout, $\text{sinc } z = \sin z/z$ with $\text{sinc } 0 = 1$.

Corpus baseline. Independence in (1134) gives

$$\Phi(t) = \prod_{j=1}^{\infty} \text{sinc} \left(\frac{t}{2^j} \right). \quad (1142)$$

The moment-generating function of X is therefore

$$M_X(s) = \mathbb{E} e^{sX} = \prod_{j=1}^{\infty} \frac{e^{s/2^j} - 1}{s/2^j} = e^{s/2} \Phi(is/2). \quad (1143)$$

Its logarithm yields Bernoulli cumulants and Bell-polynomial moments.

The complete complex zero divisor is the nonzero lattice $2\pi\mathbb{Z}$, with multiplicity $1 + \nu_2(n)$ at $2\pi n$. Equivalently,

$$\Phi(t) = \prod_{n=1}^{\infty} \left(1 - \frac{t^2}{4\pi^2 n^2} \right)^{1+\nu_2(n)}, \quad (1144)$$

where the symmetric genus-zero product converges normally. The corpus contains the associated spectral zeta, determinant, heat trace, zero jets, and sharp real-axis decay theory.

Fourier inversion gives

$$\text{up}(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi(t) e^{ity} dt, \quad (1145)$$

and symmetry gives the CDF representation

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(t) \frac{\sin((2x-1)t)}{t} dt, \quad 0 < x < 1. \quad (1146)$$

The decay is faster than every power, so these are ordinary convergent integrals rather than distributional shorthand.

The period-two extension of up has only the constant and odd Fourier modes:

$$\text{up}(y) = \frac{1}{2} + \sum_{m=0}^{\infty} \Phi((2m+1)\pi) \cos((2m+1)\pi y), \quad |y| \leq 1. \quad (1147)$$

This physical-space cosine series is established corpus material. The frequency-space odd-sample reconstruction in [Appendix 7](#) is a distinct, dual statement.

2.5 Finite splines, exact dyadic arithmetic, and limits

Corpus baseline. Put $X_N = \sum_{j=1}^N 2^{-j} U_j$ and let F_N be its CDF. Inclusion–exclusion for a weighted box gives the finite Thue–Morse spline

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{k=0}^{2^N-1} (-1)^{\text{wt}_2(k)} \left(x - \frac{k}{2^N}\right)_+^N. \quad (1148)$$

Because $0 \leq X - X_N \leq 2^{-N}$,

$$F_N(x - 2^{-N}) \leq F(x) \leq F_N(x). \quad (1149)$$

This is simultaneously a probability limit, a piecewise-polynomial limit, and a finite alternating truncated-power representation.

Every dyadic value is rational. The reciprocal-power values $V_n = F(2^{-n})$ obey

$$V_0 = 1, \quad V_n = \frac{1}{2^{\binom{n}{2}}(2^n - 1)} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}} V_k}{(n - k + 1)!}. \quad (1150)$$

The repository further supplies finite Thue–Morse sums, q -Pochhammer and Gaussian-binomial forms, an executable highest-bit recursion for arbitrary $F(a/2^e)$, and exact denominator and valuation theorems. Those exact values will become the coefficients of the new Faber and Haar representations.

2.6 Moment, orthogonal-polynomial, and endpoint forms

Corpus baseline. The even moments

$$\mu_{2r} = \int_{-1}^1 y^{2r} \text{up}(y) \, dy \quad (1151)$$

are rational and satisfy

$$\Phi(t) = \sum_{r=0}^{\infty} (-1)^r \mu_{2r} \frac{t^{2r}}{(2r)!}. \quad (1152)$$

The even cumulants are

$$\kappa_{2r} = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0, \quad (1153)$$

and complete Bell polynomials convert the cumulants into the moments. The corpus also proves an absolutely and uniformly convergent Fourier–Legendre expansion of up with exact rational coefficients.

At the flat endpoints, the natural variable is the exact lower-Lambert coordinate [15]

$$\lambda(x) = -\frac{1}{\log 2} W_{-1}(-x \log 2), \quad \lambda 2^{-\lambda} = x. \quad (1154)$$

The full endpoint expansion of F contains a nonconstant one-periodic Gamma–zeta term in this coordinate, and its coefficients are generated by Bell-polynomial algebra. The inverse function $G = F^{-1}$ inherits a phase-aware Lambert reversion. These subtle asymptotics are baseline; below we formulate only new multiresolution questions that retain, rather than average away, this periodic phase.

2.7 A compact dictionary of established representations

For later comparison, [Table 28](#) records the principal established forms in one place.

Table 28: Established representation atlas used as input.

Object	Representation	Natural use
F	CDF of $\sum 2^{-j}U_j$	Monotonicity, coupling, quantiles, stochastic integrals.
up	Density of $\sum 2^{-j}V_j$	Symmetry, convolution, support, probability.
F	$F(x/2) = \int_0^x F$	Primitive ladders and fractional Volterra calculus.
up	Infinite convolution of uniform boxes	Smoothness and finite spline approximants.
F_N	Finite Thue–Morse truncated-power sum	Exact computation and certified limits.
$F(a/2^e)$	Rational recurrence, bit recursion, q -binomial sums	Arithmetic and exact dyadic evaluation.
Φ	Infinite sinc product	Global Fourier analysis and decay.
Φ	Canonical product over $2\pi\mathbb{Z}$ with multiplicities	Zero spectrum, determinants, zeta and heat trace.
up	Fourier inversion integral	Arbitrary real arguments and analytic estimates.
F	Sine-integral CDF inversion	Arbitrary arguments and endpoint cancellation.
up	Period-two odd cosine series	Physical-space spectral approximation.
up	Fourier–Legendre series	Orthogonal approximation with rational coefficients.
M_X	Infinite exponential quotient product	Moments, cumulants, Laplace asymptotics.
F, G	Mellin, Stieltjes, beta-quantile, spacing transforms	Integral calculus and inverse-function identities.
F, G	Lower-Lambert periodic transseries	Sharp endpoint and inverse-endpoint analysis.

3 Exact rational Faber–Schauder representation

3.1 Dyadic tents and midpoint defects

Following the classical Haar–Faber–Schauder multiresolution lineage [6, 7, 8, 14], for $m \geq 0$ and $0 \leq k < 2^m$, put

$$\ell_{m,k} = \frac{k}{2^m}, \quad r_{m,k} = \frac{k+1}{2^m}, \quad \xi_{m,k} = \frac{2k+1}{2^{m+1}}, \quad h_m = 2^{-m-1}. \quad (1155)$$

The normalized tent on this cell is

$$\Lambda_{m,k}(x) = \max(1 - 2^{m+1}|x - \xi_{m,k}|, 0). \quad (1156)$$

Its support is $[\ell_{m,k}, r_{m,k}]$, its height is one, and its area is h_m . Define the midpoint defect

$$c_{m,k} = F(\xi_{m,k}) - \frac{F(\ell_{m,k}) + F(r_{m,k})}{2}. \quad (1157)$$

Every argument in (1157) is dyadic, so

$$c_{m,k} \in \mathbb{Q}. \quad (1158)$$

Theorem 3.1 (Exact rational Faber–Schauder series). *For $J \geq 0$, let*

$$S_J F(x) = x + \sum_{m=0}^{J-1} \sum_{k=0}^{2^m-1} c_{m,k} \Lambda_{m,k}(x). \quad (1159)$$

Then $S_J F$ is exactly the piecewise-linear interpolant of F on the dyadic mesh $2^{-J}\mathbb{Z} \cap [0, 1]$. Consequently

$$F(x) = x + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} \Lambda_{m,k}(x) \quad (1160)$$

uniformly on $[0, 1]$.

Proof. The level-zero affine interpolant through $(0, F(0)) = (0, 0)$ and $(1, F(1)) = (1, 1)$ is x . Passing from the mesh of level m to that of level $m+1$ leaves all old nodes fixed and introduces one midpoint in each cell. The unique correction supported on that cell, zero at the endpoints and having the required midpoint value, is precisely $c_{m,k} \Lambda_{m,k}$. Induction proves the interpolation statement. Uniform convergence follows from uniform continuity of F , since the error of linear interpolation on a mesh of diameter 2^{-J} is bounded by twice the modulus of continuity at that scale. The stronger quantitative bound in Theorem 3.6 follows from $F \in C^2$. $\square$

New deduction relative to the pinned corpus. The point of Theorem 3.1 is not the abstract existence of a Schauder expansion; it is that every coefficient is an explicitly computable rational number supplied by the repository’s exact dyadic arithmetic. The partial sums are therefore exact rational piecewise-linear functions, not floating-point projections.

Corollary 3.2 (Rvachev form). *For $|y| \leq 1$,*

$$\text{up}(y) = 1 - |y| + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} \Lambda_{m,k}(1 - |y|). \quad (1161)$$

The series is uniformly convergent on $[-1, 1]$ and is zero outside that interval after the usual compact-support extension.

Proof. Substitute $x = 1 - |y|$ in (1160) and use (1131). $\square$

3.2 Coefficient kernel, symmetry, signs, and curvature

Proposition 3.3 (Triangular-kernel formula). *The midpoint defect has the exact integral representation*

$$c_{m,k} = -\frac{1}{2} \int_{-h_m}^{h_m} (h_m - |u|) F''(\xi_{m,k} + u) \, du. \quad (1162)$$

In particular,

$$|c_{m,k}| \leq 2^{-2m}. \quad (1163)$$

Proof. The second symmetric difference satisfies

$$F(\xi + h) + F(\xi - h) - 2F(\xi) = \int_{-h}^h (h - |u|) F''(\xi + u) \, du.$$

Equation (1162) follows from the definition of $c_{m,k}$. Moreover, $\|F''\|_\infty = 8$: from $F'(x) = 2 \operatorname{up}(2x-1)$ one has $\|F'\|_\infty = 2$, and on the left half $F''(x) = 4F'(2x)$, with the reflected formula on the right half. Since the triangular kernel has integral h_m^2 , we obtain $|c_{m,k}| \leq 4h_m^2 = 2^{-2m}$. $\square$

Proposition 3.4 (Reflection and sign pattern). *For every m, k ,*

$$c_{m,2^m-1-k} = -c_{m,k}. \quad (1164)$$

Moreover, $c_{0,0} = 0$, and for $m \geq 1$,

$$c_{m,k} < 0 \quad (0 \leq k < 2^{m-1}), \quad c_{m,k} > 0 \quad (2^{m-1} \leq k < 2^m). \quad (1165)$$

Proof. Reflection follows by applying $F(1-x) = 1-F(x)$ to the three nodes in (1157). On $(0, 1/2)$, $F''(x) = 4F'(2x) > 0$, so F is strictly convex and every midpoint lies strictly below its chord. On $(1/2, 1)$ the reflected function is strictly concave. The level-zero defect vanishes because $F(1/2) = 1/2$. $\square$

Proposition 3.5 (Curvature recovery). *Uniformly in k as $m \rightarrow \infty$,*

$$-2^{2m+3} c_{m,k} = F''(\xi_{m,k}) + \mathcal{O}(4^{-m}). \quad (1166)$$

More generally, for every fixed $Q \geq 0$,

$$c_{m,k} = -\sum_{r=0}^Q \frac{F^{(2r+2)}(\xi_{m,k}) h_m^{2r+2}}{(2r+2)!} + \mathcal{O}(h_m^{2Q+4}), \quad (1167)$$

uniformly in k .

Proof. Insert the Taylor expansion of $F''(\xi + u)$ into (1162). Odd powers integrate to zero and

$$\int_{-h}^h (h - |u|) u^{2r} \, du = \frac{2h^{2r+2}}{(2r+1)(2r+2)}.$$

The compact interval and smoothness of F make the remainder uniform. $\square$

The coefficients therefore encode exact dyadic samples of curvature. This is especially suggestive near the flat endpoints, where the repository's Lambert–periodic asymptotics can be sampled through a purely rational sequence.

3.3 Sharp interpolation scale

Theorem 3.6 (Uniform error and sharp asymptotic constant). *For every $J \geq 0$,*

$$\|F - S_J F\|_\infty \leq 4^{-J}. \quad (1168)$$

The constant is asymptotically sharp:

$$\lim_{J \rightarrow \infty} 4^J \|F - S_J F\|_\infty = 1. \quad (1169)$$

Proof. On an interval of length $\Delta = 2^{-J}$, the classical linear-interpolation remainder gives

$$|F(x) - S_J F(x)| \leq \frac{\|F''\|_\infty}{8} \Delta^2 = \Delta^2 = 4^{-J}.$$

For the matching lower asymptotic, take the cell immediately to the right of $x = 1/4$ and evaluate at its midpoint. The interpolation remainder is $-F''(\eta_J)\Delta^2/8$ for some η_J in that cell. Since $\eta_J \rightarrow 1/4$ and $F''(1/4) = 8$, its absolute value divided by Δ^2 tends to one. $\square$

Remark 3.7. The estimate is stronger than a generic Faber–Schauder convergence theorem: it identifies the exact global interpolation scale and its limiting constant. The extremal curvature occurs at the dyadic points $1/4$ and $3/4$, so the sharp constant is itself compatible with exact dyadic arithmetic.

3.4 Low-level exact coefficients

The first nonzero rows illustrate both rationality and reflection:

$$(c_{1,k})_{k=0}^1 = \left(-\frac{13}{72}, \frac{13}{72}\right), \quad (1170)$$

$$(c_{2,k})_{k=0}^3 = \left(-\frac{1}{32}, -\frac{1}{32}, \frac{1}{32}, \frac{1}{32}\right), \quad (1171)$$

$$(c_{3,k})_{k=0}^7 = \frac{1}{2073600}(-3457, -28943, -28943, -3457, 3457, 28943, 28943, 3457). \quad (1172)$$

At level four, the left half is

$$-\frac{1}{147456}(5, 149, 427, 571, 571, 427, 149, 5), \quad (1173)$$

and the right half is its negative reversal. These rows were generated from (1150) and the terminating dyadic bit recursion; no numerical integration entered.

3.5 A universal kernel dictionary

Uniform convergence permits termwise integration against every $K \in L^1(0, 1)$:

$$\int_0^1 K(x)F(x) \, dx = \int_0^1 xK(x) \, dx + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} \int_0^1 K(x)\Lambda_{m,k}(x) \, dx. \quad (1174)$$

When K belongs to a parameterized locally bounded family, the same identity is locally uniform in the parameter. Thus one exact coefficient table feeds polynomial moments, Laplace transforms, Fourier transforms, Stieltjes kernels, and fractional integral kernels.

For $p \in \mathbb{N}_0$, define

$$L_p(c, h) = 2h \sum_{r=0}^{\lfloor p/2 \rfloor} \binom{p}{2r} \frac{c^{p-2r} h^{2r}}{(2r+1)(2r+2)}. \quad (1175)$$

This is exactly the moment of a tent:

$$\int_{c-h}^{c+h} x^p \left(1 - \frac{|x-c|}{h}\right) dx = L_p(c, h). \quad (1176)$$

Hence

$$\int_0^1 x^p F(x) dx = \frac{1}{p+2} + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} L_p(\xi_{m,k}, h_m). \quad (1177)$$

Every summand is rational. This does not replace the more compact moment recurrences in the corpus; it gives a new positive-scale decomposition whose truncation error is controlled directly by (1168).

Let

$$A(t) = \int_0^1 F(x) e^{-itx} dx. \quad (1178)$$

The elementary tent transform is

$$\int_0^1 \Lambda_{m,k}(x) e^{-itx} dx = h_m e^{-it\xi_{m,k}} \operatorname{sinc}^2\left(\frac{th_m}{2}\right). \quad (1179)$$

Therefore

$$A(t) = \frac{e^{-it}(1+it) - 1}{t^2} + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} h_m e^{-it\xi_{m,k}} \operatorname{sinc}^2\left(\frac{th_m}{2}\right), \quad (1180)$$

with removable interpretation at $t = 0$ and locally uniform convergence in $t \in \mathbb{C}$. Integration by parts also gives the bridge

$$\Phi(t) = e^{-it} + 2ite^{it} A(2t). \quad (1181)$$

Replacing t by $-is$ turns (1180) into an entire Laplace series with sinhc^2 kernels.

3.6 Exact-rational inverse interpolants

Let $G = F^{-1} : [0, 1] \rightarrow [0, 1]$. The inverse-frontier manuscripts in the pinned corpus develop sharp endpoint reversion, inverse derivatives, and quantile transforms. The Faber construction supplies a complementary finite exact representation.

For $J \geq 0$ and $0 \leq k \leq 2^J$, put

$$y_{J,k} = F(k/2^J), \quad d_{J,k} = y_{J,k+1} - y_{J,k} > 0 \quad (0 \leq k < 2^J). \quad (1182)$$

Every $y_{J,k}$ and $d_{J,k}$ is rational. Define G_J on $[y_{J,k}, y_{J,k+1}]$ by

$$G_J(y) = \frac{k}{2^J} + \frac{2^{-J}}{d_{J,k}}(y - y_{J,k}). \quad (1183)$$

Equivalently, with $s_{J,k} = 2^{-J}/d_{J,k}$,

$$G_J(y) = s_{J,0}y + \sum_{k=1}^{2^J-1} (s_{J,k} - s_{J,k-1})(y - y_{J,k})_+. \quad (1184)$$

Theorem 3.8 (Exact-rational inverse limit). *The map G_J is the inverse of the Faber partial sum $S_J F$. It is a strictly increasing piecewise-affine homeomorphism with rational knots, rational slopes, and the reflection law*

$$G_J(1 - y) = 1 - G_J(y). \quad (1185)$$

Moreover,

$$\boxed{G(y) = \lim_{J \rightarrow \infty} G_J(y)} \quad (1186)$$

uniformly on $[0, 1]$. If

$$\omega_G(\delta) = \sup_{|u-v| \leq \delta} |G(u) - G(v)|, \quad (1187)$$

then the explicit estimate

$$\boxed{\|G_J - G\|_\infty \leq \omega_G(4^{-J})} \quad (1188)$$

holds.

Proof. The dyadic node values of $S_J F$ are $y_{J,k}$, and strict monotonicity of F makes every chord slope positive. Thus (1183) is exactly $(S_J F)^{-1}$; (1184) is the standard slope-jump form of the same piecewise-affine map. Reflection of the dyadic node table proves (1185).

Fix y and put $x = G_J(y)$. Since $S_J F(x) = y$ and $\|S_J F - F\|_\infty \leq 4^{-J}$,

$$y - 4^{-J} \leq F(x) \leq y + 4^{-J},$$

with the endpoints clipped to $[0, 1]$. Applying the increasing map G places x between the corresponding two quantiles, which gives (1188). Uniform continuity of G completes the proof. $\square$

New deduction relative to the pinned corpus. Equations (1183) and (1184) give a finite exact-rational neural-spline or truncated-power representation of the inverse at every level. The sharp Lambert-periodic endpoint expansion already in the corpus may be inserted into $\omega_G(4^{-J})$ to resolve the non-algebraic global convergence rate.

4 Rational Haar expansions and a Haar–sinc transform

4.1 The derivative as a dyadic martingale limit

Define the unnormalized Haar step

$$H_{m,k}(x) = \mathbf{1}_{[\ell_{m,k}, \xi_{m,k})}(x) - \mathbf{1}_{[\xi_{m,k}, r_{m,k})}(x). \quad (1189)$$

Then $\Lambda'_{m,k} = 2^{m+1} H_{m,k}$ away from its three corners.

Theorem 4.1 (Exact rational Haar representation). *The derivative of the Fabius function has the Haar expansion*

$$\boxed{F'(x) = 1 + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} 2^{m+1} c_{m,k} H_{m,k}(x).} \quad (1190)$$

The partial sum through $m = J - 1$ is the derivative of $S_J F$ and equals the cell average of F' on every dyadic cell of level J . It converges to F' in $L^p(0, 1)$ for every $1 \leq p \leq \infty$, where $p = \infty$ is understood in the essential-supremum sense.

Proof. Differentiate the finite identity (1159). On a cell $[a, b]$ of the level- J mesh, the derivative of the interpolant is $[F(b) - F(a)]/(b - a)$, which is the average of F' on that cell. These conditional averages converge uniformly to the continuous function F' by uniform continuity. Values at the finitely many grid corners are irrelevant in every L^p space. $\square$

Combining (1190) with (1132) yields

$$\text{up}(2x - 1) = \frac{1}{2} + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} 2^m c_{m,k} H_{m,k}(x). \quad (1191)$$

Thus the compact C^∞ bump is an L^∞ limit of exact rational step functions. The apparent tension is useful: smoothness is encoded by strong cancellation across infinitely many dyadic scales.

4.2 Fourier transform of one Haar cell

A direct calculation gives

$$\int_0^1 H_{m,k}(x) e^{-itx} dx = e^{-itk/2^m} \frac{(1 - e^{-it/2^{m+1}})^2}{it}. \quad (1192)$$

Apply this to (1191), remembering that the centered coordinate is $y = 2x - 1$.

Theorem 4.2 (Haar-sinc² expansion of the sinc product). *For every $t \in \mathbb{C}$,*

$$\Phi(t) = \text{sinc } t + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} \frac{2^m c_{m,k}}{it} e^{it(1-2k/2^m)} \left(1 - e^{-it/2^m}\right)^2, \quad (1193)$$

$$= \text{sinc } t - \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} \frac{2^{m+2} c_{m,k}}{it} e^{-ity_{m,k}} \sin^2\left(\frac{t}{2^{m+1}}\right), \quad (1194)$$

where

$$y_{m,k} = \frac{2k+1}{2^m} - 1. \quad (1195)$$

Every apparent singularity at $t = 0$ is removable, and both series converge normally on compact subsets of $\mathbb{C}$.

Proof. Use

$$\Phi(t) = 2e^{it} \int_0^1 \text{up}(2x - 1) e^{-2itx} dx$$

and insert the finite Haar partial sums. Equation (1192) with frequency $2t$ gives (1193); factoring $(1 - e^{-ia})^2 = -4e^{-ia} \sin^2(a/2)$ gives (1194). For t in a fixed compact set, (1163) and $|\sin(t/2^{m+1})| \leq C_K 2^{-m}$ bound the entire level- m contribution by $C'_K 4^{-m}$. The Weierstrass test proves normal convergence. $\square$

Pairing reflected coefficients gives a manifestly real formula on the real axis.

Corollary 4.3 (Real paired form). *For real t ,*

$$\Phi(t) = \text{sinc } t - \sum_{m=1}^{\infty} \frac{2^{m+3}}{t} \sin^2\left(\frac{t}{2^{m+1}}\right) \sum_{k=0}^{2^{m-1}-1} c_{m,k} \sin\left(t \left[1 - \frac{2k+1}{2^m}\right]\right). \quad (1196)$$

The representation (1196) is a series-product bridge. The left side is the infinite sinc product; the right side consists of exact rational curvature coefficients multiplied by elementary sine and $\sin^2$ factors. It gives a new way to transfer arithmetic information from dyadic values of F into the Fourier image.

5 A positive rational series for the sinc energy

5.1 Haar Parseval

Let

$$A_2 = \int_0^1 F(x)^2 dx. \quad (1197)$$

The audited integral report identifies the same constant with a sinc-product energy, but leaves its arithmetic nature open. The Haar representation gives a monotone exact-rational decomposition.

Theorem 5.1 (Positive rational energy series). *One has*

$$A_2 = \frac{1}{4} + \sum_{m=0}^{\infty} 2^m \sum_{k=0}^{2^m-1} c_{m,k}^2. \quad (1198)$$

Every summand is a nonnegative rational number. Hence the partial sums

$$R_J = \frac{1}{4} + \sum_{m=0}^{J-1} 2^m \sum_{k=0}^{2^m-1} c_{m,k}^2 \quad (1199)$$

are rational and increase monotonically to A_2 .

Proof. The Haar steps are pairwise orthogonal, $\|H_{m,k}\|_2^2 = 2^{-m}$, and all have zero mean. Parseval applied to (1190) yields

$$\int_0^1 F'(x)^2 dx = 1 + \sum_{m=0}^{\infty} 2^{m+2} \sum_{k=0}^{2^m-1} c_{m,k}^2. \quad (1200)$$

On the other hand, $F'(x) = 2 \operatorname{up}(2x - 1)$ and evenness give

$$\int_0^1 F'^2 = 2 \int_{-1}^1 \operatorname{up}(y)^2 dy = 4 \int_0^1 F(x)^2 dx = 4A_2.$$

Divide (1200) by four. □

Angular Parseval and (1142) give the three-way identity

$$\frac{1}{4} + \sum_{m \geq 0} 2^m \sum_k c_{m,k}^2 = \int_0^1 F(x)^2 dx = \frac{1}{2\pi} \int_0^\infty \prod_{j=1}^\infty \operatorname{sinc}^2\left(\frac{t}{2^j}\right) dt. \quad (1201)$$

Thus an improper integral of a transcendental infinite product is represented by a positive series of exact rational squares.

Corollary 5.2 (Certified elementary tail bound). *For every $J \geq 0$,*

$$0 < A_2 - R_J \leq \frac{4}{3} 4^{-J}. \quad (1202)$$

Proof. By (1163), the level- m energy is at most

$$2^m \cdot 2^m \cdot 2^{-4m} = 4^{-m}.$$

Sum the geometric tail from $m = J$. □

5.2 Derivative norm scaling

Put

$$I_r = \int_0^1 \left(F^{(r)}(x) \right)^2 dx, \quad r \geq 1. \quad (1203)$$

The derivative tiling already present in the corpus implies the exact norm recurrence $I_r = 4^r I_{r-1}$. Since $I_1 = 4A_2$, this gives

$$I_r = 4^{r(r+1)/2} A_2. \quad (1204)$$

We use this baseline identity to evaluate every coefficient in the new energy-tail expansion.

5.3 All-orders level asymptotics

Define the contribution of one dyadic scale by

$$E_m = 2^m \sum_{k=0}^{2^m-1} c_{m,k}^2, \quad A_2 = \frac{1}{4} + \sum_{m \geq 0} E_m. \quad (1205)$$

Theorem 5.3 (Complete asymptotic expansion of one level). *As $m \rightarrow \infty$,*

$$E_m \sim A_2 \sum_{q=1}^{\infty} \beta_q 4^{-qm}, \quad (1206)$$

where

$$\beta_q = (-1)^{q+1} \frac{2^{q(q+1)+1} (2^{2q} - 1)}{(2q+2)!}. \quad (1207)$$

In particular,

$$\beta_1 = 1, \quad \beta_2 = -\frac{8}{3}, \quad \beta_3 = \frac{64}{5}, \quad \beta_4 = -\frac{139264}{945}. \quad (1208)$$

Proof. A detailed remainder proof is given in [Appendix A](#); here is the coefficient calculation. Write $\Delta = 2^{-m}$ and $h = \Delta/2$. At a cell center,

$$c_{m,k} = -(\cosh(hD) - 1)F(\xi_{m,k}), \quad D = \frac{d}{dx},$$

in the sense of an arbitrary finite Taylor expansion with uniform remainder. The midpoint Riemann sum differs from the corresponding integral by a superalgebraically small quantity because all derivatives entering the integrand are flat at both endpoints. Hence

$$E_m \sim \Delta^{-2} \int_0^1 [(\cosh(hD) - 1)F(x)]^2 dx.$$

Now

$$(\cosh z - 1)^2 = \sum_{r=2}^{\infty} \frac{2^{2r-1} - 2}{(2r)!} z^{2r}. \quad (1209)$$

Repeated integration by parts gives $\int F D^{2r} F = (-1)^r I_r$. Substituting $h = \Delta/2$, setting $q = r - 1$, and using (1204) produces (1207). $\square$

Corollary 5.4 (Leading level constant).

$$\lim_{m \rightarrow \infty} 4^m E_m = A_2. \quad (1210)$$

The same constant therefore controls the original integral, the total Haar energy, and the leading asymptotic energy at every sufficiently fine scale. This self-referential feature is a consequence of the exact derivative norm scaling.

5.4 All-orders tail and acceleration

Summing (1206) geometrically gives the tail expansion.

Corollary 5.5 (Complete rational tail expansion). *For $J \rightarrow \infty$,*

$$A_2 - R_J \sim A_2 \sum_{q=1}^{\infty} \gamma_q 4^{-qJ}, \quad (1211)$$

where

$$\gamma_q = \frac{\beta_q}{1 - 4^{-q}}. \quad (1212)$$

The first coefficients are

$$\gamma_1 = \frac{4}{3}, \quad \gamma_2 = -\frac{128}{45}, \quad \gamma_3 = \frac{4096}{315}, \quad \gamma_4 = -\frac{2097152}{14175}. \quad (1213)$$

In particular,

$$\lim_{J \rightarrow \infty} 4^J (A_2 - R_J) = \frac{4}{3} A_2. \quad (1214)$$

For any fixed truncation order Q , define

$$A_2^{[Q]}(J) = \frac{R_J}{1 - \sum_{q=1}^Q \gamma_q 4^{-qJ}}. \quad (1215)$$

Then

$$A_2^{[Q]}(J) - A_2 = \mathcal{O}(4^{-(Q+1)J}) \quad (1216)$$

for fixed Q . The numerator and denominator in (1215) are rational, so every accelerated value is itself an exact rational number before decimal conversion.

Exact/numerical check. Exact level-16 data and seven tail coefficients give

$$A_2 \approx 0.404436767983985281465621491145376633448000519787051529947065. \quad (1217)$$

The stable digits agree across neighboring values of J and Q , and the raw level-16 rational partial sum is 0.4044367678584316 . . . The numerical table and convergence plot in [Appendix 12](#) show the acceleration explicitly.

6 Triangular Bernoulli and independent-binomial layers

6.1 Regrouping the binary digits

The uniform-coordinate representation (1134) hides a second independence structure. Choose the nonterminating binary expansion only on the null set where two choices exist, and write

$$U_j = \sum_{r=1}^{\infty} B_{j,r} 2^{-r}, \quad (1218)$$

where all $B_{j,r}$ are independent Bernoulli($1/2$) variables. The weight of $B_{j,r}$ in X depends only on $n = j + r$.

Theorem 6.1 (Independent triangular binomial decomposition). *There exist independent random variables*

$$K_n \sim \text{Binomial}(n-1, \tfrac{1}{2}), \quad n \geq 2, \quad (1219)$$

such that

$$X = \sum_{n=2}^{\infty} \frac{K_n}{2^n}. \quad (1220)$$

Equivalently, if S_{n-1} is a sum of $n-1$ independent Rademacher signs and the sums are independent across n , then

$$Y = 2X - 1 = \sum_{n=2}^{\infty} \frac{S_{n-1}}{2^n}. \quad (1221)$$

Proof. Substitute (1218) into $X = \sum_{j \geq 1} 2^{-j} U_j$ and rearrange the absolutely convergent nonnegative series:

$$X = \sum_{j,r \geq 1} B_{j,r} 2^{-(j+r)} = \sum_{n \geq 2} 2^{-n} \sum_{j=1}^{n-1} B_{j,n-j}.$$

For fixed n the inner sum has the law in (1219); different n use disjoint sets of bits and are independent. Centering each binomial sum proves (1221). $\square$

This is not merely another proof of the random-series representation. It changes the elementary building blocks from one uniform variable per scale to one increasingly concentrated binomial variable per antidiagonal of the binary digit array.

6.2 A triangular cosine product

Corollary 6.2 (Product–product identity). *The moment-generating function and centered Fourier transform factor as*

$$M_X(s) = \prod_{n=2}^{\infty} \left(\frac{1 + e^{s/2^n}}{2} \right)^{n-1}. \quad (1222)$$

$$\Phi(t) = \prod_{n=2}^{\infty} \cos\left(\frac{t}{2^n}\right)^{n-1} = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right). \quad (1223)$$

Both products converge normally on compact subsets of $\mathbb{C}$.

Proof. The first two factorizations follow from independence in Theorem 6.1. For the last equality, use the classical binary product $\text{sinc } z = \prod_{r \geq 1} \cos(z/2^r)$ and regroup the normally convergent double product:

$$\prod_{j \geq 1} \text{sinc}(t/2^j) = \prod_{j,r \geq 1} \cos(t/2^{j+r}) = \prod_{n \geq 2} \cos(t/2^n)^{n-1}.$$

$\square$

The logarithmic identity

$$\log \Phi(t) = \sum_{n=2}^{\infty} (n-1) \log \cos(t/2^n) \quad (1224)$$

reorganizes every cumulant as a weighted geometric sum. It also offers a different truncation strategy: the exponent $n-1$ concentrates many tiny Rademacher increments into one layer.

Let

$$\Phi_N^\triangle(t) = \prod_{n=2}^N \cos(t/2^n)^{n-1}. \quad (1225)$$

The omitted factor is quantitatively controlled.

Proposition 6.3 (Local triangular-product tail). *For $|t| \leq 2^N$, write $\Phi(t) = \Phi_N^\triangle(t)T_N(t)$, where $T_N(t) > 0$ on the real axis. Then*

$$0 \leq -\log T_N(t) \leq \frac{(3N+1)t^2}{9 \cdot 4^N}. \quad (1226)$$

Proof. For $n > N$, $|t|/2^n \leq 1/2$, and $0 \leq -\log \cos u \leq u^2$ on $|u| \leq 1/2$. Therefore

$$-\log T_N(t) \leq t^2 \sum_{n>N} (n-1)4^{-n} = \frac{(3N+1)t^2}{9 \cdot 4^N}.$$

□

6.3 Finite combinatorial CDFs

Define

$$X_N^\triangle = \sum_{n=2}^N \frac{K_n}{2^n}, \quad \delta_N = \sum_{n>N} \frac{n-1}{2^n} = \frac{N+1}{2^N}. \quad (1227)$$

The tail lies in $[0, \delta_N]$, so

$$\mathbb{P}(X_N^\triangle \leq x - \delta_N) \leq F(x) \leq \mathbb{P}(X_N^\triangle \leq x). \quad (1228)$$

Expanding the independent binomial laws gives the finite sum

$$F_N^\triangle(x) = 2^{-N(N-1)/2} \sum_{k_2=0}^1 \cdots \sum_{k_N=0}^{N-1} \left(\prod_{n=2}^N \binom{n-1}{k_n} \right) \mathbf{1}_{\{\sum_{n=2}^N k_n 2^{-n} \leq x\}}. \quad (1229)$$

Then $F_N^\triangle(x) \rightarrow F(x)$ at every real x .

The complete mass distribution is encoded by the integer polynomial

$$P_N(q) = \prod_{n=2}^N \left(1 + q^{2^{N-n}} \right)^{n-1}. \quad (1230)$$

Its degree and total coefficient sum are

$$\deg P_N = 2^N - N - 1, \quad P_N(1) = 2^{N(N-1)/2}. \quad (1231)$$

If $p_{N,\ell} = [q^\ell]P_N(q)$, then

$$\mathbb{P}(2^N X_N^\triangle = \ell) = 2^{-N(N-1)/2} p_{N,\ell}, \quad (1232)$$

and hence

$$F(x) = \lim_{N \rightarrow \infty} 2^{-N(N-1)/2} \sum_{0 \leq \ell \leq 2^N x} p_{N,\ell}. \quad (1233)$$

This is a limit of cumulative coefficient sums of explicit sparse products. It is complementary to the alternating Thue–Morse spline (1148): one finite model is signed and polynomial in x ; the other is positive and combinatorial on a dyadic lattice.

Every factor in (1230) is palindromic, so the coefficient sequence is symmetric. Computation suggests substantially more.

Conjecture 6.4 (Triangular log-concavity). For every $N \geq 2$, the coefficients of $P_N(q)$ are log-concave:

$$p_{N,\ell}^2 \geq p_{N,\ell-1} p_{N,\ell+1} \quad (1 \leq \ell < 2^N - N - 1). \quad (1234)$$

Consequently they are unimodal.

Exact/numerical check. The bundled program verifies (1234) in exact integer arithmetic for every $2 \leq N \leq 16$. The largest checked polynomial has degree 65519 and total mass 2^{120} . This is evidence, not a proof: sparse dilation products do not inherit log-concavity from their factors by a routine closure theorem.

7 Odd-lattice reconstruction of the Fourier image

7.1 Paley–Wiener sampling

Because $\text{up} \in C_c^\infty([-1, 1])$, its transform satisfies

$$\Phi \in \text{PW}_1 \cap \mathcal{S}(\mathbb{R}) : \quad (1235)$$

Φ is entire of exponential type at most one, square-integrable on the real axis, and rapidly decreasing there. The Whittaker–Shannon cardinal sampling theorem on PW_1 [10, 11, 12] gives

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k\pi) \frac{\sin(z - k\pi)}{z - k\pi}. \quad (1236)$$

Rapid decay of the samples strengthens the usual L^2 statement to absolute and locally uniform convergence in $z \in \mathbb{C}$.

All nonzero even samples vanish because $\Phi(2\pi n) = 0$. Put

$$s_m = \Phi((2m+1)\pi) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{(2m+1)\pi}{2^j}\right). \quad (1237)$$

Theorem 7.1 (Odd-sample cardinal representation). For all $z \in \mathbb{C}$,

$$\Phi(z) = \frac{\sin z}{z} - \sin z \sum_{m \in \mathbb{Z}} \frac{\Phi((2m+1)\pi)}{z - (2m+1)\pi}, \quad (1238)$$

$$\Phi(z) = \frac{\sin z}{z} - 2z \sin z \sum_{m=0}^{\infty} \frac{s_m}{z^2 - (2m+1)^2 \pi^2}. \quad (1239)$$

At the apparent poles the expressions are interpreted by their removable limits. Both series converge normally on compact subsets of $\mathbb{C}$ after those removals.

Proof. In (1236), the term $k = 0$ is $\sin z/z$ and every nonzero even term vanishes. For odd k , $\sin(z - k\pi) = -\sin z$, giving (1238). Pair the samples at $\pm(2m+1)\pi$ and use evenness of Φ to obtain (1239). $\square$

New deduction relative to the pinned corpus. Equation (1147) reconstructs the physical bump from the same odd samples. Equation (1239) instead reconstructs the entire Fourier image at arbitrary complex arguments. Together they make the odd sample sequence a two-sided coordinate system for both domains.

For every compact $K \subset \mathbb{C}$, rapid sample decay implies the superalgebraic truncation estimate

$$\sup_{z \in K} |\Phi(z) - \Phi_M(z)| = \mathcal{O}_{K,A}(M^{-A}) \quad \text{for every } A > 0, \quad (1240)$$

where Φ_M retains $0 \leq m < M$ in the paired series.

7.2 Higher zero multiplicities as odd-sample sum rules

Factor (1239) as

$$\Phi(z) = \sin z B(z), \quad (1241)$$

where, away from its removable odd-lattice singularities,

$$B(z) = \frac{1}{z} - 2z \sum_{m=0}^{\infty} \frac{s_m}{z^2 - (2m+1)^2 \pi^2}. \quad (1242)$$

At an even lattice point $z_n = 2\pi n \neq 0$, B is analytic. The established zero order $1 + \nu_2(n)$ of Φ now becomes a hierarchy of identities among the odd samples.

Theorem 7.2 (Odd-sample zero-jet hierarchy). *For every nonzero integer n ,*

$$B^{(r)}(2\pi n) = 0 \quad (0 \leq r < \nu_2(n)), \quad B^{(\nu_2(n))}(2\pi n) \neq 0. \quad (1243)$$

In particular, for every positive even n ,

$$\sum_{m=0}^{\infty} \frac{s_m}{4n^2 - (2m+1)^2} = \frac{1}{8n^2}. \quad (1244)$$

If $4 \mid n$, then also

$$\sum_{m=0}^{\infty} \frac{s_m}{(4n^2 - (2m+1)^2)^2} = \frac{1}{32n^4}. \quad (1245)$$

Proof. The sine factor has a simple zero at $2\pi n$, while Φ has order $1 + \nu_2(n)$; hence B has order $\nu_2(n)$ there. This proves (1243). Setting $B(2\pi n) = 0$ and dividing out powers of π gives (1244). Differentiate (1242):

$$B'(z) = -\frac{1}{z^2} - 2 \sum_m \frac{s_m}{z^2 - a_m^2} + 4z^2 \sum_m \frac{s_m}{(z^2 - a_m^2)^2}, \quad a_m = (2m+1)\pi.$$

If $4 \mid n$, then $B'(2\pi n) = 0$. Substituting (1244) yields (1245). $\square$

Higher derivatives of B give explicit rational combinations of

$$\sum_{m \geq 0} \frac{s_m}{(4n^2 - (2m+1)^2)^j}, \quad j = 1, 2, \dots,$$

whenever $2^r \mid n$. This translates the 2-adic zero multiplicity into a nested family of exact cardinal-sample identities. It is a promising bridge between the spectral arithmetic already formalized in Lean and classical sampling theory.

8 Chebyshev, Gegenbauer, and Bessel–product expansions

8.1 Chebyshev coefficients

Write the Chebyshev expansion in the standard form

$$\text{up}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n T_n(x), \quad -1 \leq x \leq 1, \quad (1246)$$

where

$$a_n = \frac{2}{\pi} \int_{-1}^1 \frac{\text{up}(x) T_n(x)}{\sqrt{1-x^2}} dx. \quad (1247)$$

Since $\text{up}(\cos \theta)$ is a smooth 2π -periodic even function, the series converges absolutely and uniformly, and its coefficients decay faster than every power.

The plane-wave identity

$$\int_{-1}^1 \frac{e^{itx} T_n(x)}{\sqrt{1-x^2}} dx = \pi i^n J_n(t) \quad (1248)$$

turns the coefficients into transforms of the sinc product.

Theorem 8.1 (Chebyshev–Bessel sinc representation). *All odd coefficients vanish, and for $r \geq 0$,*

$$a_{2r} = \frac{2(-1)^r}{\pi} \int_0^\infty J_{2r}(t) \prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right) dt. \quad (1249)$$

The integral is absolutely convergent.

Proof. Insert Fourier inversion (1145) into (1247), apply Fubini, and use (1248). This gives

$$a_n = \frac{i^n}{\pi} \int_{-\infty}^{\infty} \Phi(t) J_n(t) dt.$$

Parity kills odd n and doubles the positive half-axis for even n . Rapid decay of Φ makes the integral absolute. $\square$

The first numerical coefficients are approximately

$$\begin{aligned} a_0 &= 0.681978961331094\dots, \\ a_2 &= -0.514948055352838\dots, \\ a_4 &= 0.195935940418300\dots, \\ a_6 &= -0.002356862179497\dots, \\ a_8 &= -0.028210793776951\dots, \\ a_{10} &= 0.011427972423053\dots \end{aligned} \quad (1250)$$

The sign change between a_6 and a_8 shows that a naive strict alternation conjecture would already be false.

8.2 A Gegenbauer family

Let $\lambda > 0$, let C_n^λ be the Gegenbauer polynomial, and put

$$h_n^{(\lambda)} = \int_{-1}^1 (1-x^2)^{\lambda-1/2} [C_n^\lambda(x)]^2 dx = \frac{\pi 2^{1-2\lambda} \Gamma(n+2\lambda)}{n!(n+\lambda)\Gamma(\lambda)^2}. \quad (1251)$$

Write

$$\text{up}(x) = \sum_{n=0}^{\infty} a_n^{(\lambda)} C_n^\lambda(x), \quad a_n^{(\lambda)} = \frac{1}{h_n^{(\lambda)}} \int_{-1}^1 \text{up}(x) C_n^\lambda(x) (1-x^2)^{\lambda-1/2} dx. \quad (1252)$$

Define

$$K_{n,\lambda} = 2^\lambda \sqrt{\pi} \Gamma(\lambda + \tfrac{1}{2}) \frac{(2\lambda)_n}{n!}. \quad (1253)$$

Theorem 8.2 (Gegenbauer–Bessel family). *The odd coefficients vanish, and*

$$a_{2r}^{(\lambda)} = \frac{(-1)^r K_{2r,\lambda}}{\pi h_{2r}^{(\lambda)}} \int_0^\infty \Phi(t) t^{-\lambda} J_{2r+\lambda}(t) dt. \quad (1254)$$

Proof. Use the plane-wave integral

$$\int_{-1}^1 e^{itx} C_n^\lambda(x) (1-x^2)^{\lambda-1/2} dx = i^n K_{n,\lambda} t^{-\lambda} J_{n+\lambda}(t)$$

after interpreting the right side by its entire normalized-Bessel continuation at $t = 0$. Fourier inversion and parity give (1254). $\square$

The choice $\lambda = 1/2$ recovers Legendre polynomials; standard normalization facts are as in [13]. If

$$L_{2r} = \frac{4r+1}{2} \int_{-1}^1 \text{up}(x) P_{2r}(x) dx \in \mathbb{Q} \quad (1255)$$

is the exact rational coefficient from the established Fourier–Legendre expansion, and j_n is the spherical Bessel function, then

$$L_{2r} = \frac{(4r+1)(-1)^r}{\pi} \int_0^\infty \Phi(t) j_{2r}(t) dt. \quad (1256)$$

Thus the corpus’s rational Legendre coefficients evaluate an infinite family of Bessel–sinc-product integrals exactly:

$$\int_0^\infty j_{2r}(t) \prod_{j \geq 1} \text{sinc}(t/2^j) dt = \frac{(-1)^r \pi}{4r+1} L_{2r}. \quad (1257)$$

9 Exact-rational Bernstein limits and Kummer transforms

9.1 Dyadic Bernstein polynomials for F

Bernstein’s probabilistic approximation scheme [9] starts, for $n \geq 1$, with

$$(B_n F)(x) = \sum_{k=0}^n F(k/n) \binom{n}{k} x^k (1-x)^{n-k}. \quad (1258)$$

When $n = 2^r$, every sample $F(k/n)$ is rational. Hence $B_{2^r} F$ is an exact-rational polynomial in either the Bernstein or monomial basis.

Theorem 9.1 (Rational Bernstein limit with explicit error). *For $n = 2^r$,*

$$\boxed{F(x) = \lim_{r \rightarrow \infty} (B_{2^r} F)(x)} \quad (1259)$$

uniformly on $[0, 1]$. More precisely,

$$|B_n F(x) - F(x)| \leq \frac{\|F''\|_\infty}{2n} x(1-x) \leq \frac{1}{n}. \quad (1260)$$

Proof. If $K \sim \text{Binomial}(n, x)$, then $B_n F(x) = \mathbb{E}F(K/n)$. Taylor's theorem about x , the zero mean of $K/n - x$, and $\mathbb{E}(K/n - x)^2 = x(1-x)/n$ give the first bound. Use $\|F''\|_\infty = 8$ and $x(1-x) \leq 1/4$ for the second. $\square$

This is a different exact-rational limit from the Faber expansion. Faber partial sums interpolate all dyadic nodes and converge like 4^{-J} when the mesh is 2^{-J} ; Bernstein polynomials smooth the node data by a binomial kernel and converge at the slower but shape-preserving order 2^{-J} .

9.2 Finite Kummer sums for transforms

The beta–Kummer integral is

$$\int_0^1 \binom{n}{k} x^k (1-x)^{n-k} e^{zx} dx = \frac{1}{n+1} {}_1F_1(k+1; n+2; z). \quad (1261)$$

Consequently the restricted Fabius transform $A(t)$ from (1178) has the locally uniform entire limit

$$\boxed{A(t) = \lim_{r \rightarrow \infty} \frac{1}{2^r + 1} \sum_{k=0}^{2^r} F(k/2^r) {}_1F_1(k+1; 2^r+2; -it).} \quad (1262)$$

Every coefficient in every finite sum is rational.

For the bump, put

$$g(u) = \text{up}(2u-1) = F(1-|2u-1|), \quad 0 \leq u \leq 1. \quad (1263)$$

At dyadic nodes, $g(k/2^r) \in \mathbb{Q}$. Approximate g by $B_n g$ and transport back to $[-1, 1]$. Its Fourier image is the finite sum

$$\Phi_n(t) = \frac{2e^{it}}{n+1} \sum_{k=0}^n g(k/n) {}_1F_1(k+1; n+2; -2it). \quad (1264)$$

Theorem 9.2 (Bernstein–Kummer representation of the sinc product). *Along dyadic degrees $n = 2^r$,*

$$\boxed{\prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right) = \lim_{r \rightarrow \infty} \frac{2e^{it}}{2^r + 1} \sum_{k=0}^{2^r} F(1-|2k/2^r - 1|) {}_1F_1(k+1; 2^r+2; -2it).} \quad (1265)$$

The convergence is locally uniform in $t \in \mathbb{C}$.

Proof. Bernstein's theorem gives $B_n g \rightarrow g$ uniformly. On a compact set $K \subset \mathbb{C}$, the Fourier kernels on $[-1, 1]$ are bounded by $\exp(\sup_{t \in K} |\text{Im } t|)$, so uniform approximation of the physical functions implies locally uniform convergence of their transforms. Equation (1261) evaluates each finite transform exactly. $\square$

This representation combines a limit, a finite hypergeometric series, and exact dyadic arithmetic. It is well suited to symbolic experimentation: for each fixed r the entire approximant satisfies the differential relations of a finite family of Kummer functions, while the limit is known from the corpus to be non-holonomic in much stronger senses.

10 Weighted cube volumes and hyperplane sections

10.1 Sublevel volumes

Let $a_j = 2^{-j}$ and $a^{(N)} = (a_1, \dots, a_N)$. The finite CDF in (1148) is exactly a weighted-cube sublevel volume:

$$F_N(x) = \text{vol}_N \left\{ u \in [0, 1]^N : \sum_{j=1}^N a_j u_j \leq x \right\}. \quad (1266)$$

Therefore

$$F(x) = \lim_{N \rightarrow \infty} \text{vol}_N \left\{ u \in [0, 1]^N : \sum_{j=1}^N 2^{-j} u_j \leq x \right\} \quad (1267)$$

for every real x . Equation (1148) is the inclusion–exclusion evaluation of the same polytope volume. One may also write the infinite-dimensional product-measure integral

$$F(x) = \int_{[0,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum_{j \geq 1} 2^{-j} u_j \leq x\}} \, d\mathbf{u}. \quad (1268)$$

This is the geometric form of the probability model, but it emphasizes that the Fabius function is the distribution of weighted half-spaces cutting a Hilbert cube.

10.2 Hyperplane sections and a Fourier-slice limit

Let v be uniform on $[-1, 1]^N$ and set

$$Y_N = a^{(N)} \cdot v. \quad (1269)$$

By the coarea formula, its density u_N is

$$u_N(y) = \frac{2^{-N}}{\|a^{(N)}\|_2} \mathcal{H}^{N-1} \left\{ v \in [-1, 1]^N : a^{(N)} \cdot v = y \right\}. \quad (1270)$$

The Fourier-slice theorem reduces here to Fubini:

$$\int_{\mathbb{R}} u_N(y) e^{-ity} \, dy = \prod_{j=1}^N \text{sinc}(t/2^j). \quad (1271)$$

The first three sinc factors give an integrable majorant for all later finite products, so dominated Fourier inversion yields uniform convergence $u_N \rightarrow \text{up}$. Consequently

$$\text{up}(y) = \lim_{N \rightarrow \infty} \frac{2^{-N}}{\|a^{(N)}\|_2} \mathcal{H}^{N-1} \left\{ v \in [-1, 1]^N : a^{(N)} \cdot v = y \right\}. \quad (1272)$$

Equations (1267), (1272), and (1271) form a volume–section–product triangle: sublevel volumes give F , codimension-one sections give up , and Fourier slices give the finite sinc products converging to Φ .

11 Representation synthesis

The formulas above are best viewed as a network rather than a list. Exact dyadic arithmetic supplies Faber coefficients; differentiation turns them into Haar coefficients; Haar Parseval gives a positive energy series; Haar Fourier transforms give a new sinc-product series. Independently, binary digits reorganize the same probability law into binomial layers, while compact support places the Fourier image in a sampling space. Orthogonal polynomials and Bernstein polynomials then provide two different spectral and algebraic envelopes.

Table 29: New representation families and their characteristic data.

Representation	Formula type	Exact data	Convergence/control
Faber–Schauder	$F = x + \sum c_{m,k} \Lambda_{m,k}$	Rational dyadic midpoint defects	Uniform; sharp error $\sim 4^{-J}$.
Inverse splines	$G_J = (S_J F)^{-1}$, formula (1184)	Rational knots and slopes	Uniform; error $\leq \omega_G(4^{-J})$.
Rvachev tents	$\text{up}(y) = 1 - y + \sum c \Lambda(1 - y)$	Same rational table	Uniform on support.
Haar density	$F' = 1 + \sum 2^{m+1} c H$	Rational step coefficients	L^p , including essential L^∞ .
Haar Fourier	$\Phi = \text{sinc} + \sum c e^{i \cdot} \sin^2 / (it)$	Rational coefficients, elementary kernels	Normal on compact subsets of $\mathbb{C}$.
Energy	$A_2 = 1/4 + \sum 2^m \sum c^2$	Positive rational squares	Monotone, certified geometric tail.
Energy asymptotic	$E_m \sim A_2 \sum \beta_q 4^{-qm}$	Explicit rational β_q, γ_q	Poincaré all orders.
Triangular probability	$X = \sum 2^{-n} K_n$	Independent binomial layers	Almost sure and L^p .
Triangular product	$\Phi = \prod \cos(t/2^n)^{n-1}$	Integer exponents	Normal on compacta; local tail bound.
Coefficient CDF	Cumulative coefficients of $P_N(q)$	Positive integers	Bracket width $(N+1)2^{-N}$.
Odd cardinal	Rational-kernel sum over $s_m = \Phi((2m+1)\pi)$	Explicit sinc products	Locally uniform, superalgebraic truncation.
Sample sum rules	Zero jets of $B(z)$	2-adic multiplicities	Exact convergent identities.
Chebyshev–Gegenbauer	Bessel integrals of Φ	Existing moments/Legendre rationals	Absolute integrals; uniform polynomial series.
Bernstein	Exact-rational polynomials at dyadic degrees	Dyadic values of F	Uniform; explicit $1/n$ bound for F .
Kummer Fourier	Finite ${}_1F_1$ sums, then $n \rightarrow \infty$	Rational finite weights	Locally uniform entire convergence.
Cube geometry	Sublevel volumes and hyperplane sections	Dyadic normal vector	CDF and uniform-density limits.

Several hybrid formulas deserve emphasis:

$$\text{rational wavelets} + \text{elementary oscillation} \longrightarrow \text{entire sinc product}, \quad (1273)$$

$$\text{odd sinc-product samples} + \text{rational resolvents} \longrightarrow \text{entire sinc product}, \quad (1274)$$

$$\text{rational dyadic samples} + {}_1F_1 \longrightarrow \text{entire sinc product}, \quad (1275)$$

$$\text{rational Legendre coefficients} \longleftrightarrow \text{Bessel–sinc integrals}. \quad (1276)$$

The same function is therefore reconstructed from local curvature, stepwise martingale data, odd spectral samples, binomial coefficient arrays, orthogonal-polynomial periods, and weighted-polytope geometry.

12 Numerical and exact verification

12.1 Reproducibility design

The archive contains `fabius_multiresolution_experiments.py`, a standalone, fully commented Python program. Its arithmetic and analytic checks are intentionally separated.

- *Exact layer.* Python’s `Fraction` type evaluates (1150), every $F(a/2^{16})$, all Faber coefficients below level 16, every R_J , and every coefficient of P_N for $N \leq 16$. Log-concavity is tested with integer inequalities.
- *Independent analytic layer.* The reference values of F and up come from the period-two cosine series whose coefficients are evaluated by the sinc product. Haar, triangular, cardinal, Bessel, and Kummer formulas are checked against this independent product route.
- *Outputs.* The run writes human-readable and JSON results, the TeX tables included below, and four figures. No network access or hidden data files are used.

The exact level-16 table contains 65537 rational dyadic values and is constructed by the terminating highest-bit recursion in roughly linear time in the table size. The bundled run uses 90 decimal digits for mpmath operations.

12.2 Faber convergence

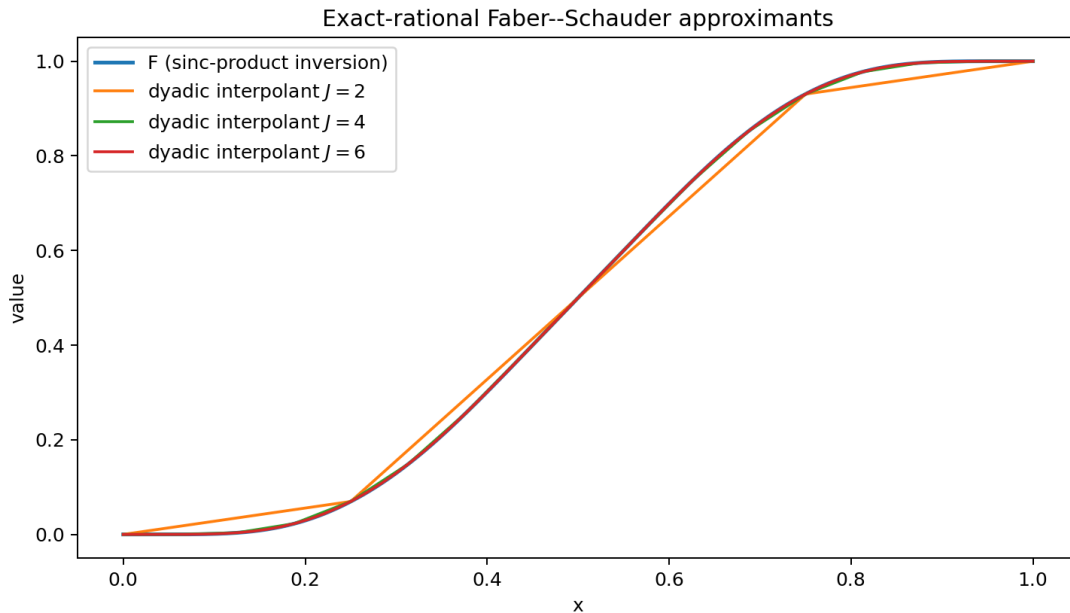


Figure 23: The exact-rational dyadic interpolants are the Faber partial sums. The level-6 curve is already visually indistinguishable from the independent sinc-product reconstruction on most of the interval.

The observed maximum errors satisfy the rigorous bound 4^{-J} and the scaled errors approach the

sharp constant one from (1169). At level $J = 10$, the maximum grid residual against the independent Fourier evaluator is approximately 9.54×10^{-7} , consistent with $4^{-10} = 9.5367 \times 10^{-7}$.

12.3 Energy partial sums and acceleration

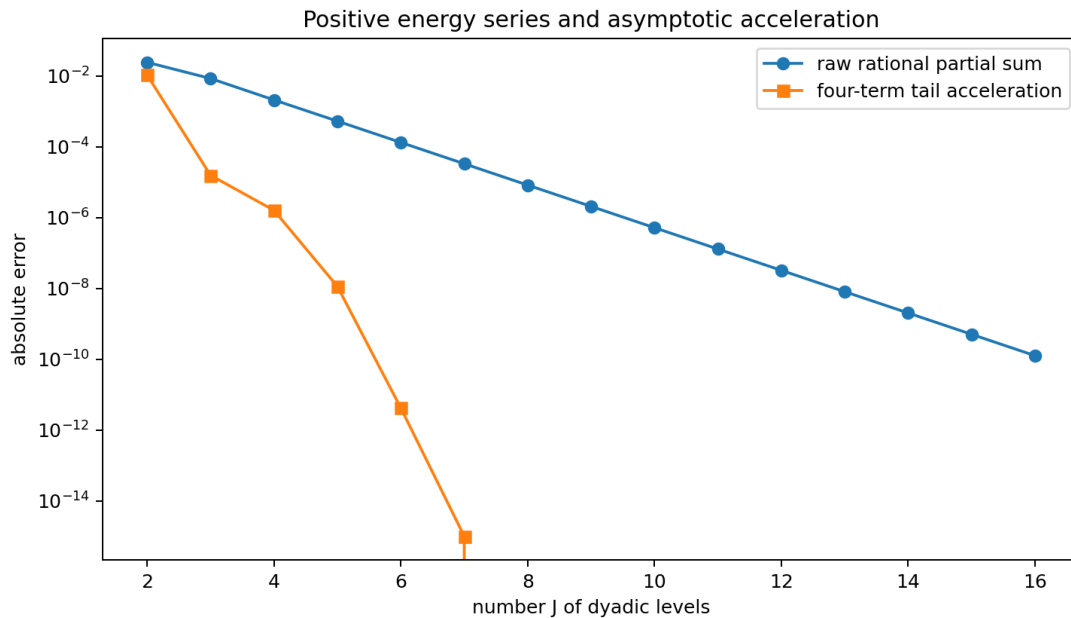


Figure 24: Raw positive rational partial sums converge geometrically. Using four terms of the all-orders tail expansion removes several additional powers of 4^{-J} . Values below machine precision relative to the numerical reference are omitted automatically from the logarithmic plot.

Table 30: Exact rational energy partial sums and accelerated values.

J	R_J (decimal)	accelerated	raw error
2	0.3804012345679012	0.414983164983165	0.024036
3	0.3960262345679012	0.4044523246650906	0.0084105
4	0.4023494663214687	0.4044383623771493	0.0020873
5	0.4039112434127894	0.4044367566681171	0.00052552
6	0.404305183886946	0.4044367679881609	0.00013158
8	0.4044285399647054	0.4044367679839853	8.228e-6
10	0.4044362537170872	0.4044367679839853	5.1427e-7
12	0.4044367358422428	0.4044367679839853	3.2142e-8
14	0.4044367659751261	0.4044367679839853	2.0089e-9
16	0.4044367678584316	0.4044367679839853	1.2555e-10

The decimal in (1217) is not advertised as a formally certified 60-digit interval: the all-orders theorem is asymptotic and the bundled run does not compute explicit constants in its remainder. It is, however, stable under changes of both level and acceleration order. For example, the five-term estimates from $J = 12, 13, 14, 15, 16$ successively stabilize through roughly 8, 12, 16, 20, and 21 additional digits beyond the raw partials. A rigorous high-precision enclosure can be obtained by combining a finite higher-order expansion with explicit derivative sup norms and an Euler–Maclaurin remainder.

Table 31: Chebyshev coefficients from cosine quadrature and the Bessel–sinc integral.

n	cosine-grid value	Bessel-product value	absolute residual
0	0.68197894345	0.681978961331	1.79e-08
2	-0.514948073175	-0.514948055353	1.78e-08
4	0.195935922808	0.195935940418	1.76e-08
6	-0.00235687932482	-0.0023568621795	1.71e-08
8	-0.028210810035	-0.028210793777	1.63e-08
10	0.0114279577063	0.0114279724231	1.47e-08
12	-0.00612048206979	-0.00612046982494	1.22e-08

Table 32: Exact tests of the triangular coefficient conjecture.

N	degree	palindromic	log-concave
4	11	yes	yes
6	57	yes	yes
8	247	yes	yes
10	1013	yes	yes
12	4083	yes	yes
14	16369	yes	yes
16	65519	yes	yes

12.4 Haar, triangular, and cardinal checks

Using levels $m < 14$, the Haar-sinc² formula (1193) has absolute residuals between 10^{-10} and 8×10^{-9} at the tested real and complex arguments. This is the expected unresolved tail size. The finite triangular product at $N = 36$ agrees with the sinc product to the precision predicted by (1226). With 40 positive odd-sample pairs, the absolute residuals are 1.2×10^{-14} to 3.9×10^{-13} at the six test points listed in `numerical_results.txt`. The explicit sum rules (1244) and (1245) can be checked from the same sample table.

Exact integer arithmetic confirms palindromicity and log-concavity for every $2 \leq N \leq 16$; selected degrees are shown in Table 32. The checks are exhaustive over every interior coefficient, not random spot tests.

12.5 Bessel and Bernstein checks

The cosine-grid and Bessel-product evaluations in Table 31 agree to about 2×10^{-8} . The residual is set by the finite cosine grid and finite Bessel integration cutoff, not by a symbolic approximation in the theorem.

For the rational Bernstein approximants to F , the observed uniform errors are:

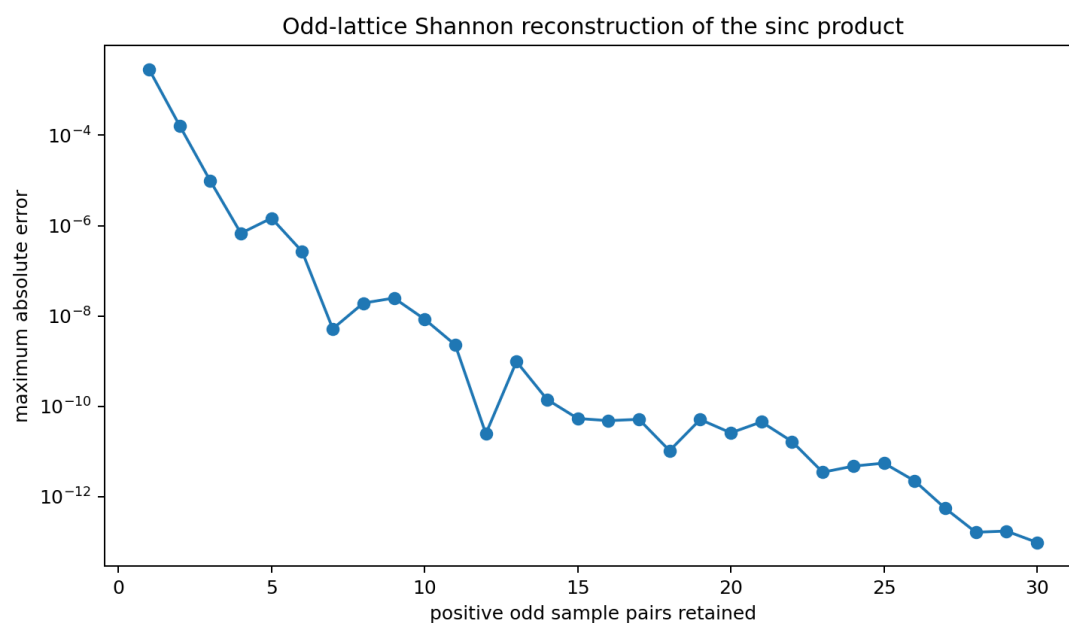


Figure 25: Maximum residual of the odd-sample cardinal formula at two real and one complex test point. The small nonmonotonic ripples reflect cancellation; the envelope decreases superalgebraically.

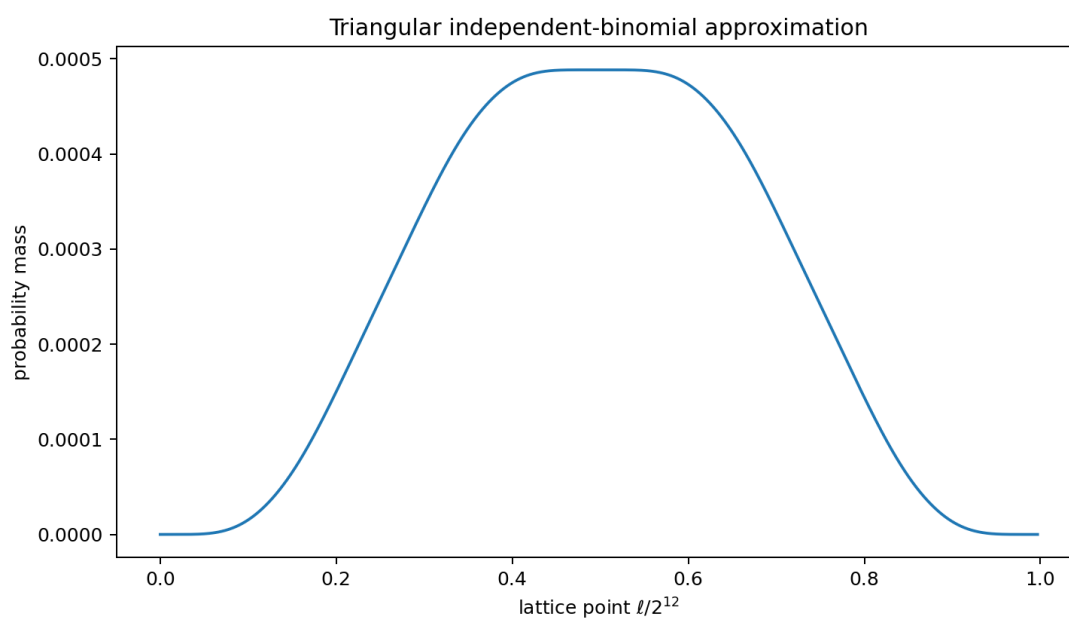


Figure 26: The positive mass distribution encoded by P_{12} . Its broad, nearly flat central region is a finite-layer effect quite different from a single binomial law.

n	observed error	rigorous bound $1/n$
8	7.1353×10^{-2}	1.25×10^{-1}
16	4.1193×10^{-2}	6.25×10^{-2}
32	2.2565×10^{-2}	3.125×10^{-2}
64	1.1906×10^{-2}	1.5625×10^{-2}
128	6.1361×10^{-3}	7.8125×10^{-3}
256	3.1214×10^{-3}	3.90625×10^{-3}

At degree 32 and $t = 2.7$, direct quadrature of the Bernstein bump and the finite Kummer sum (1264) differ by 4.5×10^{-17} .

13 Conjectures and frontier directions

13.1 Arithmetic nature of the energy constant

Conjecture 13.1 (Irrationality of the Fabius energy). The number

$$A_2 = \int_0^1 F(x)^2 dx$$

is irrational; plausibly it is transcendental.

The positive rational expansion (1198) is more arithmetically structured than the original sinc integral. A first serious attack should study the reduced denominators of R_J , their p -adic growth, and whether the tail asymptotic can be converted into rational-approximation bounds incompatible with algebraicity. The main obstacle is that an excellent asymptotic expansion by itself does not provide the denominator separation needed for an irrationality proof.

13.2 Endpoint phase tomography through rational wavelets

For a cell entirely in $(0, 1/4)$, the differential equation $F''(x) = 8F(4x)$ and (1162) give the exact scaled identity

$$-4^{m-1}c_{m,k} = \int_0^1 \min(s, 1-s) F(2^{2-m}(k+s)) ds. \quad (1277)$$

Thus every fixed- k coefficient ray is a positive triangular average of the endpoint function at a dyadically shrinking scale.

Conjecture 13.2 (Wavelet recovery of the Lambert phase). For each fixed $k \geq 0$, the logarithm of $-c_{m,k}$ has an all-orders expansion obtained by applying boundary Laplace analysis to (1277). Its leading exponential part is governed by the same lower-Lambert coordinate as F , and its bounded oscillatory part is a nonconstant translate of the repository's one-periodic endpoint correction. Distinct k determine distinct phase shifts but the same Fourier spectrum.

This would turn exact rational coefficient rays into a tomography device for the periodic endpoint phase. It is important not to replace the phase by its mean: the audited endpoint work proves that such averaging fails at the natural error scale.

13.3 The triangular coefficient problem

Conjecture 6.4 asks for a structural proof of log-concavity of a highly nonstandard sparse product. Promising approaches include:

- induction through interlacing inequalities stronger than ordinary log-concavity;
- a probabilistic proof using ultra-log-concavity under dyadically weighted Bernoulli convolution;
- total positivity of the shift-and-add operators that multiply by $(1 + q^{2^r})^a$ when the lower scales are already present;
- a stable-polynomial or Lorentzian homogenization in extra variables, followed by a carefully justified diagonal specialization.

A local central limit theorem with explicit lattice corrections for X_N^Δ would also quantify how the positive coefficient limit approaches the smooth Fabius density.

13.4 Odd samples as intrinsic spectral coordinates

The sampling formula uses all odd samples because the even samples are forced to vanish. This suggests a constrained Paley–Wiener space

$$\mathcal{V} = \{f \in \text{PW}_1 : f(2\pi n) = 0 \text{ for } n \neq 0, \text{ with the prescribed higher 2-adic jets}\}. \quad (1278)$$

Open problem 13.3 (de Branges/sampling model). Construct an intrinsic reproducing-kernel or de Branges description of the closure generated by the Fabius zero divisor, and determine whether the odd samples form a natural Riesz coordinate sequence after the higher even-zero constraints are imposed. Derive sharp weighted truncation inequalities for (1239) that exploit the exact zero multiplicities.

The identities in Theorem 7.2 are the first algebraic constraints such a model must reproduce.

13.5 Nonlinear wavelet energies

For $p = 2$, orthogonality collapses the energy to the positive diagonal sum (1198). For even $p > 2$, products of Haar functions are still finite on each dyadic tree but no longer diagonal.

Open problem 13.4 (Exact dyadic tree formulas for L^p energies). Find finite-level tree recurrences and convergent rational series for

$$A_p = \int_0^1 F(x)^p dx, \quad U_p = \int_{-1}^1 \text{up}(x)^p dx,$$

starting with $p = 3, 4$. Determine whether any rearrangement is positive and whether its level tails are again controlled by the derivative-norm spectrum.

The order-statistic formulas in the current corpus evaluate many A_p -type integrals probabilistically; the wavelet problem asks for a genuinely local, multiscale arithmetic decomposition.

13.6 Orthogonal-polynomial coefficient geometry

The Chebyshev coefficients in (1250) do not alternate forever in the simplest way. More nuanced questions remain.

Open problem 13.5 (Sign and zero laws). Study the sign changes, asymptotic phase, and zero distribution of $a_{2^r}^{(\lambda)}$ as functions of r and λ . Determine whether the Bessel integrals (1254) admit saddle descriptions tied to the sharp real-axis decay rays of Φ , and identify values of λ for which the coefficients have additional arithmetic structure.

The Legendre case is especially attractive because its coefficients are rational, making (1257) an exact bridge between orthogonal-polynomial arithmetic and oscillatory integral analysis.

13.7 Base- b and geometric-weight extensions

The repository already develops a base- b sinc-product and zero-spectrum theory. Every new construction above has a plausible geometric-weight analogue:

- b -adic Faber systems with exact node arithmetic;
- regrouping base- b digits along antidiagonals, producing multinomial rather than binomial layers;
- cardinal sampling after identifying the appropriate forced zero sublattice;
- Jacobi/Gegenbauer coefficients as Bessel transforms of the base- b product;
- Bernstein limits along degrees b^r .

The dyadic case is exceptional because binary digit regrouping and Haar multiresolution are perfectly aligned; identifying which conclusions survive for $b > 2$ should clarify which phenomena are genuinely dyadic.

13.8 Lean formalization roadmap

The new results are unusually formalization-friendly because their analytic core can be divided into finite exact layers and standard convergence lemmas. A practical order is:

1. Define $\Lambda_{m,k}$ and prove that the finite Faber sum is the dyadic linear interpolant. Import the existing exact dyadic evaluator to prove coefficient rationality.
2. Prove the C^2 interpolation bound and the sharp limit at $1/4$.
3. Differentiate finite interpolants, identify conditional averages of F' , and obtain L^2 convergence. Haar orthogonality then proves (1198) with minimal measure-theoretic overhead.
4. Formalize the coefficient kernel (1162), the uniform bound, and the first few energy-tail coefficients. The all-orders theorem can follow after an endpoint-flat Euler–Maclaurin library is in place.
5. Construct the triangular Bernoulli array on the existing infinite product probability space. Regroup nonnegative series and prove independence of antidiagonals.
6. Apply the existing Fourier product API to prove (1223); keep the finite polynomial and exact coefficient checks as executable definitions.
7. Reuse a Paley–Wiener sampling theorem from mathlib if available, or first formalize the rapidly decreasing special case needed for (1239). The sample-jet identities then reduce to differentiation of normally convergent rational series.
8. Add the beta–Kummer integral and Bernstein convergence as a separate approximation module; these require no endpoint asymptotics.

The energy identity is the best first target: it combines exact dyadic arithmetic, finite orthogonality, and a single L^2 limit, and it creates a new formally verified representation of a constant already central to the integral reports.

14 Conclusion

The Fabius–Rvachev system is often introduced through one striking identity: a compactly supported C^∞ function whose Fourier transform is an infinite product of sinc functions. The audited corpus already shows that this is only one face of a much larger structure involving uniform random series, Thue–Morse signs, q -binomial algebra, exact dyadic values, spectral multiplicities, Lambert– W endpoint phases, Bell polynomials, and a rich integral-transform calculus.

The new results add a multiresolution and sampling layer to that structure. Exact dyadic values are not merely isolated arithmetic curiosities: they are the coordinates of a rational Faber basis, and inversion of its finite interpolants gives exact-rational quantile splines. Their discrete derivatives are rational Haar coefficients of the smooth density. Orthogonality turns those coefficients into a positive rational expansion of the central sinc energy, and smooth self-similarity determines the complete tail expansion. Binary digits reorganize the same law into independent binomial layers; compact support reorganizes the Fourier image into odd cardinal samples; plane-wave identities reorganize it into Bessel transforms; Bernstein’s probabilistic polynomials reorganize it into finite Kummer sums; and finite-dimensional geometry reorganizes it into cube volumes and sections.

No one representation dominates all others. The sinc product is best for zeros and global frequency decay; the Faber series is best for exact local dyadic approximation; the Haar series is best for energy; the triangular layers are best for positive combinatorics; odd sampling is best for entire reconstruction from sparse spectral data; Bessel expansions connect to orthogonal approximation; Bernstein–Kummer limits provide finite entire approximants; and cube sections expose the underlying geometry. The most promising next step is to formalize the rational energy series and then use its exact denominators and all-order tail to investigate the arithmetic nature of A_2 .

A Remainder proof for the all-orders energy expansion

This appendix supplies a finite-order argument behind [Theorem 5.3](#). It avoids treating the nowhere-analytic function F as if its formal differential-operator series converged.

Let

$$\Delta = 2^{-m}, \quad h = \Delta/2, \quad x_k = (k + \tfrac{1}{2})\Delta. \quad (1279)$$

For an arbitrary fixed integer $M \geq 1$, Taylor’s theorem with a uniform remainder gives

$$c_{m,k} = - \sum_{r=1}^M \frac{h^{2r}}{(2r)!} F^{(2r)}(x_k) + \mathcal{O}(h^{2M+2}), \quad (1280)$$

uniformly in k . Squaring this identity and retaining terms through any prescribed power of Δ is therefore legitimate.

The level energy may be written as

$$E_m = \Delta^{-2} \left[\Delta \sum_{k=0}^{\Delta^{-1}-1} c_{m,k}^2 \right]. \quad (1281)$$

Every coefficient in the finite expansion of $c_{m,k}^2$ is a product $F^{(a)} F^{(b)}$ with $a, b \geq 2$. All derivatives of F of positive order vanish at 0 and 1. Hence each such product and all of its derivatives vanish at

both endpoints. The midpoint Euler–Maclaurin formula therefore has no boundary correction of any algebraic order: for every fixed smooth product G of this form and every $L \geq 1$,

$$\Delta \sum_{k=0}^{\Delta^{-1}-1} G(x_k) = \int_0^1 G(x) dx + \mathcal{O}_L(\Delta^L). \quad (1282)$$

Combining (1280)–(1282) shows that the Poincaré coefficients of E_m are the same as those of

$$\Delta^{-2} \int_0^1 [(\cosh(hD) - 1)F(x)]^2 dx, \quad (1283)$$

where the right side is interpreted only as a formal device for extracting any finite number of Taylor coefficients.

Since $\cosh(hD) - 1$ contains only even positive powers of D , repeated integration by parts has no boundary terms. Thus, coefficientwise,

$$\int_0^1 [(\cosh(hD) - 1)F]^2 dx = \int_0^1 F(\cosh(hD) - 1)^2 F dx \quad (1284)$$

$$\sim \sum_{r=2}^{\infty} \frac{2^{2r-1} - 2}{(2r)!} h^{2r} (-1)^r I_r. \quad (1285)$$

Now set $r = q+1$, use $h = \Delta/2$, and substitute $I_{q+1} = 2^{(q+1)(q+2)} A_2$. The coefficient of $\Delta^{2q} = 4^{-qm}$ is

$$(-1)^{q+1} \frac{2^{2q+1} - 2}{(2q+2)!} \frac{2^{(q+1)(q+2)}}{2^{2q+2}} A_2 = (-1)^{q+1} \frac{2^{q(q+1)+1}(2^{2q} - 1)}{(2q+2)!} A_2, \quad (1286)$$

which is $\beta_q A_2$.

More explicitly, for every fixed $Q \geq 1$ the preceding finite-order argument gives

$$E_m = A_2 \sum_{q=1}^Q \beta_q 4^{-qm} + \mathcal{O}_Q(4^{-(Q+1)m}). \quad (1287)$$

Summing the remainder over $m \geq J$ proves the finite-order version of (1211) and the acceleration estimate (1216).

B Exact dyadic evaluation used by the experiments

The experiments do not solve a discretized differential equation. They use the repository's exact arithmetic in the following terminating form.

Algorithm B.1 (Highest-bit evaluator). Assume the values $V_n = F(2^{-n})$ have been computed from (1150). To evaluate $F(a/2^e)$ with $0 < a < 2^e$, write

$$a = 2^b + q, \quad 0 \leq q < 2^b, \quad r = e - b, \quad y = q/2^e. \quad (1288)$$

Then

$$F(a/2^e) = \sum_{j=0}^r \frac{2^{j(j+1)/2} V_{r-j}}{j!} y^j - F(q/2^e). \quad (1289)$$

Because $q < a$, a left-to-right table of all numerators memoizes every subproblem and terminates after clearing the set bits of a .

For a fixed exponent e , the algorithm computes all $2^e + 1$ values using $\mathcal{O}(e2^e)$ rational operations. A single level-16 table then supplies all coarser Faber coefficients by exact subsampling.

C Compact formula compendium

For convenience, the principal formulas are collected here without proofs. The status column refers to the pinned corpus boundary, not to absolute historical priority.

Table 33: Formula compendium.

Object	Formula	Status
F	$F(x) = \mathbb{P}(\sum_{j \geq 1} 2^{-j} U_j \leq x)$	baseline
up	$\text{up}(y) = *_{j \geq 1} 2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]}(y)$	baseline
Φ	$\Phi(t) = \prod_{j \geq 1} \text{sinc}(t/2^j)$	baseline
Φ	$\Phi(t) = \prod_{n \geq 1} (1 - t^2/(4\pi^2 n^2))^{1+\nu_2(n)}$	baseline
F_N	$\frac{2^{N(N+1)/2}}{N!} \sum_{k < 2^N} (-1)^{\text{wt}_2(k)} (x - k/2^N)_+^N$	baseline
F	$x + \sum_{m,k} c_{m,k} \Lambda_{m,k}(x)$	new
G	$\lim_J [s_{J,0} y + \sum_k (s_{J,k} - s_{J,k-1})(y - y_{J,k})_+]$	new
up	$1 - y + \sum_{m,k} c_{m,k} \Lambda_{m,k}(1 - y)$	new
F'	$1 + \sum_{m,k} 2^{m+1} c_{m,k} H_{m,k}$	new
Φ	$\text{sinc } t + \sum_{m,k} \frac{2^m c_{m,k}}{it} e^{it(1-2k/2^m)} (1 - e^{-it/2^m})^2$	new
A_2	$\frac{1}{4} + \sum_{m \geq 0} 2^m \sum_k c_{m,k}^2$	new
E_m	$E_m \sim A_2 \sum_{q \geq 1} \beta_q 4^{-qm}$	new
X	$\sum_{n \geq 2} 2^{-n} K_n, K_n \sim \text{Bin}(n-1, 1/2)$ independent	new
Φ	$\prod_{n \geq 2} \cos(t/2^n)^{n-1}$	new
F	$\lim_N 2^{-N(N-1)/2} \sum_{\ell \leq 2^N} x[q^\ell] P_N(q)$	new
Φ	$\frac{\sin z}{z} - 2z \sin z \sum_{m \geq 0} \frac{s_m}{z^2 - (2m+1)^2 \pi^2}$	new
a_{2r}	$\frac{2(-1)^r}{\pi} \int_0^\infty J_{2r}(t) \Phi(t) dt$	new
$A(t)$	$\lim_{r \rightarrow \infty} \frac{1}{2^r + 1} \sum_k F(k/2^r) {}_1F_1(k+1; 2^r+2; -it)$	new
Φ	Dyadic Bernstein–Kummer limit (1265)	new
F	Weighted-cube sublevel-volume limit (1267)	new pack-aging
up	Weighted-cube hyperplane-section limit (1272)	new pack-aging

D Corpus and reproducibility files

The accompanying archive contains:

`Fabius_Rvachev_Multiresolution_Representations.tex`

The complete source of this report.

`Fabius_Rvachev_Multiresolution_Representations.pdf`

The compiled report.

`fabius_multiresolution_experiments.py`

Exact and numerical verification code with detailed comments.

`numerical_results.txt` **and** `numerical_results.json`

Human-readable and machine-readable output of the bundled run.

`generated_tables.tex`

The tables included by [Appendix 12](#).

`figures/`

Four plots generated by the script.

`corpus_manifest_current.txt`

The 92-path recursive TeX manifest, the five additions after the baseline snapshot, and the modified-path comparison ledger.

`README.md`, `requirements.txt`, **and** `SHA256SUMS.txt`

Reproduction instructions, pinned Python dependencies, and checksums.

The command used for the bundled numerical run is

```
python fabius_multiresolution_experiments.py \
    --max-exact-level 16 --precision 90
```

The program was executed with Python 3, NumPy, SciPy, mpmath, and Matplotlib. It does not access the network.

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Postscript: the Faber–Schauder and tent layers, formal

Two elementary layers underpinning the hat-function pictures of this volume are now kernel-verified.

Faber–Schauder midpoint defects. `FaberSchauderDefect.lean` defines, for a bounded Fabius solution F , the second difference across the level- m dyadic cell,

$$c_{m,k} = F\left(\frac{2k+1}{2^{m+1}}\right) - \frac{1}{2} \left(F\left(\frac{k}{2^m}\right) + F\left(\frac{k+1}{2^m}\right) \right)$$

(`Fabius.faberSchauderDefect`), and proves that it is *antisymmetric* under the dyadic reflection, $c_{m,2^m-1-k} = -c_{m,k}$ (`Fabius.faberSchauderDefect_reflect`). The sign is forced: $x \mapsto 1-x$ carries the strictly convex first half of F onto the strictly concave second half, so it reverses every second difference. Accordingly the defect is strictly negative on the first half (`Fabius.faberSchauderDefect_neg_succ`, `Fabius.faberSchauderDefect_neg`) and strictly positive on the second (`Fabius.faberSchauderDefect_pos_succ`), with the exact values $c_{0,0} = 0$, $c_{1,0} = -13/72$, $c_{2,0} = c_{2,1} = -1/32$ recorded as theorems.

Tent moments. `TentMoment.lean` supplies the exact polynomial moments of a dyadic tent. The symmetric single-power integral obeys a clean parity dichotomy,

$$\int_{-h}^h \left(1 - \frac{|u|}{h}\right) u^n du = \begin{cases} \frac{2h^{n+1}}{(n+1)(n+2)}, & n \text{ even,} \\ 0, & n \text{ odd,} \end{cases}$$

(`Fabius.integral_tentProfile_mul_pow`), and the shifted moment of degree p about a centre c is the corresponding binomial sum (`Fabius.integral_tentProfile_mul_pow_shift`, and `Fabius.integral_tent_mul_pow` for the truncated tent `Fabius.tent` itself). Neither the corpus nor Mathlib previously carried any tent or B-spline moment API.